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Han et al.

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(45) **Date of Patent:** **Sep. 16, 2025**

(54) **NUCLEOBASE EDITORS AND METHODS OF USE THEREOF**

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Powell, OH (US)

(73) Assignee: **Ohio State Innovation Foundation,**
Columbus, OH (US)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 629 days.

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(22) PCT Filed: **Feb. 12, 2021**

(86) PCT No.: **PCT/US2021/017868**

§ 371 (c)(1),

(2) Date: **Aug. 12, 2022**

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PCT Pub. Date: **Aug. 19, 2021**

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Related U.S. Application Data

(60) Provisional application No. 62/976,590, filed on Feb.
14, 2020.

(51) **Int. Cl.**

C12N 15/00 (2006.01)

C12N 9/22 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **C12N 15/102** (2013.01); **C12N 9/22**
(2013.01); **C12N 9/78** (2013.01); **C12N 15/11**
(2013.01);

(Continued)

(58) **Field of Classification Search**

None

See application file for complete search history.

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Primary Examiner — Aaron A Priest

(74) *Attorney, Agent, or Firm* — Meunier Carlin &
Curfman LLC

(57) **ABSTRACT**

The present disclosure relates to nucleobase editors and
methods of use thereof. Disclosed herein are fusion proteins,
systems, and compositions for editing disease-associated
mutations and methods of use thereof. In some aspects,
disclosed herein is a fusion protein comprising a Cas9
nickase and a nucleotide deaminase, wherein the Cas9
nickase comprises a first amino acid substitution at a posi-
tion selected from the group consisting of 262, 324, 409,
480, 543, 694, and 1219 when compared to SEQ ID NO: 11,
and wherein the Cas9 nickase comprises a second amino
acid substitution at a position selected from the group
consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337
when compared to SEQ ID NO: 11.

15 Claims, 63 Drawing Sheets

Specification includes a Sequence Listing.

ABE7.10

WT TadA

TadA*

miniABE

TadA*

miniABE(V82G)

TadA*(V82G)

miniABE(A56G)

TadA*(A56G)

miniABE(GG)

TadA*(GG)

- (51) **Int. Cl.** 2022/0136012 A1* 5/2022 Gaudelli C12N 9/78
C12N 9/78 (2006.01) 435/462
C12N 15/10 (2006.01) 2022/0290121 A1* 9/2022 Joung C12N 9/78
C12N 15/11 (2006.01) 2022/0401530 A1* 12/2022 Bryson C12N 9/22
C12N 15/86 (2006.01)

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- (52) **U.S. Cl.** WO 2018/218166 A1 11/2018
CPC **C12N 15/86** (2013.01); **C12N 2310/20** WO 2019/092042 A1 5/2019
(2017.05); **C12N 2750/14143** (2013.01); **C12N** WO 2019/120283 A1 6/2019
2800/40 (2013.01); **C12Y 305/04002** WO 2019/139645 A2 7/2019
(2013.01); **C12Y 305/04005** (2013.01) WO 2019/168953 A1 9/2019
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mdx^{4cv}: GAA **CAG** CAG CTG CTG CAG CAG AAC AGG AGA TAA CAG
E Q Q L Q Q N R R **X** Q
gRNA: gGTT**a**TCTCCTGTTCTGCAGC **TGT**

FIG. 1A

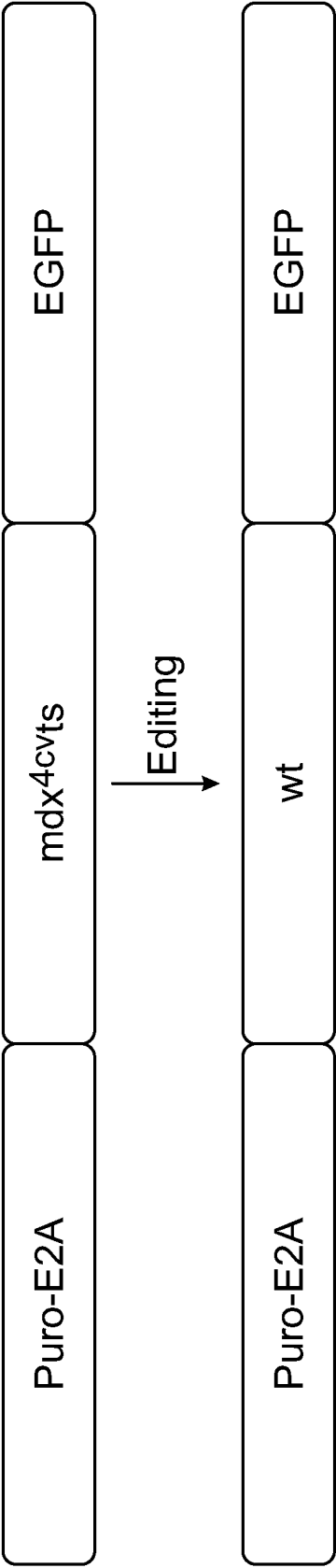


FIG. 1B

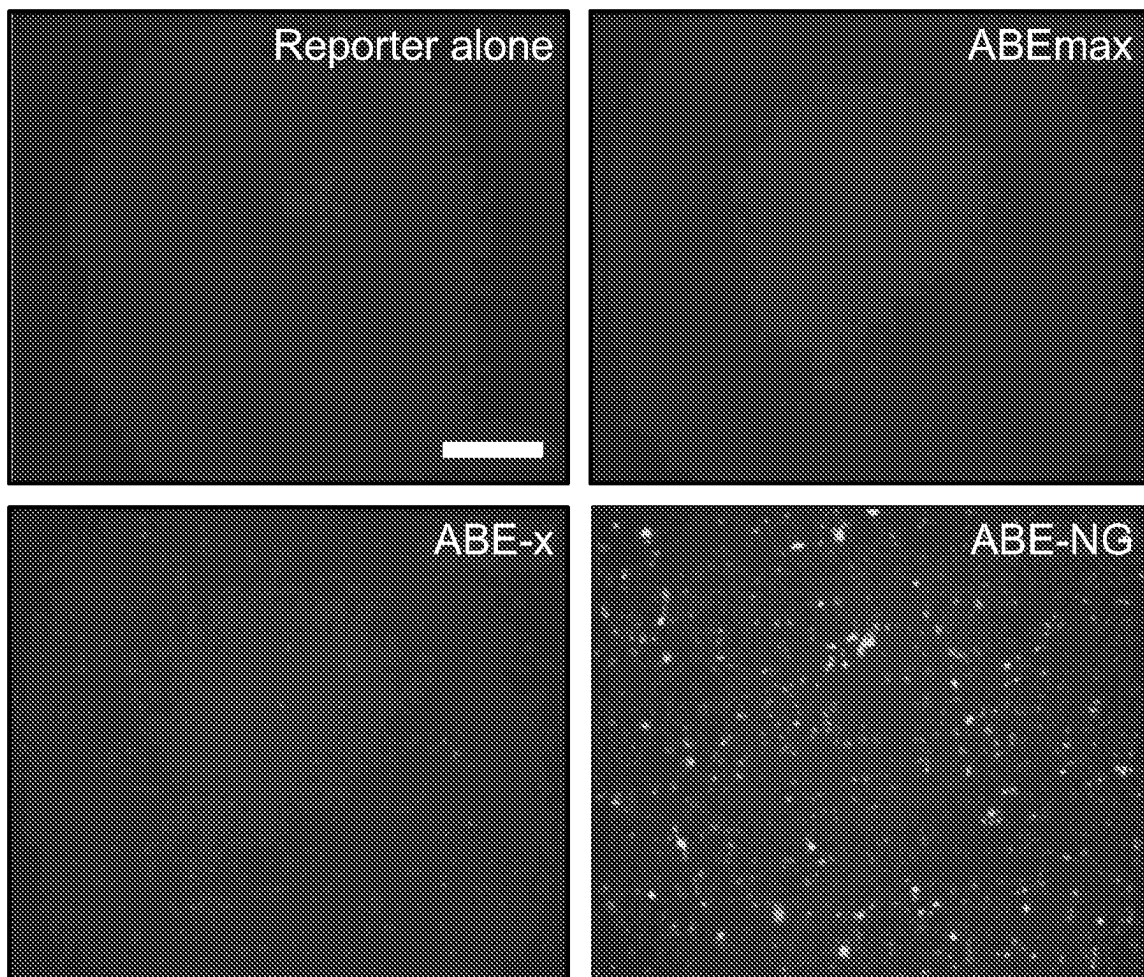


FIG. 1C

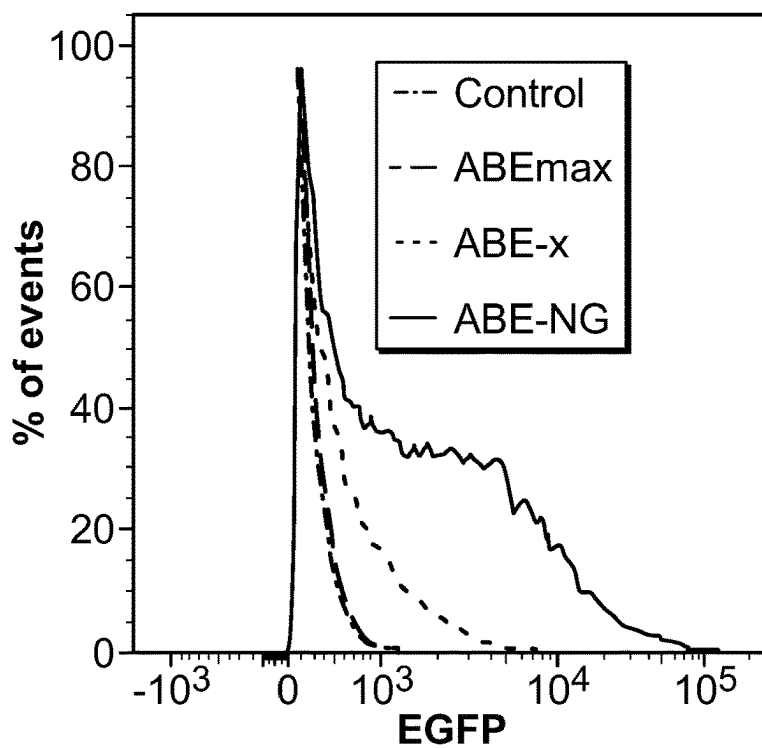


FIG. 1D

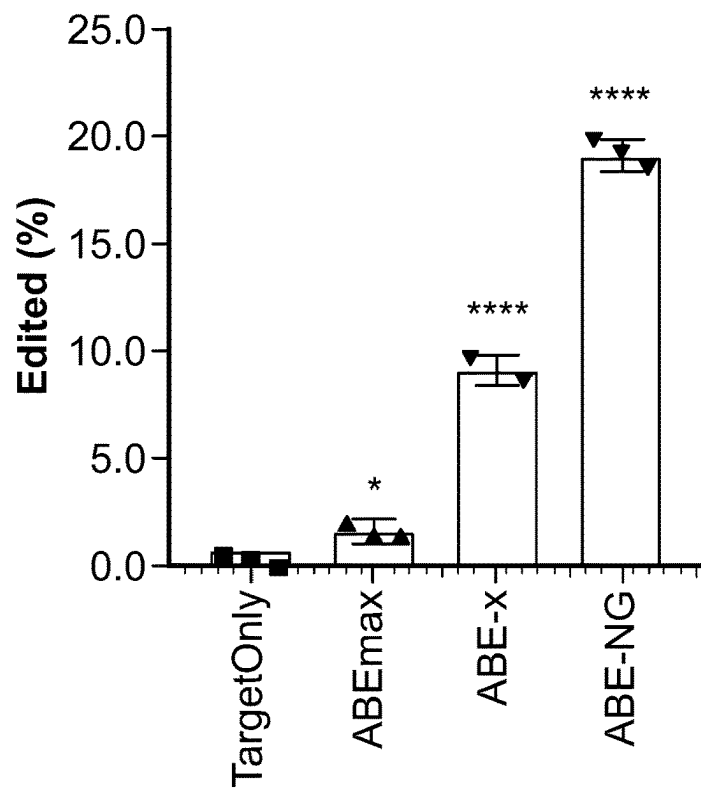
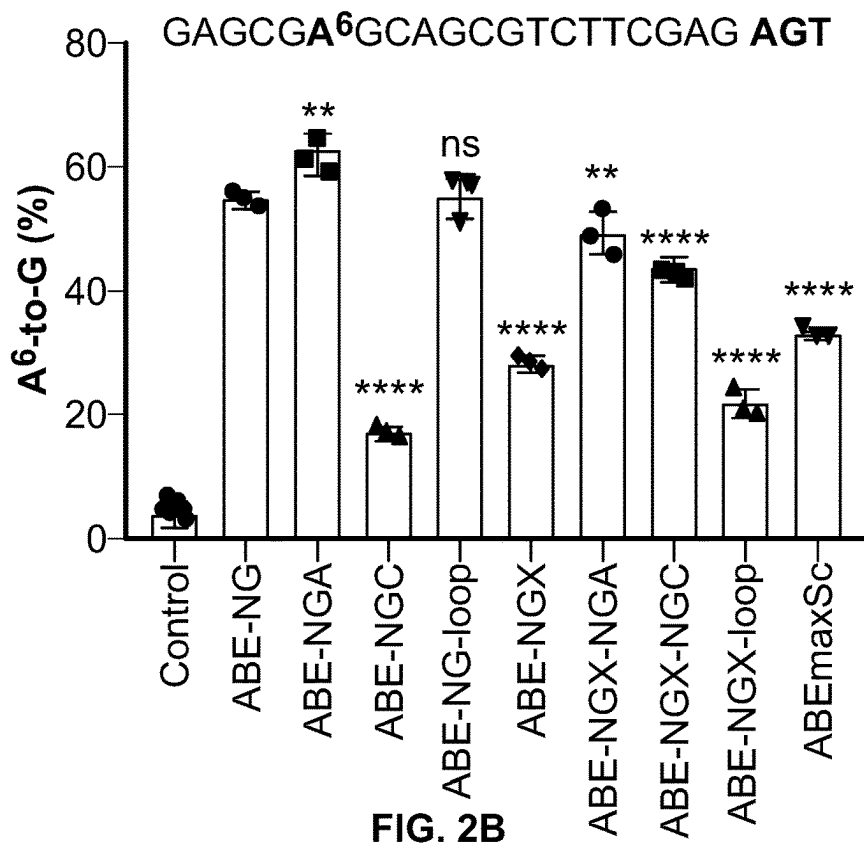
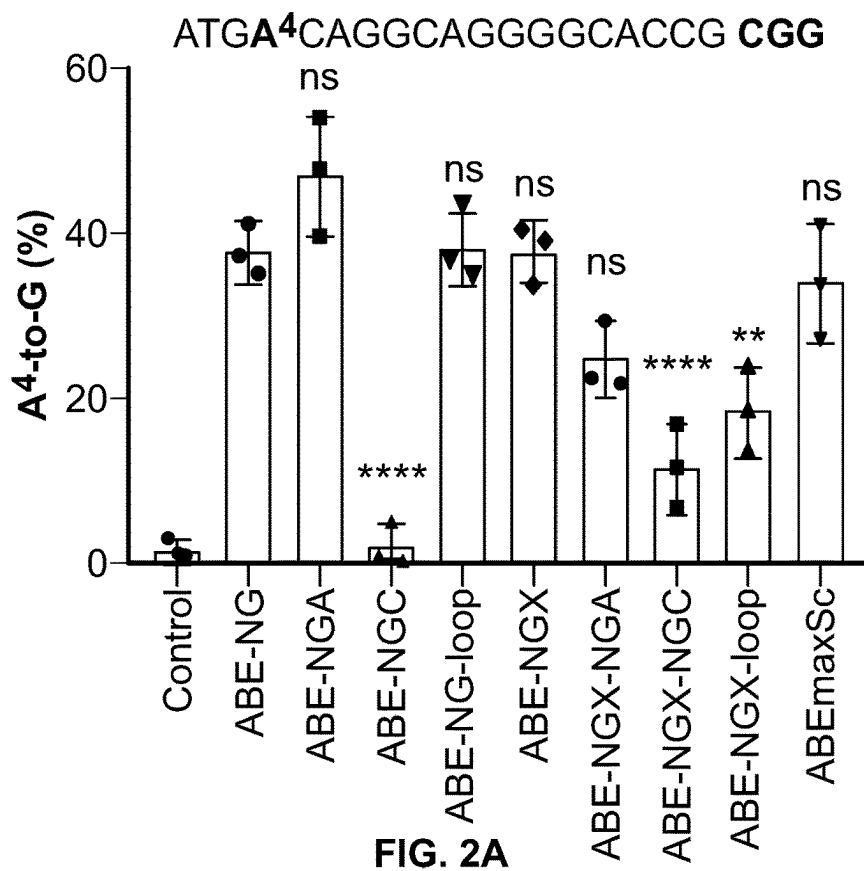
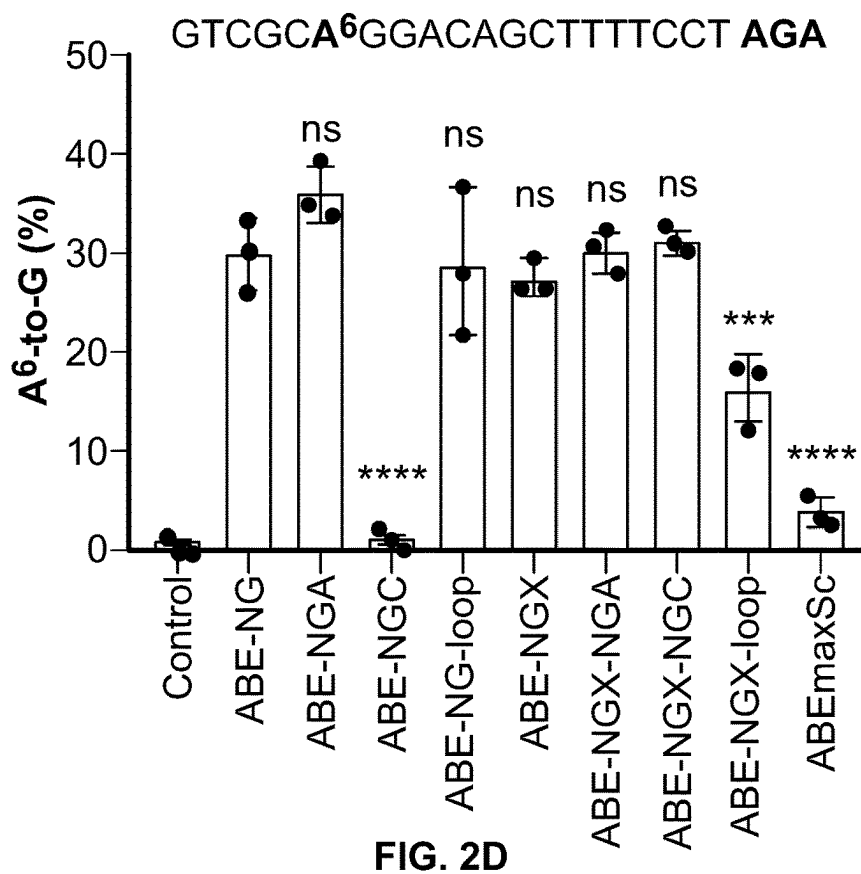
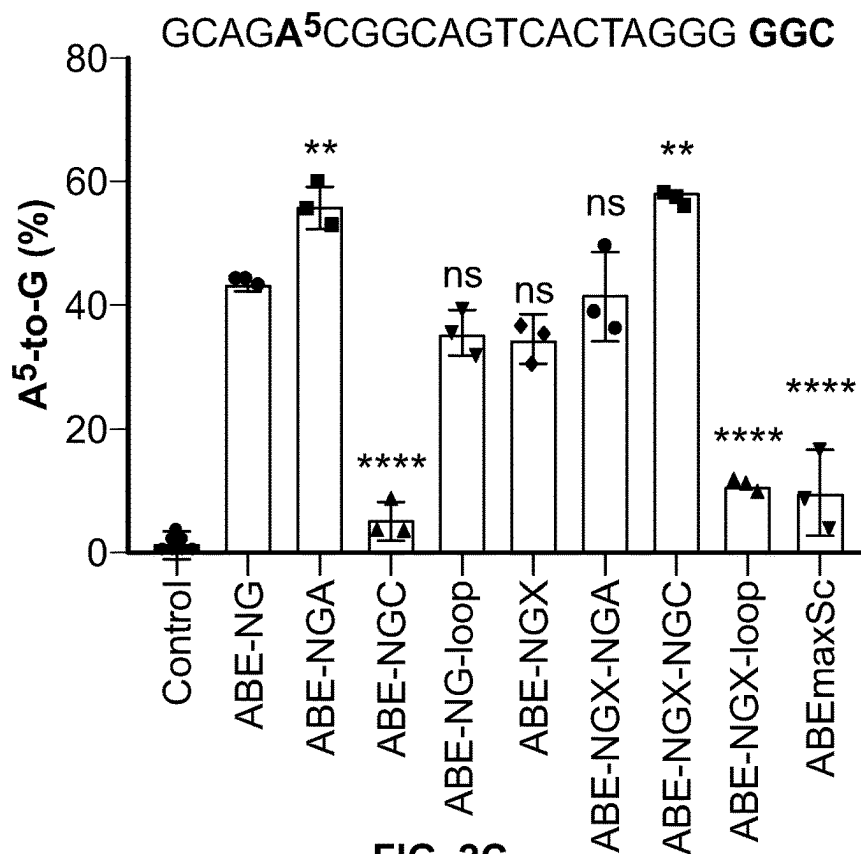


FIG. 1E





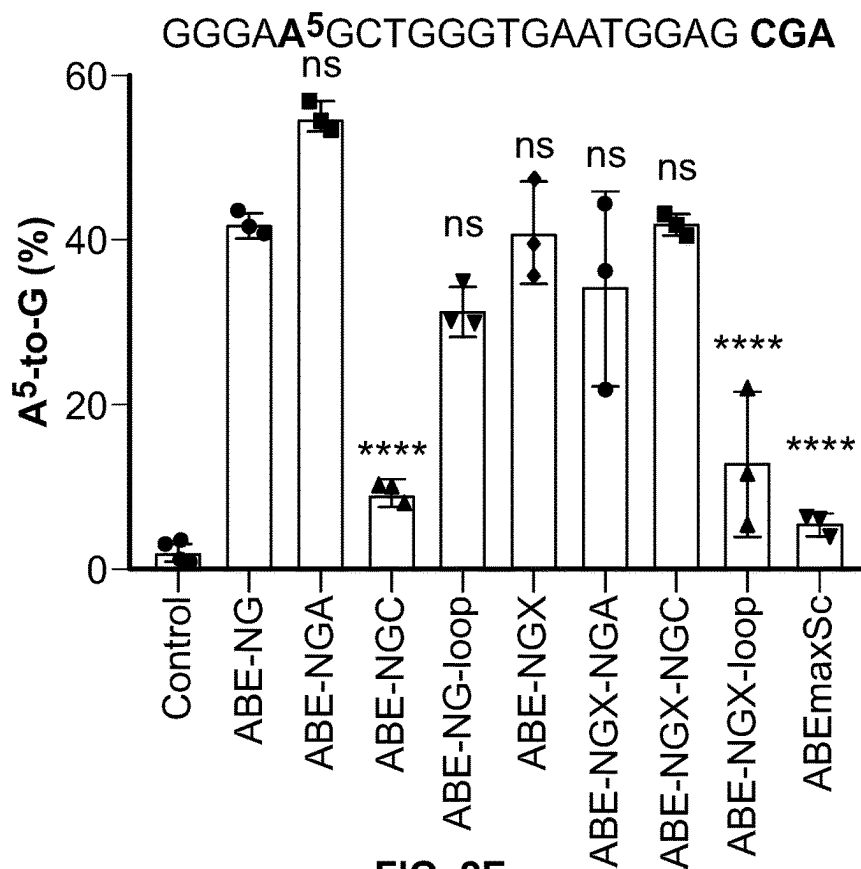


FIG. 2E

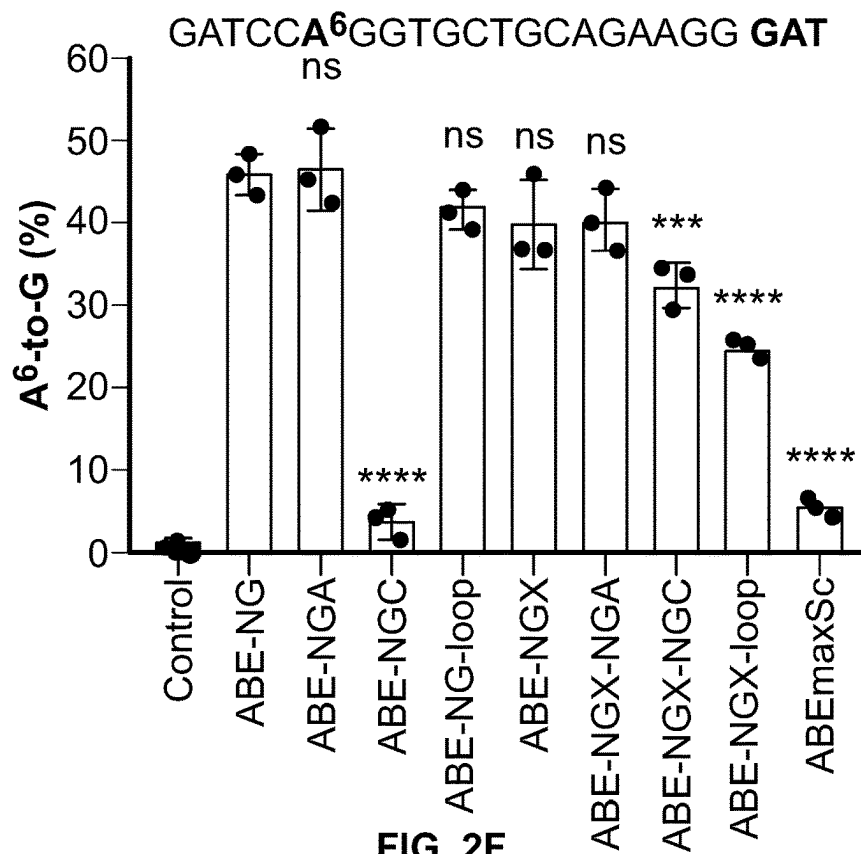


FIG. 2F

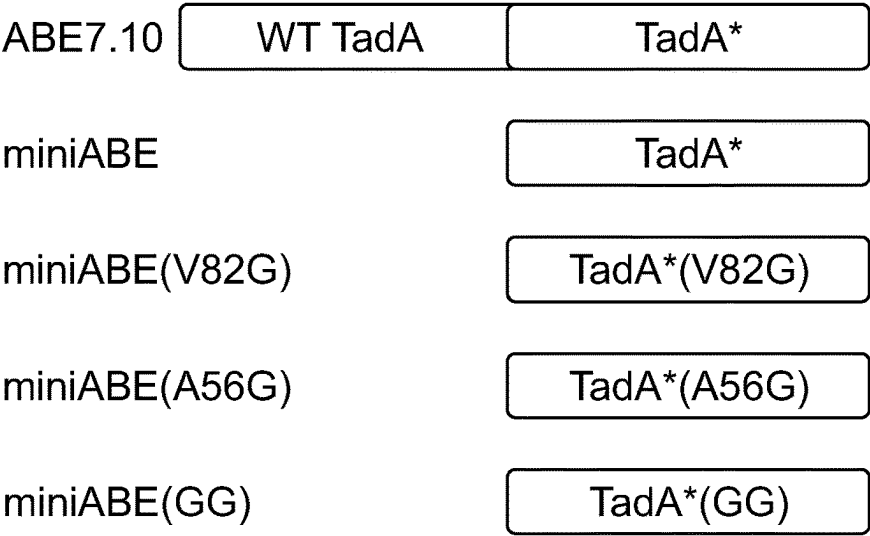


FIG. 3A

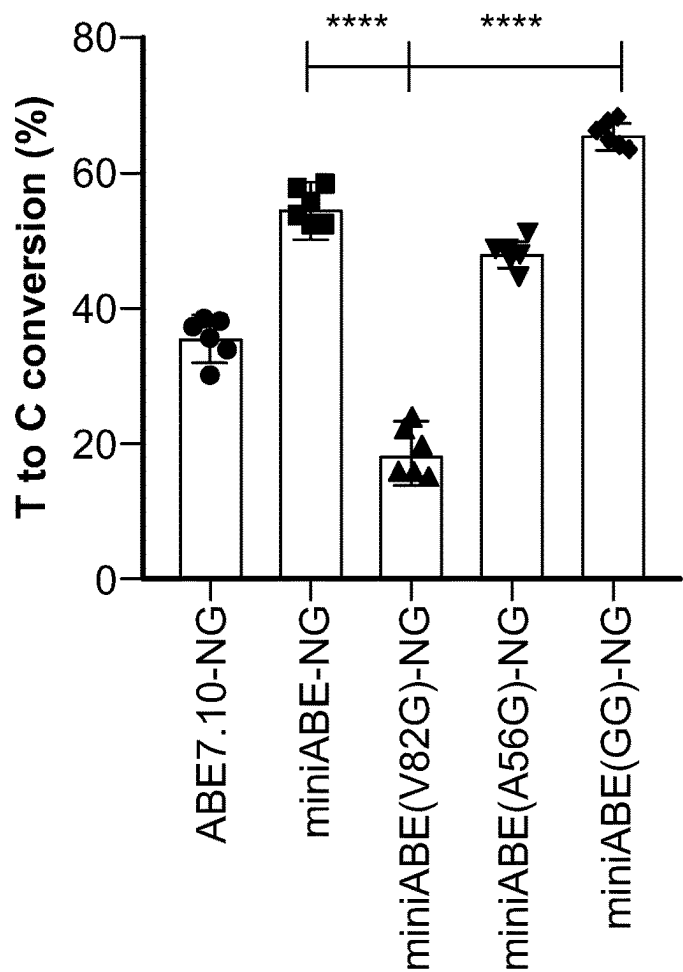


FIG. 3B

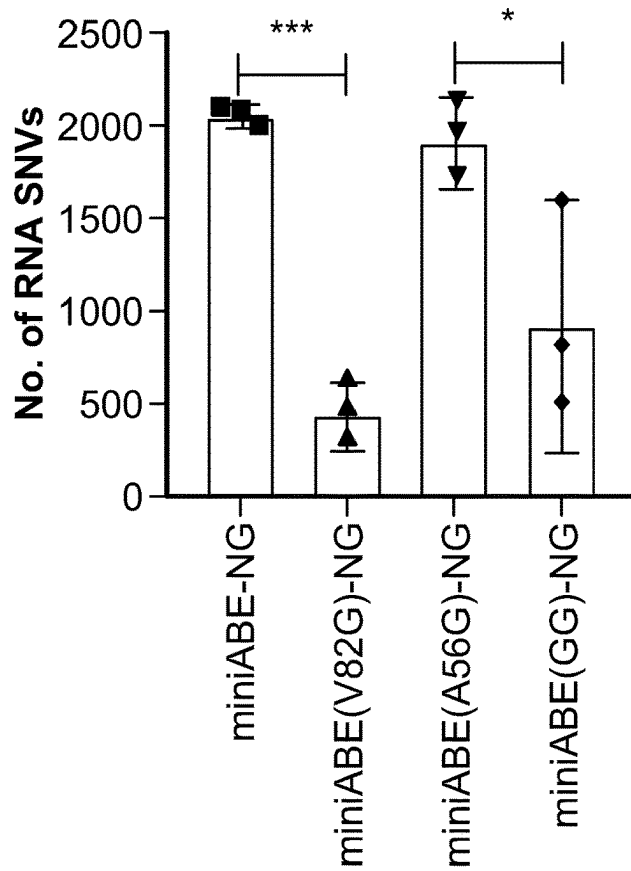


FIG. 3C

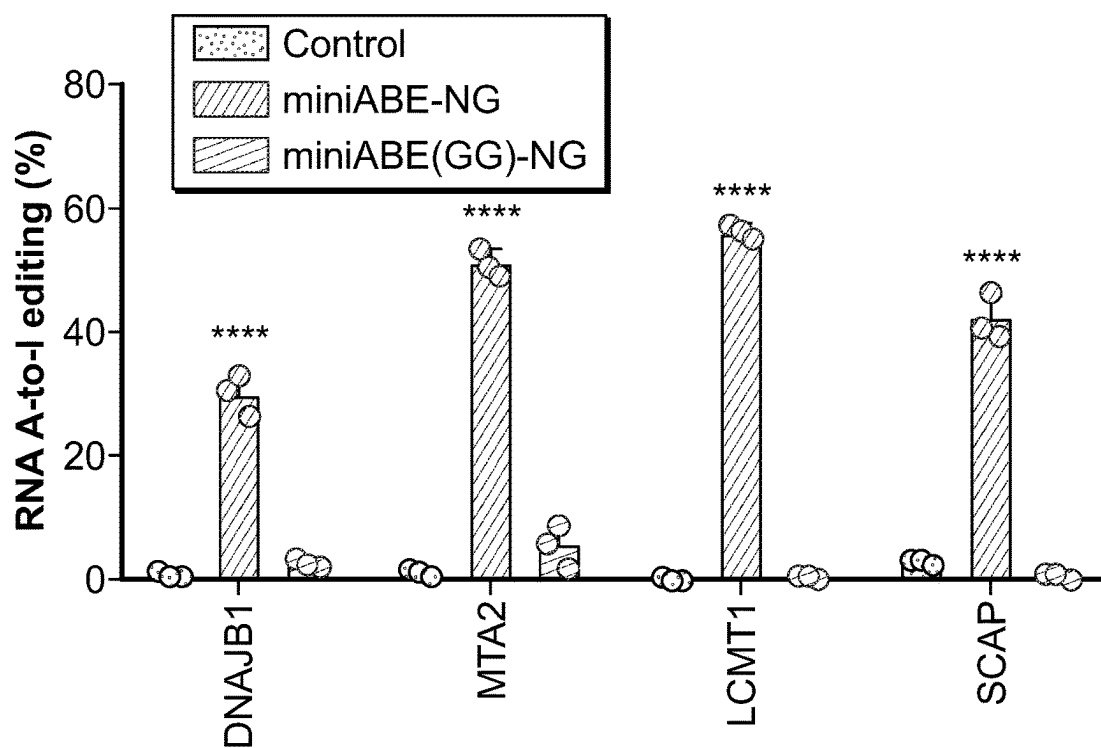


FIG. 3D

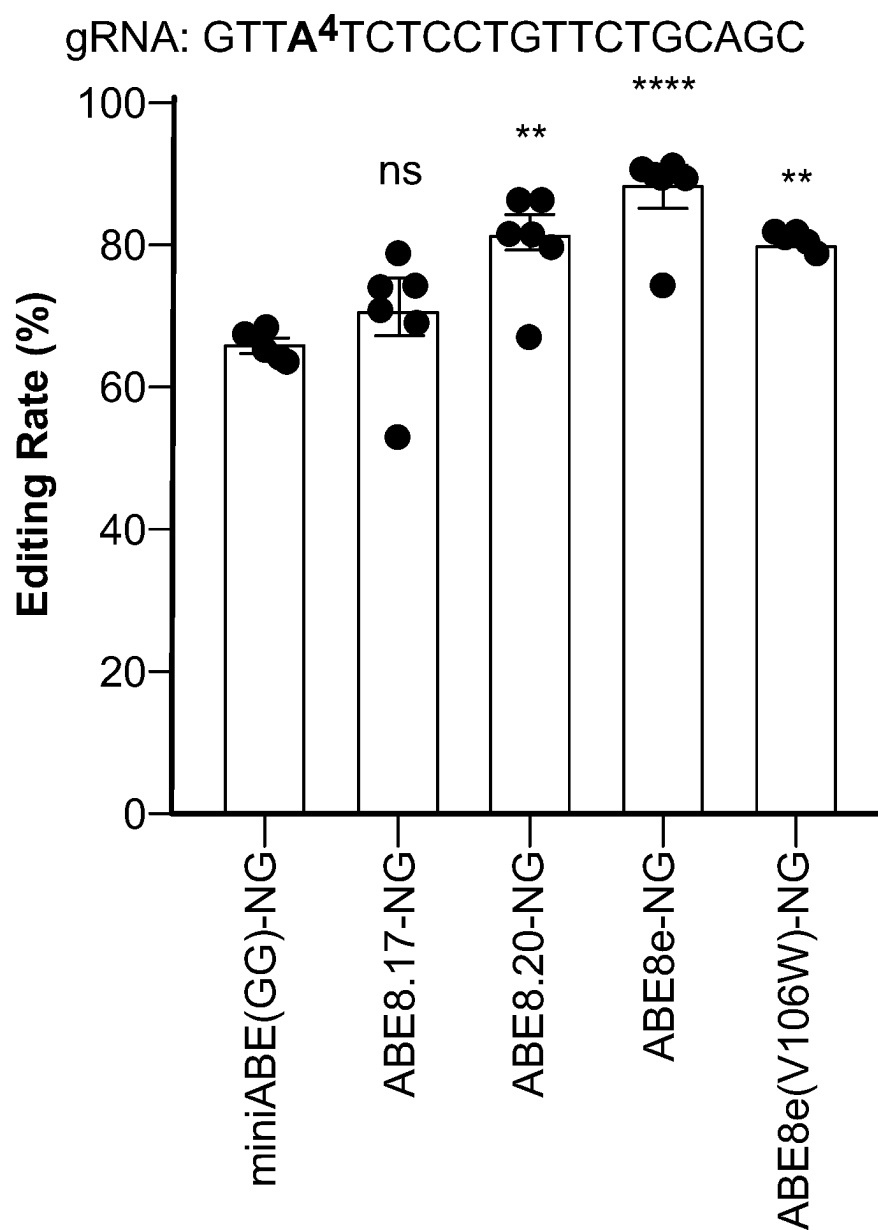


FIG. 4A

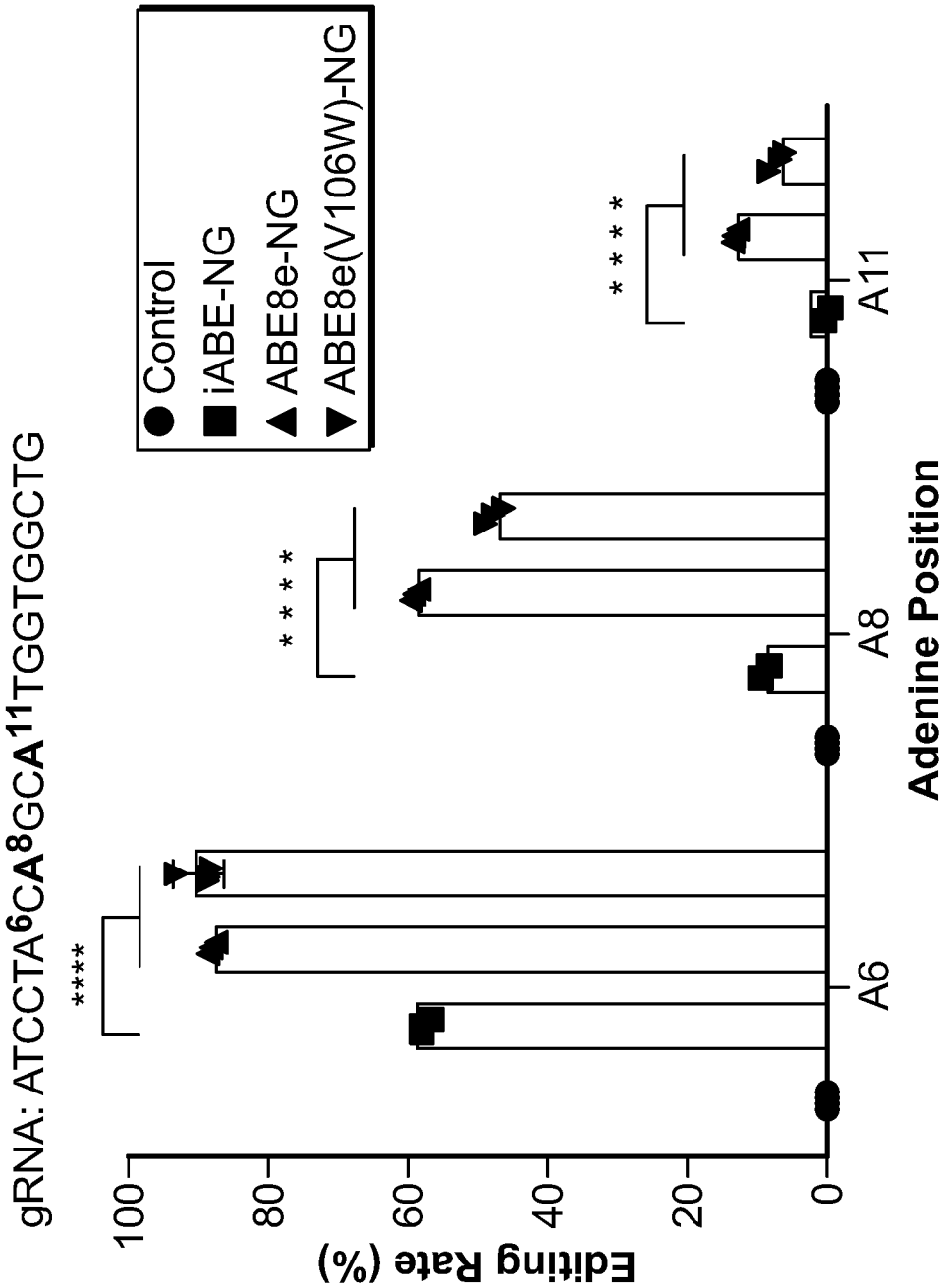


FIG. 4B

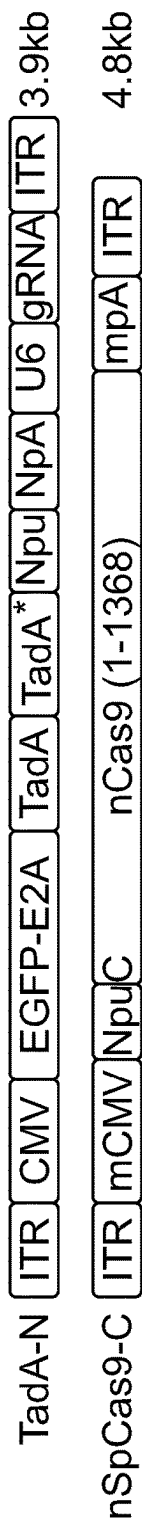


FIG. 5A

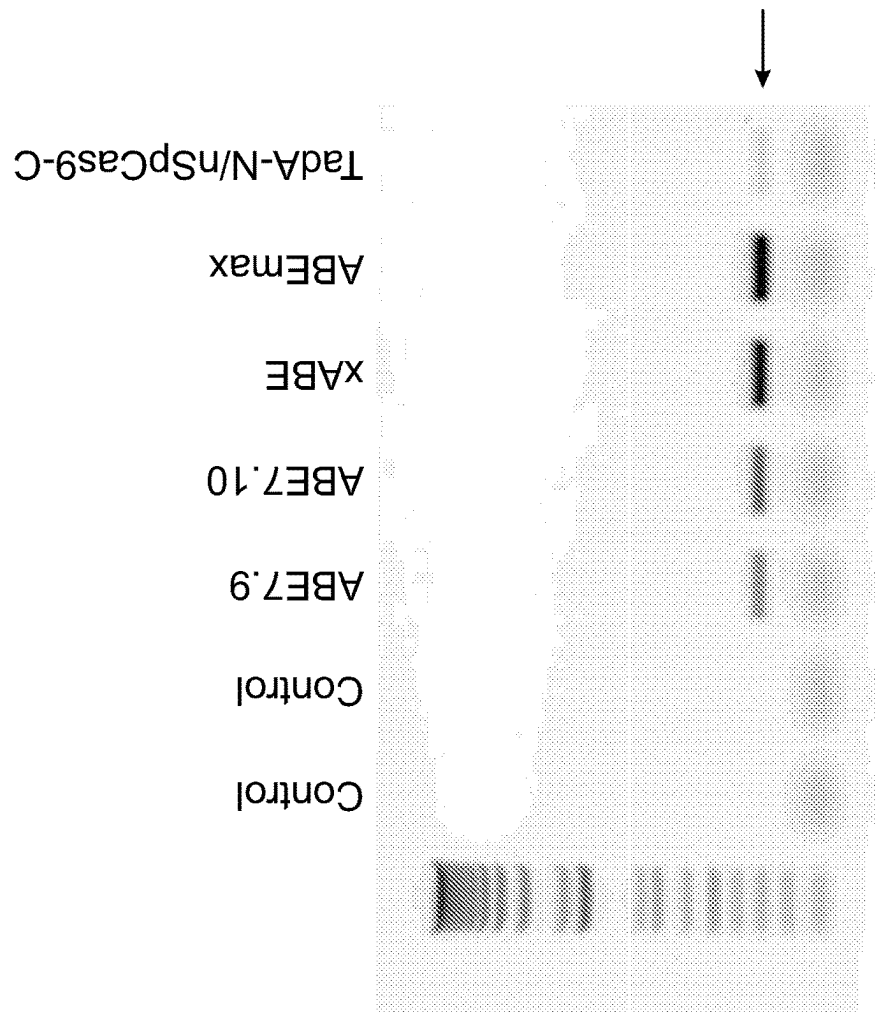


FIG. 5B

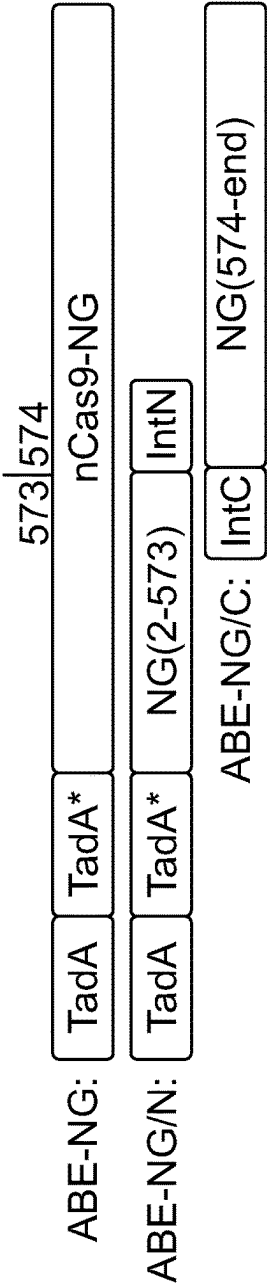


FIG. 6A

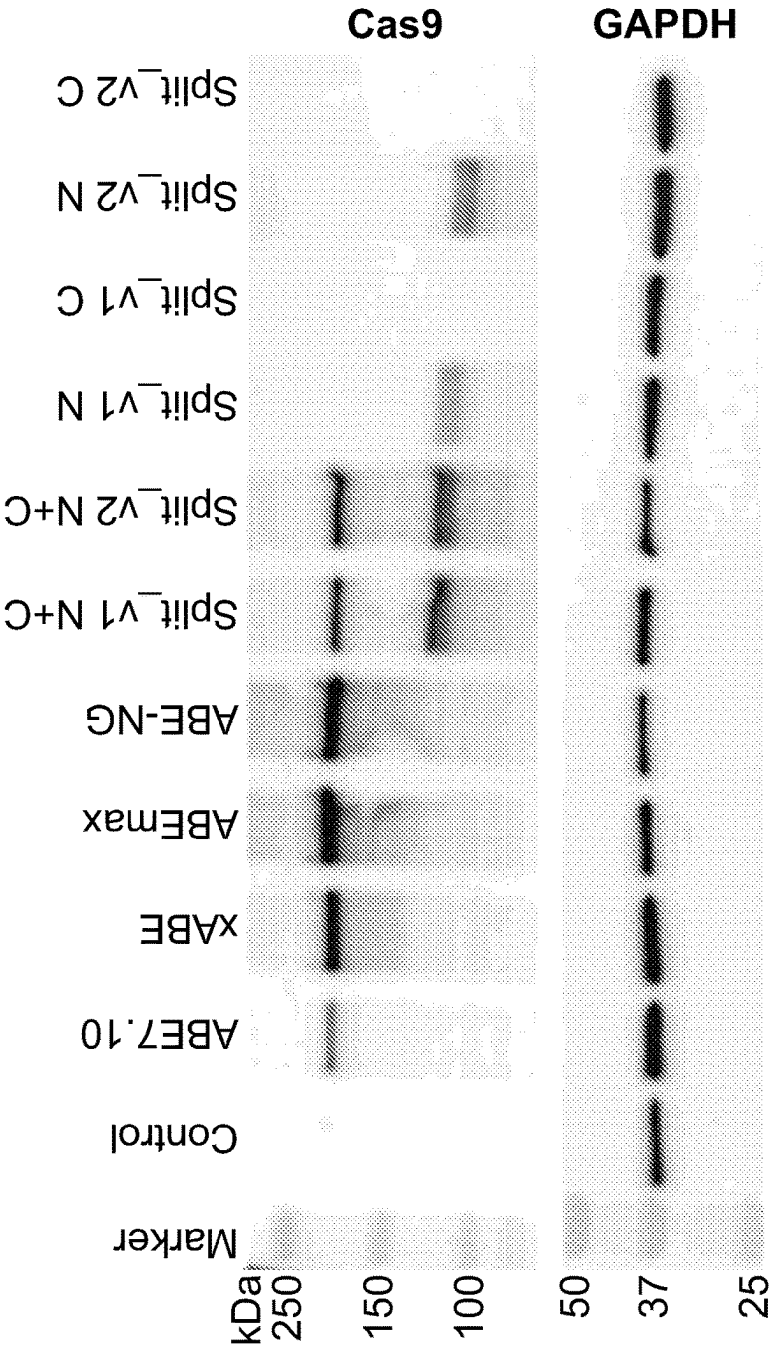


FIG. 6B

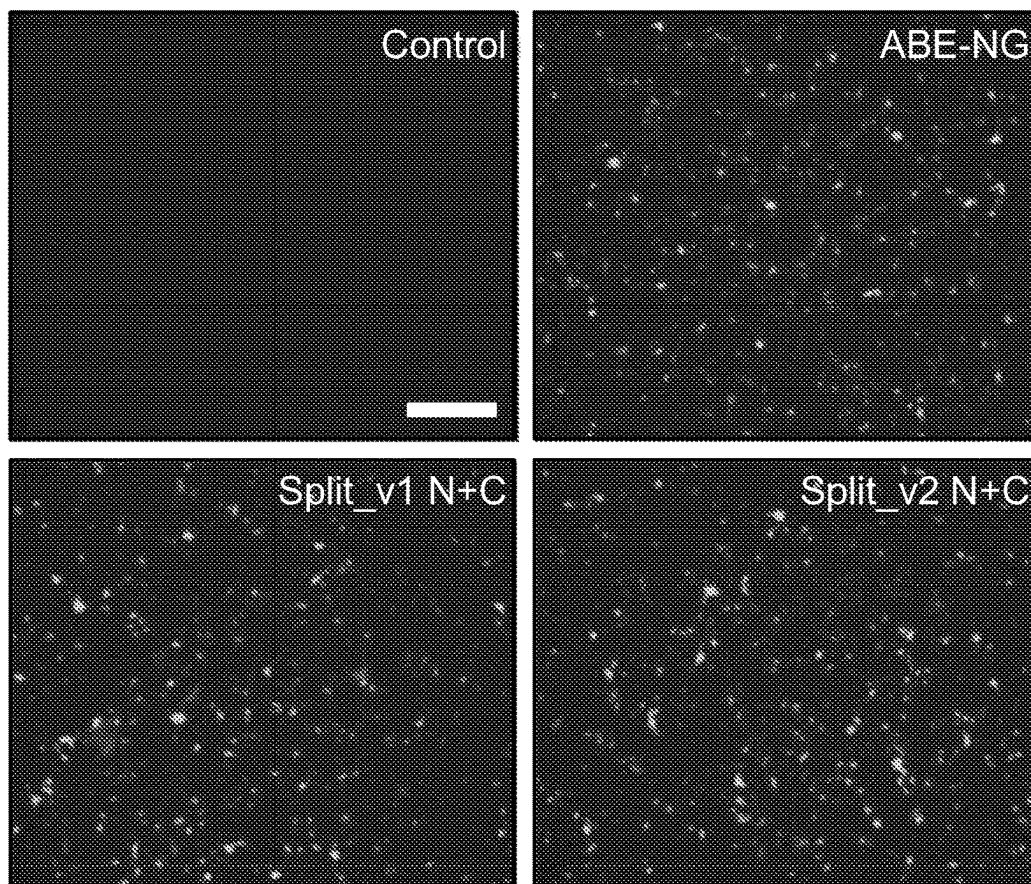


FIG. 6C

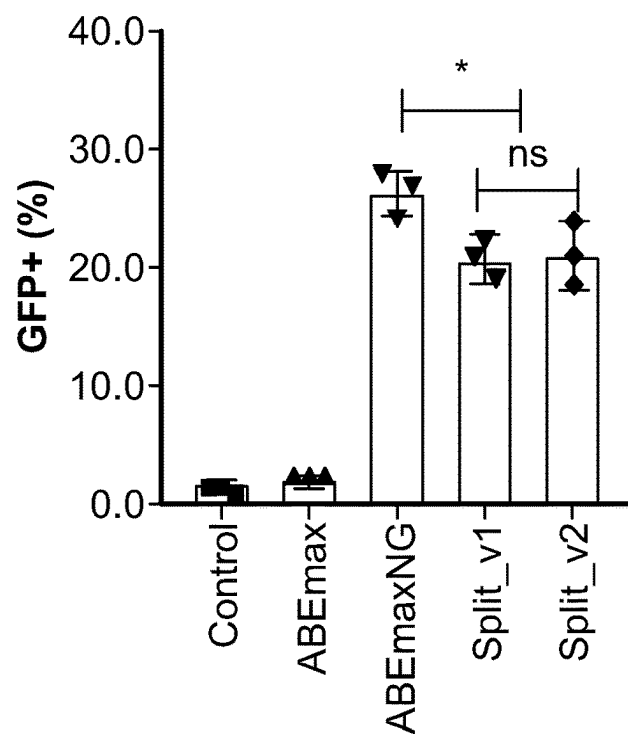


FIG. 6D

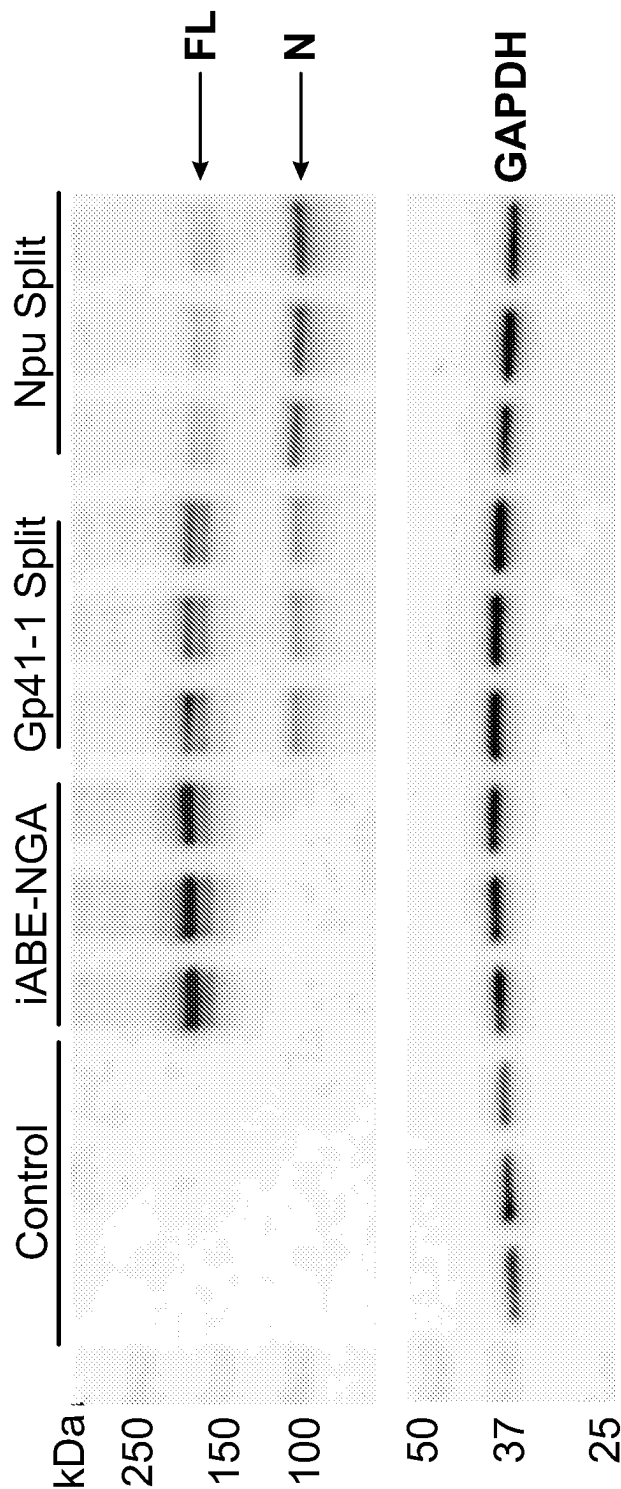


FIG. 6E

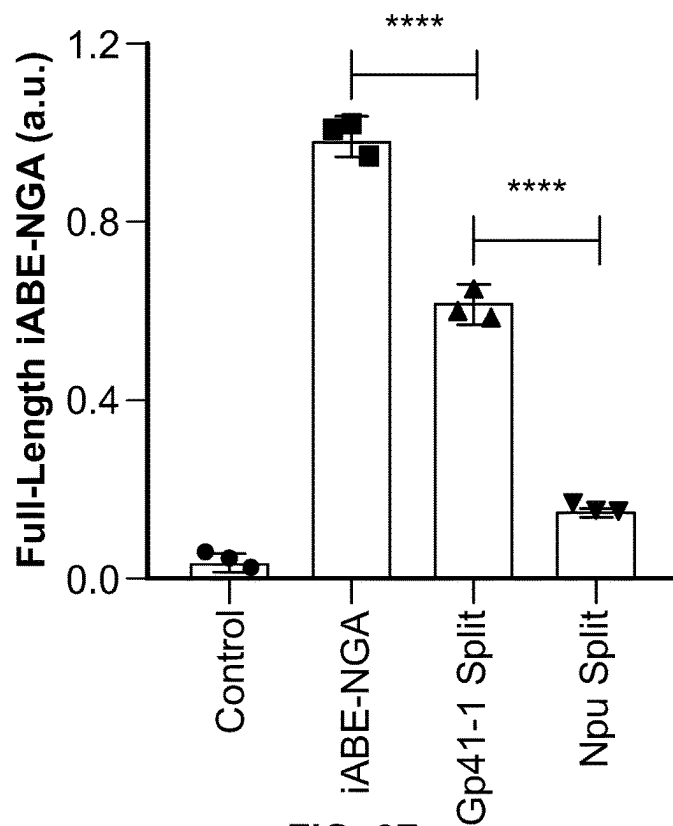


FIG. 6F

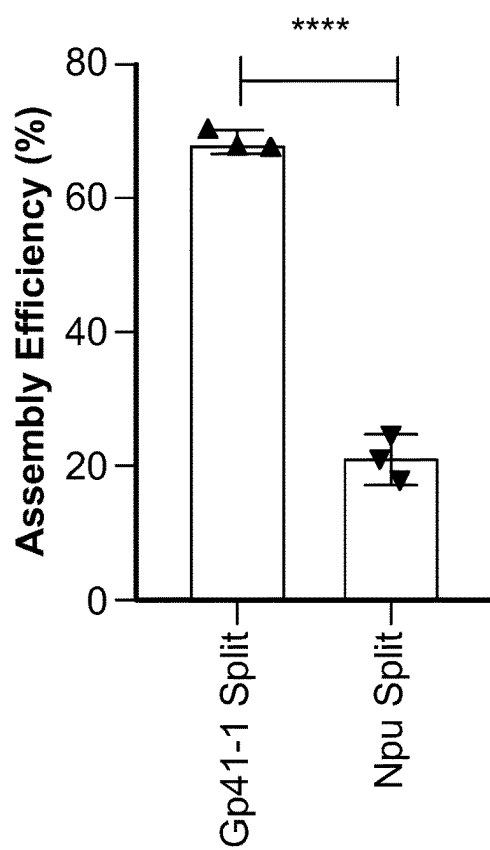


FIG. 6G

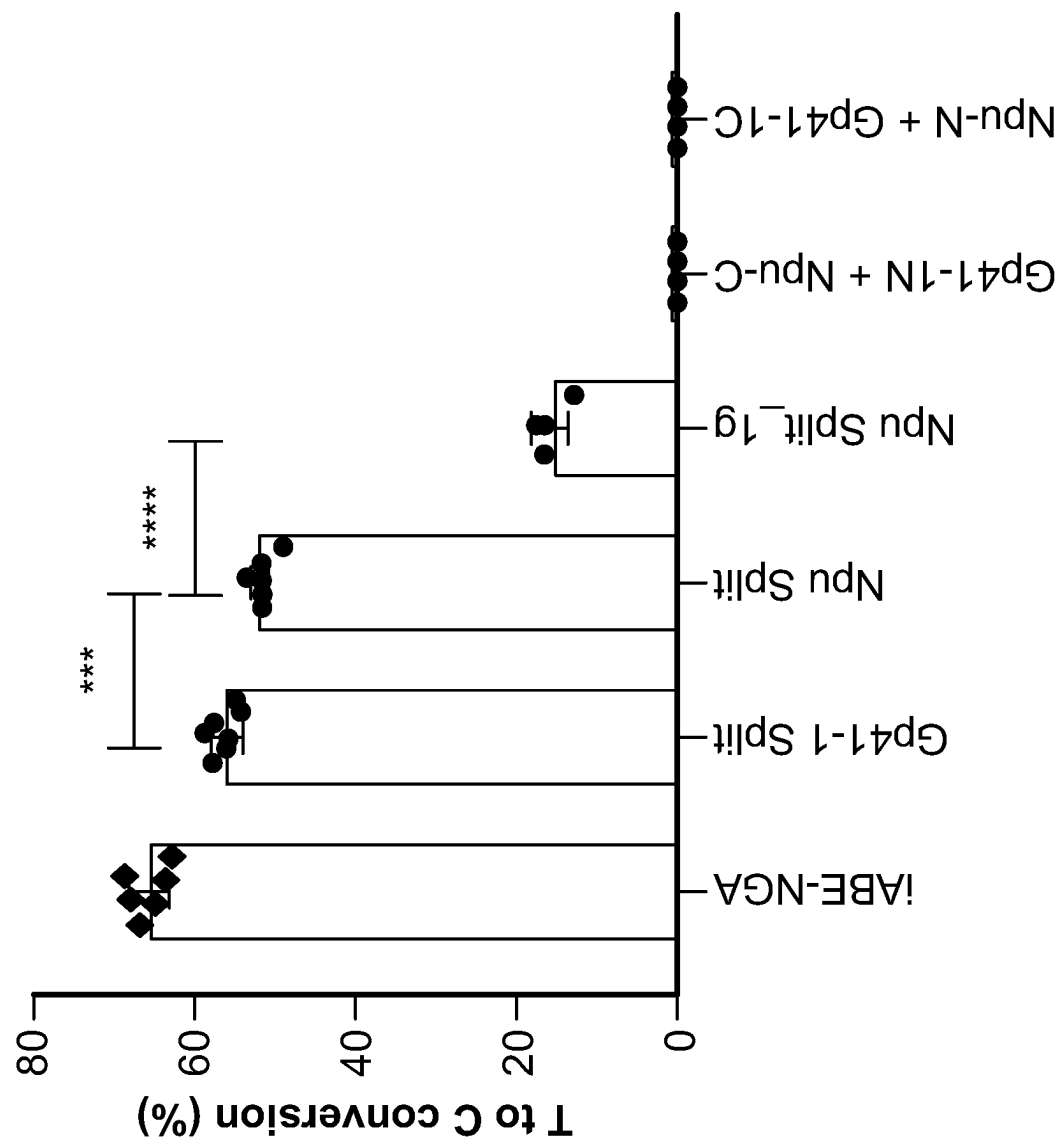


FIG. 6H

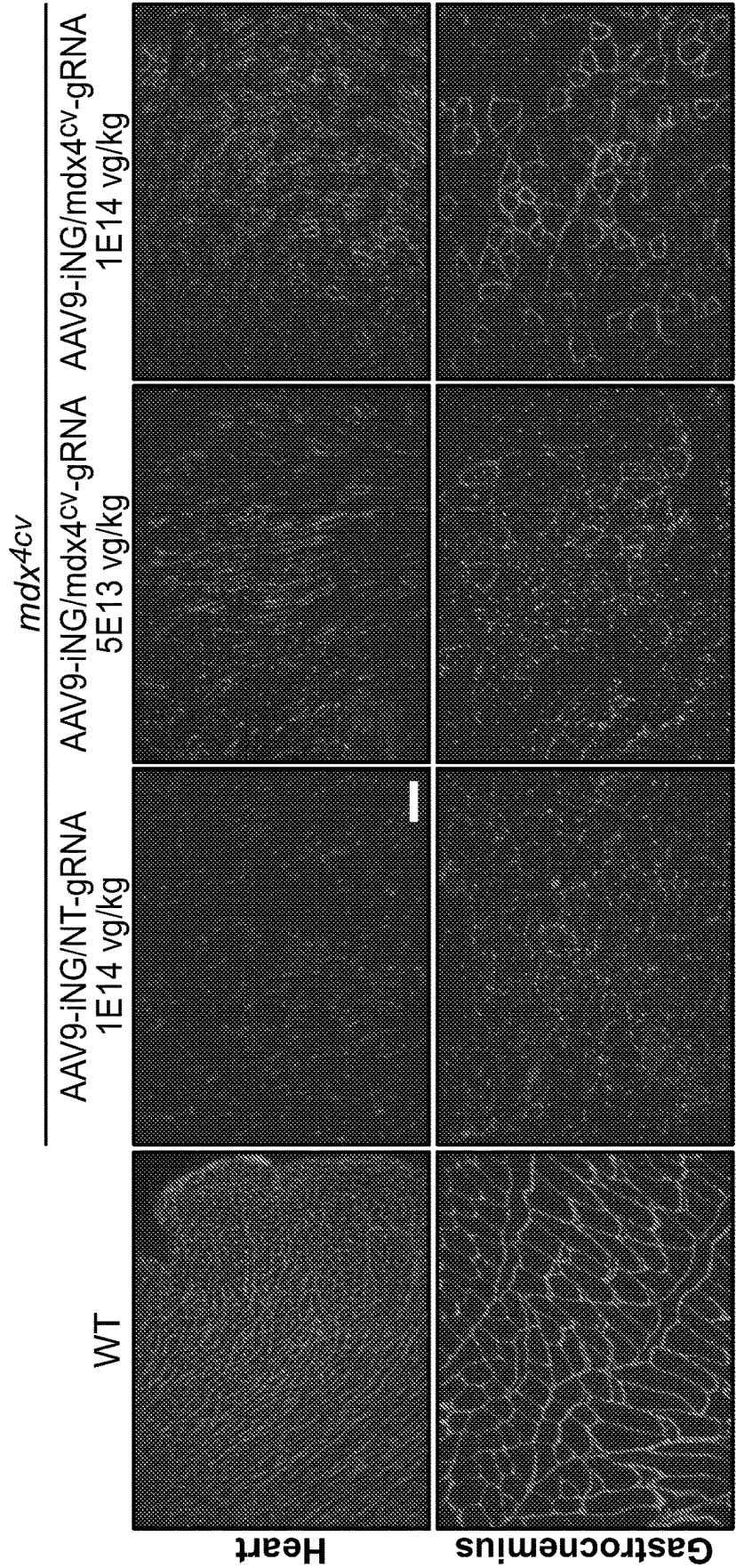


FIG. 7

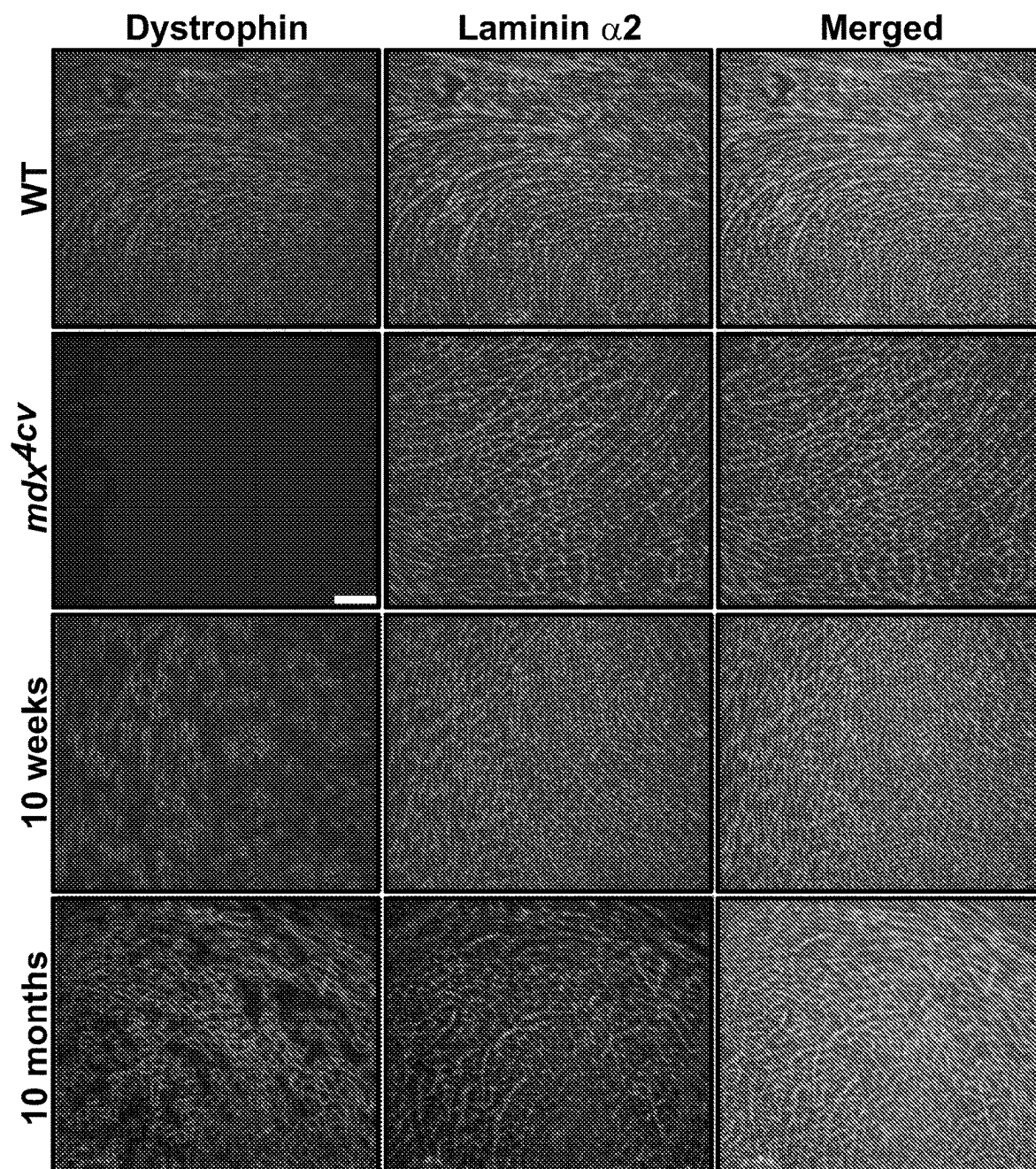
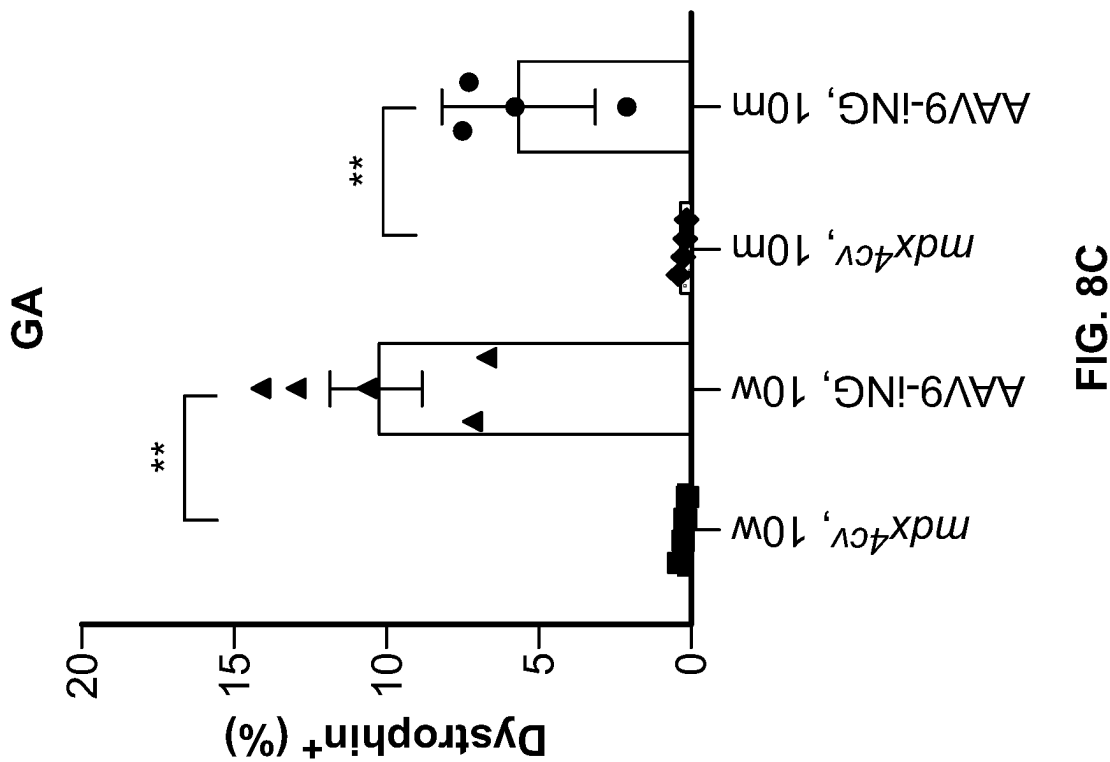
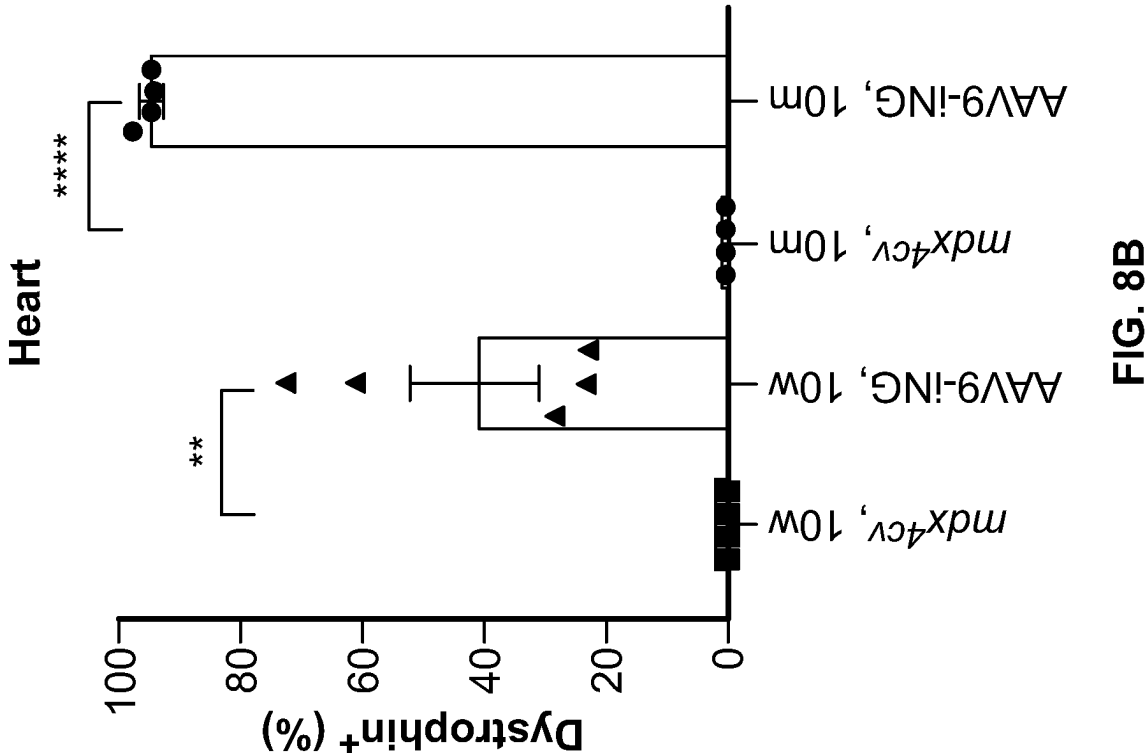


FIG. 8A



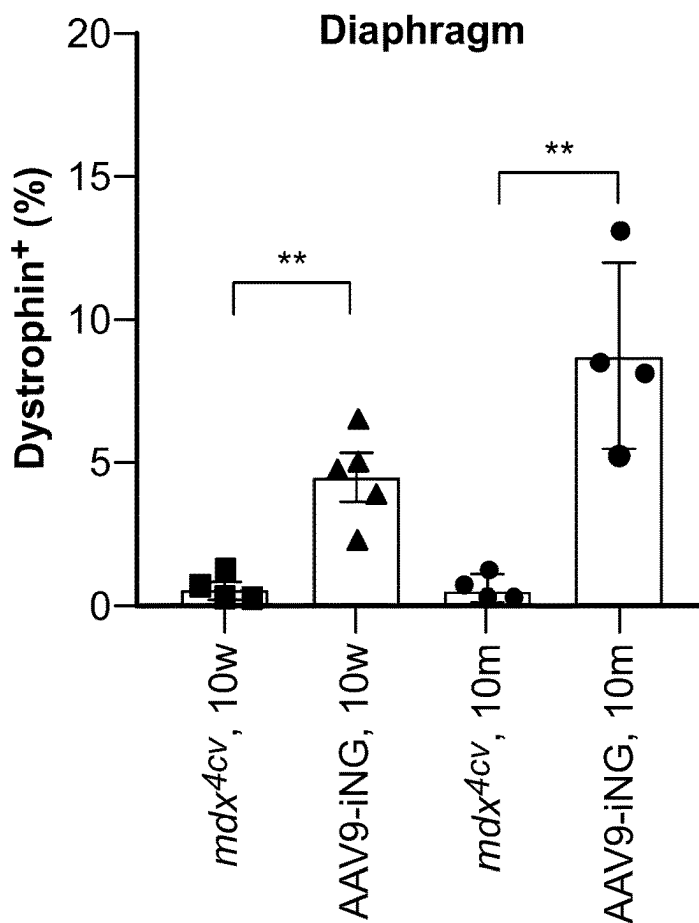


FIG. 8D

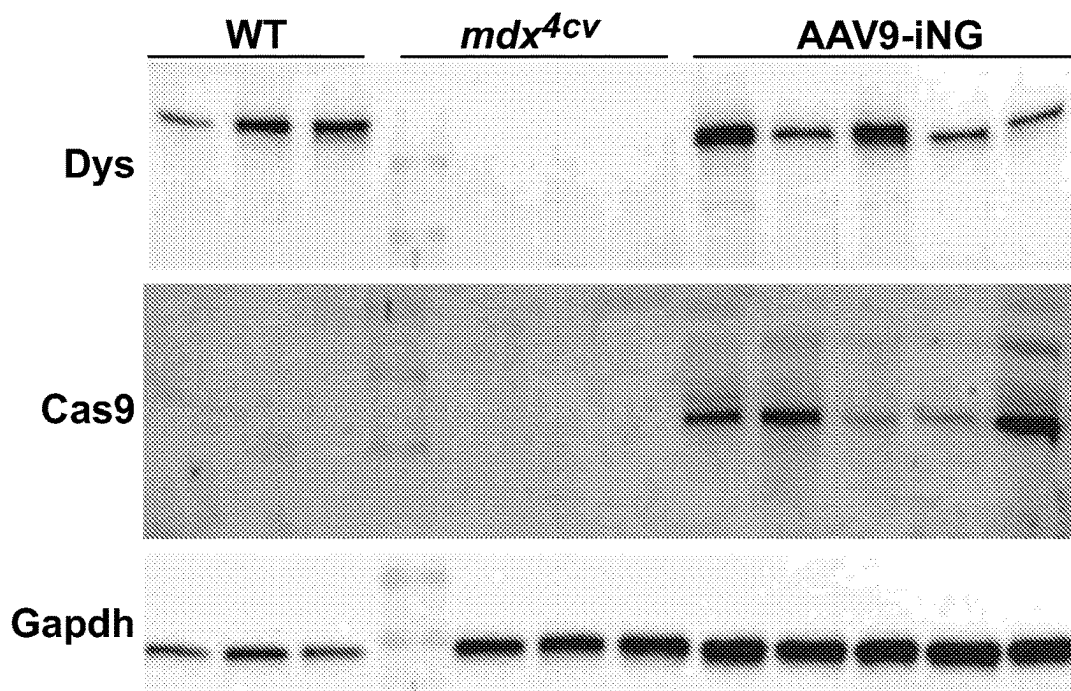


FIG. 8E

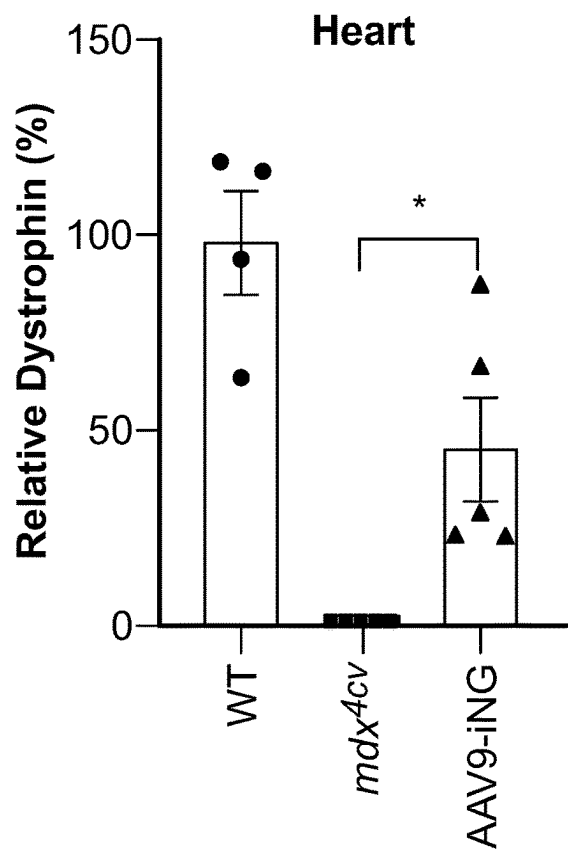


FIG. 8F

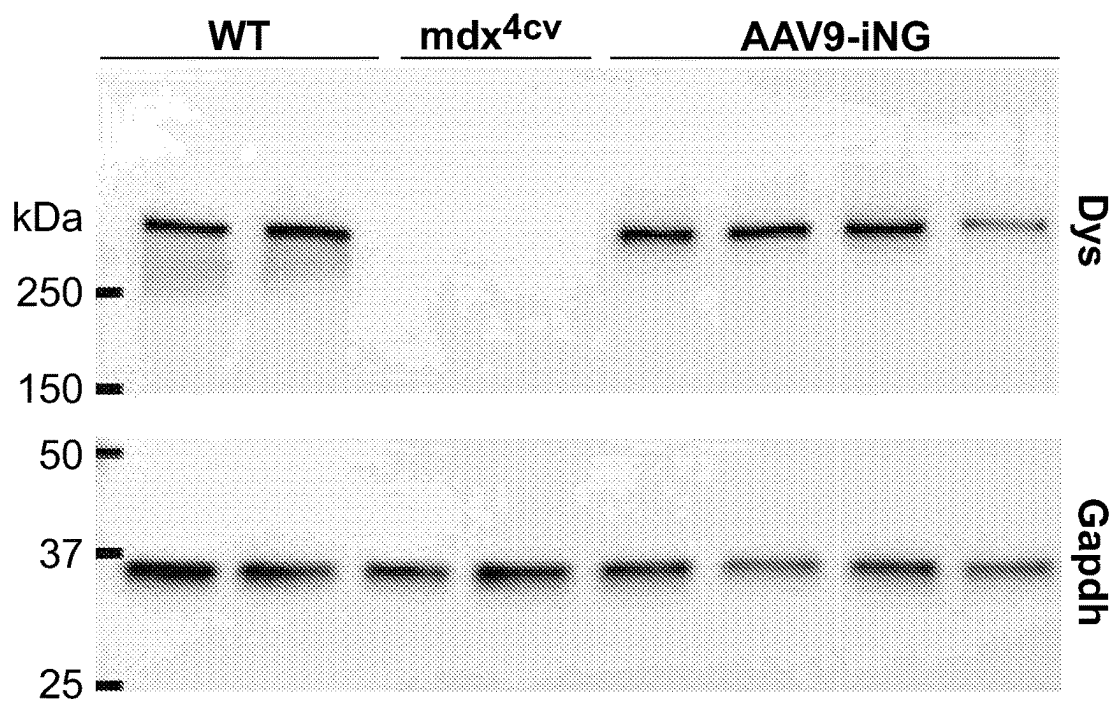


FIG. 8G

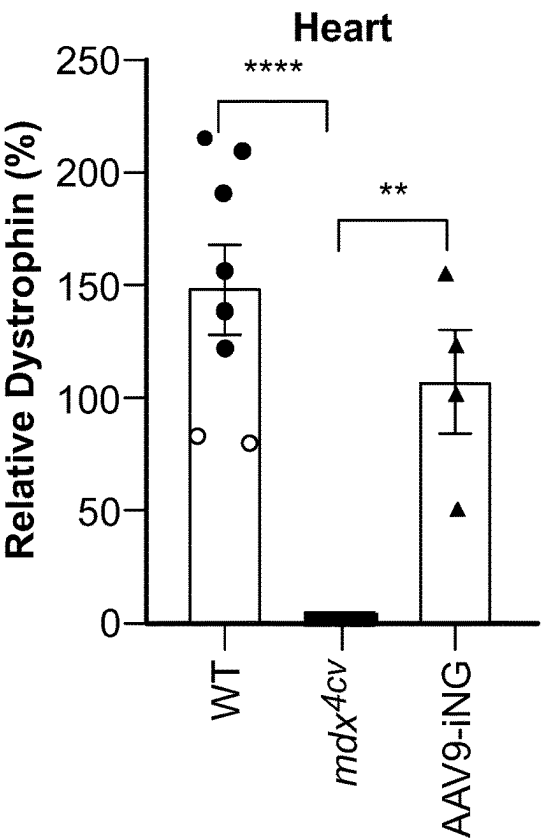


FIG. 8H

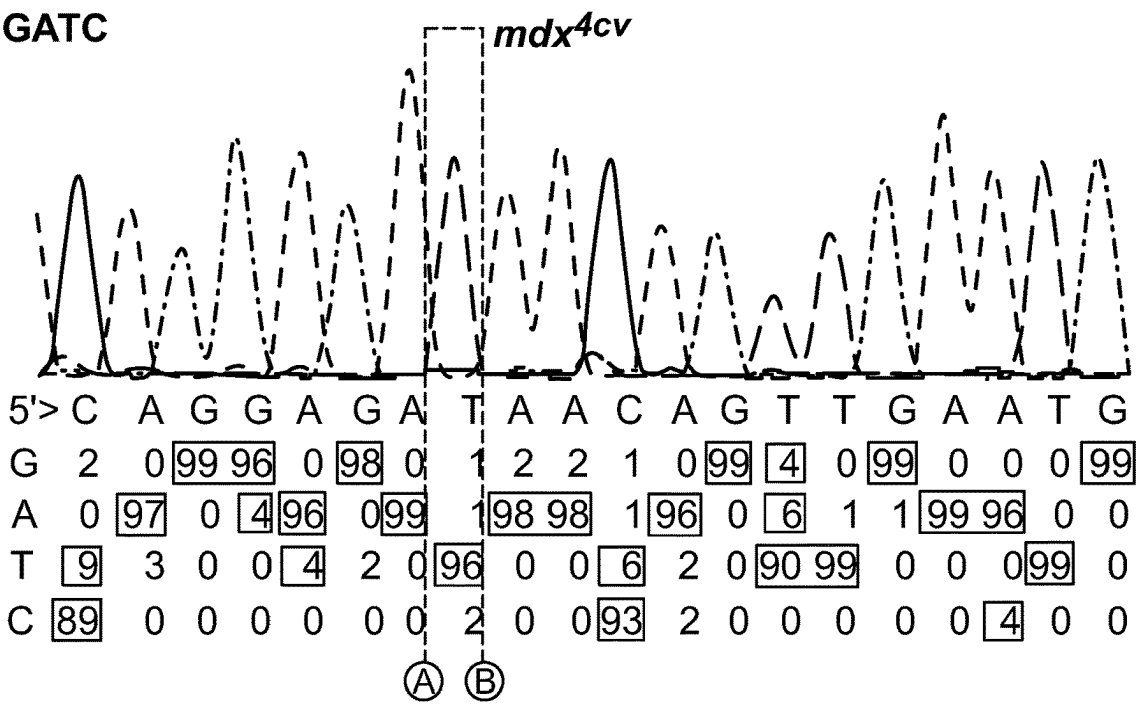


FIG. 8I

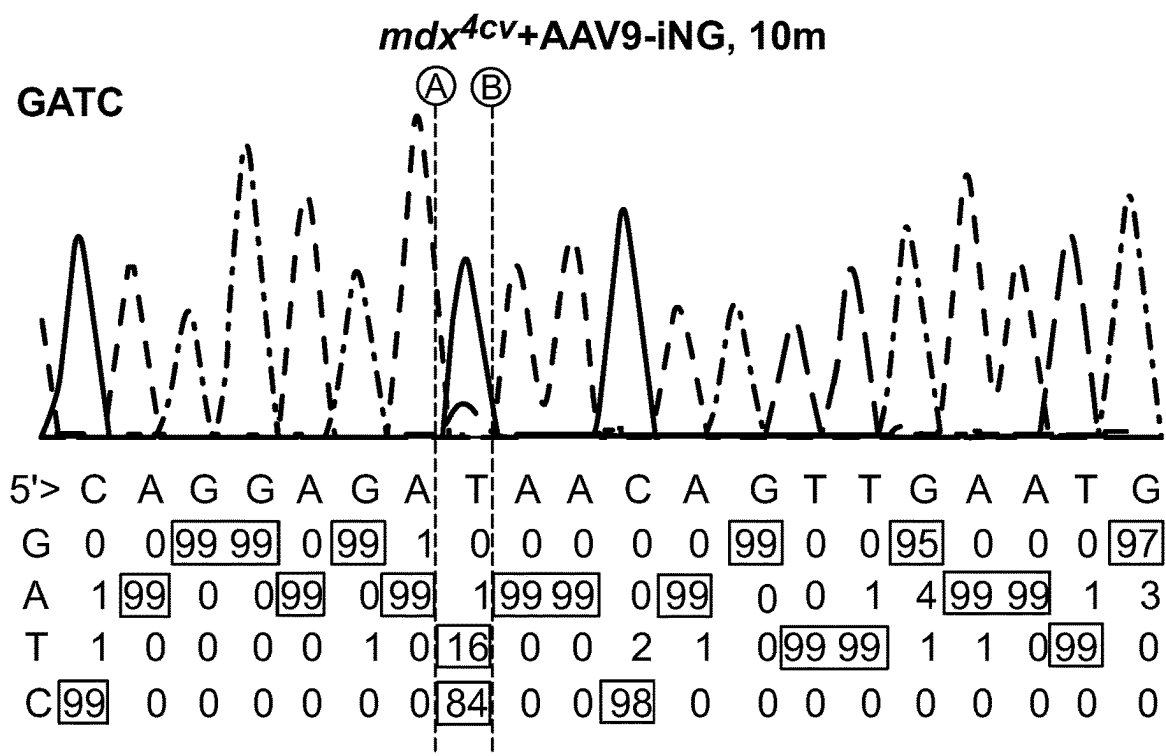
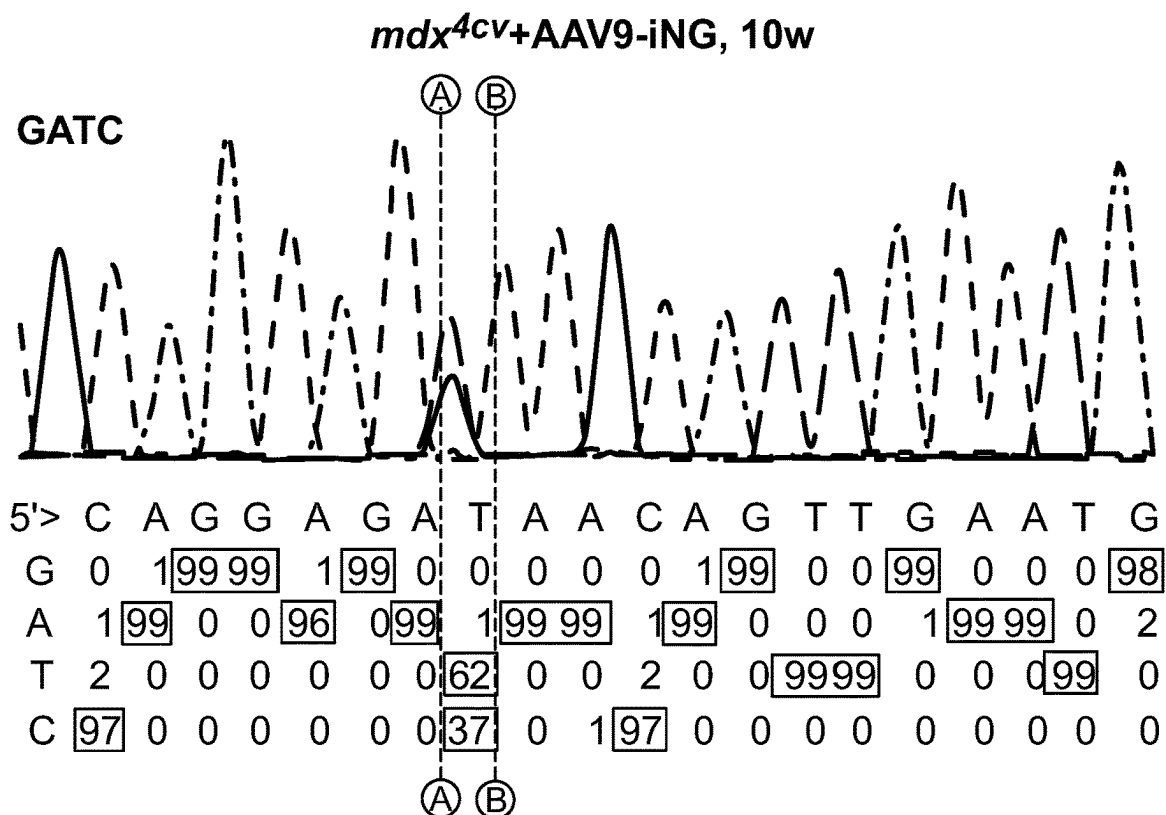


FIG. 8I
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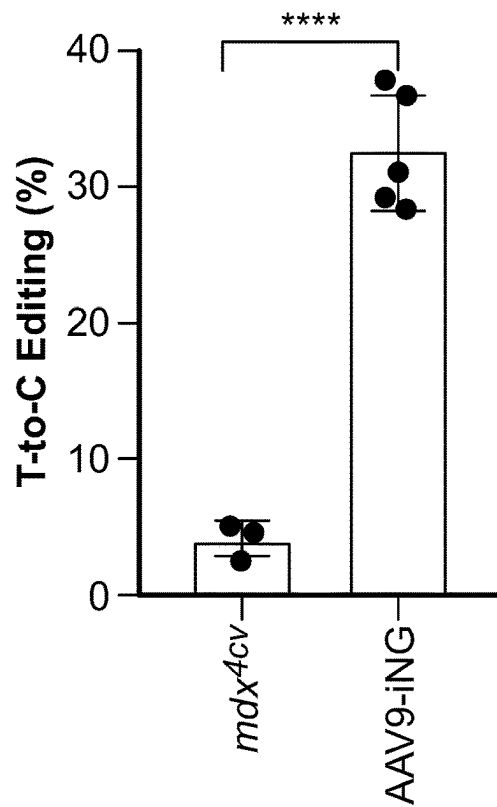


FIG. 8J

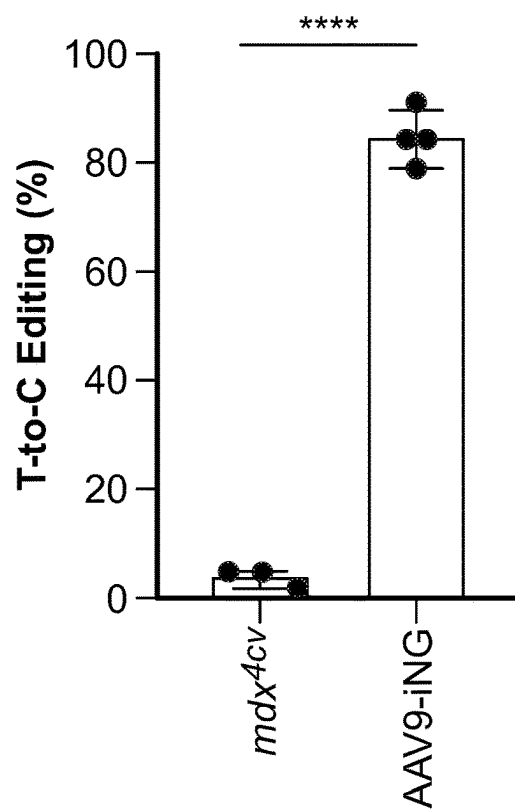


FIG. 8K

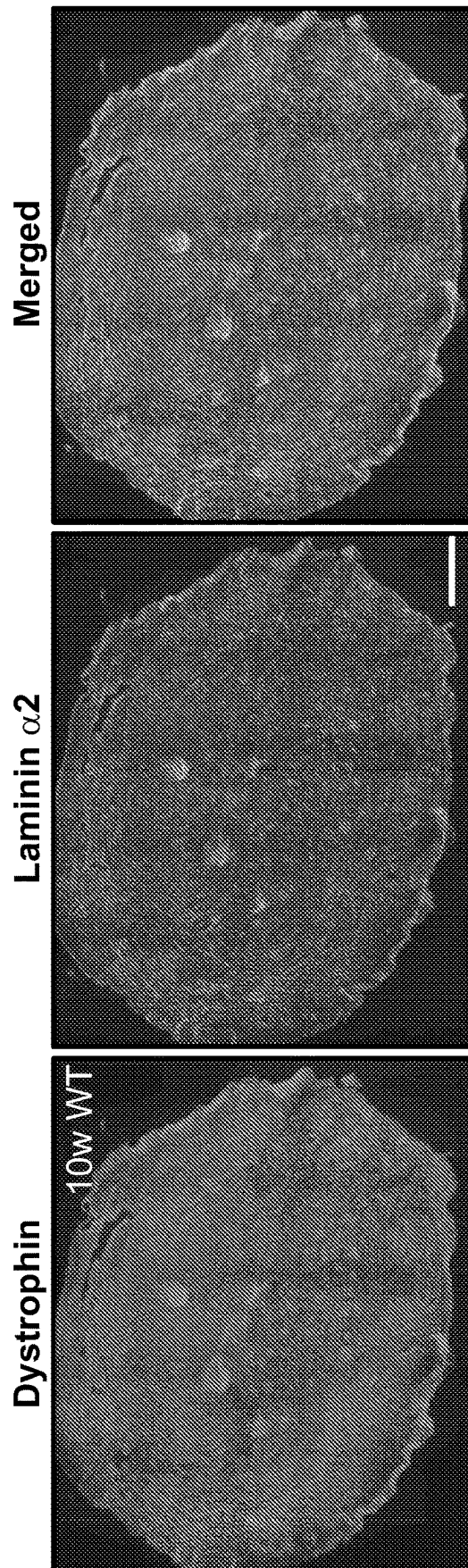


FIG. 9

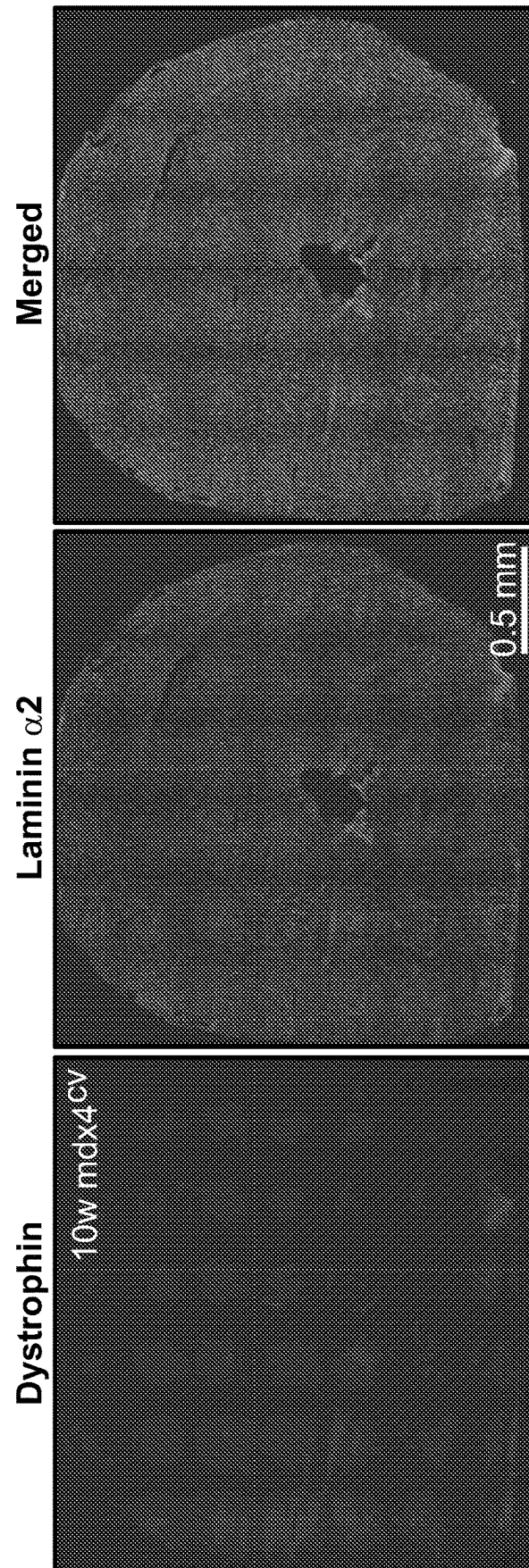


FIG. 10

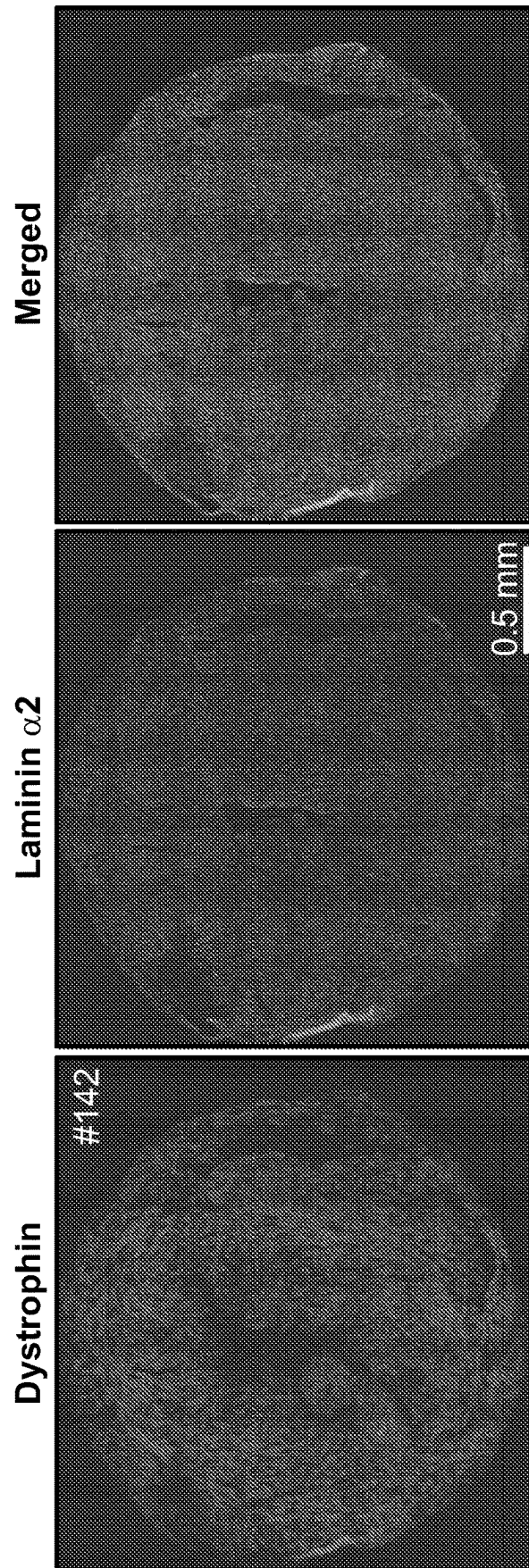


FIG. 11

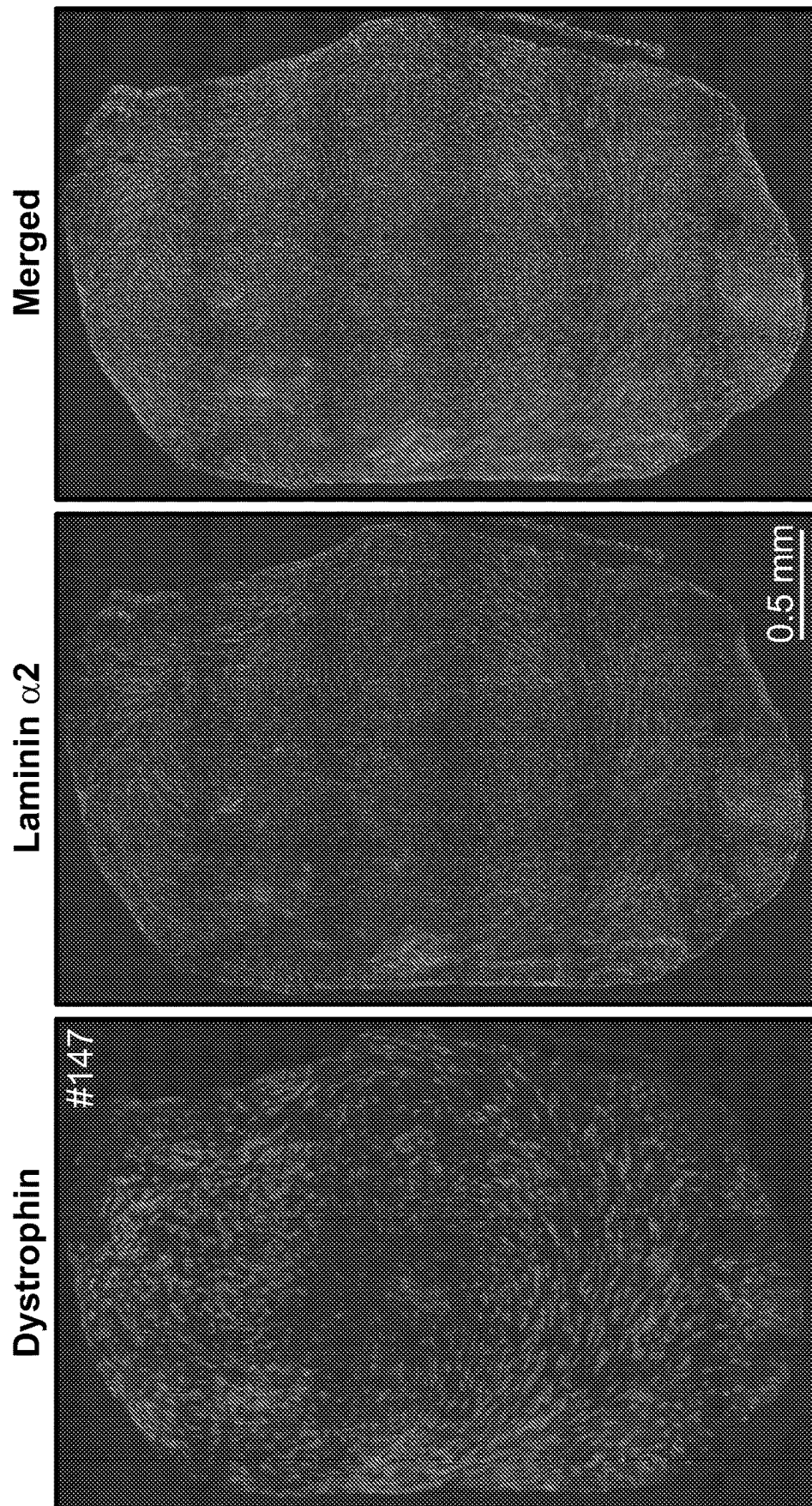


FIG. 12

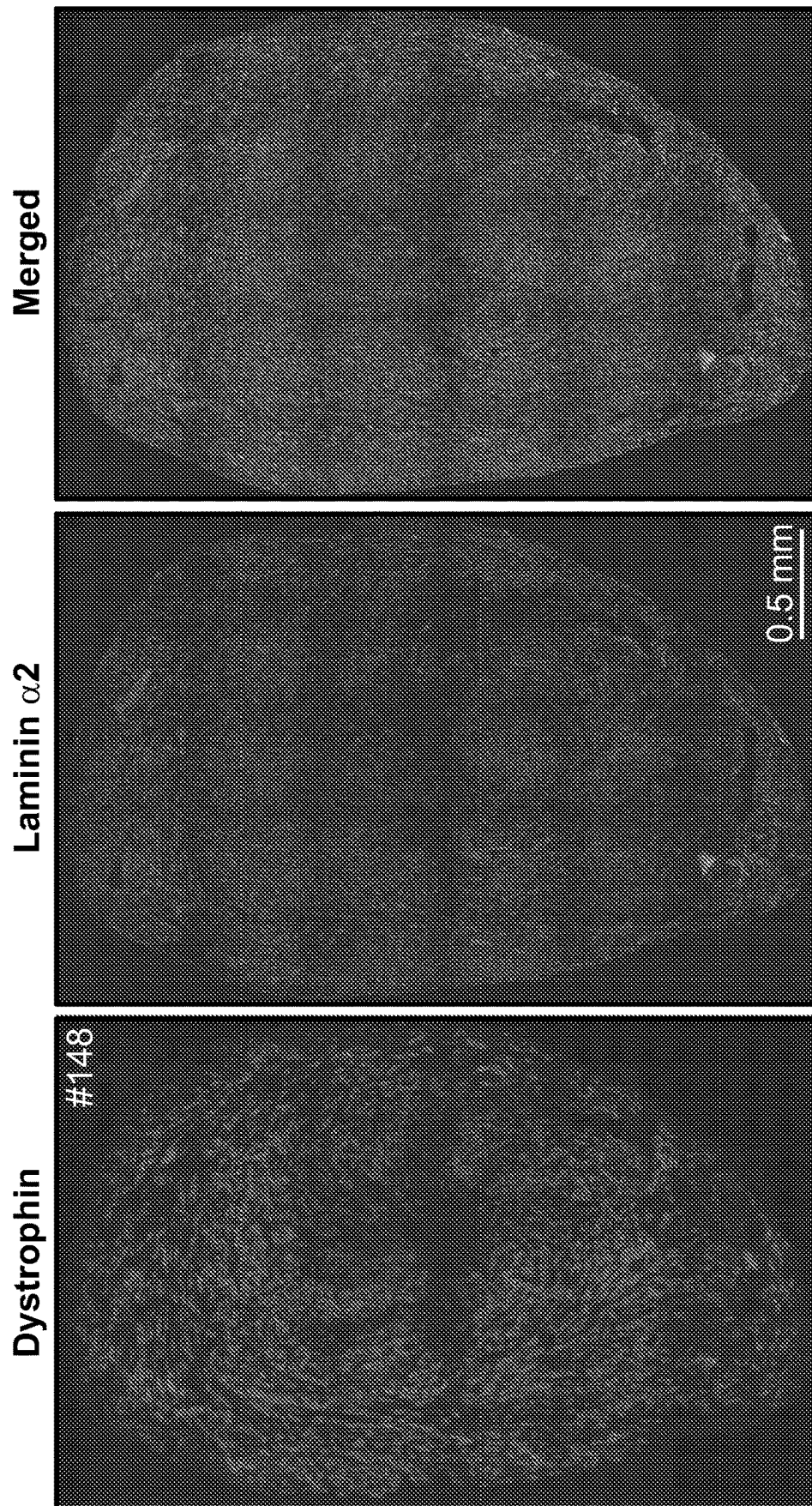
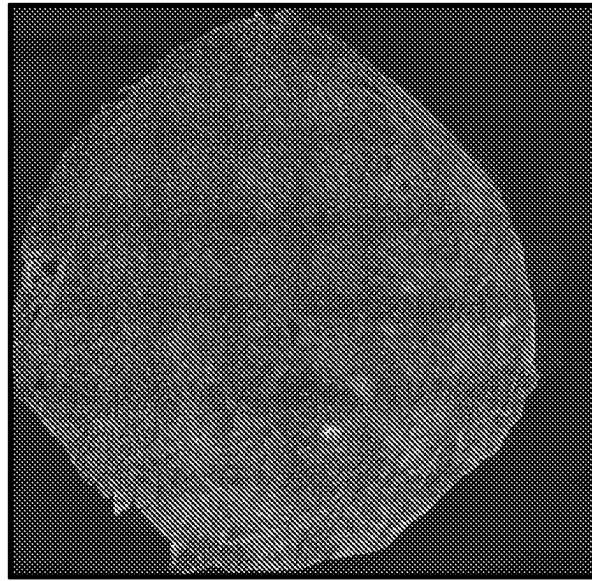


FIG. 13

Merged



Laminin $\alpha 2$



0.5 mm

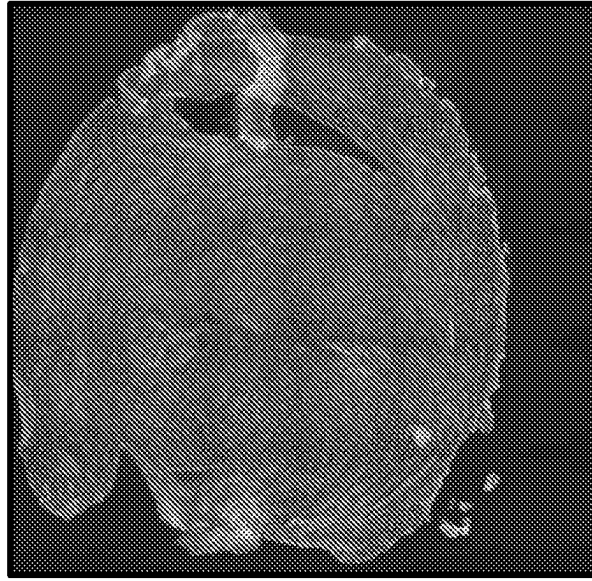
Dystrophin



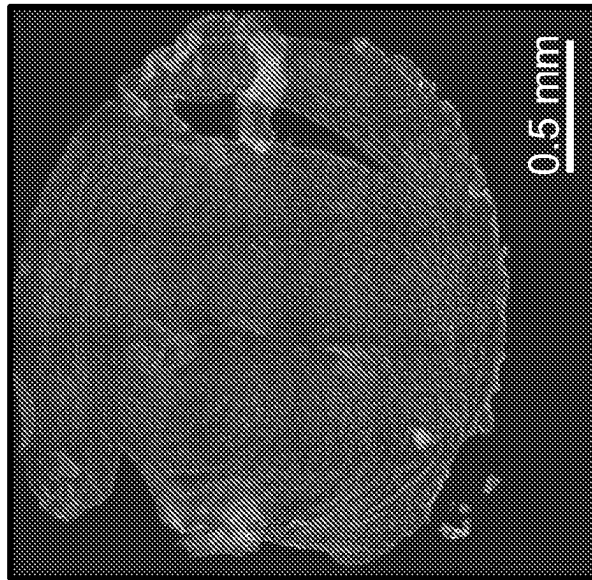
#149

FIG. 14

Merged



Laminin $\alpha 2$



Dystrophin



FIG. 15

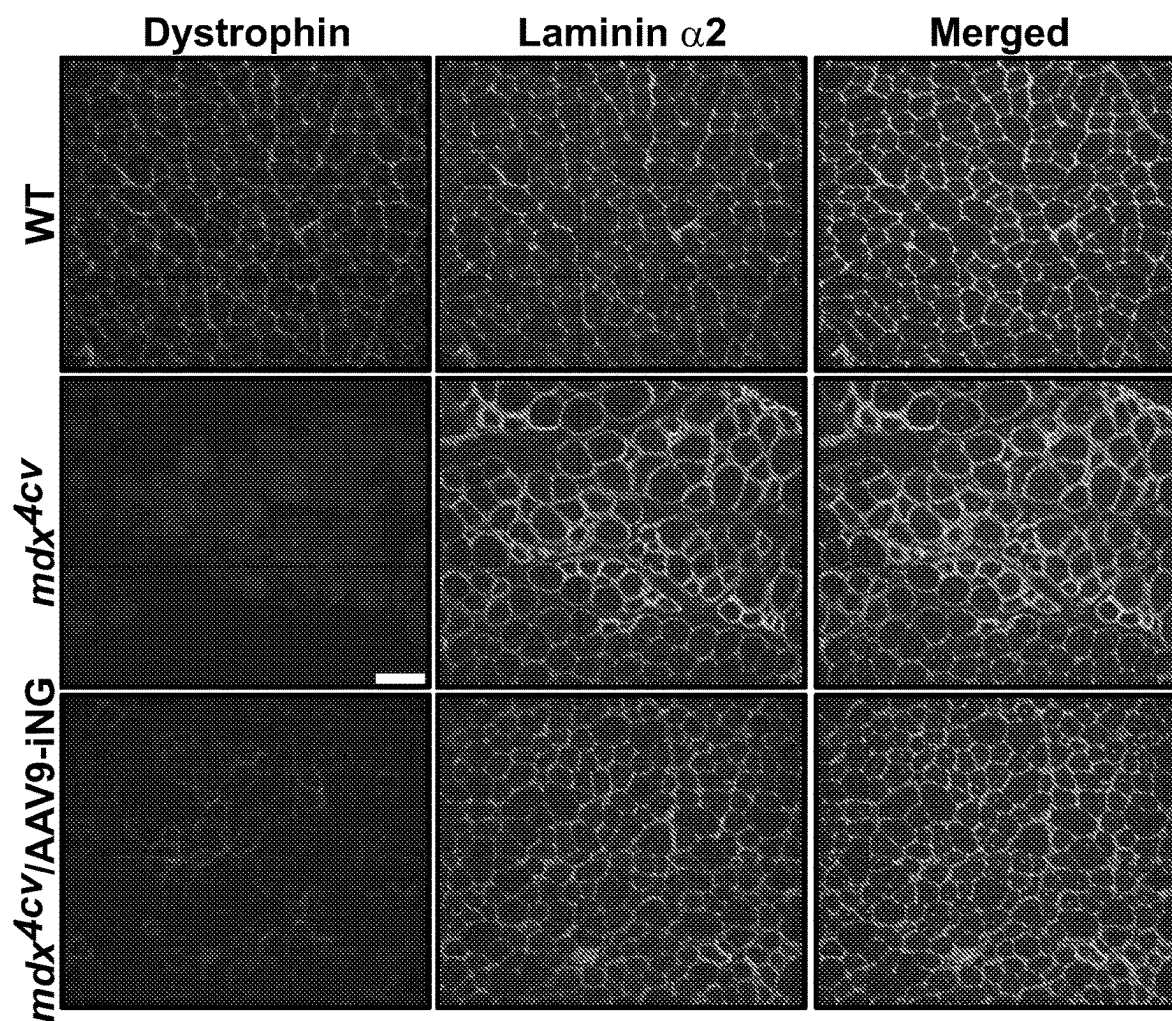


FIG. 16A

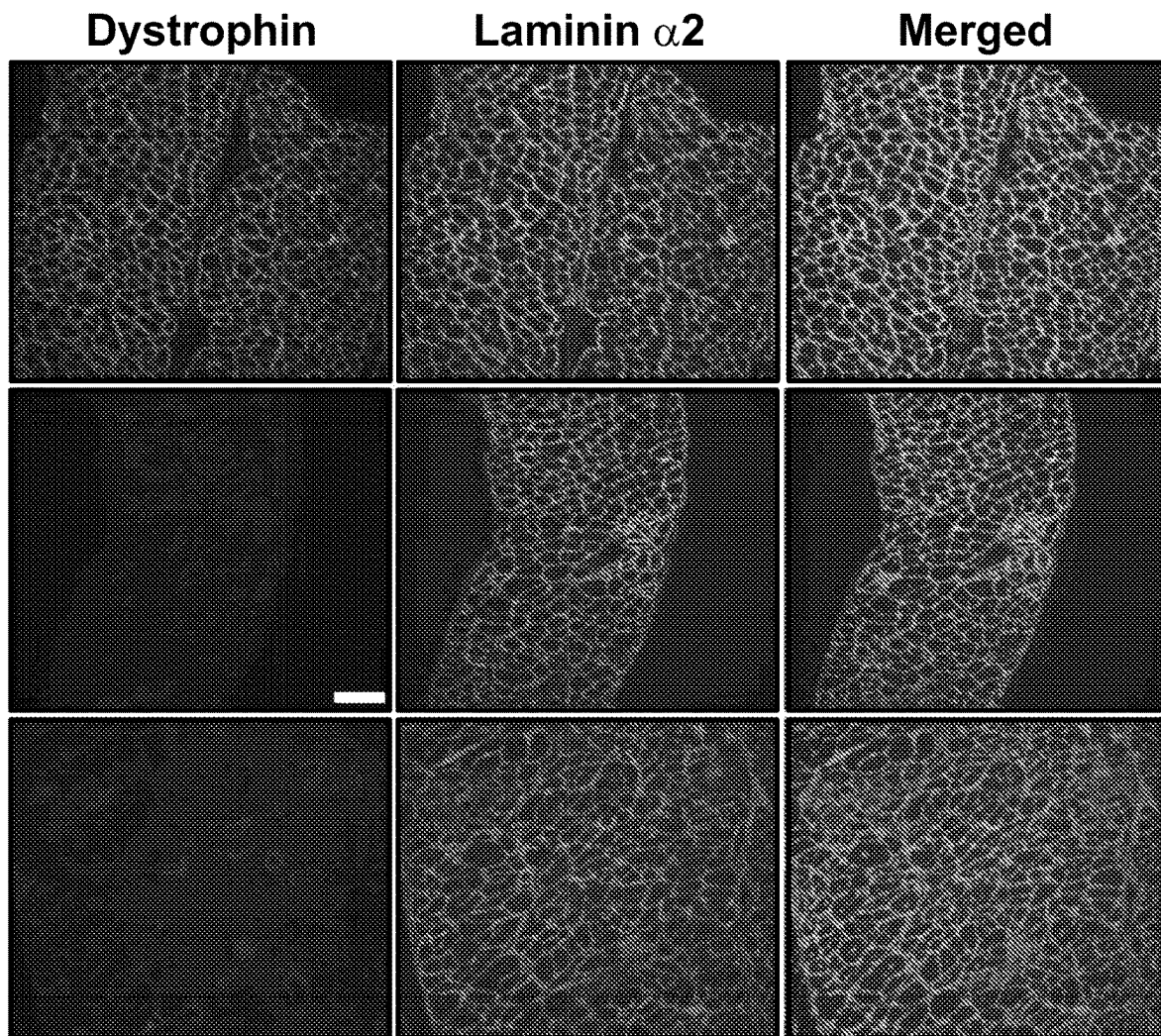


FIG. 16B

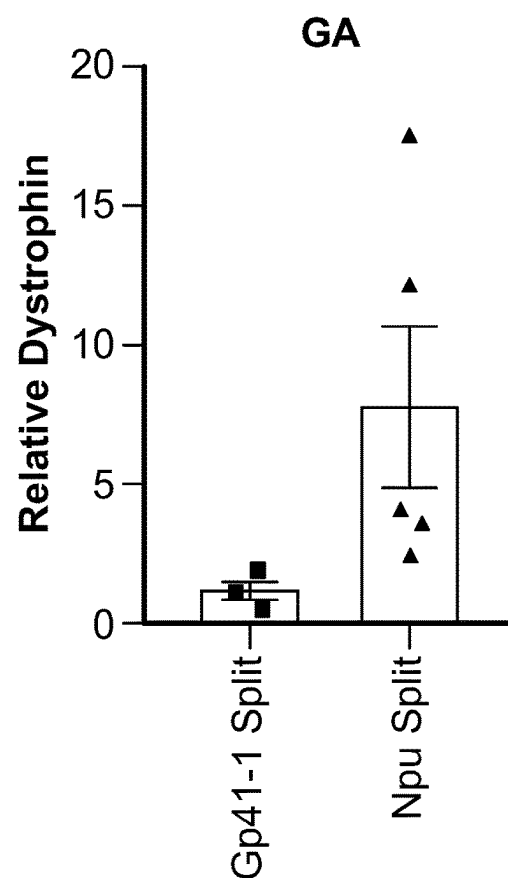
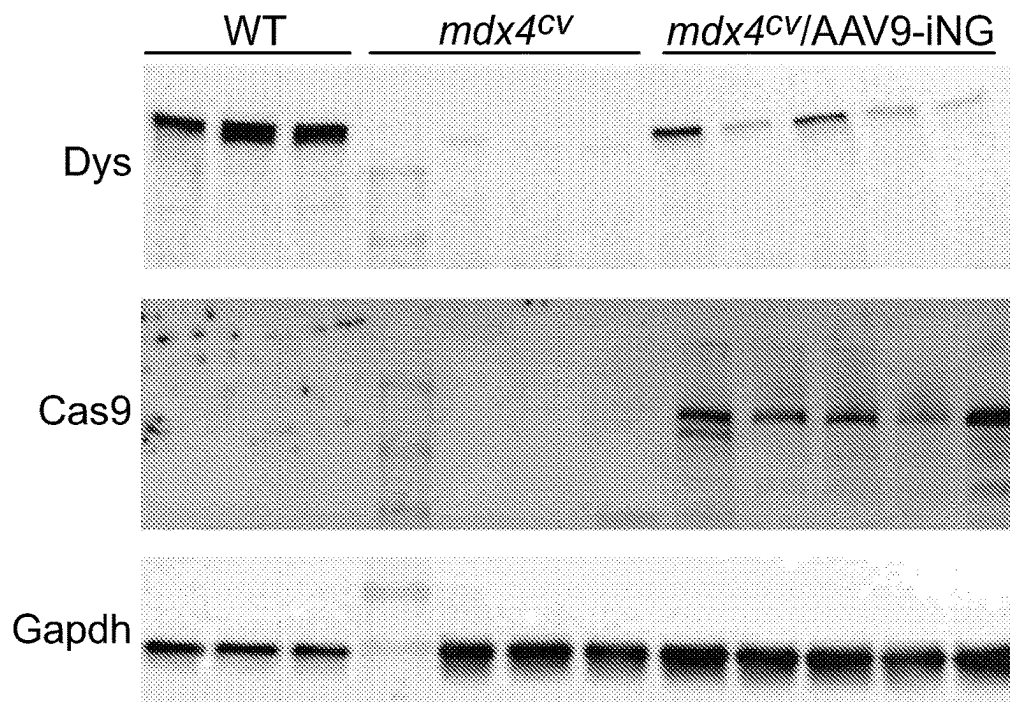


FIG. 17

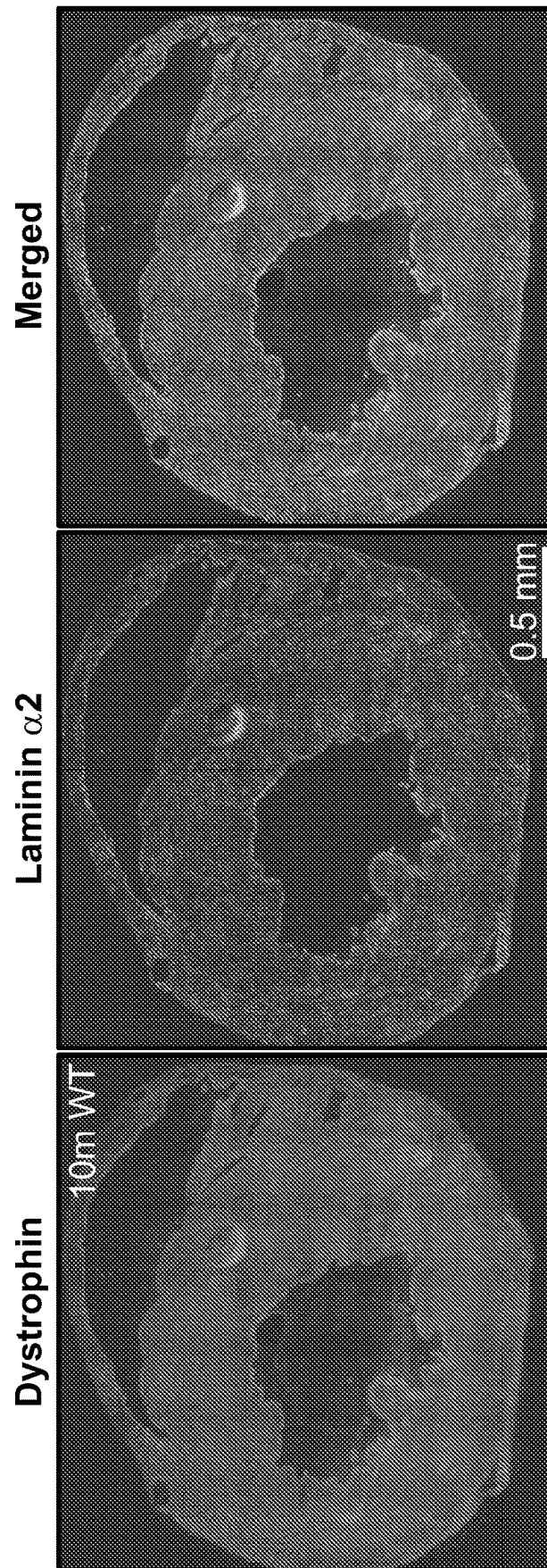


FIG. 18

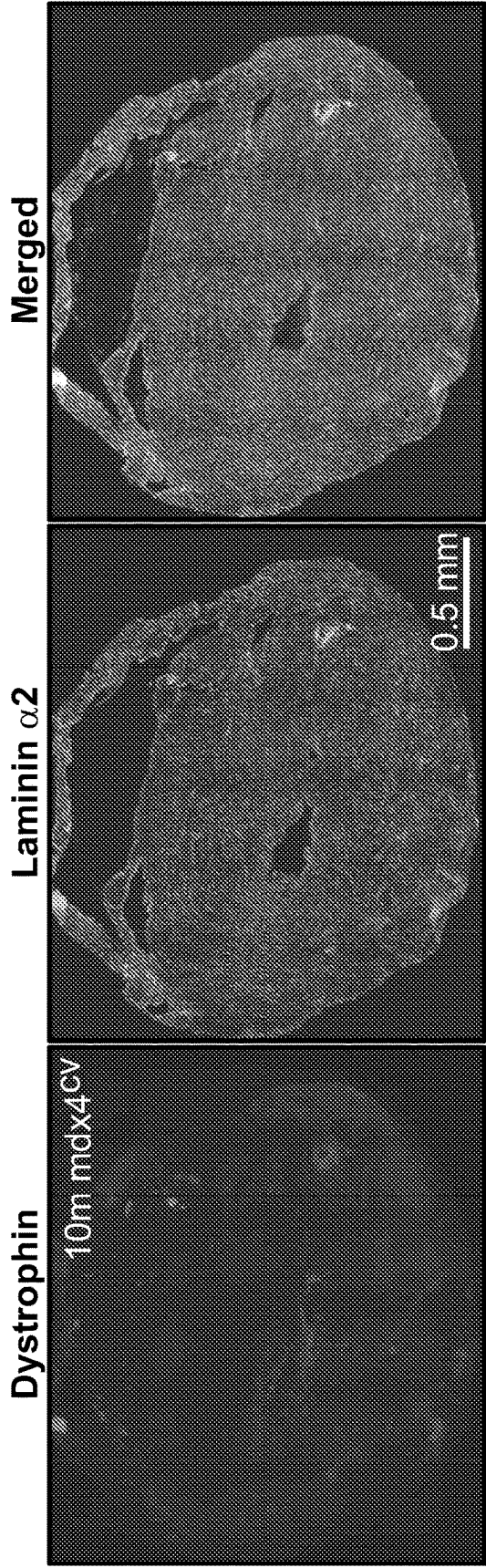


FIG. 19

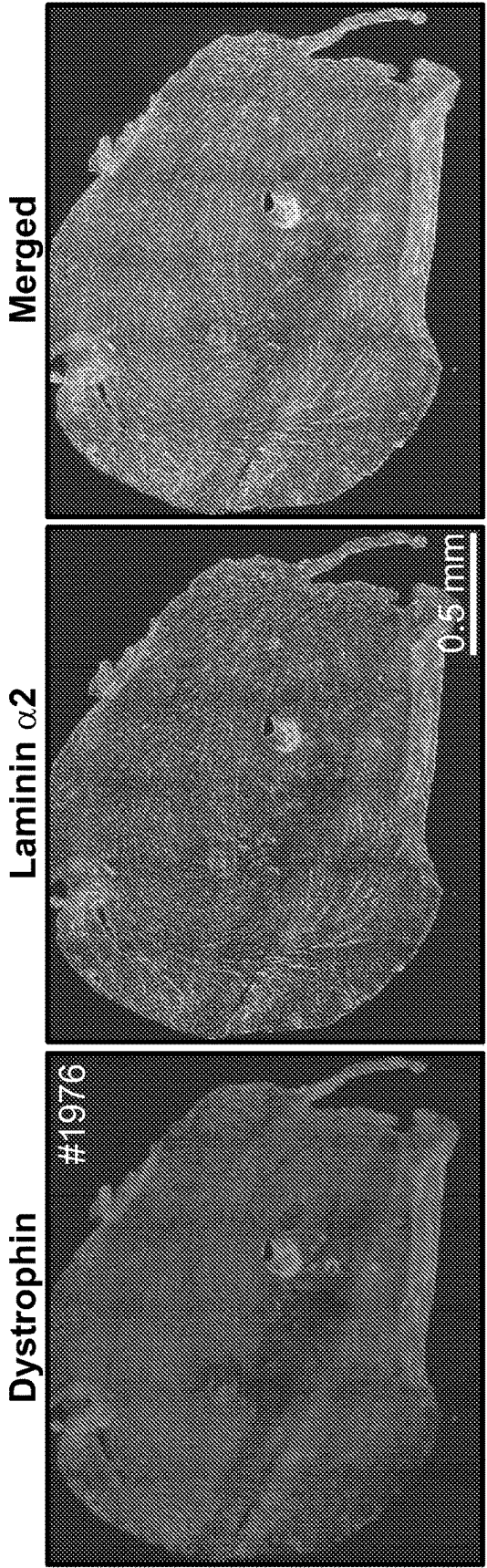


FIG. 20

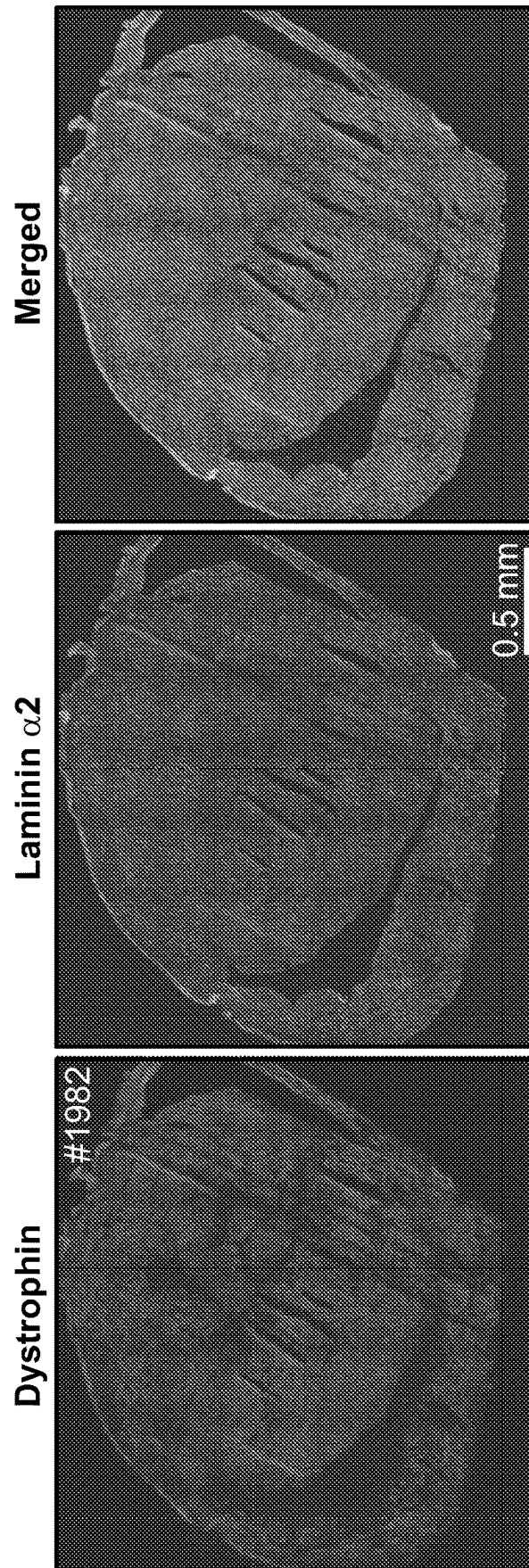


FIG. 21

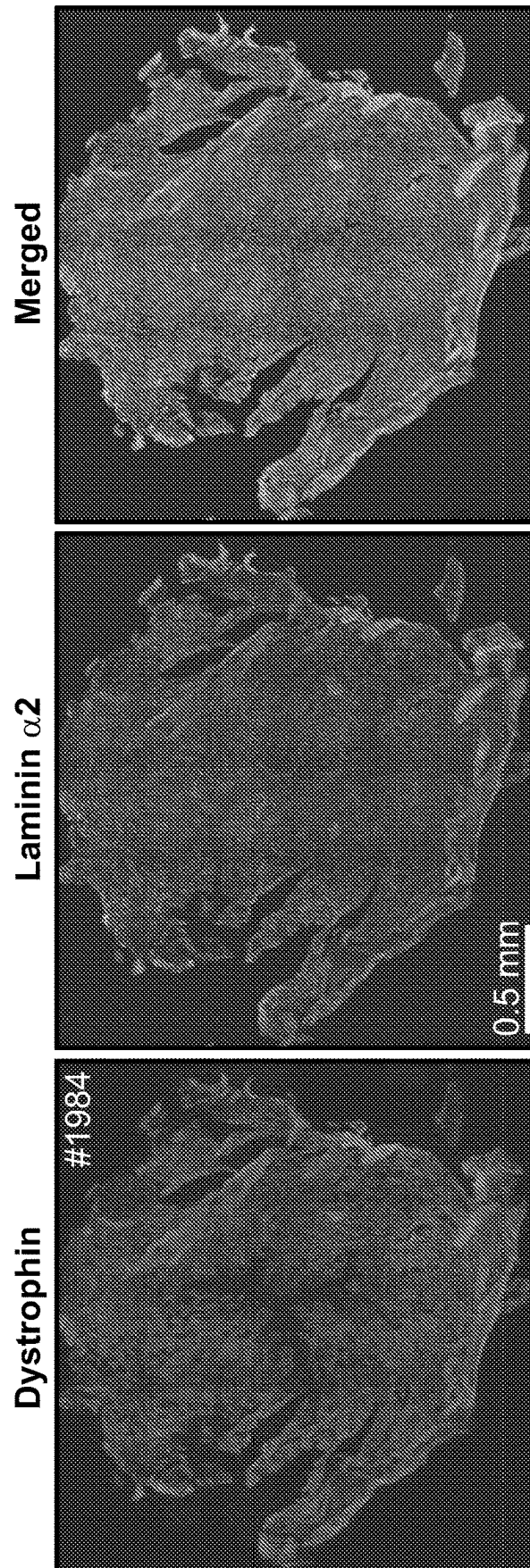


FIG. 22

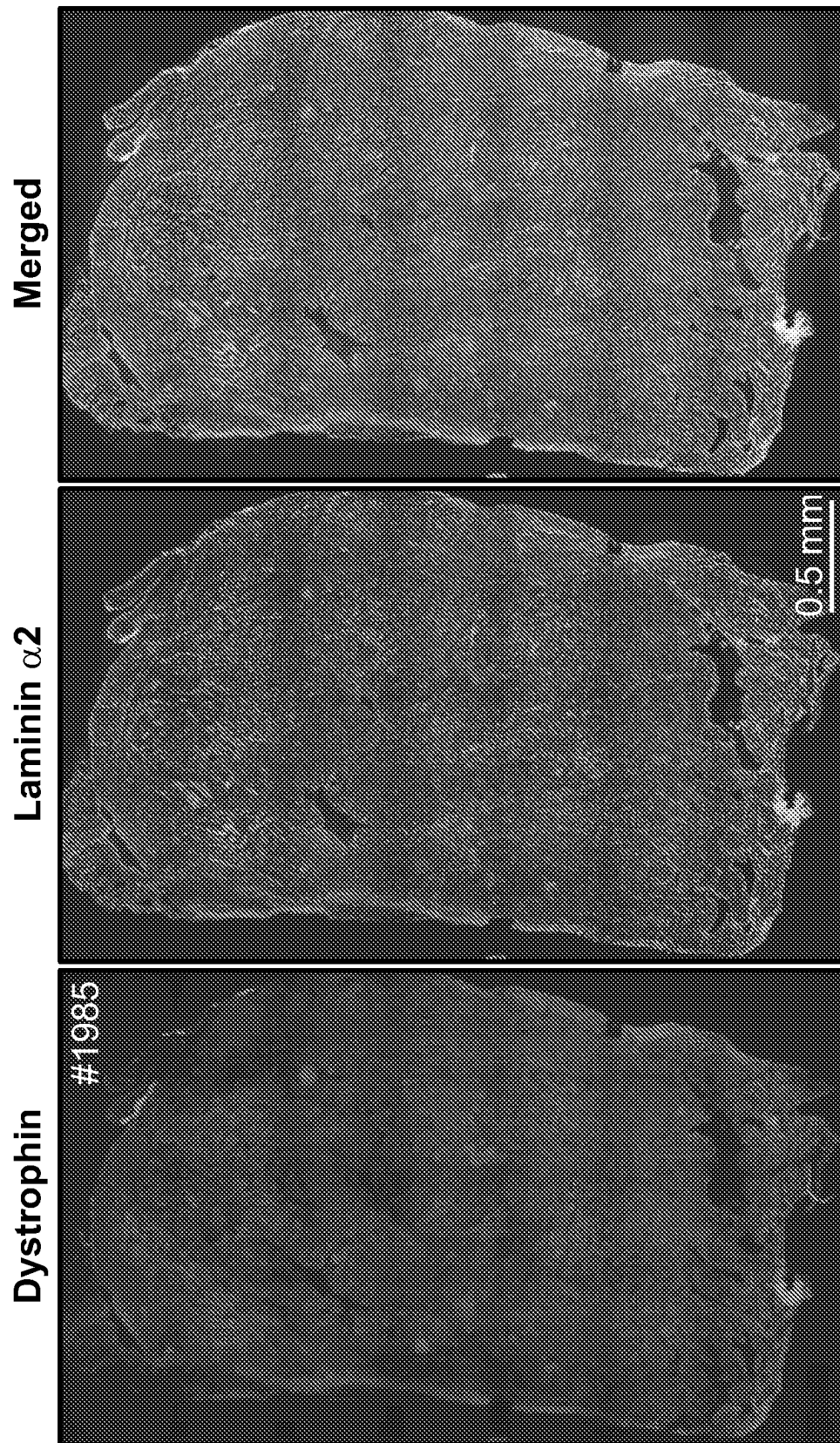


FIG. 23

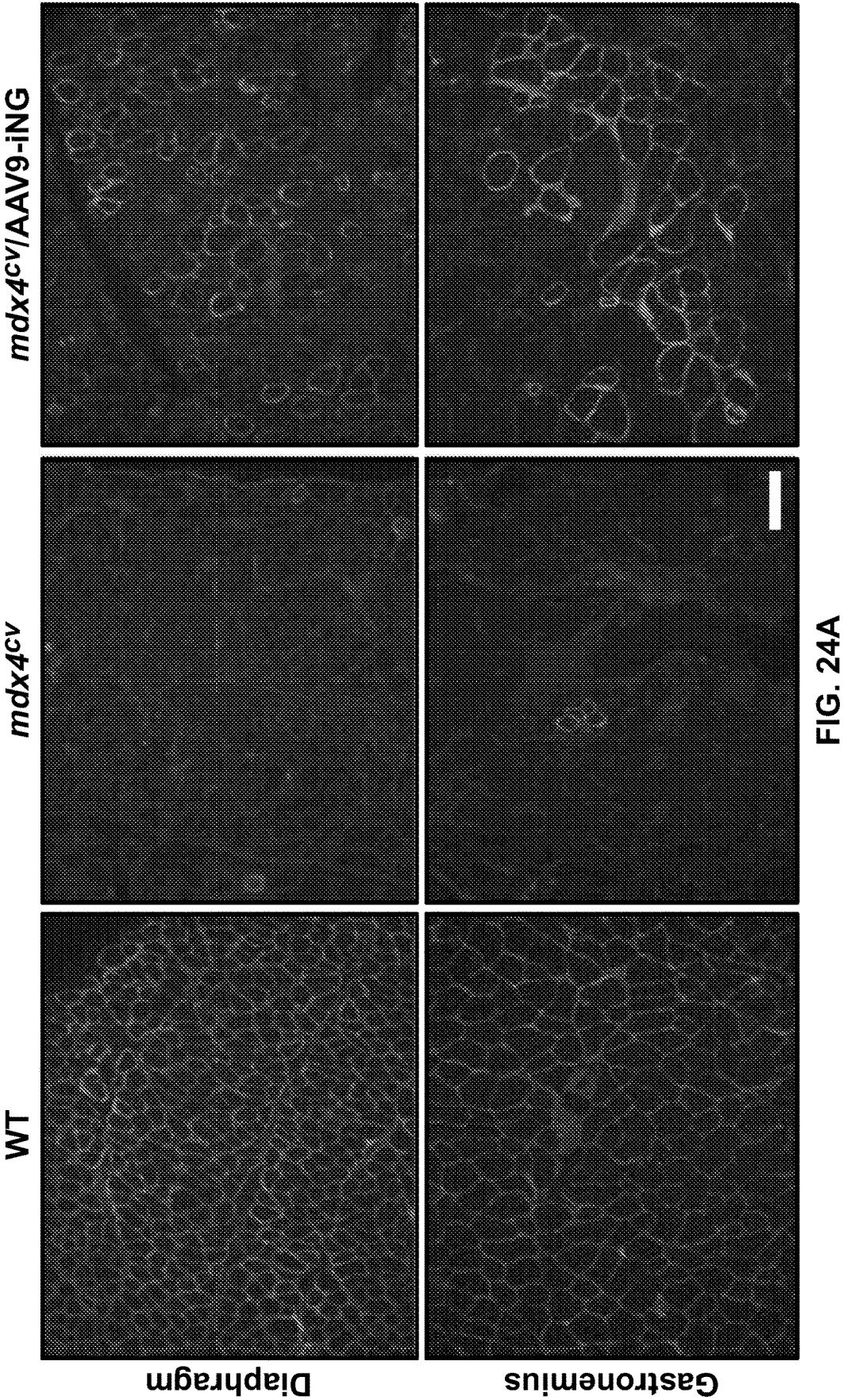


FIG. 24A

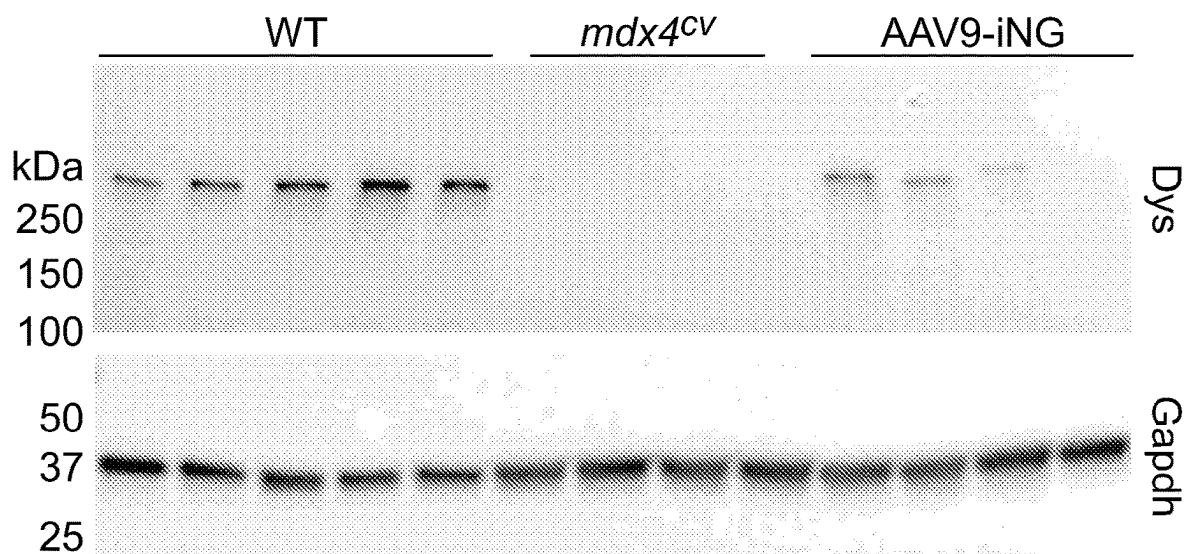


FIG. 24B

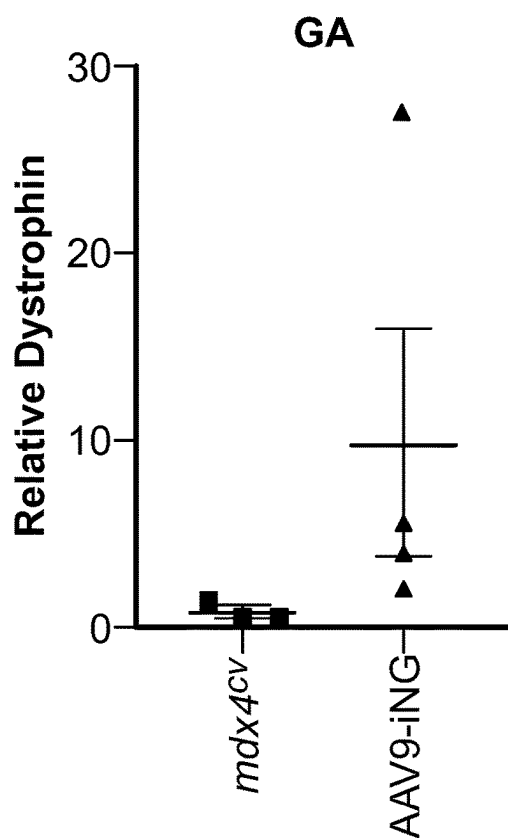


FIG. 24C

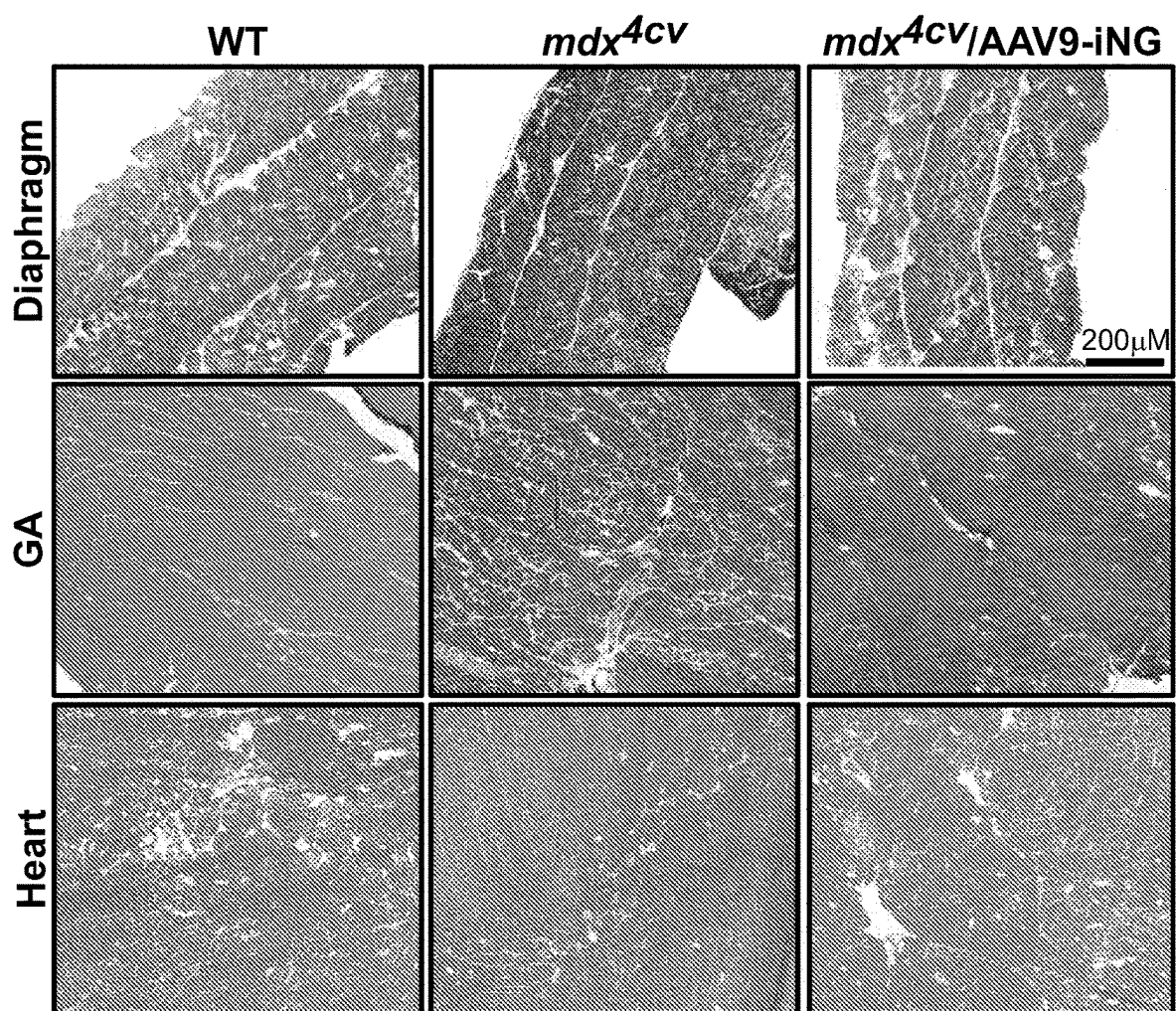


FIG. 25A

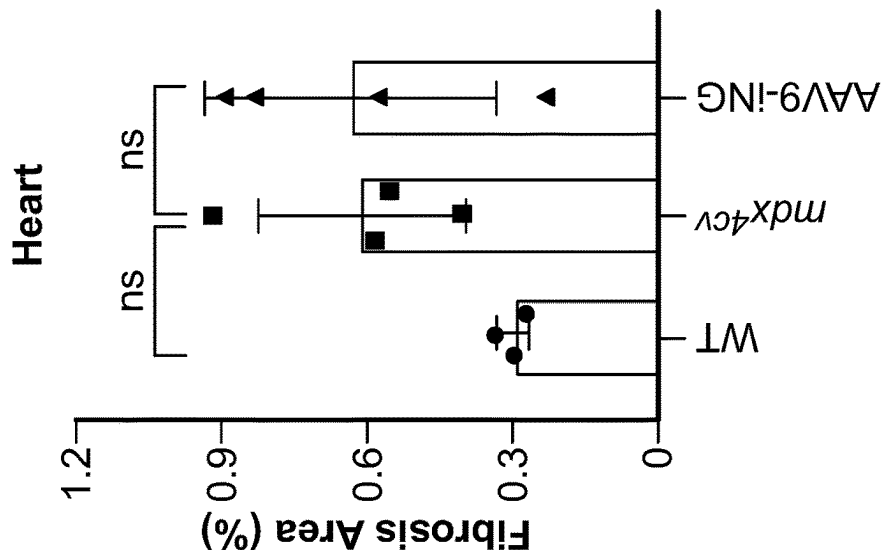


FIG. 25D

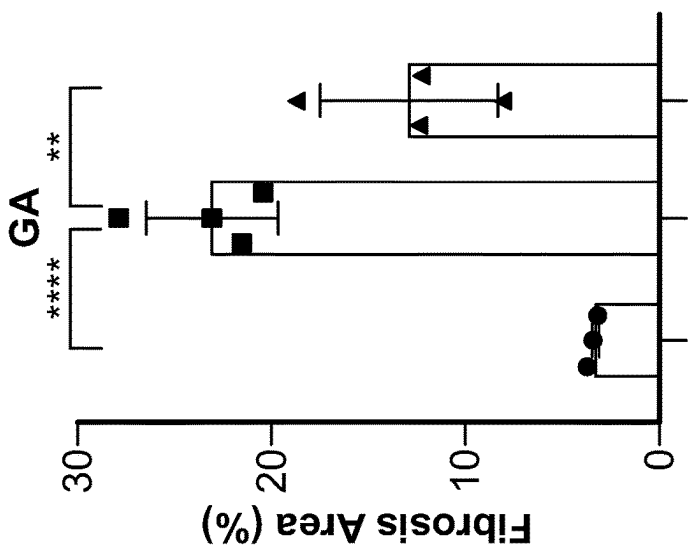


FIG. 25C

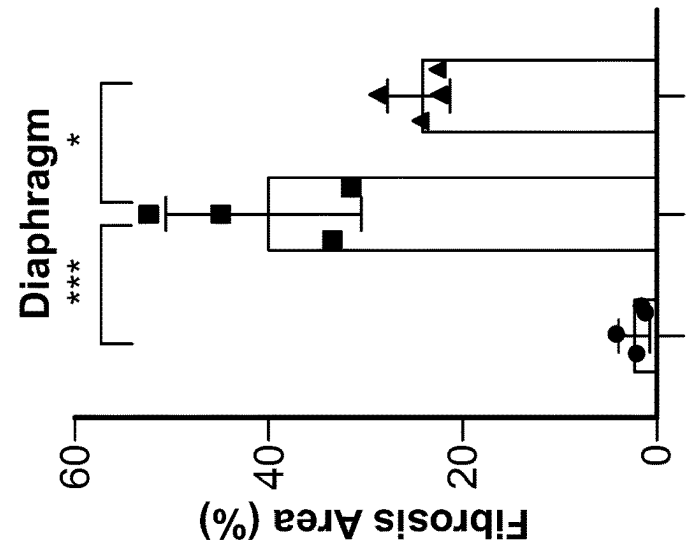
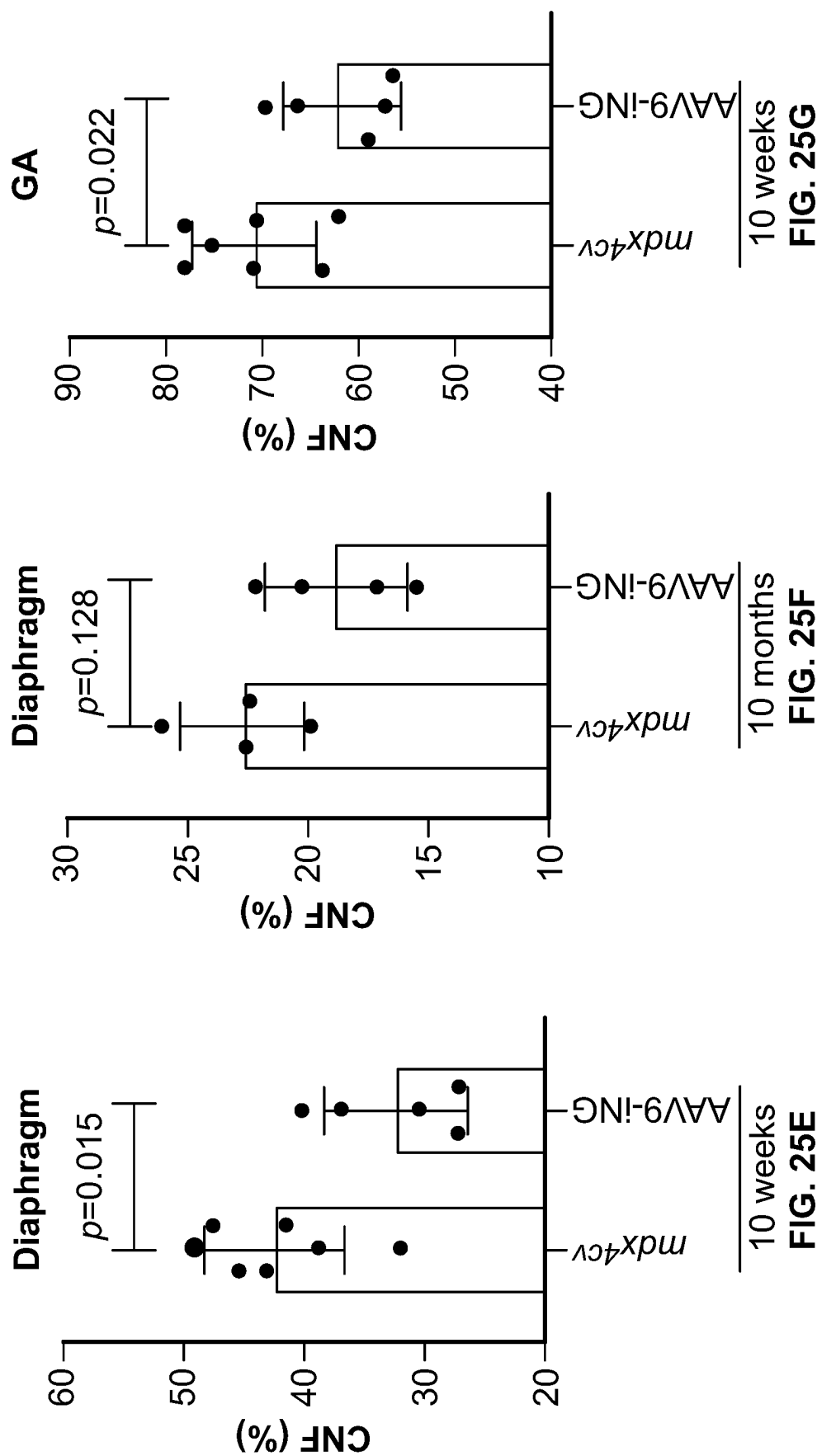


FIG. 25B



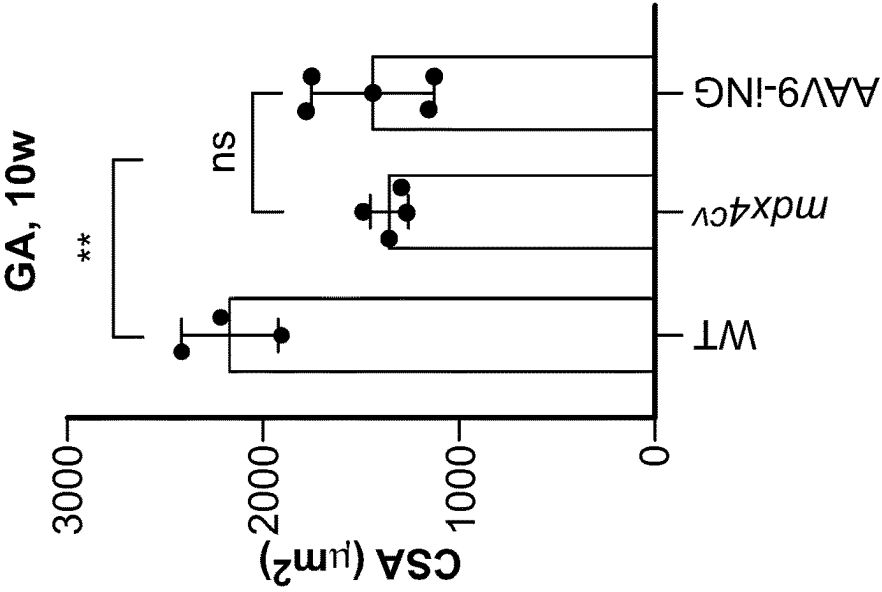


FIG. 25J

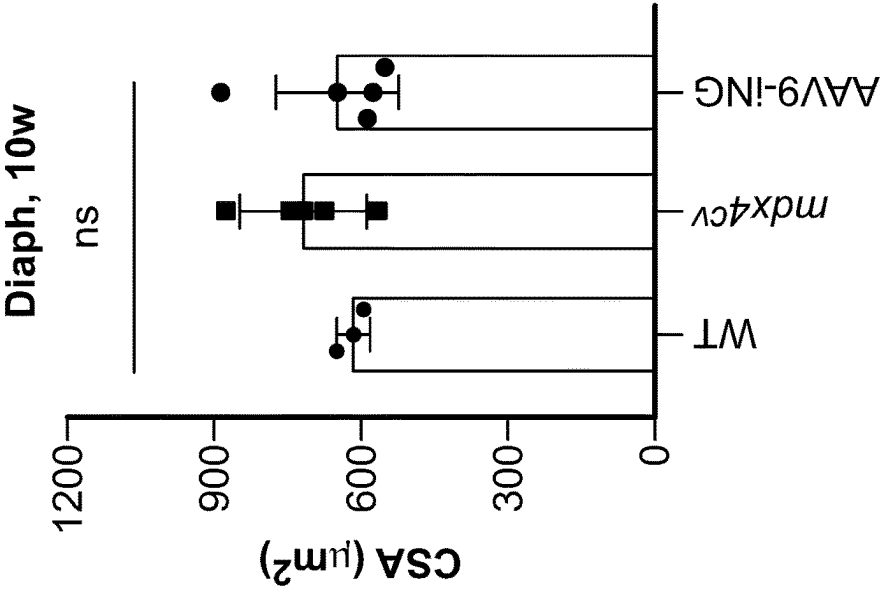
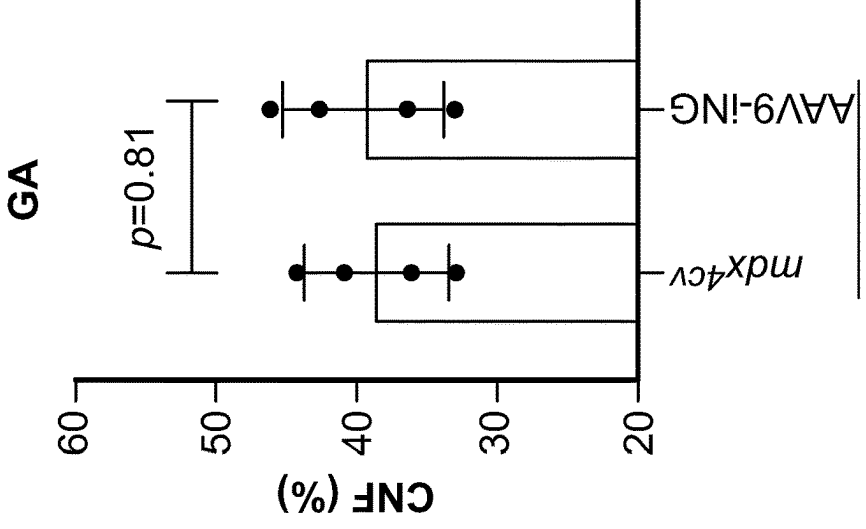


FIG. 25I



10 months
FIG. 25H

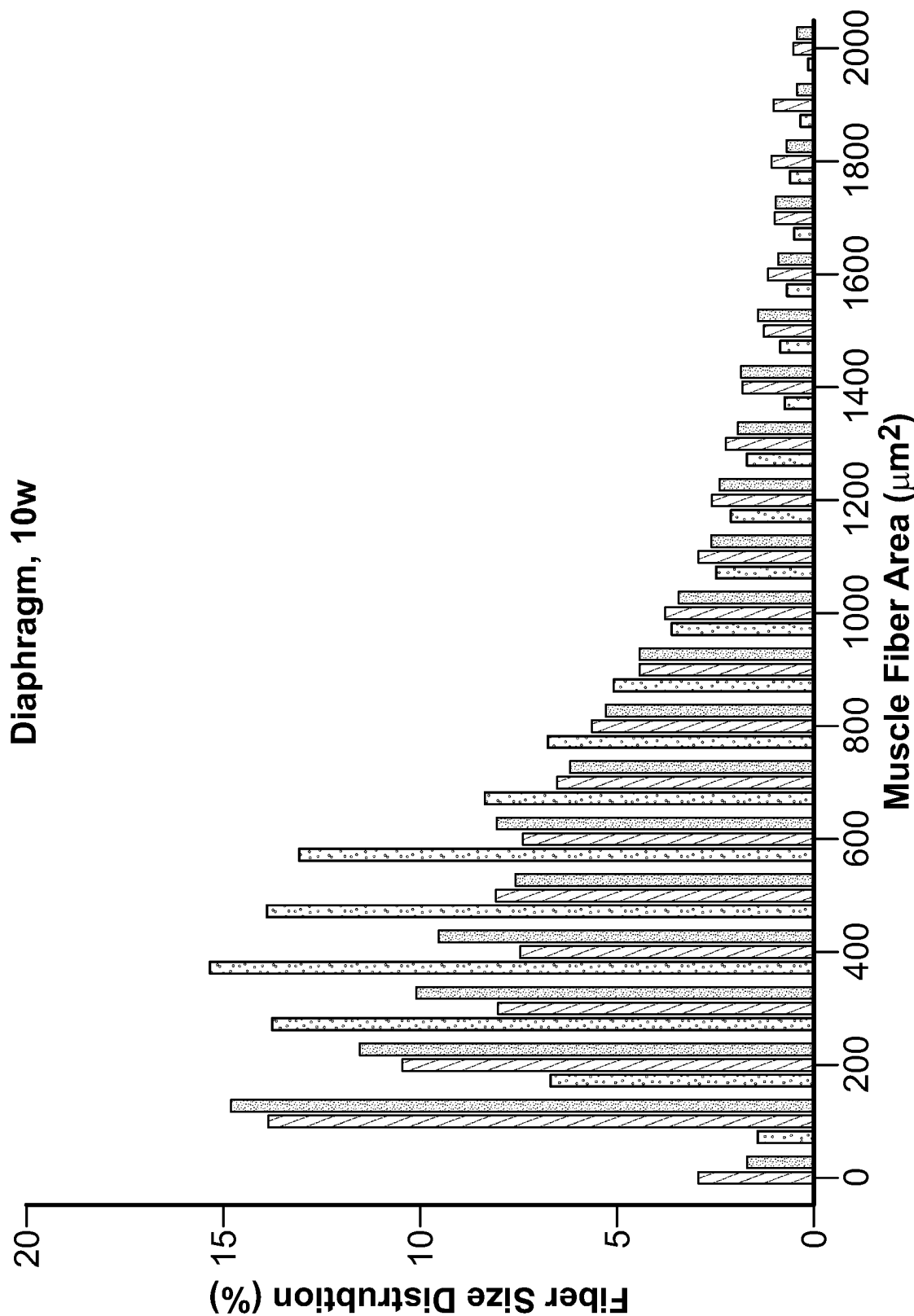


FIG. 25K

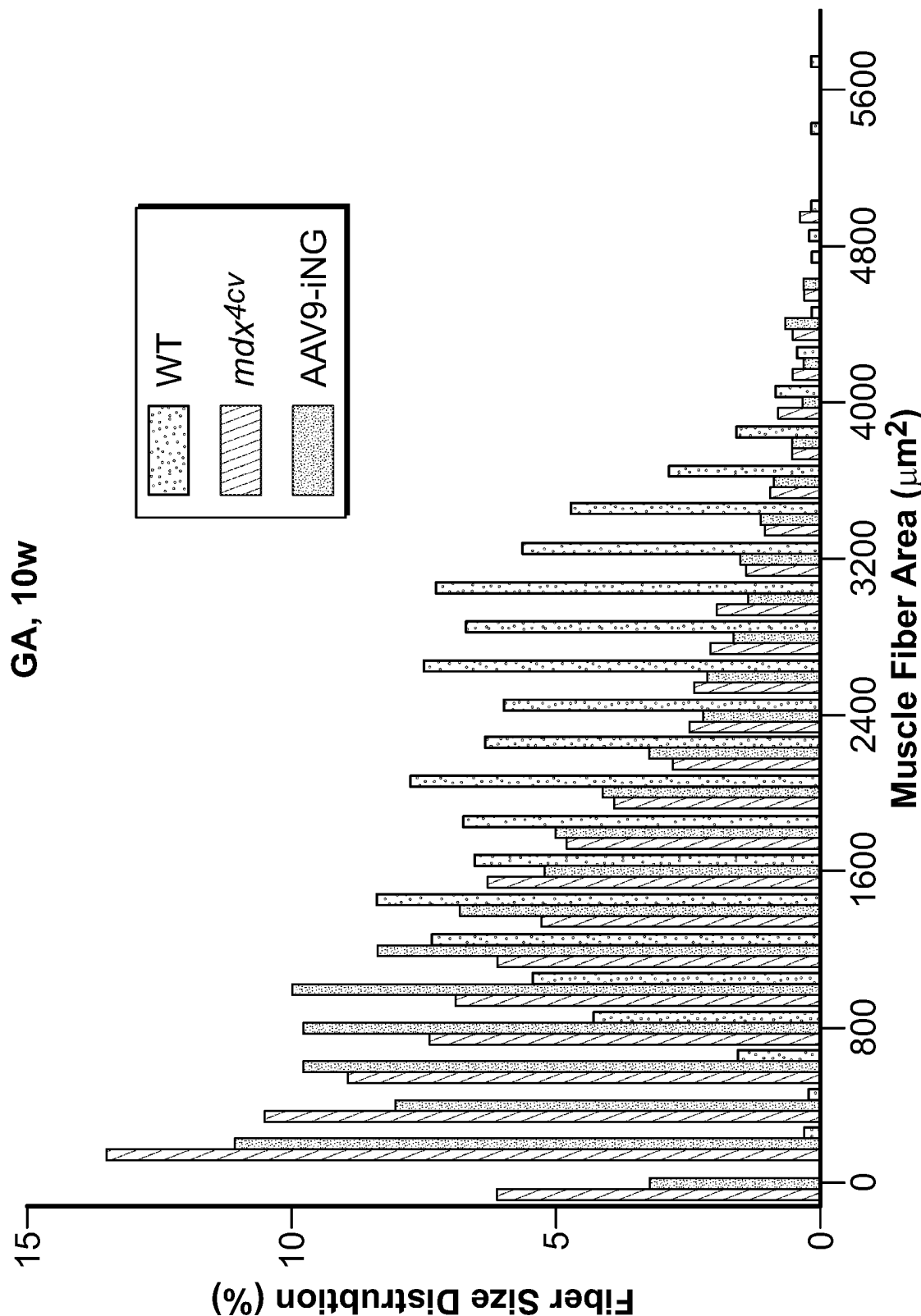


FIG. 25I

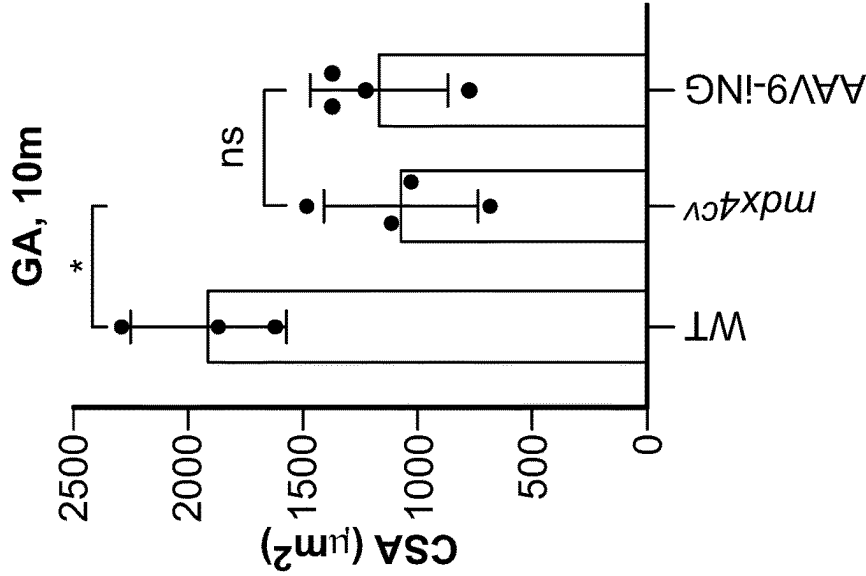


FIG. 26B

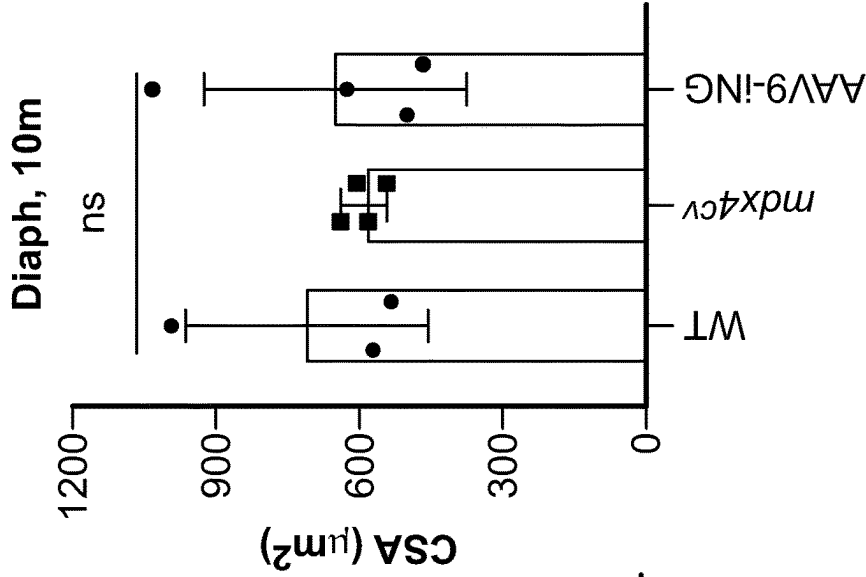


FIG. 26A

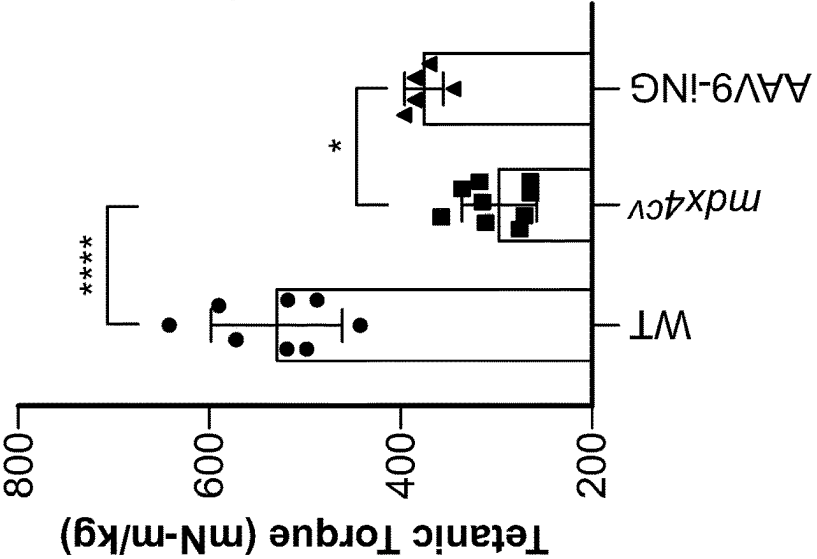


FIG. 25M

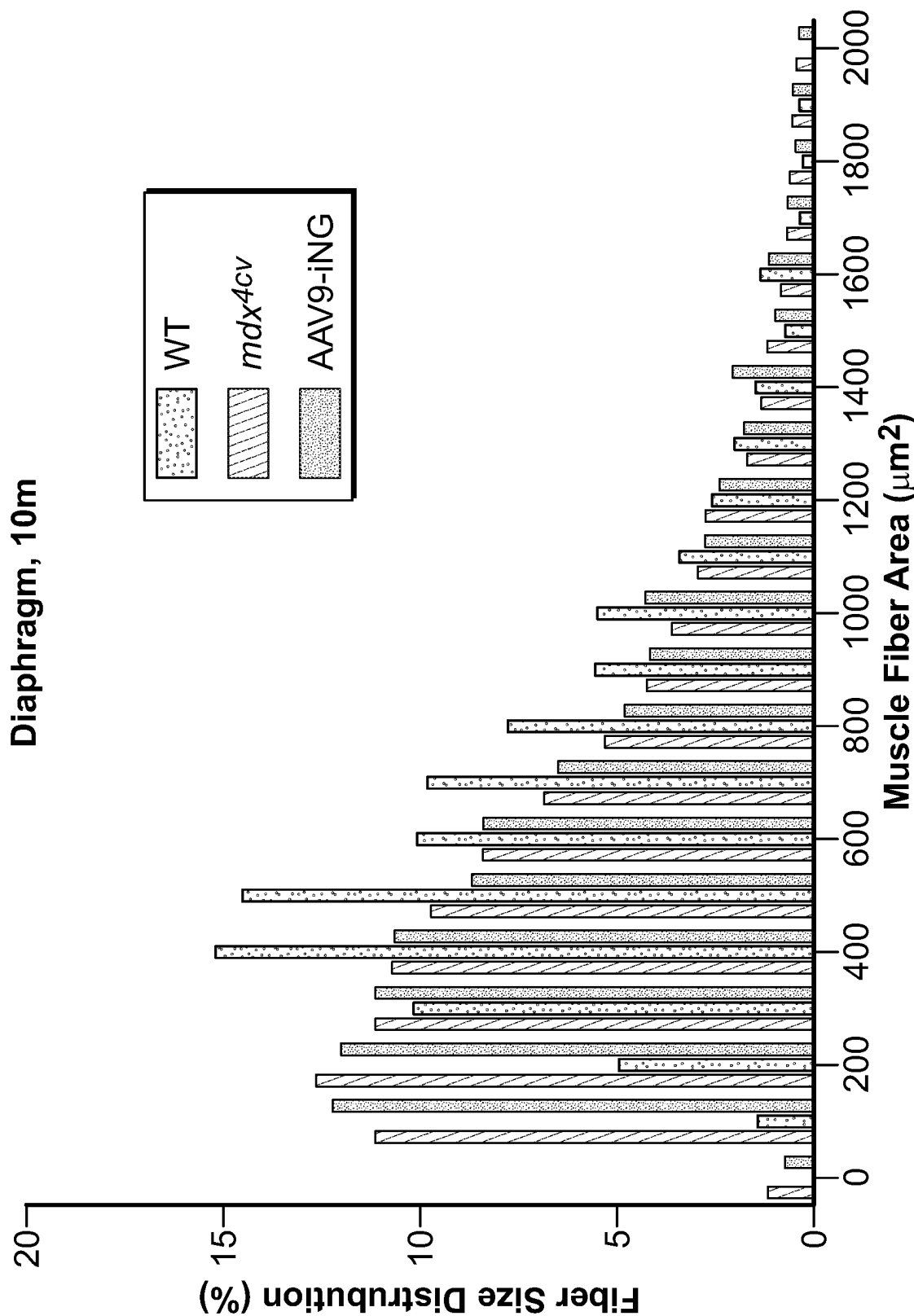


FIG. 26C

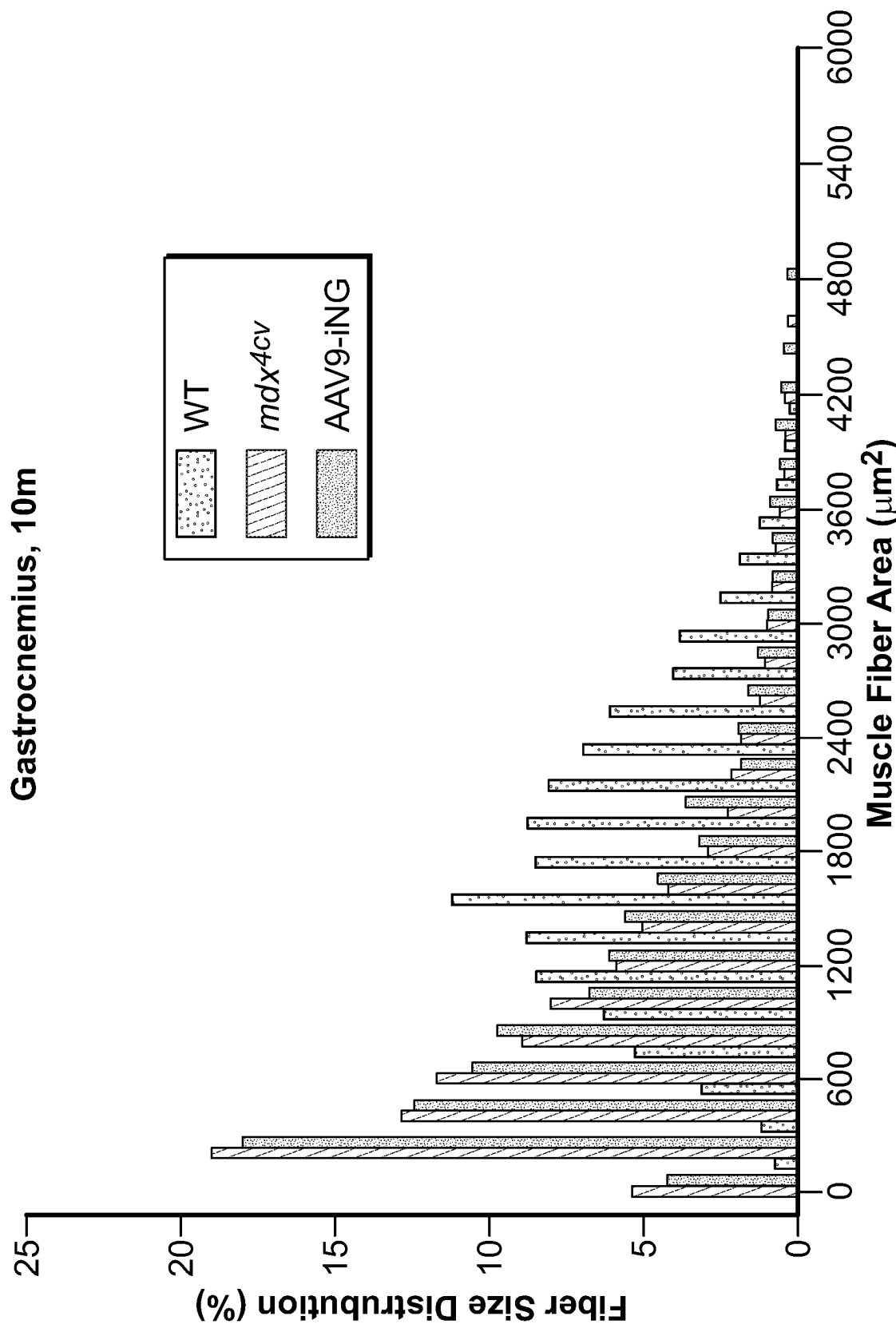
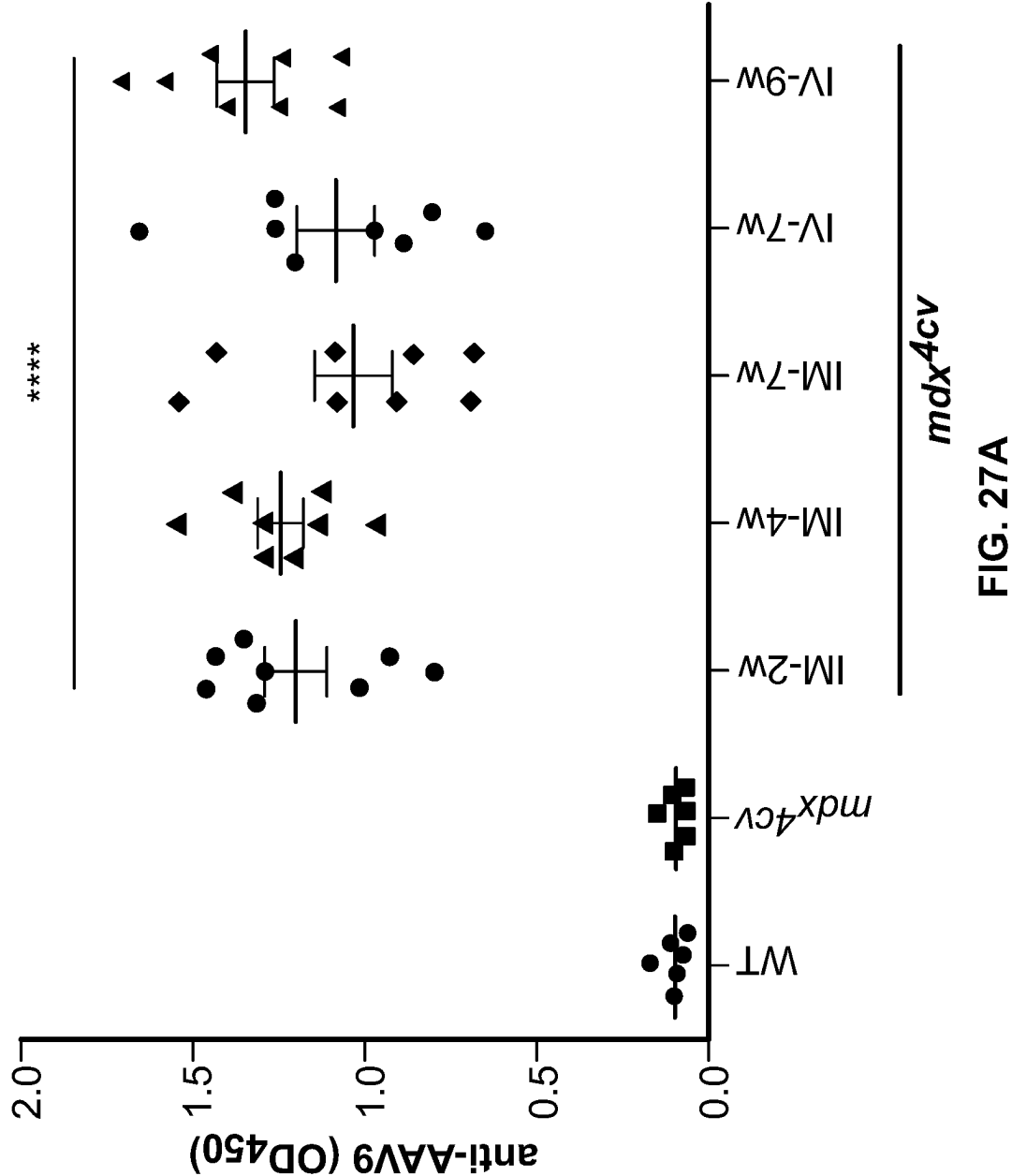


FIG. 26D



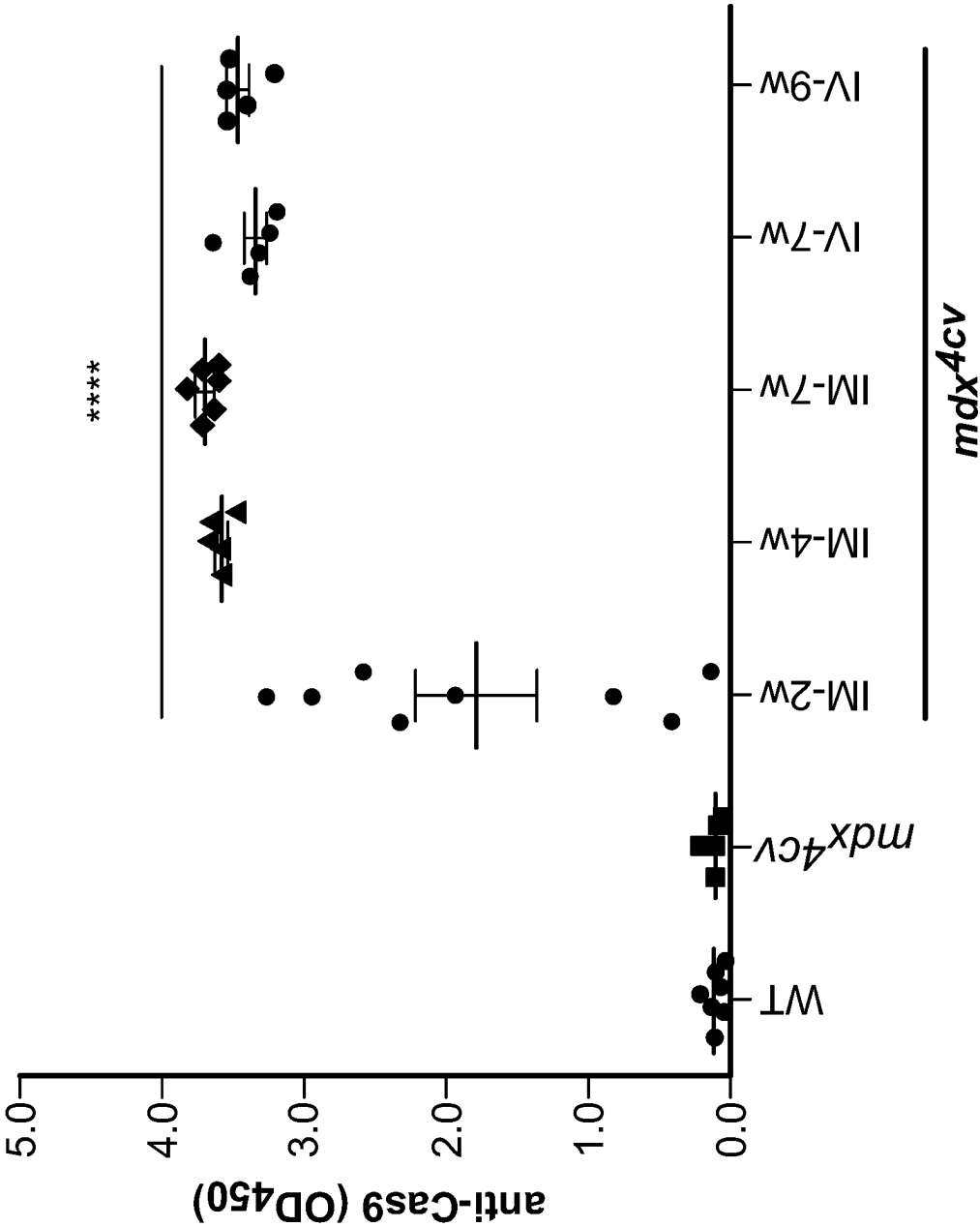


FIG. 27B

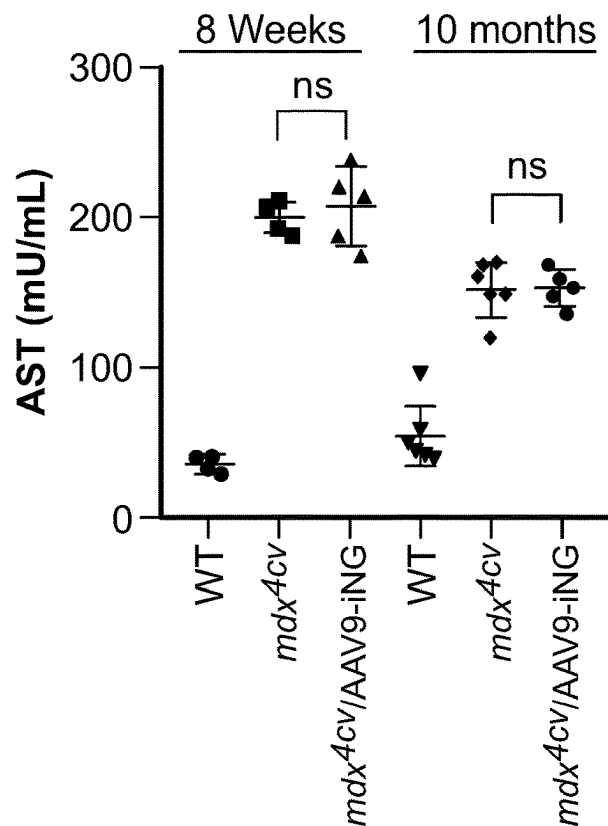


FIG. 27C

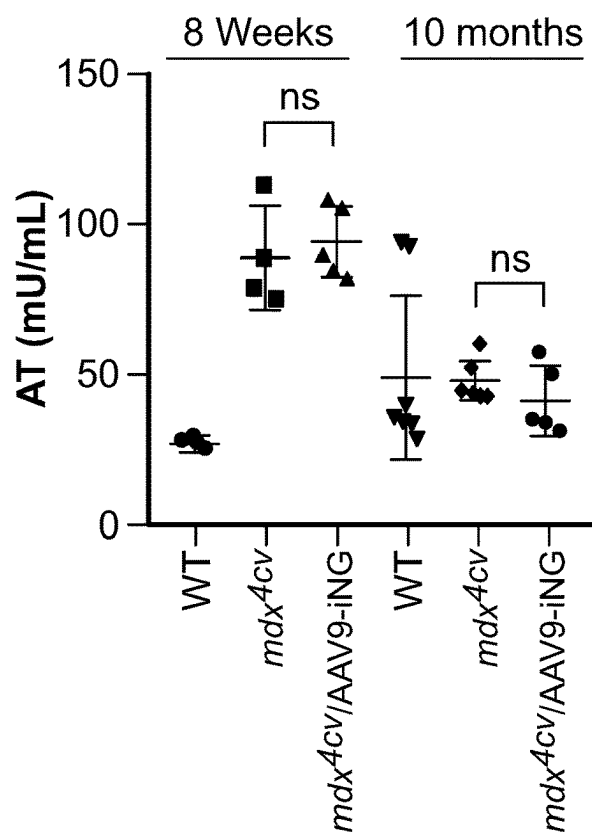


FIG. 27D

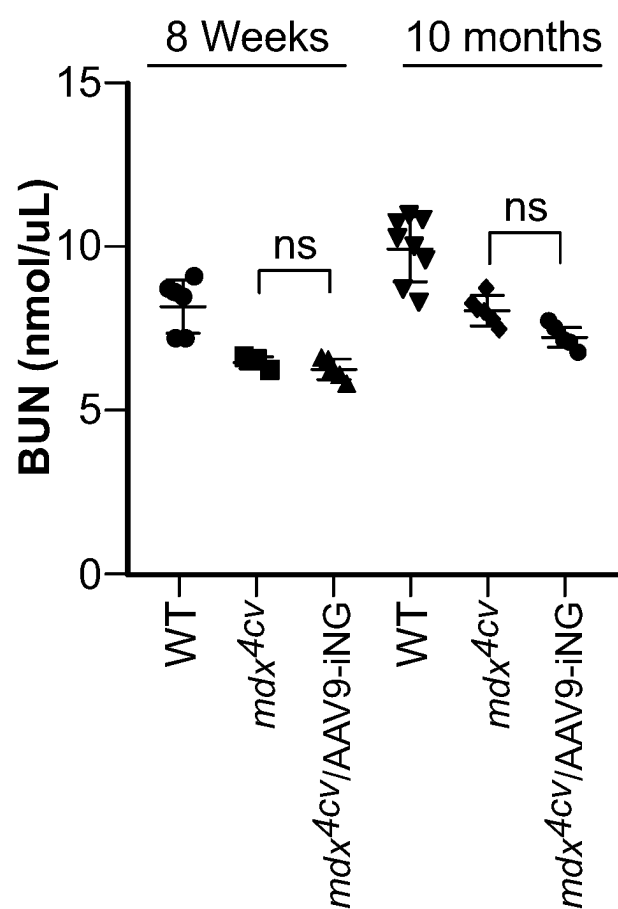


FIG. 27E

Mismatches	Number of Targets
1	1
2	2
3	55
4	562

ChrX: GTT**A**TCTCCTGTTCTGCAGC **TGT**
Chr16: GTT**A**TCTCCTG**C**TCTGCAGC **AGA**
Chr1: G**A**T**A**TCTCCTGTTCTGCAG**G** **AGA**

FIG. 27F

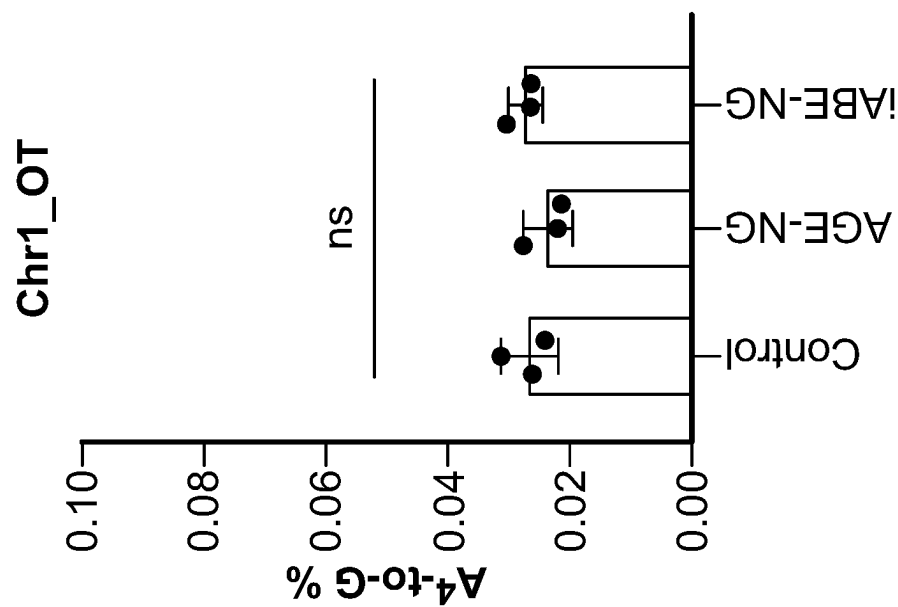


FIG. 27H

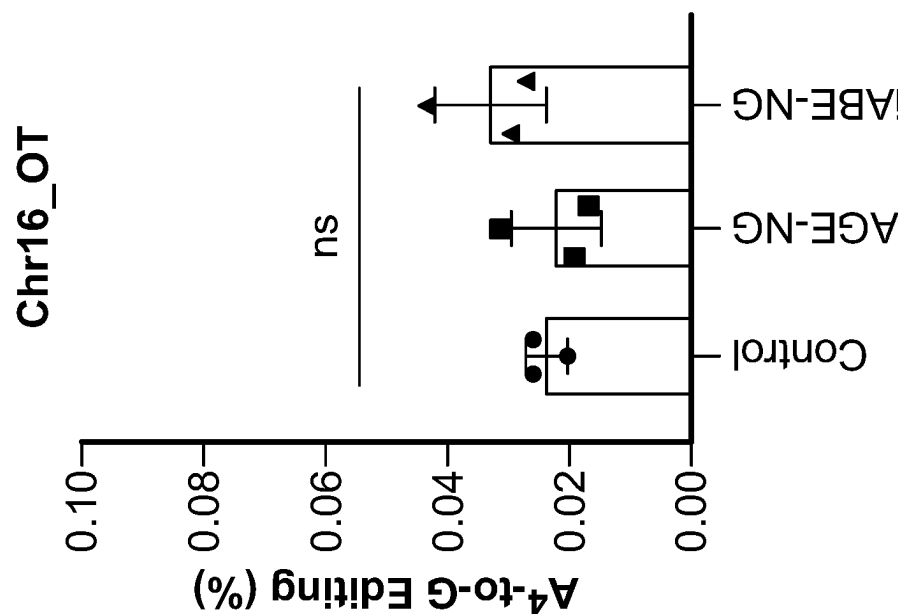
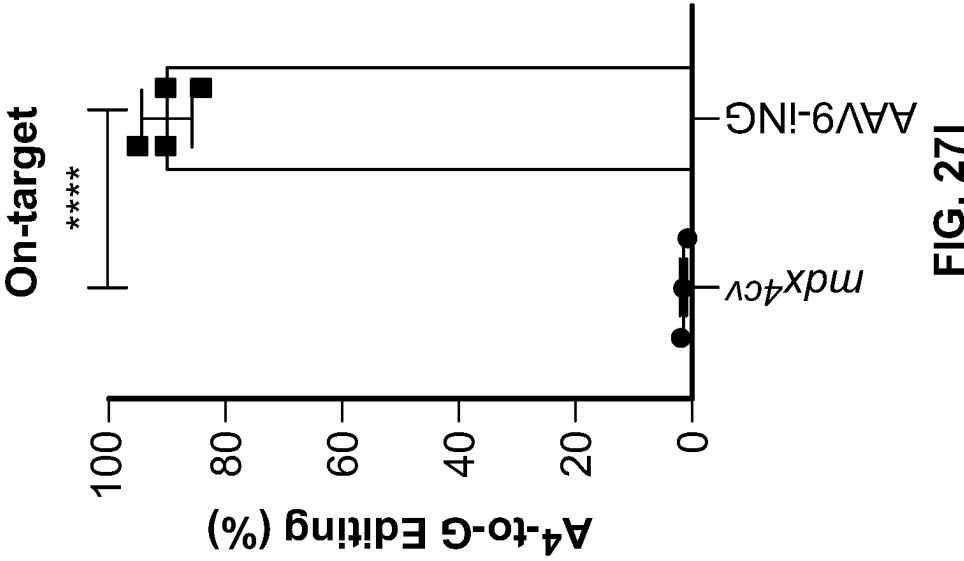
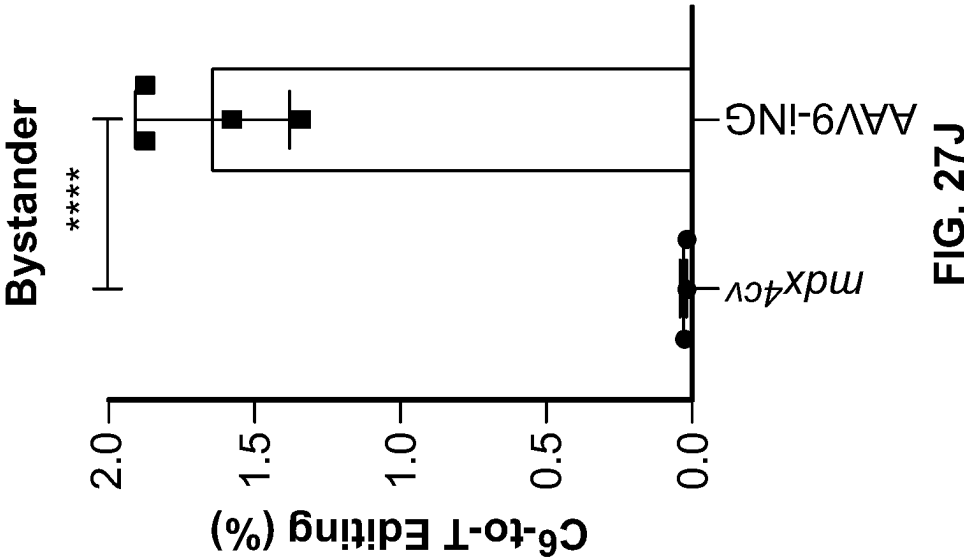


FIG. 27G



	G	T	T	A ⁴	T	C ⁶	T	C	C	T	G	T	T	C	T	G	C	A	G	C	T	G	T
Mouse1																							
A	0.05	0.01	0.00	13.64	0.01	0.03	0.03	0.05	0.02	0.02	0.02	0.01	0.01	0.00	0.03	0.08	0.01	99.90	0.02	0.01	0.01	0.03	0.01
C	0.00	0.03	0.07	0.01	0.03	97.93	0.04	99.86	99.96	0.09	0.00	0.02	0.03	99.96	0.11	0.00	99.96	0.00	0.00	99.97	0.07	0.00	0.02
G	99.95	0.00	0.00	86.34	0.00	0.19	0.00	0.02	0.00	0.01	99.97	0.00	0.00	0.00	0.00	99.92	0.00	0.08	99.97	0.00	0.05	99.96	0.00
T	0.00	99.96	99.92	0.02	99.95	1.84	99.93	0.06	0.02	99.89	0.00	99.97	99.97	0.03	99.86	0.00	0.03	0.01	0.02	0.02	99.87	0.01	99.97

	G	T	T	A ⁴	T	C ⁶	T	C	C	T	G	T	T	C	T	G	C	A	G	C	T	G	T
Mouse2																							
A	0.07	0.01	0.00	14.04	0.02	0.04	0.03	0.04	0.03	0.02	0.04	0.01	0.01	0.01	0.03	0.07	0.01	99.90	0.04	0.01	0.01	0.03	0.01
C	0.00	0.03	0.06	0.00	0.02	98.27	0.04	99.90	99.92	0.09	0.00	0.02	0.03	99.97	0.14	0.00	99.94	0.01	0.00	99.97	0.09	0.00	0.02
G	99.92	0.00	0.00	85.93	0.00	0.21	0.00	0.00	0.03	0.01	99.96	0.00	0.00	0.00	0.01	99.93	0.00	0.08	99.94	0.00	0.01	99.96	0.00
T	0.00	99.95	99.94	0.02	99.96	1.48	99.92	0.05	0.02	99.88	0.00	99.97	99.96	0.02	99.82	0.00	0.04	0.02	0.02	0.02	99.89	0.01	99.97

FIG. 28

Mouse3															
	G	T	T	T	A ⁴	T	C ⁶	T	C	C	T	G	T	T	
A	0.08	0.01	0.01		7.94	0.02	0.04	0.03	0.02		0.04	0.02	0.03	0.02	
C	0.00	0.06	0.06	0.01	0.01	0.05	97.74	0.07	99.97	99.94	0.09	0.00	0.02	0.03	
G	99.91	0.00	0.00		92.03	0.00	0.36	0.01	0.00			99.97	0.00	0.00	
T	0.00	99.93	99.93	0.02	0.02	99.93	1.86	99.89	0.01	0.02	99.89	0.00	99.96	99.95	
Mouse4															
	G	T	T	T	A ⁴	T	C ⁶	T	C	C	T	G	T	T	
A	0.10	0.01	0.00	0.00	19.63	0.02	0.03	0.04	0.03	0.04	0.02	0.05	0.01	0.01	
C	0.00	0.04	0.06	0.01	0.01	0.02	98.51	0.06	99.93	99.94	0.10	0.00	0.03	0.04	
G	99.89	0.00	0.00		80.32	0.01	0.19	0.00	0.00	0.00	0.01	99.95	0.00	0.00	
T	0.00	99.95	99.94	0.04	0.04	99.96	1.28	99.89	0.04	0.02	99.87	0.00	99.96	99.95	

FIG. 28
(Cont'd)

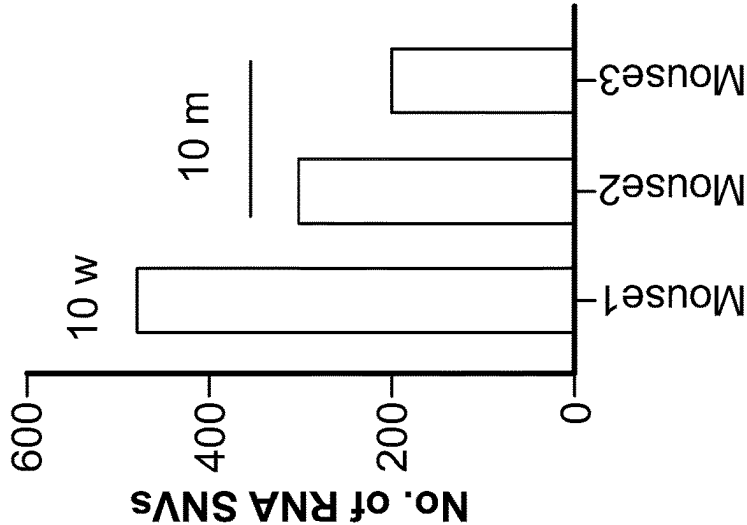


FIG. 29A

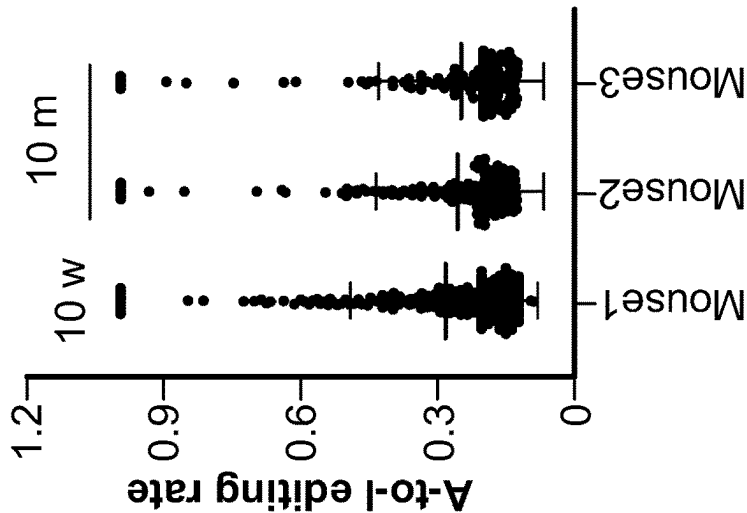


FIG. 29B

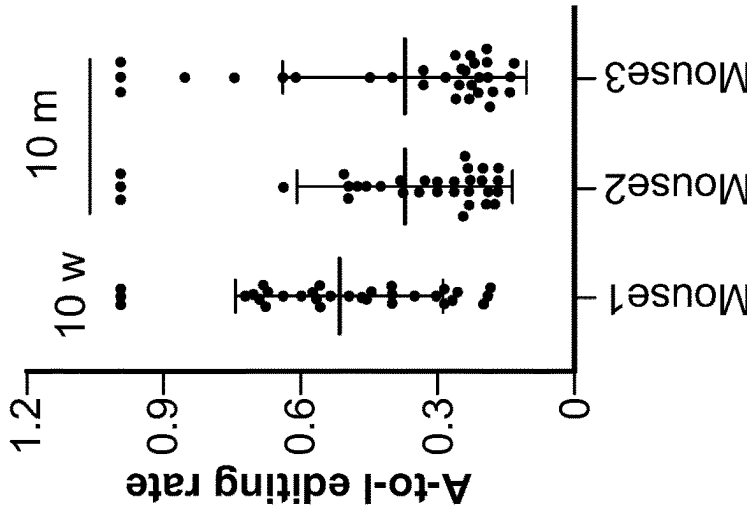


FIG. 29C

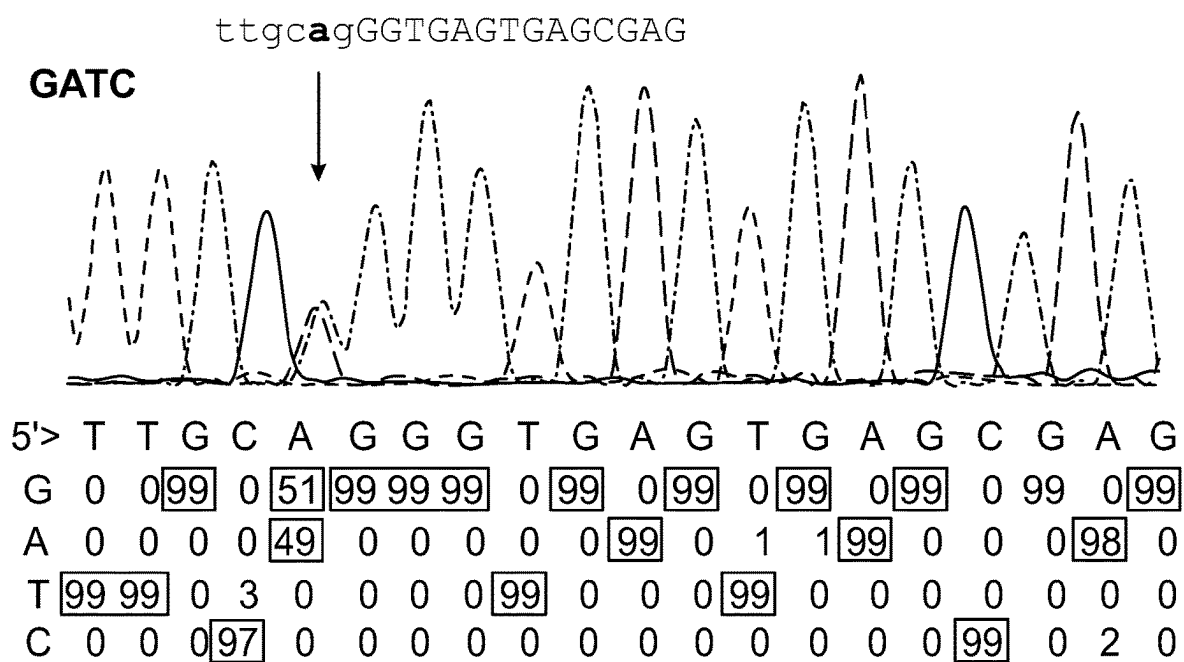


FIG. 30A

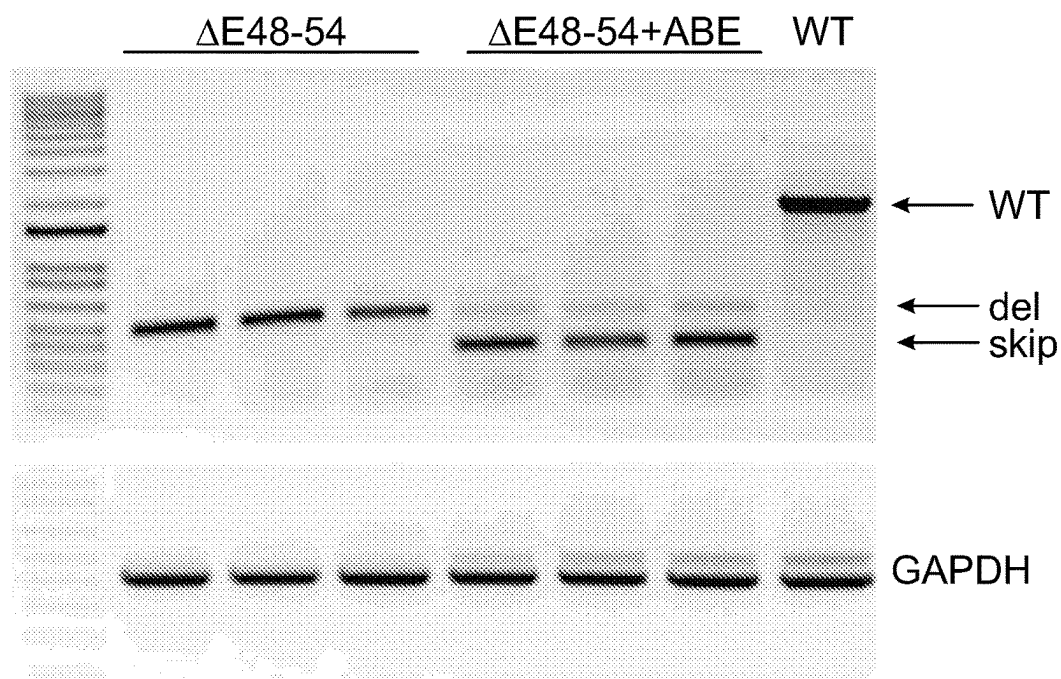


FIG. 30B

SMN1 Exon7: GGT⁶TTT**C**AGACAAAATCAAAAAGAAAGGAGGTGCT**C**ACATTCCTTAAATTAAAGGA³⁶
SMN2 Exon7: GGT⁶TTT**T**AGACAAAATCAAAAAGAAAGGAGGTGCT**C**ACATTCCTTAAATTAAAGGA
gRNA1; GTGCTC**A**CATTCCCTTAAATTAA **AGG**
gRNA2; gGCTC**A**CATTCCCTTAAATTAA **GGA**

FIG. 31A

	ABEmax	iABE-NGA	ABEmax	iABE-NGA
Control				
	gRNA1	gRNA1	gRNA2	gRNA2

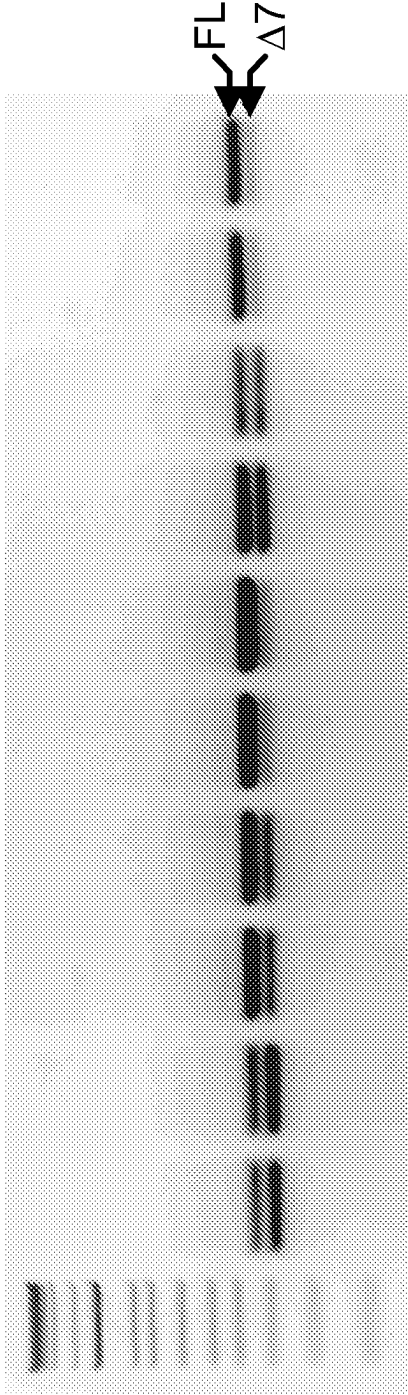


FIG. 31B

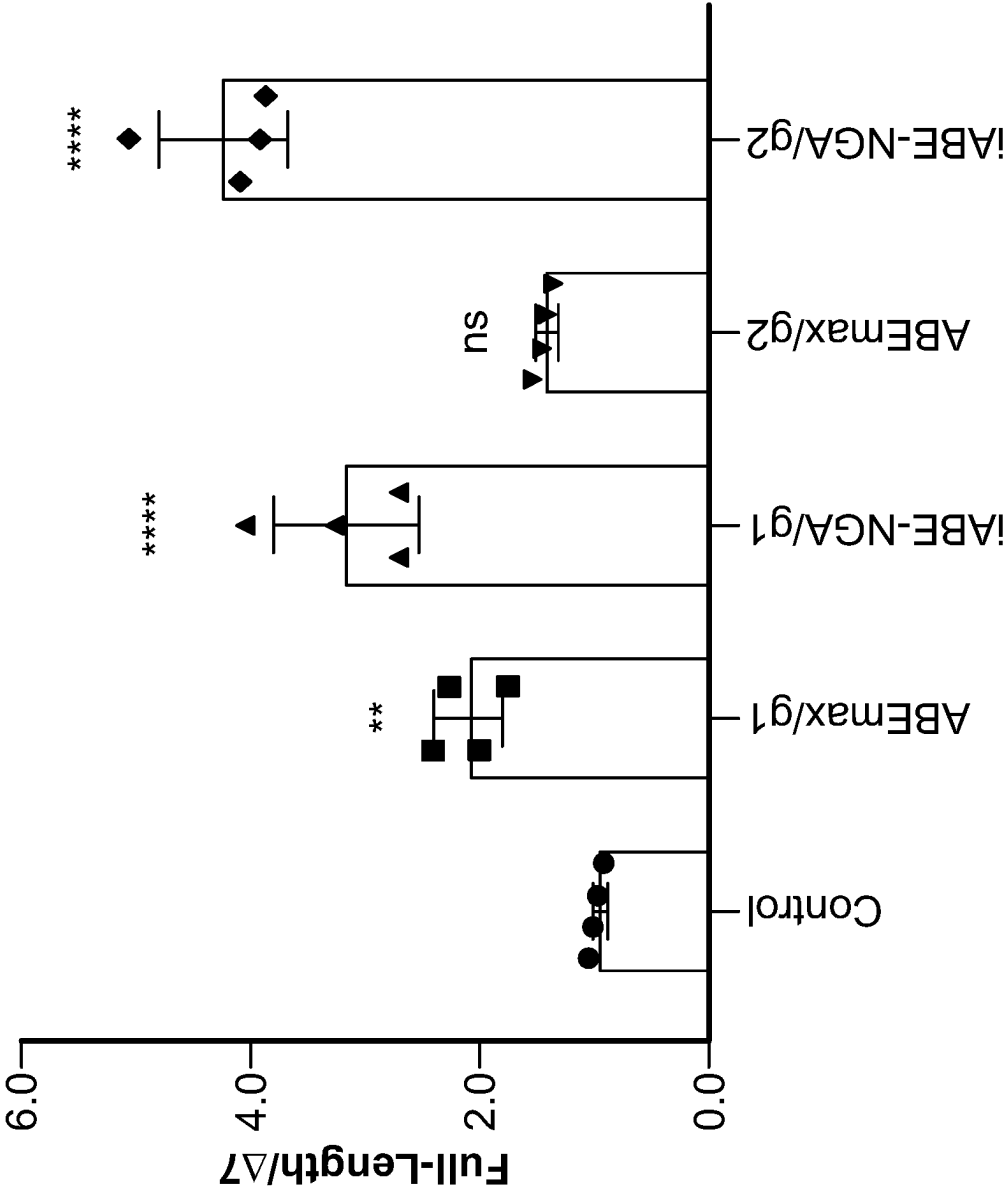
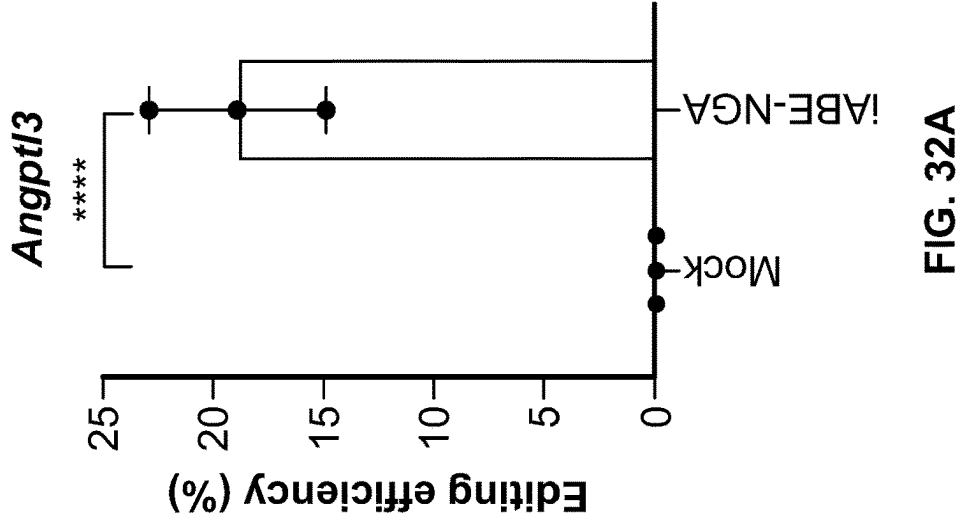
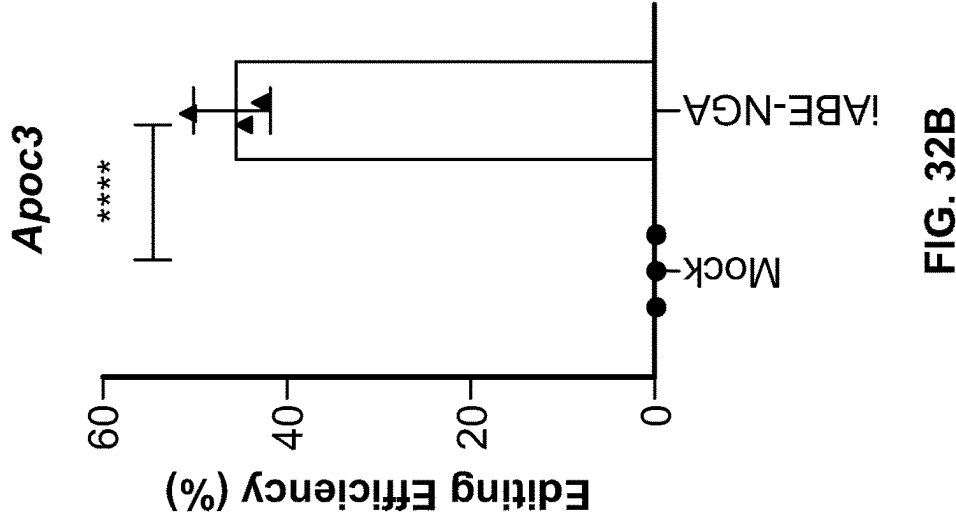
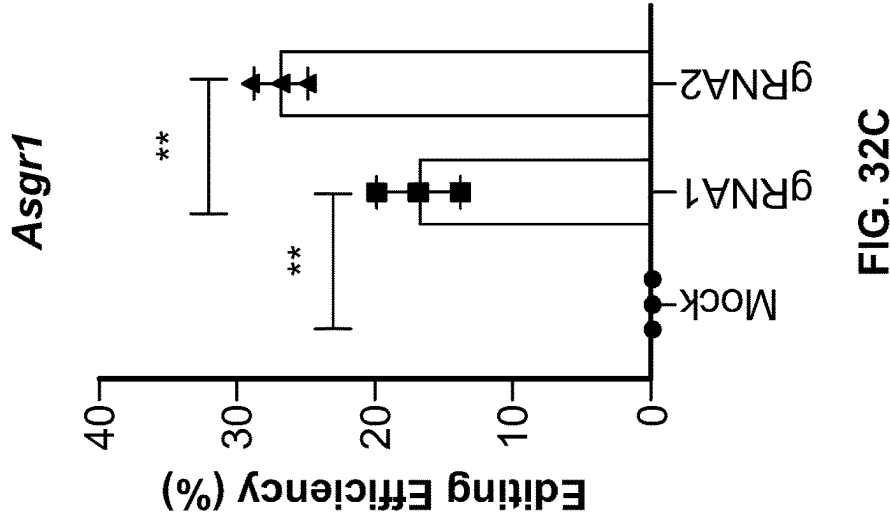


FIG. 31C



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NUCLEOBASE EDITORS AND METHODS OF USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a national stage application filed under 35 U.S.C. § 371 of PCT/US2021/017868, filed Feb. 12, 2021, which claims the benefit of U.S. Provisional Application No. 62/976,590, filed Feb. 14, 2020, which are expressly incorporated herein by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH

This invention was made with government support under Grant No. R01 HL116546 awarded by the National Institutes of Health. The government has certain rights in the invention.

FIELD

The present disclosure relates to nucleobase editors and methods of use thereof.

BACKGROUND

Many genetic diseases arise from single nucleotide alterations. Duchenne muscular dystrophy (DMD) is a fatal genetic muscle disease affecting approximately 1 in ~5000 male births worldwide, which is caused by mutations in the DMD gene. Most of the DMD mutations are due to deletions or duplications with over 500 point mutations accounting for ~10% of the cases. DMD codes for the dystrophin protein, a cytoskeletal protein that functions in the muscle force transmission and sarcolemmal stability of muscle fibers. Loss of dystrophin leads to progressive muscle weakness and wasting, loss of ambulation, respiratory impairment, cardiomyopathy, and eventual death. Previous studies showed that exon deletion through CRISPR genome editing can restore dystrophin expression and function. Although promising, this strategy has safety concerns as it relies on repair of the double strand DNA break (DSB) created by CRISPR/Cas9, which can cause unwanted large deletion and even chromosomal rearrangement.

Therefore, what is needed are compositions for precise correction of disease-associated mutations.

SUMMARY

Disclosed herein are fusion proteins, systems, and compositions for editing disease-associated mutations and methods of use thereof.

In some aspects, disclosed herein is a fusion protein comprising a Cas9 nickase and a nucleotide deaminase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

In some embodiments, the first amino acid substitution is selected from the group consisting of A262T, R324L, S409I, E480K, E543D, M694I, and E1219V when compared to SEQ ID NO: 11. In some embodiments, the second amino acid substitution is selected from the group consisting of

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L1111R, D1135V, G1218R, E1219F, A1322R, R1335V, R1335Q, R1335E, and T1337R when compared to SEQ ID NO: 11.

In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

In some embodiments, the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.

In some embodiments, the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.

In some embodiments, the nucleotide deaminase is a cytidine deaminase or an adenine deaminase. In some embodiments, the adenine deaminase is a dimeric adenine deaminase or a monomeric adenine deaminase domain. In some embodiments, the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47. In some embodiments, the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 22 or 23.

In some aspects, disclosed herein is a system for base editing comprising:

a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase;

wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-573 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20 and the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

In some embodiments, the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).

In some aspects, disclosed herein is a composition comprising:

a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

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wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ. ID NO: 11.

In some embodiments, the first nucleotide sequence comprises the sequence as set forth in SEQ ID NO: 3 or SEQ ID NO: 5. In some embodiments, the second nucleotide sequence comprises the sequence as set forth in SEQ ID NO: 4 or SEQ ID NO: 6.

In some aspects, disclosed herein is a method of treating a genetic disease or disorder in a subject, comprising administering a therapeutically effective amount of the composition of any preceding aspect. In some embodiments, the genetic disease or disorder is Duchenne muscular dystrophy, dysferlinopathy, or spinal muscular atrophy. In some embodiments, the genetic disease or disorder is due to a point mutation in a causative gene from a G:C pair to an A:T pair.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying figures, which are incorporated in and constitute a part of this specification, illustrate several aspects described below.

FIGS. 1a-1e show in vitro studies of mdx^{4cv} mutation correction using ABE-NG. FIG. 1a shows genomic DNA, encoded amino acids and guide RNA with PAM (highlighted in blue) sequences at the stop codon mutation site (red). FIG. 1b shows the reporter construct containing a puromycin resistance cassette fused with E2A peptide, mdx^{4cv} target sequence and ATG-removed, EGFP. Correction of the stop codon within the target sequence would allow EGFP expression. FIG. 1c shows fluorescence microscopy images of HEK293 cells transfected with reporter alone, or reporter, gRNA and one of the base editors (ABEmax, ABE-x and ABE-NG). Scale bar: 500 μ m. FIGS. 1d-1e shows flow cytometry analysis of EGFP expression in HEK293 cells transfected as described in FIG. 1c. * $p < 0.05$; **** $p < 0.0001$ (one-way ANOVA). The sequences in FIG. 1 are GAACAGCTGCAGAACAGGAGATAACAG (SEQ ID NO: 620), GTTASTCTCCTGTTCTG CAGCTGT (SEQ ID NO: 621).

FIGS. 2a-2f show optimization of the PAM-interacting domain to improve the editing efficiency of ABE-NG at the NGN sites. FIGS. 2a-2f show measurements of the base editing efficiencies of different ABE variants on a NGG-PAM site and five different sites with NGH or GAT PAM. The gRNA sequences are listed on the top of each graph with the PAM sequences in red and the target adenines in blue ns, not significant; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$ (one-way ANOVA). The sequences in FIG. 2 are ATGACAGGCAGGGGCACCGCGG (SEQ ID NO: 622), GAGCGAGCAGCGTCTTCGAGAGT (SEQ ID NO: 623), GCAGACGGCA GTCAGTACGGGGG (SEQ ID NO: 624), GTCGACGAGCAGCTTTCTCTAGA (SEQ ID NO: 625), GGGAGAGCTGGGTGAATGGAGCGA (SEQ ID NO: 626), GATCCAGGT GCTGCAGAAGGGAT (SEQ ID NO: 627).

FIGS. 3a-3d show re-engineering of the adenine deaminase domain to improve the efficiency and specificity of ABE-NG. FIG. 3a shows schematics of the adenine deaminase domain used in ABE-NG editors. FIG. 3b shows quantification of the editing efficiency of different ABE-NG variants with modified TadA* domain at the mdx^{4cv} target site. **** $p < 0.0001$ (one-way ANOVA). FIG. 3c shows the number of off-target RNA editing events in Neuro-2a cells transfected with different ABE-NG variants. *** $p < 0.05$; **** $p < 0.001$ (one-way ANOVA). FIG. 3d shows quantifica-

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tion of the off-target RNA editing (A-to-I) activities on four RNA adenines previously identified as being efficiently modified by ABEmax in HEK293 cells. **** $p < 0.0001$ (one-way ANOVA).

FIGS. 4a-4b show comparison of the editing efficiency of different ABEs at the mdx^{4cv} target site (FIG. 4a) and human dysferlin Q605X site (FIG. 4b). All base editors carry the SpCas9-NG and differ at the adenine deaminase domain. ns, not statistically significant; ** $p < 0.01$; **** $p < 0.0001$ (one-way ANOVA for FIG. 4a; two-way ANOVA for FIG. 4b). The sequence in FIG. 4 is GTTATCTCCTGTTCTGCAGC (SEQ ID NO: 570) and ATCCTACAGCATGGTGGCTG (SEQ ID NO: 590).

FIGS. 5a-5b show the intein split of ABEmax had relatively low editing activity. FIG. 5a shows schematics showing the two halves of intein-split ABEmax. The TadA-TadA* was fused with Npu intein N-terminal fragment and SpCas9 nickase (nSpCas9) was fused with Npu intein C-terminal fragment. FIG. 5b shows genomic DNA PCR analysis of HEK293 cells at 5 days after transfection with S2-gRNA and different versions of ABEs.

FIGS. 6a-6h show intein-mediated assembly of full-length ABE-NG. FIG. 6a shows schematics of the intein split ABE-NG. The N-terminal and C-terminal intein sequences reconstitute the full-length protein when co-expressed within cells. FIG. 6b shows Western blot analysis of HEK293 cell lysates transfected with different versions of ABEs. FIG. 6c shows fluorescence microscopy images of HEK293 cells transfected with reporter alone, or reporter, gRNA and one of the base editors (ABE-NG, split_v1 N+C or Split_v2 N+C). Scale bar: 500 μ m. FIG. 6d shows flow cytometry analysis of EGFP expression in HEK293 cells transfected as described in FIG. 6c. FIG. 6e shows Western blot analysis of HEK293 cell lysates transfected with full-length iABE-NGA, Gp41-1 or Npu split of iABE-NGA. FL, the full-length iABE-NGA band; N, the N-terminal fragment of the iABE-NGA. FIG. 6f shows densitometry quantification of the Western blot data shown in FIG. 6e. FIG. 6g shows the assembly efficiency of the Gp41-1 and Npu split of iABE-NGA (defined as the percentage of the full-length iABE-NGA bands). FIG. 6h shows quantification of the editing efficiency of full-length iABE-NGA, Gp41-1 split and Npu split of iABE-NGA at the mdx^{4cv} target site, Npu Split_1g is same as Npu Split except that only the C-terminal construct carries the gRNA. ns, not significant; * $p < 0.05$; *** $p < 0.001$; **** $p < 0.0001$; (one-way ANOVA).

FIG. 7. Immunofluorescence staining of heart and gastrocnemius muscle sections with anti-dystrophin antibody and DAPI. The mdx^{4cv} mice at 5 weeks of age received either 5×10^{13} , 1×10^{14} AAV9-iNG/ mdx^{4cv} -gRNA or 1×10^{14} vg/kg AAV9-iNG/non-targeting-gRNA (NT-gRNA) through tail vein injection, and sacrificed at 10 weeks of age for immunofluorescence staining analysis. Scale bar: 100 μ m.

FIGS. 8a-8k show widespread restoration of dystrophin expression in mdx^{4cv} mice following systemic administration of AAV9-iNG. FIG. 8a shows dystrophin and laminin- $\alpha 2$ co-immunostaining of heart sections from WT and mdx^{4cv} mice (10 weeks or 10 months of age) with or without tail vein injection of AAV9-iNG (a total of 1×10^{14} vg/kg, 1:1 of the N and C-terminal halt). Scale bar: 100 μ m. FIGS. 8b-8d show quantification of dystrophin-positive fibers in the heart (FIG. 8b), gastrocnemius (GA, FIG. 8c) and diaphragm (Diaph, FIG. 8d) muscles. 10 w, 10 weeks old; 10 m, 10 months old. ** $p < 0.01$; **** $p < 0.0001$ (Student's t-test). FIG. 8e shows Western blot analysis of heart homogenates from 10-week-old mice with anti-dystrophin, Cas9 and Gapdh antibodies. The WT muscle lysates were loaded

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at 5 $\mu\text{g}/\text{lane}$ while the $\text{mdx}^{4\text{cv}}$ muscle lysates were loaded at 25 $\mu\text{g}/\text{lane}$. FIG. 8f shows densitometry quantification of Western blot data shown in FIG. 8e. * $p < 0.05$ (one-way ANOVA). FIG. 8g shows Western blot analysis of heart homogenates from 10-month-old mice with anti-dystrophin and Gapdh antibodies. FIG. 8h shows densitometry quantification of Western blot data in FIG. 8g, ** $p < 0.01$; *** $p < 0.0001$ (one-way ANOVA). FIG. 8i shows representative sequencing trace of dystrophin transcripts of WT and $\text{mdx}^{4\text{cv}}$ mouse hearts (10 weeks or 10 months old) with or without AAV9-iNG treatment. FIGS. 8j and 8k show quantification of the targeted T-to-C editing efficiency in the $\text{mdx}^{4\text{cv}}$ mouse hearts (FIG. 8j, 10 weeks; FIG. 8k, 10 months) as assayed by sequencing of dystrophin transcripts. *** $p < 0.0001$ (Student's t-test).

FIG. 9 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections of a WT mouse at 10 weeks of age. Scale bars: 0.5 mm.

FIG. 10 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections of a control $\text{mdx}^{4\text{cv}}$ mouse at 10 weeks of age. Scale bars: 0.5 mm.

FIG. 11 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #142 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 12 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #147 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 13 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #148 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 14 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #149 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 15 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #150 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIGS. 16a-16b show immunofluorescence staining of dystrophin and laminin $\alpha 2$ in the gastrocnemius (FIG. 16a) and diaphragm (FIG. 16b) muscles from WT and $\text{mdx}^{4\text{cv}}$ (10 weeks of age) treated with or without tail vein injection of AAV9-iNG. Scale bar: 100 μm .

FIG. 17 shows Western blot analysis of gastrocnemius muscles from WT and $\text{mdx}^{4\text{cv}}$ (10 weeks of age) treated with or without tail vein injection of AAV9-iNG.

FIG. 18 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections of a WT mouse at 10 months of age. Scale bars: 0.5 mm.

FIG. 19 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections of a control $\text{mdx}^{4\text{cv}}$ mouse at 10 months of age. Scale bars: 0.5 mm.

FIG. 20 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #1976 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow, Scale bars: 0.5 mm.

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FIG. 21 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #1982 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 22 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #1985 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

FIG. 23 shows stitched large images showing dystrophin and laminin- $\alpha 2$ immunostaining of the entire heart sections $\text{mdx}^{4\text{cv}}$ mouse #1984 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow, Scale bars: 0.5 mm.

FIGS. 24a-24c show restoration of dystrophin expression in the skeletal muscles of 10-month-old $\text{mdx}^{4\text{cv}}$ mice after tail vein injection of AAV9-iNG at 5 weeks of age. FIG. 24a shows immunofluorescence staining of dystrophin in diaphragm and gastrocnemius muscles of WT and $\text{mdx}^{4\text{cv}}$ mice with or without systemic AAV9-iNG delivery. FIG. 24b shows Western blot of dystrophin expression in gastrocnemius muscles. FIG. 24c shows quantification of Western blot data.

FIGS. 25a-25m show systemic delivery of AAV9-iNG improved histopathology and contractility in $\text{mdx}^{4\text{cv}}$ mice. FIG. 25a shows trichrome staining of muscle and heart sections showing the extensive fibrosis in diaphragm and gastrocnemius (GA) muscles of $\text{mdx}^{4\text{cv}}$ mice (10 months of age), which were substantially reduced following systemic AAV9-iNG delivery. The $\text{mdx}^{4\text{cv}}$ mouse heart had little fibrosis at 10 months of age. Scale bar: 200 μm . FIGS. 25b-25d show quantification of fibrotic area of the diaphragm, gastrocnemius and heart muscles. ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$ (one-way ANOVA). FIGS. 25e-25h show measurement of CNF in the diaphragm (FIGS. 25e and 25f) and gastrocnemius (FIGS. 25g and 25h) muscles of $\text{mdx}^{4\text{cv}}$ mice with or without AAV9-iNG treatment at 10 weeks (FIGS. 25e and 25g) or 10 months (FIGS. 25f and 25h) of age (two-tailed, unpaired Student's t-test). FIGS. 25i-25l show muscle fiber size measurement and distribution in diaphragm and gastrocnemius muscles of the mice (WT and $\text{mdx}^{4\text{cv}}$ with or without AAV9-iNG) at 10 weeks of age. ns, not significant; ** $p < 0.01$ (one-way ANOVA). FIG. 25m shows tetanic torque measurements of the posterior compartment muscles. * $p < 0.05$; **** $p < 0.0001$ (one-way ANOVA).

FIGS. 26a-26d show measurement of muscle fiber size in diaphragm and gastrocnemius muscles at 10 months of age. FIGS. 26a-26b show average CSA of diaphragm (FIG. 26a) and gastrocnemius (FIG. 26b) muscle fibers. c, d, Distribution of diaphragm (FIG. 26c) and gastrocnemius (FIG. 26d) muscle fiber areas. ns, not significant; * $p < 0.05$ (one-way ANOVA).

FIGS. 27a-27j show host immune response, toxicity and off-target studies of AAV9-iNG therapy in $\text{mdx}^{4\text{cv}}$ mice. FIGS. 27a and 27b show host immune response to AAV9 capsid and the base editor transgene (anti-Cas9). IM, intramuscular injection; IV, intravenous injection. **** $p < 0.0001$ (one-way ANOVA) compared to WT or $\text{mdx}^{4\text{cv}}$. FIGS. 27c-27e show measurements of serum AST (FIG. 27c), ALT (FIG. 27d) and BUN (FIG. 27e) of mice treated with or without AAV9-iNG. ns, not statistically significant (one-way ANOVA). FIG. 27f shows predicted off-target sites with different number of mismatches from the $\text{mdx}^{4\text{cv}}$ target gRNA, and the sequences of two most similar off-target sites located on chromosome 16 and 1, respectively. FIGS. 27g

and 27h show quantification of deep sequencing reads of the genomic DNA PCR amplicons of the chromosome 16 off-target site (FIG. 27g) or the chromosome 1 off-target site (FIG. 27h) from Neuro-2a cells transfected with ABE-NG, iABE-NGA or control plus the gRNA. ns, not significant (one-way ANOVA). FIG. 27i shows quantification of the A⁴-to-G editing in mdx^{4cv} mice treated with or without AAV9-iNG. ****p<0.0001 (student's t-test). FIG. 27j shows quantification of the bystander C⁶-to-T editing in mdx^{4cv} mice treated with or without AAV9-iNG. ***p<0.001 (student's t-test). The sequences in FIG. 27 are GTTATCTCCTGTTCTGCAGCTGT (SEQ ID NO: 621), GTTATCTCCTGCTCTGCAGCAGA (SEQ ID NO: 628), GATATCTCCTGT TCTGCAGGAGA (SEQ ID NO: 629).

FIG. 28 shows off-target activities of AAV9-iNG. The nucleotide frequency at the on-target site of the four mdx^{4cv} mice at 10 months after treatment with AAV9-iNG. The desired edit at A⁴ is highlighted in green and the bystander C⁶ edit in red. The sequence in FIG. 28 is GTTATCTCCTGTTCTGCAGCTGT (SEQ ID NO: 621),

FIGS. 29a-29c show off-target RNA editing activities in the mdx^{4cv} mouse hearts (10 weeks and 10 months of age) induced by systemic delivery of AAV9-iNG. FIG. 29a shows the number of RNA SNVs in three AAV9-iNG treated mouse hearts after filtering the RNA SNVs in three control mouse hearts. FIG. 29b shows the A-to-I editing rate of all RNA SNVs. FIG. 29c shows the A-to-I editing rate of the common RNA SNVs in the three AAV9-iNG treated mouse hearts.

FIGS. 30a-30b show iABE-NGA editing-mediated exon skipping of human DMD exon 55 in DMD-hiPSC-derived cardiomyocytes. FIG. 30a shows Sanger sequencing showing that iABE-NGA mediated efficient conversion of A to G at the splice acceptor site of human DMD exon 55. FIG. 30b shows RT-PCR analysis showing that iABE-NGA editing led to skipping of exon 55 in ΔE48-54 DMD-hiPSC-derived cardiomyocytes. The WT, ΔE48-54 (green arrow) and E55 skipped transcript (red arrow) were clearly discernable.

FIGS. 31a-31c show iABE-NGA editing-mediated exon 7 inclusion of human SMN2. FIG. 31a shows sequences of SMN1 and SMN2 exon 7 with the different nucleotides at position 6 highlighted in red. Two gRNAs were designed to edit A36 to G with the PAM highlighted in purple. FIG. 31b shows RT-PCR analysis of Neuro-2a cells transfected with a SMN2 reporter gene plus or minus the base editors and gRNAs. The exon 7 inclusion (FL) or exclusion (Δ7) transcripts were measured. FIG. 31c shows quantification of the RT-PCR products. **p<0.01; ****p<0.0001; ns, not significant when compared with the control group, using one-way ANOVA with Turkey post test. The sequences in FIG. 31a are GGTTCAGACAAAATCA AAAAGAAGGAAG-GTGCTCACATTCCTTAAATTAAGGA (SEQ ID NO: 632), GGTTCAGACAAAATCA AAAAGAAGGAAG-GAAGGTGCTCACATTCCTTAAATTAAGGA (SEQ ID NO: 633), GTGCTCACATTCCTTAAATTAAGG (SEQ ID NO: 634), gGCTCACATTCCTTA AATTAAGGA (SEQ ID NO: 635),

FIGS. 32a-32c show the iABE-MGA editing of mouse Angptl3 (FIG. 32a), Apoc3 (FIG. 32b) and Asgr1 (FIG. 32c) in Neuro-2a cells. **p<0.01; ***p<0.001; ****p<0.0001

DETAILED DESCRIPTION

Fusing the CRISPR-Cas9 nickase with nucleobase deaminases (e.g. cytidine or adenine deaminase), a new paradigm-shifting class of genome editing technology, termed "base editors", have recently been developed. DNA base editors,

via catalyzing the conversion of one base to another, directly and precisely install point mutations into chromosomal DNA without making DSBs. Therefore, base editing can be developed as promising therapeutics to correct the genetic diseases without DNA cleavage. In particular, the adenine base editors (ABEs) show remarkable fidelity in mouse embryos and rice as compared to cytosine base editors (CBEs), making them highly attractive in therapeutic development. Moreover, nearly half of the point mutations causing human diseases are G-to-A or C-to-T, highlighting the potential of ABEs in correcting a large number of human diseases. In particular, 174 out of 508 pathogenic point mutations for DMD are due to G:C to A:T conversion (Table 5), which can be targeted by ABE editing.

In vivo base editing can correct a custom-made mouse model of Duchenne muscular dystrophy (DMD), which carries a nonsense mutation in exon 20 with a classical 5'-TGG protospacer adjacent motif (PAM) sequence in the noncoding strand for recognition by the Cas9 from *Streptococcus pyogenes* (SpCas9). In silico analysis of the ClinVar database showed that about 42.8% of the 53469 human disease-causing mutations can be potential targets for base editing correction; however, the majority (~72.4%) of these potential targets cannot be suitable for SpCas9 base editing due to the lack of the 5'-NGG PAM sequence within the suitable distance from the mutations. Several variants of SpCas9 have recently been engineered with relaxed PAM (such as xCas9-3.7, SpCas9-NG and ScCas9) and non-G PAM. These enzymes greatly increase the target scope for correcting human mutations. However, their performance to correct genetic mutations in preclinical animal models remains to be determined. Here, the efficacy of correcting a commonly used mouse model of DMD, mdx^{4cv} mice were explored, using NG-targeting base editors.

The present disclosure provides fusion proteins, systems, and compositions for editing disease-associated mutations and uses thereof for treating a disease (for example, Duchenne muscular dystrophy).

Reference will now be made in detail to the embodiments of the invention, examples of which are illustrated in the drawings and the examples. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs.

Terminology

Terms used throughout this application are to be construed with ordinary and typical meaning to those of ordinary skill in the art. However, Applicant desires that the following terms be given the particular definition as defined below.

As used herein, the article "a," "an," and "the" means "at least one," unless the context in which the article is used clearly indicates otherwise.

The term "comprising" and variations thereof as used herein is used synonymously with the term "including" and variations thereof and are open, non-limiting terms. Although the terms "comprising" and "including" have been used herein to describe various embodiments, the terms "consisting essentially of" and "consisting of" can be used in place of "comprising" and "including" to provide for more specific embodiments and are also disclosed.

As used herein, the terms “may,” “optionally,” and “may optionally” are used interchangeably and are meant to include cases in which the condition occurs as well as cases in which the condition does not occur. Thus, for example, the statement that a formulation “may include an excipient” is meant to include cases in which the formulation includes an excipient as well as cases in which the formulation does not include an excipient.

The terms “about” and “approximately” are defined as being “close to” as understood by one of ordinary skill in the art. In one non-limiting embodiment, the terms are defined to be within 10%. In another non-limiting embodiment, the terms are defined to be within 5%. In still another non-limiting embodiment, the terms are defined to be within 1%.

The term “Cas9” or “Cas9 nuclease” refers to an RNA-guided nuclease comprising a Cas9 protein, or a fragment thereof (e.g., a protein comprising an active, inactive, or partially active DNA cleavage domain of Cas9, and/or the gRNA binding domain of Cas9). A Cas9 nuclease is also referred to sometimes as a casn1 nuclease or a CRISPR (clustered regularly interspaced short palindromic repeat)-associated nuclease. CRISPR is an adaptive immune system that provides protection against mobile genetic elements (viruses, transposable elements and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (rnc) and a Cas9 protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently, Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer. The target strand not complementary to crRNA is first cut endonucleolytically, then trimmed 3'-5' exonucleolytically. In nature, DNA-binding and cleavage typically requires protein and both RNAs. However, single guide RNAs (“sgRNA”, or simply “gRNA”) can be engineered so as to incorporate aspects of both the crRNA and tracrRNA into a single RNA species. See, e.g., Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of which is hereby incorporated by reference. Cas9 recognizes a short motif in the CRISPR repeat sequences (the PAM or protospacer adjacent motif) to help distinguish self versus non-self. Cas9 nuclease sequences and structures are well known to those of skill in the art (see, e.g., “Complete genome sequence of an M1 strain of *Streptococcus pyogenes*.” Ferretti et al., J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663(2001); “CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III.” Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607(2011); and “A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity.” Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of each of which are incorporated herein by reference). Cas9 orthologs have been described in various species, including, but not limited to, *S. pyogenes* and *S. thermophiles*. Additional suitable Cas9 nucleases and sequences will be apparent to those of skill in the art based on this disclosure, and

such Cas9 nucleases and sequences include Cas9 sequences from the organisms and loci disclosed in Chylinski, Rhun, and Charpentier, “The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems” (2013) RNA Biology 10:5, 726-737; the entire contents of which are incorporated herein by reference. In some embodiments, a Cas9 nuclease has an inactive (e.g., an inactivated) DNA cleavage domain, that is, the Cas9 is a nickase.

A “composition” is intended to include a combination of active agent and another compound or composition, inert (for example, a fusion protein, nucleic acid, or virus) or active, such as an adjuvant.

Dystrophinopathies are a group of muscular dystrophies resulting from mutations in the dystrophin gene, located on the short arm of the X chromosome in the Xp21 region (Kunkel et al. 1985; Monaco et al. 1985; Ray et al. 1985). Of these, “Duchenne muscular dystrophy” or “DMD” is the most common dystrophinopathy resulting from complete absence of the dystrophin gene product, the subsarcolemmal protein dystrophin (Hoffman et al. 1987a; Koenig et al. 1987; Hoffman et al. 1988). While dystrophin deficiency can be a primary cause of DMD, multiple secondary pathways are responsible for the progression of muscle necrosis, abnormal fibrosis and failure of regeneration that results in a progressively worsening clinical status. There is evidence supporting oxidative radical damage to myofibers (Rando 2002), inflammation (Spencer and Tidball 2001; Porter et al. 2002), abnormal calcium homeostasis (Allen 2010; Millay 2009), myonuclear apoptosis (Rando 2001b; Sandri et al. 2001; Tews 2002), abnormal fibrosis and failure of regeneration (Rando 2001b; Bernasconi 1995); (Melone 2000; Morrison 2000; Luz 2002). This body of literature has been validated by cross sectional genome-wide approaches that allow an overall analysis of multiple defective mechanisms in DMD (Chen et al. 2000; Porter 2003). The main symptom of DMD is muscle weakness associated with muscle wasting first with the voluntary muscles, e.g., the hips, pelvic area, thighs, shoulders, and calf muscles.

As used herein, the term “effective amount” refers to an amount of a composition necessary or sufficient to realize a desired biologic effect. An effective amount of the composition would be the amount that achieves a selected result, and such an amount could be determined as a matter of routine experimentation by a person skilled in the art. For example, an effective amount of the composition could be that amount necessary for preventing, treating and/or ameliorating Duchenne muscular dystrophy in a subject. The term is also synonymous with “sufficient amount.”

The term “subject” is defined herein to include animals such as mammals, including, but not limited to, primates (e.g., humans), cows, sheep, goats, horses, dogs, cats, rabbits, rats, mice and the like. In some embodiments, the subject is a human.

“Pharmaceutically acceptable carrier” (sometimes referred to as a “carrier”) means a carrier or excipient that is useful in preparing a pharmaceutical or therapeutic composition that is generally safe and non-toxic, and includes a carrier that is acceptable for veterinary and/or human pharmaceutical or therapeutic use. The terms “carrier” or “pharmaceutically acceptable carrier” can include, but are not limited to, phosphate buffered saline solution, water, emulsions (such as an oil/water or water/oil emulsion) and/or various types of wetting agents.

As used herein, the term “carrier” encompasses any excipient, diluent, filler, salt, buffer, stabilizer, solubilizer, lipid, stabilizer, or other material well known in the art for use in pharmaceutical formulations. The choice of a carrier

for use in a composition will depend upon the intended route of administration for the composition. The preparation of pharmaceutically acceptable carriers and formulations containing these materials is described in, e.g., *Remington's Pharmaceutical Sciences*, 21st Edition, ed. University of the Sciences in Philadelphia, Lippincott, Williams R. Wilkins, Philadelphia, PA, 2005. Examples of physiologically acceptable carriers include saline, glycerol, DMSO, buffers such as phosphate buffers, citrate buffer, and buffers with other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEEN™ (ICI, Inc.; Bridge-water, NJ), polyethylene glycol (PEG), and PLURONICS™ (BASF; Florham Park, NJ). To provide for the administration of such dosages for the desired therapeutic treatment, compositions disclosed herein can advantageously comprise between about 0.1% and 99% by weight of the total of one or more of the subject compounds based on the weight of the total composition including carrier or diluent.

A gRNA is a component of the CRISPR/Cas system. A "gRNA" (guide ribonucleic acid) herein refers to a fusion of a CRISPR-targeting RNA (crRNA) and a trans-activation crRNA (tracrRNA), providing both targeting specificity and scaffolding/binding ability for Cas9 nuclease. A "crRNA" is a bacterial RNA that confers target specificity and requires tracrRNA to bind to Cas9. A "tracrRNA" is a bacterial RNA that links the crRNA to the Cas9 nuclease and typically can bind any crRNA. The sequence specificity of a Cas DNA-binding protein is determined by gRNAs, which have nucleotide base-pairing complementarity to target DNA sequences. The native gRNA comprises a Specificity Determining Sequence (SDS), which specifies the DNA sequence to be targeted. At least a portion of the target DNA sequence is complementary to the SDS of the gRNA. For Cas9 to successfully bind to the DNA target sequence, a region of the target sequence is complementary to the SDS of the gRNA sequence and is immediately followed by the correct protospacer adjacent motif (PAM) sequence (e.g., NGG or NG for Cas9 used herein). In some embodiments, an SDS is 100% complementary to its target sequence. In some embodiments, the SDS sequence is less than 100% complementary to its target sequence and is, thus, considered to be partially complementary to its target sequence. For example, a targeting sequence may be 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% complementary to its target sequence.

The term "linker," as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., two domains of a fusion protein, such as, for example, a Cas9 domain and a nucleic acid editing domain (e.g., a deaminase domain). In some embodiments, a linker joins a gRNA binding domain of an RNA-programmable nuclease, including a Cas9 nuclease domain, and the catalytic domain of a nucleic acid editing protein. In some embodiments, a linker joins a Cas9 and a nucleic acid editing protein. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodi-

ments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 5-100 amino acids in length, for example, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated.

The term "mutation," as used herein, refers to a substitution of a residue within a sequence, e.g., a nucleic acid or amino acid sequence, with another residue, or a deletion or insertion of one or more residues within a sequence. Mutations are typically described herein by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. Various methods for making the amino acid substitutions (mutations) provided herein are well known in the art, and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)).

The term "nickase" as used herein, refers to a nuclease that cleaves only a single DNA strand, either due to its natural function or because it has been engineered to cleave only a single DNA strand. Jinek et al., "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity" *Science* 337(6096):816-821 (2012) and Cong et al. *Multiplex genome engineering using CRISPR/Cas systems* *Science* 339(6121):819-823 (2013).

The term "nucleic acid editing domain," as used herein refers to a protein or enzyme capable of making one or more modifications (e.g., deamination of a cytidine residue) to a nucleic acid (e.g., DNA or RNA). Exemplary nucleic acid editing domains include, but are not limited to a deaminase, a nuclease, a nickase, a recombinase, a methyltransferase, a methylase, an acetylase, an acetyltransferase, a transcriptional activator, or a transcriptional repressor domain. In some embodiments the nucleic acid editing domain comprises a deaminase (e.g., a cytidine deaminase or an adenine deaminase).

An "adenine deaminase" is an enzyme involved in purine metabolism. It is needed for the breakdown of adenosine from food and for the turnover of nucleic acids in tissues. Its primary function in humans is the development and maintenance of the immune system. An adenine deaminase catalyzes hydrolytic deamination of adenosine (forming inosine, which base pairs as G) in the context of DNA.

The term "recombinant" as used herein in the context of proteins or nucleic acids refers to proteins or nucleic acids that do not occur in nature, but are the product of human engineering. For example, in some embodiments, a recombinant protein or nucleic acid molecule comprises an amino acid or nucleotide sequence that comprises at least one, at least two, at least three, at least four, at least five, at least six, or at least seven mutations as compared to any naturally occurring sequence.

As used herein, the terms "treating" or "treatment" of a subject includes the administration of a drug to a subject with the purpose of curing, healing, alleviating, relieving, altering, remedying, ameliorating, improving, stabilizing or affecting a disease or disorder, or a symptom of a disease or disorder. The terms "treating" and "treatment" can also refer to reduction in severity and/or frequency of symptoms, elimination of symptoms and/or underlying cause, and improvement or remediation of damage. For example, Duchenne muscular dystrophy, may result in e.g., a slowing of muscle degeneration, decreased fatigue, increased muscle strength, reduced blood levels of creatine kinase (CK),

decreased difficulty with motor skills, decreased muscle fiber deformities, decreased inflammation or fibrotic tissue infiltration in the muscle, stabilization of the progression of the disease (e.g., by halting progressive muscle weakness) etc.

“Therapeutically effective amount” or “therapeutically effective dose” of a composition (e.g. a fusion protein, a nucleic acid, or virus) refers to an amount that is effective to achieve a desired therapeutic result. In some embodiments, a desired therapeutic result is the prevention of Duchenne muscular dystrophy. In some embodiments, a desired therapeutic result is the treatment of Duchenne muscular dystrophy. Therapeutically effective amounts of a given therapeutic agent will typically vary with respect to factors such as the type and severity of the disorder or disease being treated and the age, gender, and weight of the subject. The term can also refer to an amount of a therapeutic agent, or a rate of delivery of a therapeutic agent (e.g., amount over time), effective to facilitate a desired therapeutic effect, such as coughing relief. The precise desired therapeutic effect will vary according to the condition to be treated, the tolerance of the subject, the agent and/or agent formulation to be administered (e.g., the potency of the therapeutic agent, the concentration of agent in the formulation, and the like), and a variety of other factors that are appreciated by those of ordinary skill in the art. In some instances, a desired biological or medical response is achieved following administration of multiple dosages of the composition to the subject over a period of days, weeks, or years.

A “vector” is a composition of matter which comprises an isolated nucleic acid and which can be used to deliver the isolated nucleic acid to the interior of a cell. Numerous vectors are known in the art including, but not limited to, linear polynucleotides, polynucleotides associated with ionic or amphiphilic compounds, plasmids, and viruses. Thus, the term “vector” includes an autonomously replicating plasmid or a virus. The term should also be construed to include non-plasmid and non-viral compounds which facilitate transfer of nucleic acid into cells, such as, for example, polylysine compounds, liposomes, and the like. Examples of viral vectors include, but are not limited to, lentiviral vectors, adenoviral vectors, adeno-associated virus vectors, retroviral vectors, and the like.

An “adeno-associated virus” or “AAV” is a virus which infects humans and some other primate species. The wild-type AAV genome is a single-stranded deoxyribonucleic acid (ssDNA), either positive- or negative-sensed. The genome comprises two inverted terminal repeats (ITRs), one at each end of the DNA strand, and two open reading frames (ORFs): rep and cap between the ITRs. The rep ORF comprises four overlapping genes encoding Rep proteins required for the AAV life cycle. The cap ORF comprises overlapping genes encoding capsid proteins: VP1, VP2 and VP3, which interact together to form the viral capsid. VP1, VP2 and VP3 are translated from one mRNA transcript, which can be spliced in two different manners: either a longer or shorter intron can be excised resulting in the formation of two isoforms of mRNAs: a ~2.3 kb- and a ~2.6 kb-long mRNA isoform. The capsid forms a supranuclear assembly of approximately 60 individual capsid protein subunits into a non-enveloped, T-1 icosahedral lattice capable of protecting the AAV genome. The mature capsid is composed of VP1, VP2, and VP3 (molecular masses of approximately 87, 73, and 62 kDa respectively) in a ratio of about 1:1:10.

The term “nucleic acid” as used herein means a polymer composed of nucleotides, deoxyribonucleotides or ribonucleotides.

The terms “ribonucleic acid” and “RNA” as used herein mean a polymer composed of ribonucleotides.

The terms “deoxyribonucleic acid” and “DNA” as used herein mean a polymer composed of deoxyribonucleotides.

The term “oligonucleotide” denotes single- or double-stranded nucleotide multimers of from about 2 to up to about 100 nucleotides in length. Suitable oligonucleotides may be prepared by the phosphoramidite method described by Beaucage and Carruthers, *Tetrahedron Lett.*, 22: 1859-1862 (1981), or by the triester method according to Matteucci, et al., *J. Am. Chem. Soc.*, 103:3185 (1981), both incorporated herein by reference, or by other chemical methods using either a commercial automated oligonucleotide synthesizer or VLSIPSTTM technology. When oligonucleotides are referred to as “double-stranded,” it is understood by those of skill in the art that a pair of oligonucleotides exist in a hydrogen-bonded, helical array typically associated with, for example, DNA. In addition to the 100% complementary form of double-stranded oligonucleotides, the term “double-stranded,” as used herein is also meant to refer to those forms which include such structural features as bulges and loops, described more fully in such biochemistry texts as Stryer, *Biochemistry*, Third Ed., (1988), incorporated herein by reference for all purposes.

The term “polynucleotide” refers to a single or double stranded polymer composed of nucleotide monomers.

The term “polypeptide” refers to a compound made up of a single chain of D- or L-amino acids or a mixture of D- and L-amino acids joined by peptide bonds.

The terms “identical” or percent “identity,” in the context of two or more nucleic acids or polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same (i.e., about 60% identity, preferably 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or higher identity over a specified region when compared and aligned for maximum correspondence over a comparison window or designated region) as measured using a BLAST or BLAST 2.0 sequence comparison algorithms with default parameters described below, or by manual alignment and visual inspection (see, e.g. NCBI web site or the like). Such sequences are then said to be “substantially identical.” This definition also refers to, or may be applied to, the complement of a test sequence. The definition also includes sequences that have deletions and/or additions, as well as those that have substitutions. As described below, the preferred algorithms can account for gaps and the like. Preferably, identity exists over a region that is at least about 10 amino acids or 20 nucleotides in length, or more preferably over a region that is 10-50 amino acids or 20-50 nucleotides in length. As used herein, percent (%) nucleotide sequence identity is defined as the percentage of amino acids in a candidate sequence that are identical to the nucleotides in a reference sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN, ALIGN-2 or Megalign (DNASTAR) software. Appropriate parameters for measuring alignment,

including any algorithms needed to achieve maximal alignment over the full-length of the sequences being compared can be determined by known methods.

For sequence comparisons, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Preferably, default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters.

One example of an algorithm that is suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al. (1977) *Nuc. Acids Res.* 25:3389-3402, and Altschul et al. (1990) *J. Mol. Biol.* 215:403-410, respectively. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology information (www.ncbi.nlm.nih.gov). This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al. (1990) *J. Mol. Biol.* 215:403-410). These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) or 10, M=5, N=-4 and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength of 3, and expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff and Henikoff (1989) *Proc. Natl. Acad. Sci. USA* 89:10915) alignments (B) of 50, expectation (E) of 10, M=5, N=-4, and a comparison of both strands.

The BLAST algorithm also performs a statistical analysis of the similarity between two sequences (see, e.g., Karlin and Altschul (1993) *Proc. Sci. USA* 90:5873-5787). One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a nucleic acid is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid to the reference nucleic acid is less than about 0.2, more preferably less than about 0.01.

The term "increased" or "increase" as used herein generally means an increase by a statically significant amount;

for the avoidance of any doubt, "increased" means an increase of at least 10% as compared to a reference level, for example an increase of at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, or at least about a 2-fold, or at least about a 3-fold, or at least about a 4-fold, or at least about a 5-fold or at least about a 10-fold increase, or any increase between 2-fold and 10-fold or greater as compared to a reference level.

The term "reduced", "reduce", "reduction", or "decrease" as used herein generally means a decrease by a statistically significant amount. However, for avoidance of doubt, "reduced" means a decrease by at least 10% as compared to a reference level, for example a decrease by at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% decrease (i.e. absent level as compared to a reference sample), or any decrease between 10-100% as compared to a reference level.

Throughout this application, various publications are referenced. The disclosures of these publications in their entireties are hereby incorporated by reference into this application in order to more fully describe the state of the art to which this pertains. The references disclosed are also individually and specifically incorporated by reference herein for the material contained in them that is discussed in the sentence in which the reference is relied upon.

Compositions

In some aspects, disclosed herein is a fusion protein comprising a Cas9 nickase and a nucleotide deaminase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11. In some embodiments, the Cas9 nickase and the nucleotide deaminase are operably linked.

In some embodiments, the first amino acid substitution is selected from the group consisting of A262T, R324L, S409I, E480K, E543D, M694I, and E1219V when compared to SEQ ID NO: 11. In some embodiments, the second amino acid substitution is selected from the group consisting of L1111R, D1135V, G1218R, E1219F, A1322R, R1335V, R1335Q, R1335E, and T1337R when compared to SEQ ID NO: 11.

In some embodiments, the fusion protein comprises a Cas9 nickase fused to a deaminase and further fused to a Uracil DNA glycosylase inhibitor (UGI) domain.

In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

In some embodiments, the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.

In some embodiments, the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.

In some embodiments, the nucleotide deaminase is a cytidine deaminase or an adenine deaminase. In some embodiments, the nucleotide deaminase is a cytidine deaminase. In some embodiments, the nucleotide deaminase is an adenine deaminase. In some embodiments, the adenine deaminase is a dimeric adenine deaminase or a monomeric adenine deaminase domain. In some embodiments, the dimeric adenine deaminase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NOs: 22. In some embodiments, the monomeric adenine deaminase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ NOs: 23. In some embodiments, the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47. In some embodiments, the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 22 or 23.

In some embodiments, the adenine deaminase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 41 or 42. In some embodiments, the adenine deaminase is encoded by the nucleotide sequence as set forth in SEQ ID NO: 41 or 42.

In some embodiments, the Cas9 nickase comprising the monomeric adenine deaminase domain has a lower off-target RNA editing activity than a Cas9 nickase comprising a dimeric adenine deaminase domain (e.g., at least 10% lower, at least 20% lower, at least 30% lower, at least 40% lower, at least 50% lower, at least 60% lower, at least 70% lower, at least 80% lower, at least 90% lower, at least 95% lower, at least 99% lower, or at least 2 times lower, at least 3 times lower, at least 4 times lower, at least 5 times lower, at least 6 times lower, at least 7 times lower, at least 8 times lower, at least 9 times lower, at least 10 times lower, at least 20 times lower, at least 50 times lower, at least 100 times lower, at least 150 times lower, at time 200 times lower, at least 500 times lower, or at least 1000 times lower).

In some aspects, disclosed herein is a construct comprising a nucleotide sequence encoding the fusion protein of any preceding aspect. In some embodiments, the construct further comprises a promoter operably linked to the nucleotide sequence, wherein the promoter is a CMV promoter. Accordingly, in some embodiments, the nucleotide sequence of any preceding aspect is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 1 or SEQ ID NO: 2.

In some aspects, disclosed herein is a system for base editing comprising:

a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase;

wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

Accordingly, in some embodiments, the Cas9 nickase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39. In some embodiments, the Cas9 nickase is encoded by a nucleotide sequence selected from the group consisting of SEQ ID NOs:

In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-560, 2-561, 2-562, 2-563, 2-564, 2-565, 2-566, 2-567, 2-568, 2-569, 2-570, 2-571, 2-572, 2-573, 2-574, 2-575, 2-576, 2-577, 2-578, 2-579, or 2-580 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

An “intein” is a segment of a protein that is able to excise itself and join the remaining portions (the exteins) with a peptide bond in a process known as protein splicing. Inteins are also referred to as “protein introns.” The process of an intein excising itself and joining the remaining portions of the protein is herein termed “protein splicing” or “intein-mediated protein splicing.” In some embodiments, an intein of a precursor protein (an intein containing protein prior to intein-mediated protein splicing) comes from two genes. Such intein is referred to herein as a split intein. The inteins used herein can be, for example, Npu DnaE intein, Cfa, DnaE intein or gp41-1 intein. The intein encoded by the DnaE-n gene is herein referred to as “intein-N.” The intein encoded by the DnaE-c gene is herein referred to as “intein-C.”

Other intein systems may also be used. For example, a synthetic intein based on the dnaE intein, the Cfa-N and Cfa-C intein pair, has been described (e.g., in Stevens et al., J Am Chem Soc. 2016 Feb. 24; 138(7):2162-5, incorporated herein by reference). Non-limiting examples of intein pairs that may be used in accordance with the present disclosure include: Ssp GyrB intein, Ssp DnaX intein, Ter DnaE3 intein, Ter ThyX intein, Rma DnaB intein and Cne Prp8 intein (e.g., as described in U.S. Pat. No. 8,394,604, incorporated herein by reference).

In some embodiments, the intein-N is a Cfa intein-N or a gp41-1 intein-N. Accordingly, in some embodiments, the intein-N comprises the amino acid sequence of SEQ ID NO: 24 or 26. In some embodiments, the intein-N is encoded by the nucleotide sequence of SEQ ID NO: 43 or 45. Accordingly, the first nucleotide sequence of any preceding aspects comprises SEQ ID NO: 43 or 45.

In some embodiments, the intein-C is a Cfa intein-C or a gp41-1 intein-C. Accordingly, in some embodiments, the intein-C comprises the amino acid sequence of SEQ ID NO: 25 or 27. In some embodiments, the intein-C is encoded by the nucleotide sequence of SEQ ID NO: 44 or 46. Accord-

ingly, the second nucleotide sequence of any preceding aspects comprises SEQ ID NO: 44 or 46.

In some embodiments, the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).

In some embodiments, the first nucleotide sequence comprises an N terminal portion of SEQ ID NO: 1 or 2. In some embodiments, the first nucleotide comprises a portion of SEQ ID NO: 1 or 2.

In some embodiments, the second nucleotide sequence comprises a C terminal portion of SEQ ID NO: 1 or 2. In some embodiments, the second nucleotide comprises a portion of SEQ ID NO: 1 or 2.

In some aspects, disclosed herein is a composition comprising:

a first recombinant viral particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second recombinant viral particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

Accordingly, in some embodiments, the Cas9 nickase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39. In some embodiments, the Cas9 nickase is encoded by a nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39.

In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-560, 2-561, 2-562, 2-563, 2-564, 2-565, 2-566, 2-567, 2-568, 2-569, 2-570, 2-571, 2-572, 2-573, 2-574, 2-575, 2-576, 2-577, 2-578, 2-579, or 2-580 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

In some embodiments, the intein-N is a Cfa intein-N or a gp41-1 intein-N. Accordingly, in some embodiments, the intein-N comprises the amino acid sequence of SEQ ID NO: 24 or 26. In some embodiments, the intein-N is encoded by the nucleotide sequence of SEQ ID NO: 43 or 45. Accordingly, the first nucleotide sequence of any preceding aspects comprises SEQ ID NO: 43 or 45.

In some embodiments, the intein-C is a Cfa intein-C or a gp41-1 intein-C. Accordingly, in some embodiments, the intein-C comprises the amino acid sequence of SEQ ID NO: 25 or 27. In some embodiments, the intein-C is encoded by the nucleotide sequence of SEQ ID NO: 44 or 46. Accordingly, the second nucleotide sequence of any preceding aspects comprises SEQ ID NO: 44 or 46.

A number of viral based systems have been developed for gene transfer into mammalian cells. For example, retroviruses provide a convenient platform for gene delivery systems. A selected gene can be inserted into a vector and packaged in retroviral particles using techniques known in the art. The recombinant virus can then be isolated and delivered to cells of the subject either in vivo or ex vivo. A number of retroviral systems are known in the art. In some embodiments, adenovirus vectors are used. A number of adenovirus vectors are known in the art. In one embodiment, lentivirus vectors are used.

AAV is a preferred vector because it can infect many cell types and is nonpathogenic to humans. AAV type vectors can transport about 4 to 5 kb and wild type AAV is known to stably insert into chromosome 19. Vectors which contain this site-specific integration property are preferred. In some embodiments, the AAV vector of any preceding aspect further a herpes simplex virus thymidine kinase gene, HSV-tk, and/or a marker gene, such as the gene encoding the green fluorescent protein, GFP.

Accordingly, in some aspects, disclosed herein is a composition comprising:

a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

In some embodiments, the AAV contains a pair of inverted terminal repeats (ITRs) which flank at least one cassette containing a promoter which directs cell-specific expression operably linked to a heterologous gene. Heterologous in this context refers to any nucleotide sequence or gene which is not native to the AAV or B19 parvovirus. Typically the AAV and B19 coding regions have been deleted, resulting in a safe, noncytotoxic vector. The AAV ITRs, or modifications thereof, confer infectivity and site-specific integration. U.S. Pat. No. 6,261,834 is herein incorporated by reference for material related to the AAV vector.

The inserted genes in viral and retroviral usually contain promoters, and/or enhancers to help control the expression of the desired gene product. A promoter is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. A promoter contains core elements required for basic interaction of RNA polymerase and transcription factors, and may contain upstream elements and response elements.

The spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. Depending on the promoter, it appears that individual elements can function either cooperatively or independently to activate transcription.

The term "promoter" as used herein is defined as a DNA sequence recognized by the synthetic machinery of the cell, or introduced synthetic machinery, required to initiate the specific transcription of a polynucleotide sequence. One example of a suitable promoter is the immediate early cytomegalovirus (CMV) promoter sequence. This promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. Another example of a suitable promoter is Elongation Growth Factor-1 alpha (EF-1 α). However, other promoter sequences may also be used, including, but not limited to the simian virus 40 (SV40), early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, promoter, PGK-1 promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter as well as synthetic protein, such as a CAG promoter. Further, the invention should not be limited to the use of constitutive promoters, inducible promoters are also contemplated as part of the invention. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. In some embodiments, the AAV of any preceding aspect comprises a constitutive promoter or a muscle tissue specific promoter, e.g., a muscle-specific MHP1 promoter.

In some embodiments, the promoter is a CMV promoter. In some embodiments, the CMV promoter comprises the nucleotide sequence of SEQ ID NO: 28. Accordingly, in some embodiments, the first recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 3 or SEQ ID NO: 5, wherein the second recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 4 or SEQ ID NO: 6.

In some embodiments, the promoter is a MHP1 promoter. In some embodiments, the MHP1 promoter comprises the nucleotide sequence of SEQ ID NO: 29. Accordingly, in some embodiments, the first recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 7, wherein the second recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 8.

In some embodiments, viral vector is a lentivirus vector. Accordingly, disclosed herein is a composition comprising:

a first recombinant lentivirus particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second recombinant lentivirus particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase;

wherein first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and

wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

Methods of introducing and expressing genes into a cell are known in the art. In the context of an expression vector, the vector can be readily introduced into a host cell, e.g., mammalian, bacterial, yeast, or insect cell by any method in the art. For example, the expression vector can be transferred into a host cell by physical, chemical, or biological means. See, e.g., WO2012079000A1, incorporated by reference herein in their entireties.

Reporter genes are used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a polypeptide whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assayed at a suitable time after the DNA has been introduced into the recipient cells. Suitable reporter genes may include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (e.g., Ui-Tei et al., 2000 FEBS Letters 479: 79-82). Suitable expression systems are well known and may be prepared using known techniques or obtained commercially. In general, the construct with the minimal 5' flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions may be linked to a reporter gene and used to evaluate agents for the ability to mod late promoter-driven transcription. In some embodiments, the recombinant nucleic acid of any preceding aspect further comprises a reporter gene. In some embodiments, the reporter gene in invention is GFP.

In some embodiments, the fusion protein, nucleotide, system, or composition of any preceding aspect can be further formulated in a pharmaceutically acceptable carrier. Genetic Diseases and Methods of Treatment

It is estimated that over 10,000 human diseases are caused by genetic disorders, which are abnormalities in genes or chromosomes, See, e.g., McClellan, J. and M. C. King, Genetic heterogeneity in human disease. Cell. 141(2): p. 210-7; Leachman, S. A., et al., Therapeutic siRNAs for dominant genetic skin disorders including pachyonychia congenita. J Dermatol Sci, 2008. 51(3): p. 151-7. The compositions disclosed herein can be used to treat a number of these genetic disorders.

In some aspects, disclosed herein is a method of treating a genetic disease in a subject, comprising administering to the subject a therapeutically effective amount of the fusion protein, system, or composition of any preceding aspect.

In some embodiments, the genetic disease is a muscular dystrophy. Muscular dystrophies are a group of muscle diseases caused by mutations in a person's genes. Over time, muscle weakness decreases mobility, making everyday tasks difficult. The methods and compositions disclosed herein can be used for treating a muscular dystrophy, including, for example, Duchenne muscular dystrophy, Becker muscular dystrophy, myotonic dystrophy, limb-girdle muscular dystrophy, facioscapulohumeral dystrophy, congenital muscular dystrophy, distal muscular dystrophy, oculopharyngeal muscular dystrophy, or Emery-Dreifuss muscular dystrophy.

In some embodiments, the genetic disease or disorder is Duchenne muscular dystrophy, dysferlinopathy, or spinal muscular atrophy. In some embodiments, the methods and compositions disclosed herein can be used for treating Duchenne muscular dystrophy. In some embodiments, the genetic disease is spinal muscular atrophy.

In some embodiments, the genetic disease or disorder is hypercholesterolemia. For treating hypercholesterolemia, iABE-NGA is used to create a protective silent mutation in functional genes, for example ANGPTL3, APOC3, PCSK9, or ASGR1, to lower plasma cholesterol levels.

In some embodiments, the genetic disease or disorder is due to a point mutation in a causative gene from a G:C pair to an A:T pair.

In some aspects, disclosed herein is a method of treating a genetic disease in a subject, comprising administering to the subject a therapeutically effective amount of the fusion protein, system, and composition of any preceding aspect.

The disclosed methods can be performed any time prior to the onset of a genetic disease, even prior to the apparent of any symptom. In one aspect, the disclosed methods can be employed 60, 59, 58, 57, 56, 55, 54, 53, 52, 51, 50, 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 years; 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 months; 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, or 3 days; 60, 48, 36, 30, 24, 18, 15, 12, 10, 9, 8, 7, 6, 5, 4, 3, or 2 hours prior to the onset of the genetic disease or any symptom thereof; or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15, 18, 24, 30, 36, 48, 60 hours; 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 45, 60, 90 or more days; 4, 5, 6, 7, 8, 9, 10, 11, 12 or more months; 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 45, or 60 years after the onset of the genetic disease or any symptom thereof.

Dosing frequency for the composition of any preceding aspects, includes, but is not limited to, at least once every year, once every two years, once every three years, once every four years, once every five years, once every six years, once every seven years, once every eight years, once every nine years, once every ten years, at least once every two months, once every three months, once every four months, once every five months, once every six months, once every seven months, once every eight months, once every nine months, once every ten months, once every eleven months, at least once every month, once every three weeks, once every two weeks, once a week, twice a week, three times a week, four times a week, five times a week, six times a week, daily, two times per day, three times per day, four times per day, five times per day, six times per day, eight times per day,

nine times per day, ten times per day, eleven times per day, twelve times per day, once every 12 hours, once every 10 hours, once every 8 hours, once every 6 hours, once every 5 hours, once every 4 hours, once every 3 hours, once every 2 hours, once every hour, once every 40 min, once every 30 min, once every 20 min, once every 10 min, once every 5 min, or once per min. Administration can also be continuous and adjusted to maintaining a level of the compound within any desired and specified range.

The compositions of the present invention can be administered to the appropriate subject in any manner known in the art, e.g., orally, intramuscularly, intravenously, sublingual mucosal, intraarterially, intrathecally, intradermally, intraperitoneally, intranasally, intrapulmonarily, intraocularly, intravaginally, intrarectally, subcutaneously, or through by inhalation. Parenteral administration, if used, is generally characterized by injection. Injectables can be prepared in conventional forms, either as liquid solutions or suspensions, solid forms suitable for solution or suspension in liquid prior to injection, or as emulsions.

EXAMPLES

The following examples are set forth below to illustrate the compounds, systems, methods, and results according to the disclosed subject matter. These examples are not intended to be inclusive of all aspects of the subject matter disclosed herein, but rather to illustrate representative methods and results. These examples are not intended to exclude equivalents and variations of the present invention which are apparent to one skilled in the art.

Example 1. In Vitro Reporter Assay Demonstrates the Feasibility to Correct the mdx^{4cv} Mutation Using ABE-NG

The mdx^{4cv} mouse carries a premature stop codon (CAA-to-TAA) in the exon 53 of Dmd gene, which disrupts the expression of dystrophin and leads to the development of muscular dystrophy. Targeting the noncoding strand with ABEs can correct this nonsense mutation. However, in the noncoding strand, there is a lack of 5'-NGG sequence at the downstream of this mutation within the suitable editing window, but a 5'-TGT PAM is present with the mutated A located at position 4 in the guide RNA (gRNA) (FIG. 1a), making it feasible to correct the stop codon with the NG-targeting base editors in this widely used mouse model of DMD. A reporter plasmid with the targeting sequence from the mdx^{4cv} mice was first constructed (FIG. 1b). The nonsense mutation in the mdx^{4cv} targeting sequence disrupts the expression of downstream EGFP and successful editing of the nonsense mutation is indicated by the restoration of EGFP expression. As shown in FIG. 1c, transfection with the reporter alone resulted in minimal background fluorescence. Similarly, co-transfection with the reporter, mdx^{4cv}-gRNA and ABE_{max} failed to restore EGFP expression. However, ABE-NG (based on SpCas9-NG) successfully restored EGFP expression in this reporter assay. In contrast, ABE-x (based on xCas9-3.7) was found to be less efficient in restoring EGFP expression even though xCas9-3.7 was also engineered to target 5'-NG PAM, consistent with previous reports that xCas9-3.7 is generally less efficient than SpCas9-NG. FACS analysis showed that ABE-x and ABE-NG restored EGFP expression in 10% and 20% cells, respectively (FIGS. 1d, 1e). These in vitro studies showed that ABE-NG corrects the nonsense mdx^{4cv} mutation.

Example 2. Improvements in the Editing Efficiency and Specificity of ABE-NG

The relative low efficiency of ABE-NG, together with the recently reported off-target RNA editing activity, prompted the re-design of ABE-NG in order to improve the editing efficiency and specificity. First, the targeting efficiency of ABE-NG at the sites with 5'-NG PAM can be improved by optimizing the PAM-interacting domain. The targeting property of ABE-NG can be modified by combining the mutations in SpCas9-NG (R1335V/L1111R/D1135V/G1218R/E1219F/A1322R/T1337R) with other mutations designed to target different PAM sequences such as those in xCas9(3.7) (A262T/R324L/S409I/E480K/E543D/M694I/E1219V), VQR (D1135V/R1335Q/T1337R), VRER (D1135V/G1218R/R1335E/T1337R) and the loop sequence in ScCas9 (amino acids 367-376). Seven new ABE variants were generated with different combinations of the aforementioned variants (Table 1) and compared their base editing activities at six different loci with those of ABE-NG and ABEmaxSC. While all variants except ABE-NGC (containing all NG mutations plus R1335E) performed similarly at the NGG site (FIG. 2a), it was observed that ABE-NGA (carrying all NG mutations plus R1335Q) had a small improvement in editing the NGH sites as compared to ABE-NG (FIGS. 2b-2e). ABE-NGA and ABE-NGX-NGC (carrying the xCas9(3.7) mutations, NG mutations and R1335E) worked equally well at the NGC site (FIG. 2c). The ABE-NG and ABE-NGA also edited the site containing a 5'-GAT PAM with high efficiency (FIG. 2f), with similar efficiency as compared to the ABE-NGX variant carrying both the xCas9(3.7) mutations and ABE-NG mutations, which was previously reported to have the broadest targeting scope active in plant. However, the efficiency of ABE-NGX at the NGH sites was lower than ABE-NGA (FIG. 2b). Since ABE-NGA is generally superior to other variants tested at NGH sites, ABE-NGA was chosen for further in vitro and in vivo studies.

Previous studies showed that the deaminase domain in the ABEs can elicit transcriptome-wide RNA off-target editing activity, and that the off-target RNA editing activity can be substantially reduced by removing the WT ecTadA domain and mutating the evolved ecTadA domain. The dimeric adenine deaminase domain (ecTadA-ecTadA*) in ABE-NG was replaced with the originally evolved ecTadA* monomer or its high-fidelity version (ecTadA*-V82G) (FIG. 3a) in order to minimize the off-target RNA editing activity. The miniABE-NG (the monomeric TadA* fused with SpCas9-NG nickase) performed slightly better at the mdx^{4cv} target site as compared to ABE7.10-NG (FIG. 3b). However, the on-target DNA editing activity of miniABE(V82G)-NG was remarkably reduced by over 50% when compared to ABE7.10-NG (FIG. 3b). Then the next experiment attempted to improve the on-target DNA editing efficiency of the high-fidelity mini ABE(V82G)-NG without compromising its low off-target RNA editing activity. The V82G is one of the 26 amino acid residue positions in ecTadA that reside near the enzymatic pocket around the substrate tRNA, inferred from the *S. aureus* TadA-tRNA co-crystal structure. The V82G mutation does not only affect the non-specific affinity to RNA substrates, but can also reduce its affinity to the DNA substrates. It was noticed that the A56G mutation (which is also inferred to lie near the enzymatic pocket around the tRNA substrate) had higher on-target DNA editing activity without affecting the off-target RNA editing activity as compared to miniABEmax. Installing the A56G mutation into miniABE(V82G)-NG can improve its on-

target DNA editing activity without compromising its off-target RNA editing profile. Indeed, it was observed that adding the A56G mutation into miniABE(V82G)-NG (named miniABE(GG)-NG) completely restored its on-target DNA editing activity (FIG. 3b).

RNA-seq was used to compare the transcriptome-wide off-target RNA editing activities of miniABE(GG)-NG to other ABE variants in mouse Neuro2a cells. These studies were performed in triplicate. Edited RNA adenines were identified from RNA-seq experiments by filtering out background editing observed with read-count-matched controls. MiniABE-NG and miniABE(A56G)-NG induced much higher numbers of adenine editing as compared to miniABE(V82G)-NG or miniABE(GG)-NG (FIG. 3c). There was no statistical difference between miniABE(V82G)-NG and miniABE(GG)-NG. To further verify the off-target RNA editing activity of mini ABE(GG)-NG, four RT-PCR amplicons were amplified and sequenced, which were shown to be highly modified by ABEmax in human cells. Transfection of HEK293 cells with mini ABE-NG induced high levels of A-to-I RNA editing in all these transcripts (FIG. 3d); however, such A-to-I RNA editing was essentially eliminated in cells transfected with miniABE(GG)-NG (FIG. 3d). Taken together, these results showed that miniABE(GG)-NG does not only have increased DNA editing activity, but also inherits the high fidelity of miniABE(V82G)-NG. Hereafter, the improved ABE-NG carrying the miniABE(GG) domain and Cas9-NGA nickase was referred to as iABE-NGA.

A new generation of ABEs were developed through directed evolution, namely, ABE8s (such as ABE8.17 and ABE8.20) and ABE8e. To directly compare miniABE(GG) with ABE8.17, ABE8.20 and ABE8e, each of them was fused with SpCas9-NG and tested their activities for editing the mdx^{4cv} target site using the reporter assay in Neuro2A cells. All these editors showed above 60% editing efficiency with the ABE8e-NG exhibiting the highest activity (FIG. 4a). ABE8e can have also increased bystander activity than miniABE(GG). Since the mdx^{4cv} target site has no extra adenine within the editing window, the bystander editing activity of ABE8e and miniABE(GG) was compared by testing their performance to edit a nonsense mutation in human DYSF gene (encoding dysferlin) that causes limb girdle muscular dystrophy. The target A⁶ was edited with ~58% and 88% efficiency by iABE-NG and ABE8e-NG, respectively (FIG. 4b). The two bystander adenines at positions 8 and 11 were also edited at substantially higher (quantitative numbers) rates by ABE8e-NG than miniABE(GG)-NG. Even the ABE8e with V106W mutation still displayed very high bystander editing activity at position 8. Thus, for in vivo applications that require not only efficiency but also precision, the use of miniABE(GG) was recommended.

Example 3. Intein-Split Allows Efficient Assembly of Full-Length ABE-NG and Editing

The large size of the ABE-NG and other base editors poses a major challenge for viral packaging and in vivo delivery. A dual trans-splicing adeno-associated virus (AAV) approach was used to deliver ABE and a dual protein trans-splicing (PTS) approach using the split-intein moiety from *Nostoc punctiforme* (Npu) was used to deliver CBE. The PTS approach was adopted to deliver ABE. The ABE was split between the ecTad-ecTadA* and the Cas9 nickase with Npu intein moieties, and this split renders low editing efficiency (FIG. 5). To improve the editing efficiency of the split ABE, the amino acid position 573 and 574 of the Cas9

nickase were chosen as the splitting site because previous studies showed that 573/574 split Cas9 exhibited near the full-length Cas9 activity. Moreover, split at this site would produce a roughly equal size of the two halves for AAV packaging (FIG. 4a). The split ABE can be further improved by using inteins with fast rate of PTS. Two inteins were selected with the remarkably fast rate of PTS: Cfa ($t_{1/2}$ =20 s at 30° C.) and Gp41-1 ($t_{1/2}$ =5 s at 37° C.), which are ~2.5-fold and ~10-fold faster than the rate reported for the Npu DnaE intein ($t_{1/2}$ =50 s at 37° C.), respectively. Transfection of both split versions into HEK293 cells resulted in robust expression of full-length ABEs as detected by the anti-Cas9 antibody (FIG. 6b), although the expression level was generally lower than the ABEmax but higher than the original ABE7.10. Co-transfection with the split ABH-NG, mdx^{4cv}-gRNA and the mdx^{4cv} reporter restored EGFP expression to a similar level as the full-length ABE-NG (FIG. 6c). There was no significant difference between the Cfa and Gp41-1 intein splits (FIG. 6d). The Gp41-1 version was chosen for further studies.

It was shown that the Npu intein split of ABE worked well in vivo and in vivo. The Gp41-1 split and Npu split was directly compared. While both the Gp41-1 split and Npu split allowed the assembly of full-length iABE-NGA, Western blotting analysis showed that the Gp41-1 split rendered significantly more full-length iABE-NGA protein as compared to the Npu split (FIGS. 6e, 6f). The assembly efficiency (as measured by the percentage of the full-length band) of the Gp41-1 split was about 70%, while the Npu split resulted in only 21% (FIG. 6g). To further compare the editing efficiency of the Gp41-1 split and Npu split, the T-to-C conversion of the mdx^{4cv} stop codon was quantified in Neuro2A cells using the reporter assay. As compared to the full-length iABE-NGA, the Gp41-1 split and Npu split retained about 85.5% and 78.8% of its activity, respectively (FIG. 6h). The difference in the editing efficiency between the Gp41-1 split and Npu split was small but statistically significant ($p=0.009$) (FIG. 6h). In the Gp41-1 split and Npu split shown herein, each half carries a U6-gRNA expression cassette, while only the C-terminal half of the Npu split reported by the Liu group carries the U6-gRNA expression cassette. To test if the double U6-gRNA cassette has higher editing activity than a single U6-gRNA cassette, the U6-gRNA cassette was removed from the N-terminal construct of the Npu split used herein, which resulted in greatly reduced editing (FIG. 6h), indicating that the gRNA dosage is a limiting factor for ancient Cas9-mediated editing. Moreover, to test the specificity of intein-mediated assembly of iABE-NGA, the N and C-terminal fragments of the Gp41-1 and Npu splits were swapped, and observed no editing (FIG. 6h), indicating that the intein-mediated protein splicing and assembly of full-length iABE-NGA are required for efficient editing.

Example 4. Systematic Delivery of AAV9-iNG Leads to Widespread Dystrophin Restoration

The two Gp41 intein split halves of the iABE-NGA were packaged into AAV9 (hereafter referred to as AAV9-iNG) and tested if in vivo delivery of iABE-NG-A could correct the mutation in mdx^{4cv} mice. A truncated MHCK7 promoter was used to drive the expression of two halves of iABE-NGA. A preliminary testing of two dosages (a total of 5×10^{13} or 1×10^{14} vg/kg, 1:1 of the N and C-terminal half) showed that the higher dose appeared to increase the dystrophin-positive myocytes in the mdx^{4cv} mouse heart (FIG. 7). The higher dose (1×10^{14} vg/kg, 1:1 of the N and

C-terminal half) was chosen for the rest of the study. In addition, the study also showed that injection of AAV9-iNG carrying a non-targeting gRNA failed to induce dystrophin rescue (FIG. 7).

A cohort of nine mdx^{4cv} mice were treated with AAV9-iNG (a total of 1×10^{14} vg/kg, 1:1 of the N and C-terminal half) through a single tail vein injection at 5 weeks of age. A subset of the mice was sacrificed at 5 weeks after AAV9-iNG administration. Dystrophin was found to be widely rescued in mdx^{4cv} heart (FIG. 8a and FIGS. 9-15). Quantification of the entire heart sections showed that $41.9 \pm 10.5\%$ cardiomyocytes of mdx^{4cv} mice became dystrophin positive at 10 weeks of age after systematic AAV9-iNG treatment ($N=5$) while the control mdx^{4cv} hearts were essentially dystrophin negative ($0.03 \pm 0.02\%$, $N=4$) (FIG. 8b). Dystrophin was also rescued in skeletal muscles (diaphragm and gastrocnemius) of mdx^{4cv} mice treated with AAV9-iNG, albeit the recovery was less efficient as compared to that in the heart (FIGS. 8c, 8d, and FIG. 16). Western blot analysis showed that dystrophin was rescued in mdx^{4cv} mouse heart to $45.9 \pm 11.7\%$ of the WT level following systemic AAV9-iNG treatment (FIGS. 8e, 8f). Consistent with the immunofluorescence data, Western blot showed dystrophin was restored to about $8.0 \pm 2.6\%$ of the WT level in the gastrocnemius muscle of mdx^{4cv} mice (FIG. 17).

A group of mdx^{4cv} mice treated with intravenous administration of AAV9-iNG at 5 weeks of age were kept for 10 months to study the long-term impact of systemic ABE editing therapy. A near complete dystrophin restoration was observed in the hearts of all four treated mdx^{4cv} mice at 10 months of age (FIGS. 8a, 8b and FIGS. 18-23). Dystrophin was also rescued in the skeletal muscles of these older animals with a similar percentage of dystrophin-positive muscle fibers as analyzed at the 10 weeks of age (FIGS. 5c, 5d and FIG. 24). Western blot analysis showed near WT levels of dystrophin expression in the hearts of the 10-month-old mdx^{4cv} mice treated with AAV9-iNG (FIGS. 8g, 8h).

The heart and muscle tissues contain many different types of cells, which makes it challenging to precisely determine the DNA editing efficiency in myocytes. To estimate the editing efficiency of the Dmd gene, the total RNA was extracted from the heart tissues treated with or without AAV9-iNG, amplified the target region by RT-PCR, and analyzed the resulting amplicons by Sanger sequencing and BEAT program. The AAV9-iNG treated mdx^{4cv} hearts showed an average $32.6 \pm 2.0\%$ T-to-C editing at 10 weeks of age (FIGS. 8i, 8j) and $84.6 \pm 2.6\%$ at 10 months of age (FIGS. 8i, 8k).

Repeated cycles of muscle degeneration and regeneration in muscular dystrophy result in muscle fibrosis. To examine if systemic AAV9-iNG delivery can improve the histopathology of mdx^{4cv} mice, Trichrome staining was performed in 10-month-old mice. As compared to WT mice, the mdx^{4cv} mice showed significantly elevated fibrosis in both diaphragm and gastrocnemius muscles and the fibrotic areas in these muscles were significantly reduced in the mdx^{4cv} mice treated with AAV9-iNG (FIGS. 25a-25c). Consistent with previous studies that the mdx mice do not develop overt cardiomyopathy before one year old, there were no significant changes in cardiac fibrosis in mdx^{4cv} mice with or without AAV9-iNG treatment at 10 months of age as compared to the WT controls (FIG. 25d). AAV9-iNG treatment also significantly reduced the percentage of centrally nucleated fibers (CNF) in both diaphragm and gastrocnemius muscles at 10 weeks of age (FIGS. 25e, 25g). By 10 months of age, the effects of AAV9-iNG treatments on CNF were

blunted (FIGS. 25f, 25h). Although a significant difference in cross-sectional area (CSA) of muscle fibers was not observed following AAV9-iNG treatment (FIGS. 25i, 25j, and FIGS. 26a, 26b), the AAV9-iNG treatment appeared to shift the fiber size distribution towards those of the WT muscles (FIGS. 25k, 25l, and FIGS. 26c, 26d), particularly in gastrocnemius muscles at 10 weeks of age.

To test if systemic AAV9-iNG treatment can improve the muscle function, the muscle contractility was measured using an in vivo muscle test system. Maximum plantarflexion tetanic torque was measured during supramaximal electric stimulation of the tibial nerve at 150 Hz. While the mdx^{4cv} mice produced significantly reduced torque as compared to the WT controls, systemic delivery of AAV9-iNG significantly increased the tetanic torque in mdx^{4cv} mice (FIG. 25m).

Example 5. The Safety Profile and Off-Target Activity of AAV9-iNG Treatment

Previous studies showed that AAV-mediated delivery of CRISPR/Cas9 into neonatal mice resulted in humoral immune responses to AAV capsid but not Cas9. In contrast, AAV-mediated delivery of CRISPR/Cas9 into adult mice evoked robust anti-Cas9 immunity. Serum samples were collected to analyze the host immune responses to the AAV9 capsid and the base editor iABE-NGA. Intramuscular injection of AAV9-iNG into 5-6 weeks old mdx^{4cv} mice produced robust anti-AAV9 capsid (FIG. 27a) and anti-Cas9 antibodies (FIG. 27b) at 2 weeks after injection. The anti-AAV9 titers were similar at different time points from 2 to 7 weeks post intramuscular injection and from 7 to 9 weeks post intravenous injection (FIG. 27a). The anti-Cas9 antibody titers showed a large variation among mice at 2 weeks after intramuscular injection, but all increased to peak by 4 weeks (FIG. 27b).

The liver toxicity of AAV9-iNG treatment was examined by measuring serum aspartate aminotransferase (AST) and alanine aminotransferase (ALT), and kidney toxicity by measuring blood urine nitrogen (BUN). As compared to WT mice, the mdx^{4cv} mice showed elevated AST (FIG. 27c) and ALT (FIG. 27d). However, treatment of mdx^{4cv} mice with AAV9-iNG did not further increase the serum levels of AST and ALT at either 8 weeks or 10 months of age. Measurement of BUN did not find significant changes in the treated or untreated mdx^{4cv} mice (FIG. 27e).

One concern with ABE-mediated gene correction is the off-target activities such as gRNA mismatch tolerance, bystander editing, and off-target RNA editing. Previous studies showed that ABE can tolerate 1-2 mismatches between the gRNA and its target sites. Prediction by Cas-OFFinder showed that one site on chromosome 16 (Chr16_OT) has only one mismatch, two other sites have two mismatches and 55 sites have three mismatches (FIG. 27f). The Chr16_OT differs from the mdx^{4cv} target sequence by only one C at position 12. Neuro-2a cells were transfected with ABE-NG or iABE-NG plus the gRNA, amplified the Chr16_OT by PCR and subjected the amplicon to next generation sequencing (NGS). As shown in FIG. 27g, no significant editing of the A4 in either ABE-NG or iABE-NG transfected cells was observed. Similarly, the off-target site on chromosome 1 (Chr1_OT) was analyzed, which differs from the mdx^{4cv} target sequence by an A at position 2 and a G at position 20. Again, it was found that ABE-NG or iABE-NG did not edit the A4 at Chr1_OT (FIG. 27h).

Next, the bystander editing at the on-target mdx^{4cv} locus was analyzed in the mice treated with AAV9-iNG. Since the

10-month treated mouse hearts showed a high level of dystrophin rescue, the on-target editing efficiency was first determined in these mouse hearts by NGS. As mouse hearts contain multiple different cell types, analysis of the genomic DNA PCR products can significantly underestimate the editing efficiency. To verify this, NGS of the genomic DNA PCR products was performed from two mouse hearts receiving AAV9-iNG and exhibiting high dystrophin rescue, and an up to 11% edits at A4 was detected. Thus, the RT-PCR products were sequenced to estimate the editing efficiency at the on-target mdx^{4cv} locus. The A at position 4 (corresponding to the T within the premature stop codon in the coding strand) was converted to G with high efficiency from all four mouse hearts (FIG. 28). On average, 86.2±2.4% A-to-G conversion was measured (FIG. 27i). At the mdx^{4cv} target site, there was only one A within the editing window of 4-8, disallowing us to analyze the bystander A-to-G editing at this particular site. Another type of undesired ABE-mediated genome edits at an on-target locus is ABE-dependent cytosine-to-uracil conversion resulting in C·G to T·A mutation at that site. It was found that C6 at the mdx^{4cv} target site was edited above background with an average efficiency of 1.6±0.1% (FIG. 27j).

Finally, RNA-seq was performed to characterize the transcriptome-wide RNA off-target RNA editing induced by AAV9-iNG in the mdx^{4cv} mouse heart samples. After filtering the confident variants from control mdx^{4cv} heart samples, a few hundred RNA editing events were found in the three AAV9-iNG treated heart samples with only 32 shared by all of them (FIG. 29).

The present study has improved the split ABE-NG for AAV-mediated in vivo delivery by engineering a new NG PAM-interacting domain variant, a new adenine deaminase domain with higher on-target DNA editing efficiency without compromising the high fidelity of ABE-V82G, and a Gp41-1 intein split that mediates higher efficiency of protein splicing and editing. Together, these improvements allowed to achieve widespread dystrophin rescue and functional improvement in dystrophic mice. The editing efficiency in the heart was extraordinarily high in mdx^{4cv} mice following systemic delivery of AAV9-iNG and over 90% of cardiomyocytes were corrected to express dystrophin in mdx^{4cv} hearts at 10 months of age after a single intravenous administration of AAV9-iNG at 5 weeks old. There was no obvious toxicity detected following AAV9-iNG treatment, despite the host immune response to the AAV9 capsid and ABE. This has tremendous implication for base correction of genetic cardiomyopathies.

It was shown herein that the ecTadA* domain with the V82G mutation had significantly reduced on-target DNA editing activity as compared to the original ecTadA*. By adding the A56G mutation into the V82G variant of ecTadA*, the on-target DNA editing activity was dramatically improved without compromising the high fidelity of the V82G variant in terms of its low off-target RNA editing activity. Although the A56G_V82G variant was less efficient than the recently reported ABE8e, ABE8e had also significantly higher bystander editing activity than the A56G_V82G variant. It is essential to consider both the on-target DNA editing efficiency and the bystander DNA editing as well as off-target RNA editing activity for in vivo applications. The editors have high on-target DNA editing but induce minimal bystander DNA editing and off-target RNA editing events. The A56G_V82G variant offers a balance between the editing efficiency and the editing precision.

The iABE-NG and iABE-NGA can be broadly applied to correct DMD mutations and many other disease-causing mutations. Analysis of the ClinVar database showed that over 100 of the 174 total G>A or T>C point mutations for DMD can be targeted for repair by at least one of the ABEs (iABE-NGA). Describe all other cases including targeting the splicing sites for exon skipping and other diseases.

The recent advances in engineering Cas9 variants with non-G PAM further increases targeting capacity. Moreover, the ABE editing can be designed to induce skipping of mutant exons via targeting the canonical splicing donor or acceptor, thus further broadening the applicability of ABE editing therapy for a larger population of DMD.

The mice at ten months after AAV9-iNG delivery showed significantly higher dystrophin rescue than the mice at 10 weeks after the treatment. One explanation is that the DMD cardiomyocytes with restored dystrophin expression can gain advantage for selective survival and regeneration during the development stages after delivery of AAV9-iNG. Additionally, transduced cardiomyocyte-derived extracellular vesicles can deliver genetic materials such as transcripts encoding iABE-NG into proximal un-transduced cardiomyocytes and confer base editing in those cells.

This study has also shown that systemic delivery of AAV9-iNG resulted in dystrophin restoration in skeletal muscles and functional improvement. As compared to cardiomyocytes, the editing efficiency in skeletal muscles was substantially lower. This can be attributed to the observation that AAV9 has higher tropism towards cardiomyocytes than skeletal muscles. However, other mechanisms can also be responsible for the lower editing efficiency in skeletal muscles. For example, the dystrophic and inflammatory microenvironment in skeletal muscles can pose further constraints on AAV9 delivery and base editing. In addition, targeting muscle satellite cells can be required to improve the overall editing outcomes in skeletal muscle as they are constantly activated to replace injured skeletal muscle in DMD. Although AAV9 has been shown to transduce muscle satellite cells, the efficiency is relatively low. Moreover, the use of a muscle-specific promoter can further reduce the base editing in muscle satellite cells in the present study.

Improvements in these examples show exceptionally high editing efficiency in comparison to other approaches. First, the intein used in the current study (Gp41-1) has superfast kinetics, which allows more efficient assembly of full-length ABE (FIG. 6). Second, each half of the AAVs carries a gRNA-expressing cassette in the present study, while in the other study, gRNA is present in only the C-terminal half of the Npu intein split constructs. The data (FIG. 6h) showed that the gRNA dosage affects the editing efficiency. Third, the promoters used in these studies were also different, which can drive different expression levels of ABE in heart tissues. Finally, the intrinsic difference in the gRNAs and ABE variants can have impacts on the overall editing outcomes. Thus, the exceptionally high editing efficiency achieved in adult dystrophic mice indicates that the AAV9-iNG vectors disclosed herein are useful for clinical applications.

Example 6. Methods

Mice. Mice (C57BL/6J and B6Ros.Cg-Dmd^{mdx-4^{cv}}/J) were purchased from the Jackson Laboratory and maintained at The Ohio State University Laboratory Animal Resources in accordance with animal use guidelines. All the experimental procedures were approved by the Animal Care, Use, and Review Committee of the Ohio State University.

Plasmid construction. The pCMV-ABE7.10, pCMV-ABE-xCas9(3.7) and pCMV-ABEmax were obtained from Addgene. The NG mutations were introduced by fusion PCR of pCMV-ABEmax and subcloned into pCMV-ABEmax to make pCMV-ABEmaxNG. The A56G and V82G mutations were introduced into TadA* domain by fusion PCR and cloned into pCMV-ABEmaxNG to generate pCMV-iABEmaxNG. The CfaN minigene was synthesized by IDTdna and fused at the amino acid 573 of SpCas9-max through PCR amplification. The TadA-TadA*-SpCas9max(2-573)-CfaN fragment was PCR amplified and subcloned into pAAV under the control of meCMV promoter to generate pAAV-ABEmaxN-temp. The hU6 promoter with mdx^{4^{cv}}-targeting gRNA was PCR amplified and cloned into pAAV-ABEmaxN-temp to make pAAV-ABEmaxN. The CfaC fused with SpCas9max(574-end) was generated by PCR and cloned into pAAV-ABEmaxN-temp to make pAAV-ABEmaxC. Similarly pAAV-ABEmaxN2 and pAAV-ABEmaxC2NG with the Gp41-1 intein, and pAAV-ABEmaxN3 and pAAV-ABEmaxC3NG with the Npu intein were constructed. The mdx^{4^{cv}} gRNA and other gRNA oligos (listed in Table 2) were annealed and ligated into pLentiogRNA. The mdx^{4^{cv}} reporter oligos were annealed and ligated into pLKO-puro-2A-mdx^{4^{cv}}-EGFP. All plasmids used in this study are listed in Table 3.

Generation of AAV particles. AAV vectors were produced at the viral vector core of the Nationwide Children's Hospital as previously described. The Gp41-1 intein split of iABE-NGA and the gRNA targeting mdx^{4^{cv}} mutation (GT-TaTCTCTGTTCTGCAGC TGT (SEQ ID NO: 621); note: the underlined PAM sequences were not included in the gRNA) or a non-targeting gRNA (GTTTGTGTCACCAGAGTAAC (SEQ ID NO: 571), the different nucleotides are highlighted in blue) expression cassettes were packaged into AAV9 capsid using the standard triple transfection protocol. A quantitative PCR-based titration method was used to determine an encapsulated vector genome titer utilizing a Prism 7500 Fast Taqman detector system (PE Applied Biosystems Grand Island, NY USA). The following primers/probes were used: 5'-GGATTTC-CAAGTCTCCACCC-3' (SEQ ID NO: 630) and 5'-TCC-CACCGTACACGCCTAC-3' (SEQ ID NO: 631) for titrating AAV9-NG, and AAV9-iNG was titrated using digital droplet PCR. Titers are expressed as DNase resistant particles per ml (DRP/ml) and rAAV titers used for injection in mice were 8.9×10^{12} DRP/ml (AAV9-NG) and 3.0×10^{13} DRP/ml (AAV9-iNG).

Cell culture and transfection. HEK293 cells were cultured in Dulbecco's modified eagle's medium (DMEM) (Corning, Manassas, VA) containing 10% fetal bovine serum (FBS) and 1% 100x penicillin-streptomycin (10,000 U/ml, invitrogen). Cells were plated in 6-well plates and transfected with the 2 µg plasmids (0.5 µg reporter, 0.75 µg gRNA and 0.75 µg ABE) per well unless specified otherwise by polyethylenimine (PEI) as previously described.

Flow cytometry. At 72 hour post transfection, HEK293 cells transfected with ABE plasmids were collected from 6-well plate and analyzed on Becton Dickinson LSR II (BD Biosciences) to determine GFP-positive cells. A total of 100,000 cell events were collected and data analysis was performed using the FlowJo software (Tree Star, Ashland, OR, USA).

Intramuscular and intravenous administration of AAV/9 particles. AAV9-iNG viral particles (2×10^{11} vg, 25 µl) were injected into the right gastrocnemius compartment of the male mdx^{4^{cv}} mice at 5-6 weeks of age or day 3. For systematic delivery, the male mdx^{4^{cv}} mice at 5-6 weeks of

age were administered with AAV9-NG, AAV9-iNG or AAV9-GFP viral particles (1×10^{14} vg/kg) via tail vein injection.

Serological analysis. Blood samples were collected at various time points after intramuscular or intravenous injection. The blood samples were allowed to clot for 15 min to 30 min and centrifuged at 5000 rpm for 10 min in room temperature. The supernatant was collected as serum and stored at -80°C . for the biochemical assays. Measurement of ALT (BioVision Incorporated), AST (BioVision Incorporated), BUN (Arbor Assays, Michigan, USA) and cardiac Troponin I (Life Diagnostics, Inc) were performed according to the manufacturer's protocols.

Antibody ELISA. Antibodies against AAV9 and SpCas9 were detected by adapting previously published protocols. In brief, recombinant AAV9 (2×10^9 vg/well) and SpCas9 protein (0.27 μg /well) were diluted in $1 \times$ Coating Buffer A (BioLegend) and used to coat a 96-well Nunc MaxiSorp plate. Proteins were incubated overnight at 4°C . to adsorb to the plate. Plates were washed four times 5 min each with PBS plus 0.05% Tween-20 and then blocked with $1 \times$ Assay Diluent A (BioLegend) for 1 h at room temperature. The anti-AAV2 (A20, cat. # 03-65155, American Research Products, Inc) and anti-SpCas9 antibody (Diagenode C15310258) was used as positive control for detection of anti-AAV9 and anti-SpCas9 antibodies, respectively. Serum samples were added in 1:50 dilution and plates were incubated for 2 h at room temperature with shaking. Plates were washed four times 5 min each and 100 μl of blocking solution containing goat anti-mouse IgG (Sigma 1:3,000) was added to each well and incubated at 1 h at room temperature. Plates were washed four times 5 min each, 100 μl of freshly mixed TMB Substrate Solution (BioLegend) was added to each well, and incubated in the dark for 20 min. The reaction was stopped by adding 100 μl 2N H_2SO_4 Stop Solution. Optical density at 450 nm was measured with a plate reader.

Muscle contractility measurements. At 5 weeks after intramuscular AAV9-NG or intravenous AAV9-iNG injection, muscle contractility was measured using an in vivo muscle test system (AuroraScientific Inc). Mice were anesthetized with 3% (w/v) isoflurane and anesthesia was maintained by 1.5% isoflurane (w/v) during muscle contractility measurement. Maximum plantarflexion tetanic torque was measured during a train of supramaximal electric stimulations of the tibial nerve (pulse frequency 150 Hz, pulse duration 0.2 ms).

Histopathological assessment of tissues. Mice were sacrificed at various time points, and tissues (heart, lung, diaphragm, spleen, kidney, liver, quadriceps and gastrocnemius) were harvested for histological, histochemical, biochemical and molecular analyses. For immunohistological examinations, tissues were embedded in optimal cutting temperature (OCT, Sakura Finetek, Netherlands) compound and snap-frozen in cold isopentane for cryosectioning. The tissues were stored at -80°C . and processed for biochemical analysis and histology assessment. Frozen cryosections (7 μm) were fixed with 4% paraformaldehyde for 15 minutes at room temperature. After washing with PBS, the slides were blocked with 3% BSA for 1 hour. The slides were incubated with primary antibodies against dystrophin (ab15277, 1:100, Abcam) and laminin- $\alpha 2$ (ALX-804-190-C100, 1:100, Enzo) at 4°C . for 1 hour. After that, the slides were washed extensively with PBS and incubated with secondary antibodies (Alexa Fluor 488 goat anti-rat IgG, Invitrogen, Carlsbad, CA or Alexa Fluor 568 donkey anti-rabbit IgG, Invitrogen) for 1 hour at room temperature. The slides were

sealed with VECTASHIELD Antifade Mounting Medium with DAPI (Vector Laboratory, Burlingame, CA). All images were taken under a Nikon Ti-E fluorescence microscope (magnification 200 $\times$) (Nikon, Melville, NY). Laminin- $\alpha 2$ -positive and dystrophin-positive muscle fibers were counted using NIS-Elements AR version 4.50 (Nikon, Melville, NY). The amount of dystrophin positive muscle fibers is represented as a percentage of total laminin- $\alpha 2$ -positive muscle fibers.

For trichrome staining, Masson's 2000 Trichrome Kit was used (American MasterTech, Lodi, CA). The muscle and heart sections were fixed with 4% paraformaldehyde for 1 hour at room temperature. After washing with PBS, the tissue sections were stained with Masson's trichrome reagent following the manufacturer's instruction.

Western blot analysis. Mouse tissues from mdx^{4cv} mice treated with or without AAV9-NG or AAV9-iNG were lysed with cold RIPA buffer supplemented with protease inhibitors and extracted protein samples were separated by SDS-PAGE (BioRad, 4-15%) and transferred onto Nitrocellulose membranes (0.45 μm). The rabbit polyclonal anti-dystrophin (E2660, 1:500, Spring Bioscience, Pleasanton, CA), rabbit polyclonal anti-Cas9 (C15310258-100, 1:1000, Diagenode, Denville, NJ) and rabbit monoclonal anti-Gapdh (#2118, 1:2000, Cell Signaling Technology, Danvers, MA) antibodies were used for immunoblotting analysis. HRP conjugated goat anti-mouse (1:4000) and goat anti-rabbit (1:4000) secondary antibodies were obtained from Cell Signaling Technology, Danvers, MA. The membranes were developed using ECL western blotting substrate (Pierce Biotechnology, Rockford, IL) and scanned by ChemiDoc XRS+ system (BioRad, Hercules, CA). Western blots were quantified using Image Lab 6.0.1 software (Bio-Rad Laboratories, Hercules, CA) according to the manufacturer's instruction.

Extraction of genomic DNA and total RNA, PCR and Sanger sequencing. Genomic DNA from mouse tissues and cultured HEK293 cells were extracted using DNeasy Blood & Tissue Kit (Qiagen, Germantown, MD). Total RNA was extracted from mouse tissues and HEK293 cells using Quick-RNA MiniPrep Kit (ZYMO Research, Irvine, CA). Five μg of treated RNA was used as template for first-strand cDNA synthesis by using RevertAid RT Reverse Transcription Kit (Life Technologies, Carlsbad, CA). Aliquots of the RT product were used for RT-PCR analysis of dystrophin editing. PCR reactions were carried out with 100 ng genomic DNA or cDNA in the GoTaq Master Mix (Promega) according to the manufacturer's instruction. The primers used for RT-PCR of the reporter genes and PCR of endogenous loci were listed in Table 2. The PCR products were purified using the Wizard SV Gel and PCR Clean-up System (Promega). Purified genomic DNA and RT PCR products (100 ng) were subjected to Sanger sequencing at the Genomics Shared Resource of the Ohio State University Comprehensive Cancer Center. The sequencing data were analyzed by BEAT program.

Targeted deep sequencing. The on-target and off-target loci were first amplified by genomic DNA PCR and/or RT-PCR using gene-specific primers with Illumina adapters (primers are provided in Table 4). The first PCR products were purified using a commercial purification kit (Promega, Madison, WI, USA), diluted, pooled, and subjected to a second round PCR with primers including the index sequences. The final PCR products were electrophoresed on an agarose gel, showing a single sharp peak. The quality and quantity were assayed using an Agilent Bioanalyzer 2100 (Genomics Shared Resource, Ohio State University Comprehensive Cancer Center). The purified amplicons were

pooled and sent for sequencing using a MiSeq nano-scale flow cell (paired-end 300 base-pair reads) at The Genomics Services Laboratory of Nationwide Children's Hospital. The FASTQ files were analyzed using CRISPResso2 with default parameters.

RNA-seq experiments. RNA library preparation was performed using NEBNext® Ultra™ II Directional (stranded) RNA Kit for Illumina (NEB #E7760L New England Biolabs) with an initial input of 100 ng ng extracted RNA per sample, measured using Qubit RNA HS reagents (#Q32852 Invitrogen) for fragmentation, cDNA synthesis and amplification. Depletion of ribosomal RNA (rRNA) was carried out with NEBNext rRNA Depletion Kit (human, mouse, rat) from New England Biolabs (#E6310X). NEBNext Multiplex oligos indexes kits (E7335L, E7500L and E7710L) from New England Biolabs were used to barcode each library following the manufacturer protocol. RNA-seq libraries were examined using an Agilent 2100 Bioanalyzer and a High Sensitivity DNA kit (Agilent Technologies, Inc). RNA-seq libraries were sequenced on Novaseq SP Paired-End 150 bp format at The Genomics Services Laboratory of Nationwide Children's Hospital.

RNA sequence variant calling and variant filtering. Illumina paired-end fastq sequencing reads were processed according to GATK Best Practices for RNA-seq variant calling. In brief, reads were aligned to the mouse mm10 reference genome using STAR version 1.5.2 in two-pass mode with the parameters implemented by the ENCODE project. Picard tools (version 2.19.0) was then applied to sort and mark duplicates of the mapped BAM files. The refined BAM files were subject to split reads that spanned splice junctions, local realignment, base recalibration and variant calling with SplitNCigarReads, IndelRealigner, BaseRecalibrator and HaplotypeCaller tools from GATK (version 4.1.2.0), respectively. Known variants in dbSNP version 142 were used during base quality recalibration. From all called variants, downstream analyses focused solely on single-nucleotide variants (SNVs) on canonical (1-22, X, Y and M) chromosomes. To identify variants with high confidence, clusters of at least five SNVs were filtered that were within a window of 35 bases and variants with Fisher strand values >30.0, qual by depth values <2.0 and sequencing depth <10. Base edits labelled as A-to-I comprise A-to-I edits called on the positive strand as well as T-to-C edits sourced from the negative strand, since the RNAs were converted into cDNA before sequencing, both the nucleotide and its complementary base can be sequenced. Results obtained with this pipeline can underestimate the actual number of RNA edits occurring in cells because of the high stringency of the variant calling pipeline and potential under-representation of intronic and intergenic RNA in our experiments.

Any confident variants found in wild-type Neuro2a cells were considered to be SNPs and were filtered out from the base-editor-transfected groups for off-target analysis. Similarly, any confident variants found in control mdx^{4cv} heart samples were filtered out from the AAV9-iNG group for off-target analysis. The editing rate was calculated as the number of mutated reads divided by the sequencing depth for each site.

ClinVar database analysis. The ClinVar data was converted into a tab-delimited flat file. A python script (clinvar.py) was written to process the tab-delimited flat file of ClinVar data.

Statistical analysis. The data were expressed as mean±S.E.M. and analyzed with GraphPad Prism 8.0.1 software (San Diego, USA). Statistical significance was determined using one-way ANOVA followed by Bonferroni

post hoc-tests for multiple groups or student's t-test for two groups. A P value of less than 0.05 is regarded as significant.

Data availability. The sequencing data have been deposited in the NCBI SRA under project accession numbers (PRJNA673243).

Example 7. Use of the System for Treatment of Spinal Muscular Atrophy

The iABE-NGA system can be broadly used to rescue dystrophin expression in DMD and treat other conditions, in addition to correcting point mutations in DMD and dysferlinopathy. For example, by targeting the splice sites, iABE-NGA can mediate exon skipping and rescue dystrophin expression in a human induced pluripotent stem cell-derived cardiomyocytes with a large deletion from exon 48 through 54 in DMD gene (FIG. 30).

Mutations in the survival motor neuron 1 (SMN1) gene cause spinal muscular atrophy (SMA). There is a paralogous gene WM in human, present in almost all SMA patients. The SMN2 is different from SMN1 in exon 7, with position 6 converted from C to T in SMN2, which results in the skipping of exon 7 in SMN2 mRNA and non-functional SMN2 protein. Using iABE-NGA-mediated editing of A36 showed that the exon 7 of SMN2 was spliced in (FIG. 31), and thus this can be a therapy for SMA.

Moreover, by targeting the splice sites, one can generate loss-of-functions in the therapeutic targets such as those involved in the regulation of plasma cholesterol levels. Previous studies identified a number of protective loss-of-functions in genes such as ANGPTL3, APOC3 and ASGR1, which reduce the plasma low cholesterol levels and renders cardioprotection. iABE-NGA can install such loss-of-function mutations in the aforementioned genes, which can be used as a one-time cure for dyslipidemia (FIG. 32).

TABLE 1

List of ABE variants engineered in this study.

Name	Description
ABE-NG	ABEmax with SpCas9-NG mutations R1335V/L1111R/D1135V/G1218R/E1219F/A1322R/T1337R
ABE-NGA	ABE-NG with R1335Q mutation
ABE-NGC	ABE-NG with R1335E mutation
ABE-NG-loop	ABE-NG with the loop sequence from ScCas9 (amino acids 367-376) inserted
ABE-NGX	ABE-NG with A262T/R324L/S409I/E480K/E543D/M694I mutations
ABE-NGX-NGA	ABE-NGX with R1335Q
ABE-NGX-NGC	ABE-NGX with R1335E
ABE-NGX-loop	ABE-NGX with the loop sequence from ScCas9 (amino acids 367-376) inserted
ABEmaxSc	ABEmax with SpCas9 nickase replaced with ScCas9 nickase
ABE-NGm	ABE-NG with the dimeric TadA-TadA* replaced with monomeric TadA* containing two additional mutations A56G and V82G
iABE-NGA	ABE-NGA with the dimeric TadA-TadA* replaced with monomeric TadA* containing two additional mutations A56G and V82G

TABLE 2

List of gRNA target sequences and primers for PCR in this study.			
Name	Sequence	SEQ ID NOs	
Mdx4cv-gRNA	GTTATCTCCTGTTCTGCAGC	SEQ ID NO: 570	
NT-gRNA	GTTTATGTCACCAGAGTAAC	SEQ ID NO: 571	
mDMD-i52-F	GAGGTAATAGAGCCAAGCCCT	SEQ ID NO: 572	
mDMD-i53-R	GCAAGAATTCCACTTTTCACTTCCT	SEQ ID NO: 573	
mDMD-E51-F	CTGTCATCTCCAACTAGAAATGC	SEQ ID NO: 574	
mDMD-E55-R	GCAGCCTCTTGCTCACTTACTC	SEQ ID NO: 575	
S1-gRNA	GATGACAGGCAGGGCACCG	SEQ ID NO: 576	
S1-F	TTCCAGTGGTTCAATGGTCA	SEQ ID NO: 577	
S1-R	CTTTCAACCCGAACGGAGAC	SEQ ID NO: 578	
VEGFA-S5-gRNA	GAGCGAGCAGCGTCTTCGAG	SEQ ID NO: 579	
VEGFA-S12-gRNA	GCAGACGGCAGTCACTAGGG	SEQ ID NO: 580	
VEGFA-S14-gRNA	GGGAAGCTGGGTGAATGGAG	SEQ ID NO: 581	
VEGFA-F	AGCTGTTTGGGAGGTCAGAA	SEQ ID NO: 582	
VEGFA-R	AGGGAGCAGGAAAGTGAGGT	SEQ ID NO: 583	
Site13-gRNA	GTCGCAGGACAGCTTTTCCT	SEQ ID NO: 584	
Site13-F	TGTAGCTACGCCTGTGATGG	SEQ ID NO: 585	
Site13-R	TGCCCTGAGATCTTTTCCTC	SEQ ID NO: 586	
FANCF-gRNA	GATCCAGGTGCTGCAGAAGG	SEQ ID NO: 587	
FANCF-F	CTCTTGCCCTCCACTGGTTGT	SEQ ID NO: 588	
FANCF-R	TCGGTAGGATGCCCTACATC	SEQ ID NO: 589	
Q623X-gRNA	ATCCTACAGCATGGTGGCTG	SEQ ID NO: 590	
Puro-F	AGTGGTCTCCGAAACCTCCGCGCCCC GCAAC	SEQ ID NO: 591	
GFP-R	TCCTTGAAGAAGATGGTGCG	SEQ ID NO: 592	

TABLE 3

List of plasmids used in this study.		
ID	Name	Description
pXL-0570	pCMV_ABE _{max} (Addgene #112095)	Expressing ABE _{max} ; used in FIG. 1c-e; FIG. 4b, d; Suppl. FIG. S2
pXL-0550	pCMV_xCas9_3.7_-ABE_7.10 (Addgene #108382)	Expressing ABE-x; used in FIG. 1c-e; FIG. 6b; FIG. 5.
pXL-0645	pCMV_ABE _{max} NG	Expressing ABE-NG; used in FIG. 1c-e; FIG. 2a-f; FIG. 3b; FIG. 6b-6d;
pXL-0752	pLKO-puro-2A-mdx4cv-GFP	mdx ^{4cv} reporter; used in FIG. 1c-e; FIG. 6c, 6d, h; FIG. 4.
pXL-0631	pLenti-puro-OgRNA_mdxE53	gRNA targeting mdx ^{4cv} mutation; used in FIG. 1c-e; FIG. 3b-d; FIG. 6c-6h; FIG. 27g, 27h; FIG. 4.
pXL-0858	pCMV_ABE _{max} NG-NGA	Expressing ABE-NGA; used in FIG. 2a-f.
pXL-0869	pCMV_ABE-NGC	Expressing ABE-NGC; used in FIG. 2a-f.
pXL-0872	pCMV_ABE-NG-loop	Expression ABE-NG-loop; used in FIG. 2a-f.

TABLE 3-continued

List of plasmids used in this study.		
ID	Name	Description
pXL-0712	pCMV-ABEmaxNGX	Expressing ABE-NGX; used in FIG. 2a-f.
pXL-0868	pCMV-ABEmaxNGX-NGA	Expressing ABE-NGX-NGA; used in FIG. 2a-f.
pXL-0875	pCMV-ABEmaxNGX-NGC	Expressing ABE-NGX-NGC; used in FIG. 2a-f.
pXL-0877	pCMV-ABEmaxNGX-loop	Expressing ABE-NGX-loop; used in FIG. 2a-f.
pXL-0723	pCMV-ABEmaxSC	Expressing ABEmaxSc; used in FIG. 2a-f.
pXL-0627	pLenti-puro-S1OgRNA	S1 targeting gRNA; used in FIG. 2a.
pZC0009	pLenti-Q2440X-ogRNA	Non-targeting gRNA; used in FIG. 7.
pXL-0796	pLenti-VEGFA-S5	gRNA targeting VEGFA Site5; used in FIG. 2b.
pXL-0797	pLenti-VEGFA-S12	gRNA targeting VEGFA Site 12; used in FIG. 2c.
pXL-0798	pLenti-VEGFA-S14	gRNA targeting VEGFA Site14; used in FIG. 2d.
pXL-0800	pLenti-Site13	gRNA targeting Site13; used in FIG. 2e.
pXL-0801	pLenti-FANCF-ogRNA	gRNA targeting FANCF; used in FIG. 2f.
pZC0103	pCMV__miniABE__NG(V82G)	Expressing miniABE(82G)-NG; used in FIG. 3b, 3c.
pZC0104	pCMV__miniABE-NG	Expressing miniABE-NG; used in FIG. 3b-3d.
pXL-0853	pCMV__miniABEmax-NG(A56G)	Expressing miniABE(A56G)-NG; used in FIG. 3b, 3c.
pXL-0854	pCMV__miniABEmaxNG(GG)	Expressing miniABE(GG)-NG; used in FIG. 3b-3d, FIG. 4.
pXL-0420	pCMV__ABE7.10 (Addgene #102919)	Expressing ABE7.10; used in FIG. 6b, FIG. 5
pXL-0670	pBac-rAAV-ABEmaxN-E53 OgRNA	Expressing Cfa Split_N of ABE and mdx4cv gRNA; used in FIG. 6b-6d.
pXL-0671	pBac-rAAV-ABEmaxC-NG-E53 OgRNA	Expressing Cfa Split_C of ABE-NG and mdx4cv gRNA; used in FIG. 6b-6d.
pXL-0672	pBac-rAAV-ABEmaxN2-E53 OgRNA	Expressing Gp41-1 Split_N of ABE and mdx4cv gRNA; used in FIG. 6b-6d.
pXL-0673	pBac-rAAV-ABEmaxC2-NG-E53 OgRNA	Expressing Gp41-1 Split_C of ABE-NG and mdx4cv gRNA; used in FIG. 6b-6d.
pZC0117	pX601-mhCMV- ABEmaxNGA-C3-E53ogRNA	Expressing Npu Split_C of iABE-NGA and mdx4cv gRNA; used in FIG. 6e-6h.
pZC0118	pX601-mhCMV-miniABEmax-N3-E53ogRNA	Expressing Npu Split_N of iABE-NGA and mdx4cv gRNA; used in FIG. 6e-6h.
pZC0119	pX601-mhCMV-miniABEmax-N3-zeo	Expressing Npu Split_N of iABE-NGA; used in FIG. 6e-6h.
pZC0031	pX601-mhCMV-miniABEmax-N2-E53ogRNA	Expressing Gp41-1 Split_N of iABE-NGA and mdx4cv gRNA; used in FIG. 6e-6h.
pZC0033	pX601-mhCMV-ABEmaxNGA-C2-E53ogRNA	Expressing Gp41-1 Split_C of iABE-NGA and mdx4cv gRNA; used in FIG. 6e-6h.
pXL-0855	pX601-MHP1-miniABEmaxNG-N2-(GG)	Expressing Gp41-1 Split_N of iABE-NGA and mdx4cv gRNA, MHP1 promoter; used for AAV9 production in FIG. 8, 27.
pXL-0706	pX601-MHP1-ABEmaxC2-NG-E53 ogRNA	Expressing Gp41-1 Split_C of iABE-NGA and mdx4cv gRNA, MHP1 promoter; used for AAV9 production in FIG. 8, 27.
pYZ1059	pCMV__NG-ABE8e	Expressing ABE8e-NG; used in FIG. 4
pZC0105	pCMV__NG-ABE8.17	Expressing ABE8.17-NG; used in FIG. 4
pZC0106	pCMV__NG-ABE8.20	Expressing ABE8.20-NG; used in FIG. 4
pZC0111	pCMV__SpG-ABE8e(V106W)	Expressing ABE8e-SpG(V106W); used in FIG. 4

TABLE 3-continued

List of plasmids used in this study.		
ID	Name	Description
pPW-0007	pLKO-puro-2A-Q623X-GFP	Q623X reporter; used in FIG. 4b.
pPW-0010	pLenti-Q623X-ogRNA	Q623X targeting gRNA; used in FIG. 4b.
pXL-0419	pCMV-ABE7.9 (Addgene #102918)	Expressing ABE7.9; used in FIG. 8.
pXL-0419	pCMV-ABE7.9 (Addgene #102918)	Expressing ABE7.9; used in FIG. 8.
pYG9010	pBac-rAAV-IntC-SpCas9n	Expressing Npu_N-SpCas9 nickase; used in FIG. 8.
pYG9011	pBac-rAAV-ABE7.10v3	Expressing GFP-TadA-TadA*-Npu_C; used in FIG. 8.

TABLE 4

List of primers used for NGS in this study.		
Name	Sequence	
Mdx4cv-E52-F	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGAACTCAT TACTGCTGCCCAAGA	SEQ ID NO: 593
Mdx4cc-E53-R	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGACCTGTT CGGCTTCTTCCCTTA	SEQ ID NO: 594
Mdx4cv-i52-F	ACACTCTTTCCCTACACGACGCTCTTCCGATCTAAATTTC ACTGTCTTCTCTTGAGT	SEQ ID NO: 595
Mdx4cv-i53-R	GTGACTGGAGTTCAGACGTGTGGTCTTCCGATCGCTTGCCCT CTGACCTGTCTTAT	SEQ ID NO: 596
mChr16-OT-F	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGTGACTAG GGGCAAAGCAAGAT	SEQ ID NO: 597
mChr16-OT-R	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCCTTCCAAA CTTTCTGCCCATTC	SEQ ID NO: 598
mChr10-T-F	ACACTCTTTCCCTACACGACGCTCTTCCGATCTAACACAGC GTGCTCTTTCCCTTAC	SEQ ID NO: 599
mChr10-T-R	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGTTTCAGAA GAACATCCCGTTGAC	SEQ ID NO: 600
NGS-final-F	AATGATACGGCGACACCGAGATCTACACTCTTTCCCTACA CGAC	SEQ ID NO: 601
NGS-final-R1	CAAGCAGAAGACGGCATACGAGATCTTGTAGTGACTGGAG TTCAGACGT	SEQ ID NO: 602
NGS-final-R2	CAAGCAGAAGACGGCATACGAGATCAGATCGTGACTGGA GTTTCAGACGT	SEQ ID NO: 603
NGS-final-R3	CAAGCAGAAGACGGCATACGAGATCCGTCCGTGACTGGAG TTCAGACGT	SEQ ID NO: 604
NGS-final-R4	CAAGCAGAAGACGGCATACGAGATATGTGAGTGACTGGAG TTCAGACGT	SEQ ID NO: 605
NGS-final-R5	CAAGCAGAAGACGGCATACGAGAT GTCCGC GTGACTGGAGTTCAGACGT	SEQ ID NO: 606
NGS-final-R6	CAAGCAGAAGACGGCATACGAGAT TTAGGC GTGACTGGAGTTCAGACGT	SEQ ID NO: 607
NGS-final-R7	CAAGCAGAAGACGGCATACGAGAT CGATGT GTGACTGGAGTTCAGACGT	SEQ ID NO: 608
NGS-final-R8	CAAGCAGAAGACGGCATACGAGAT TGACCA GTGACTGGAGTTCAGACGT	SEQ ID NO: 609
NGS-final-R9	CAAGCAGAAGACGGCATACGAGAT AGTCAA GTGACTGGAGTTCAGACGT	SEQ ID NO: 610

TABLE 4-continued

List of primers used for NGS in this study.		
Name	Sequence	
NGS-final-R10	CAAGCAGAAGACGGCATAACGAGAT AGTTCC GTGACTGGAGTTCAGACGT	SEQ ID NO: 611
NGS-final-R11	CAAGCAGAAGACGGCATAACGAGAT GATCAG GTGACTGGAGTTCAGACGT	SEQ ID NO: 612
NGS-final-R12	CAAGCAGAAGACGGCATAACGAGAT ACAGTG GTGACTGGAGTTCAGACGT	SEQ ID NO: 613
NGS-final-R13	CAAGCAGAAGACGGCATAACGAGAT TATACT GTGACTGGAGTTCAGACGT	SEQ ID NO: 614
NGS-final-R14	CAAGCAGAAGACGGCATAACGAGAT CAACAA GTGACTGGAGTTCAGACGT	SEQ ID NO: 615
NGS-final-R15	CAAGCAGAAGACGGCATAACGAGAT GTTGTT GTGACTGGAGTTCAGACGT	SEQ ID NO: 616
NGS-final-R16	CAAGCAGAAGACGGCATAACGAGAT TCGGTT GTGACTGGAGTTCAGACGT	SEQ ID NO: 617
NGS-final-R17	CAAGCAGAAGACGGCATAACGAGAT AGTATT GTGACTGGAGTTCAGACGT	SEQ ID NO: 618
NGS-final-R18	CAAGCAGAAGACGGCATAACGAGAT TCTTGT GTGACTGGAGTTCAGACGT	SEQ ID NO: 619

TABLE 5

chr	pos	hgvs_c	hgvs_p	all_traits	variants	variant type	re-pair editor	reference_seq	sgRNA_repair
X	31173588	NM_004020.3:c.2843 + 5081C > T	NP_003997.1:p.Gln3427Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	intron variant	ABE	SEQ ID NO: 48	SEQ ID NO: 222; SEQ ID NO: 223
X	32464674	NM_004006.2:c.3188G > A	NP_003997.1:p.Trp1063Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	C > T	nonsense	ABE	SEQ ID NO: 49	SEQ ID NO: 224; SEQ ID NO: 225; SEQ ID NO: 226; SEQ ID NO: 227
X	32217037	NM_004006.2:c.6317G > A	NP_003997.1:p.Trp2106Ter	not provided; Not Provided	C > T	nonsense	ABE	SEQ ID NO: 50	SEQ ID NO: 228; SEQ ID NO: 229
X	32362826	NM_004006.2:c.5287C > T	NP_003997.1:p.Arg1763Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 51	SEQ ID NO: 230; SEQ ID NO: 231; SEQ ID NO: 232

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31178668	NM_004006.2:c.10223 + 1G > A	NP_004010.1:p.Thr340=	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	synon- ymous variant	ABE	SEQ ID NO: 52	SEQ ID NO: 233; SEQ ID NO: 234; SEQ ID NO: 235; SEQ ID NO: 236
X	31180423	NM_004006.2:c.10033C > T	NP_003997.1:p.Arg3345Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy; not provided not specified	G > A	nonsense	ABE	SEQ ID NO: 53	SEQ ID NO: 237; SEQ ID NO: 238
X	31478163	NM_004006.2:c.8880G > A	NP_003997.1:p.Trp2960Ter		C > T	nonsense	ABE	SEQ ID NO: 54	SEQ ID NO: 239
X	31479043	NM_004006.2:c.8608C > T	NP_003997.1:p.Arg2870Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy 3B	G > A	nonsense	ABE	SEQ ID NO: 55	SEQ ID NO: 240
X	32697947	NM_004006.2:c.883C > T	NP_003997.1:p.Arg295Ter	Becker muscular dystrophy; Duchenne muscular dystrophy; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 56	SEQ ID NO: 241; SEQ ID NO: 242; SEQ ID NO: 243; SEQ ID NO: 244; SEQ ID NO: 245
X	32573766	NM_004006.2:c.1683G > A	NP_003997.1:p.Trp561Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 57	SEQ ID NO: 246; SEQ ID NO: 247; SEQ ID NO: 248; SEQ ID NO: 249
X	32484918	NM_004006.2:c.2803 + 1G > A		Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	C > T	splice donor variant	ABE	SEQ ID NO: 58	SEQ ID NO: 250
X	32573744	NM_004006.2:c.1704 + 1G > A		Becker muscular dystrophy; Duchenne muscular dystrophy; not specified	C > T	splice donor variant	ABE	SEQ ID NO: 59	SEQ ID NO: 251; SEQ ID NO: 252; SEQ ID NO: 253; SEQ ID NO: 254
X	32816541	NM_004006.2:c.457C > T	NP_003997.1:p.Gln153Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 60	SEQ ID NO: 255; SEQ ID NO: 256; SEQ ID NO: 257; SEQ ID NO: 258
X	32348501	NM_004006.2:c.5353C > T	NP_003997.1:p.Gln1785Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 61	SEQ ID NO: 259; SEQ ID NO: 260; SEQ ID NO: 261

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32411811	NM_004006.2:c.4174C > T	NP_003997.1:p.Gln1392Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 62	SEQ ID NO: 262; SEQ ID NO: 263; SEQ ID NO: 264; SEQ ID NO: 265; SEQ ID NO: 266
X	31182861	NM_004006.2:c.9851G > A	NP_003997.1:p.Trp3284Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	nonsense	ABE	SEQ ID NO: 63	SEQ ID NO: 267; SEQ ID NO: 268
X	32809577	NM_004006.2:c.565C > T	NP_003997.1:p.Gln189Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 64	SEQ ID NO: 269
X	32216981	NM_004006.2:c.6373C > T	NP_003997.1:p.Gln2125Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 65	SEQ ID NO: 270
X	32573812	NM_004006.2:c.1637G > A	NP_003997.1:p.Trp546Ter	Motor delay; Muscle weakness; Muscle cramps; EMG abnormality; EMG; myopathic abnormalities; Calf muscle hypertrophy	C > T	nonsense	ABE	SEQ ID NO: 66	SEQ ID NO: 271; SEQ ID NO: 272; SEQ ID NO: 273; SEQ ID NO: 274; SEQ ID NO: 275
X	31507314	NM_004006.2:c.8357G > A	NP_003997.1:p.Trp2786Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 67	SEQ ID NO: 276; SEQ ID NO: 277; SEQ ID NO: 278; SEQ ID NO: 279; SEQ ID NO: 280
X	32463458	NM_004006.2:c.3413G > A	NP_003997.1:p.Trp1138Ter	Becker muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 68	SEQ ID NO: 281; SEQ ID NO: 282; SEQ ID NO: 283
X	31223071	NM_004006.2:c.9337C > T	NP_003997.1:p.Arg3113Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 69	SEQ ID NO: 284; SEQ ID NO: 285

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31929602	NM_004006.2:c.6906G > A	NP_003997.1:p.Trp2302Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	5 prime UTR variant	ABE	SEQ ID NO: 70	SEQ ID NO: 286
X	32595765	NM_004006.2:c.1594C > T	NP_003997.1:p.Gln532Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 71	SEQ ID NO: 287; SEQ ID NO: 288; SEQ ID NO: 289; SEQ ID NO: 290
X	32573834	NM_004006.2:c.1615C > T	NP_003997.1:p.Arg539Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 72	SEQ ID NO: 291; SEQ ID NO: 292; SEQ ID NO: 293; SEQ ID NO: 294; SEQ ID NO: 295
X	31679565	NM_004006.2:c.7682G > A	NP_003997.1:p.Trp2561Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 73	SEQ ID NO: 296; SEQ ID NO: 297
X	31679564	NM_004006.2:c.7683G > A	NP_003997.1:p.Trp2561Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 74	SEQ ID NO: 298
X	32485057	NM_004006.2:c.2665C > T	NP_003997.1:p.Arg889Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 75	SEQ ID NO: 299; SEQ ID NO: 300; SEQ ID NO: 301; SEQ ID NO: 302; SEQ ID NO: 303
X	32849781	NM_004006.2:c.133C > T	NP_003997.1:p.Gln45Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	5 prime UTR variant	ABE	SEQ ID NO: 76	SEQ ID NO: 304; SEQ ID NO: 305; SEQ ID NO: 306; SEQ ID NO: 307
X	32501767	NM_004006.2:c.2368C > T	NP_003997.1:p.Gln790Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 77	SEQ ID NO: 308; SEQ ID NO: 309; SEQ ID NO: 310
X	32545250	NM_004006.2:c.2077C > T	NP_003997.1:p.Gln693Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 78	SEQ ID NO: 311; SEQ ID NO: 312; SEQ ID NO: 313

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31679429	NM_004006.2:c.7818G > A	NP_003997.1:p.Trp2606Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	nonsense	ABE	SEQ ID NO: 79	SEQ ID NO: 314; SEQ ID NO: 315
X	31679492	NM_004006.2:c.7755G > A	NP_003997.1:p.Trp2585Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 80	SEQ ID NO: 316; SEQ ID NO: 317; SEQ ID NO: 318; SEQ ID NO: 319; SEQ ID NO: 320
X	31875331	NM_004006.2:c.6955C > T	NP_003997.1:p.Gln2319Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	5 prime UTR variant	ABE	SEQ ID NO: 81	SEQ ID NO: 321
X	32342105	NM_004006.2:c.5917C > T	NP_003997.1:p.Gln1973Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 82	SEQ ID NO: 322
X	31774193	NM_004006.2:c.7310 - 1G > A		Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 83	SEQ ID NO: 323; SEQ ID NO: 324
X	31177932	NM_004006.2:c.10262C > T	NP_003997.1:p.Ala3421Val	Becker muscular dystrophy; BECKER MUSCULAR DYSTROPHY; not specified	G > A	500B down-stream variant	ABE	SEQ ID NO: 84	SEQ ID NO: 325; SEQ ID NO: 326
X	31658123	NM_004006.2:c.7894C > T	NP_003997.1:p.Gln2632Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 85	SEQ ID NO: 327
X	32468683	NM_004006.2:c.2977C > T	NP_003997.1:p.Gln993Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 86	SEQ ID NO: 328; SEQ ID NO: 329
X	32699111	NM_004006.2:c.831 + 1G > A		Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 87	SEQ ID NO: 330; SEQ ID NO: 331; SEQ ID NO: 332
X	32411772	NM_004006.2:c.4213C > T	NP_003997.1:p.Gln1405Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	nonsense	ABE	SEQ ID NO: 88	SEQ ID NO: 333; SEQ ID NO: 334

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31178721	NM_004006.2:c.10171C > T	NP_003997.1:p.Arg3391Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 89	SEQ ID NO: 335; SEQ ID NO: 336
X	31478983	NM_004006.2:c.8668G > A	NP_003997.1:p.Glu2890Lys	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	missense variant	ABE	SEQ ID NO: 90	SEQ ID NO: 337; SEQ ID NO: 338
X	31180369	NM_004006.2:c.10086 + 1G > A		Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 91	SEQ ID NO: 339; SEQ ID NO: 340; SEQ ID NO: 341
X	32364647	NM_004006.2:c.5089C > T	NP_003997.1:p.Gln1697Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 92	SEQ ID NO: 342
X	31496892	NM_004006.2:c.8443C > T	NP_003997.1:p.Gln2815Ter	Dilated cardiomyopathy 3B; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 93	SEQ ID NO: 343; SEQ ID NO: 344; SEQ ID NO: 345; SEQ ID NO: 346
X	32411868	NM_004006.2:c.4117C > T	NP_003997.1:p.Gln1373Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 94	SEQ ID NO: 347; SEQ ID NO: 348
X	31819974	NM_004006.2:c.7309 + 1G > A		Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 95	SEQ ID NO: 349
X	31444621	NM_004006.2:c.8944C > T	NP_003997.1:p.Arg2982Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 96	SEQ ID NO: 350; SEQ ID NO: 351; SEQ ID NO: 352; SEQ ID NO: 353

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31627852	NM_004006.2:c.8038C > T	NP_003997.1:p.Arg2680Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 97	SEQ ID NO: 354; SEQ ID NO: 355; SEQ ID NO: 356; SEQ ID NO: 357
X	32501833	NM_004006.2:c.2302C > T	NP_003997.1:p.Arg768Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 98	SEQ ID NO: 358; SEQ ID NO: 359; SEQ ID NO: 360
X	32365049	NM_004006.2:c.4996C > T	NP_003997.1:p.Arg1666Ter	Dilated cardiomyopathy 3B; not provided; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 99	SEQ ID NO: 361; SEQ ID NO: 362
X	32345999	NM_004006.2:c.5530C > T	NP_003997.1:p.Arg1844Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 100	SEQ ID NO: 363; SEQ ID NO: 364; SEQ ID NO: 365; SEQ ID NO: 366
X	32468573	NM_004006.2:c.3087G > A	NP_003997.1:p.Trp1029Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	nonsense	ABE	SEQ ID NO: 101	SEQ ID NO: 367; SEQ ID NO: 368
X	32518098	NM_004006.2:c.2202G > A	NP_003997.1:p.Trp734Ter	Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 102	SEQ ID NO: 369; SEQ ID NO: 370; SEQ ID NO: 371; SEQ ID NO: 372
X	31820055	NM_004006.2:c.7229G > A	NP_003997.1:p.Trp2410Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	5 prime UTR variant	ABE	SEQ ID NO: 103	SEQ ID NO: 373; SEQ ID NO: 374
X	32650983	NM_004006.2:c.961 – 5831C > T		Dilated cardiomyopathy 3B; Duchenne muscular dystrophy	G > A	intron variant	ABE	SEQ ID NO: 104	SEQ ID NO: 375; SEQ ID NO: 376; SEQ ID NO: 377; SEQ ID NO: 378; SEQ ID NO: 379
X	31478129	NM_004006.2:c.8914C > T	NP_003997.1:p.Gln2972Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 105	SEQ ID NO: 380; SEQ ID NO: 381; SEQ ID NO: 382; SEQ ID NO: 383; SEQ ID NO: 384; SEQ ID NO: 385

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32389605	NM_004006.2:c.4414C > T	NP_003997.1:p.Gln1472Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	nonsense	ABE	SEQ ID NO: 106	SEQ ID NO: 386; SEQ ID NO: 387; SEQ ID NO: 388; SEQ ID NO: 389
X	32595855	NM_004006.2:c.1504C > T	NP_003997.1:p.Gln502Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 107	SEQ ID NO: 390
X	32699219	NM_004006.2:c.724C > T	NP_003997.1:p.Gln242Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	nonsense	ABE	SEQ ID NO: 108	SEQ ID NO: 391; SEQ ID NO: 392; SEQ ID NO: 393; SEQ ID NO: 394
X	33211304	NM_004006.2:c.9G > A	NP_003997.1:p.Trp3Ter	Becker muscular dystrophy; BECKER MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B; Duchenne muscular dystrophy; not provided; Not Provided	C > T	nonsense	ABE	SEQ ID NO: 109	SEQ ID NO: 395; SEQ ID NO: 396; SEQ ID NO: 397
X	32454833	NM_004006.2:c.3433 - 1G > A		Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	splice acceptor variant	ABE	SEQ ID NO: 110	SEQ ID NO: 398; SEQ ID NO: 399; SEQ ID NO: 400; SEQ ID NO: 401; SEQ ID NO: 402
X	32472252	NM_004006.2:c.2861G > A	NP_003997.1:p.Trp954Ter	Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 111	SEQ ID NO: 403; SEQ ID NO: 404; SEQ ID NO: 405
X	32389614	NM_004006.2:c.4405C > T	NP_003997.1:p.Gln1469Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 112	SEQ ID NO: 406; SEQ ID NO: 407; SEQ ID NO: 408
X	32438240	NM_004006.2:c.4071 + 1G > A		Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 113	SEQ ID NO: 409; SEQ ID NO: 410; SEQ ID NO: 411; SEQ ID NO: 412; SEQ ID NO: 413; SEQ ID NO: 414
X	32491492	NM_004006.2:c.2407C > T	NP_003997.1:p.Gln803Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 114	SEQ ID NO: 415

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31206581	NM_004006.2:c.9649 + 1G > A		Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 115	SEQ ID NO: 416; SEQ ID NO: 417; SEQ ID NO: 418
X	32390175	NM_004006.2:c.4240C > T	NP_003997.1:p.Gln1414Ter	Duchenne muscular dystrophy; not specified	G > A	nonsense	ABE	SEQ ID NO: 116	SEQ ID NO: 419; SEQ ID NO: 420; SEQ ID NO: 421; SEQ ID NO: 422; SEQ ID NO: 423
X	32491387	NM_004006.2:c.2512C > T	NP_003997.1:p.Gln838Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 117	SEQ ID NO: 424; SEQ ID NO: 425; SEQ ID NO: 426; SEQ ID NO: 427
X	32342154	NM_004006.2:c.5868G > A	NP_003997.1:p.Trp1956Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 118	SEQ ID NO: 428; SEQ ID NO: 429; SEQ ID NO: 430; SEQ ID NO: 431
X	32816509	NM_004006.2:c.489G > A	NP_003997.1:p.Trp163Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	C > T	nonsense	ABE	SEQ ID NO: 119	SEQ ID NO: 432; SEQ ID NO: 433
X	32454685	NM_004006.2:c.3580C > T	NP_003997.1:p.Gln1194Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 120	SEQ ID NO: 434; SEQ ID NO: 435
X	32573529	NM_004006.2:c.1812 + 1G > A		Becker muscular dystrophy; Muscular dystrophy, Becker; Duchenne muscular dystrophy; not provided; Not Provided	C > T	splice donor variant	ABE	SEQ ID NO: 121	SEQ ID NO: 436
X	31223046	NM_004006.2:c.9361 + 1G > A		Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 122	SEQ ID NO: 437
X	32545310	NM_004006.2:c.2017C > T	NP_003997.1:p.Gln673Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	nonsense	ABE	SEQ ID NO: 123	SEQ ID NO: 438; SEQ ID NO: 439

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32343232	NM_004006.2:c.5641C > T	NP_003997.1:p.Gln1881Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 124	SEQ ID NO: 440; SEQ ID NO: 441; SEQ ID NO: 442; SEQ ID NO: 443
X	31679430	NM_004006.2:c.7817G > A	NP_003997.1:p.Trp2606Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 125	SEQ ID NO: 444
X	32809559	NM_004006.2:c.583C > T	NP_003997.1:p.Arg195Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 126	SEQ ID NO: 445; SEQ ID NO: 446; SEQ ID NO: 447; SEQ ID NO: 448
X	32573786	NM_004006.2:c.1663C > T	NP_003997.1:p.Gln555Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 127	SEQ ID NO: 449; SEQ ID NO: 450; SEQ ID NO: 451; SEQ ID NO: 452
X	31507313	NM_004006.2:c.8358G > A	NP_003997.1:p.Trp2786Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	C > T	nonsense	ABE	SEQ ID NO: 128	SEQ ID NO: 453; SEQ ID NO: 454; SEQ ID NO: 455; SEQ ID NO: 456; SEQ ID NO: 457; SEQ ID NO: 458
X	32491463	NM_004006.2:c.2436G > A	NP_003997.1:p.Trp812Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	C > T	nonsense	ABE	SEQ ID NO: 129	SEQ ID NO: 459; SEQ ID NO: 460; SEQ ID NO: 461
X	31658118	NM_004006.2:c.7899G > A	NP_003997.1:p.Trp2633Ter	not specified	C > T	nonsense	ABE	SEQ ID NO: 130	SEQ ID NO: 462; SEQ ID NO: 463; SEQ ID NO: 464; SEQ ID NO: 465; SEQ ID NO: 466
X	32448495	NM_004006.2:c.3747G > A	NP_003997.1:p.Trp1249Ter	Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 131	SEQ ID NO: 467; SEQ ID NO: 468; SEQ ID NO: 469; SEQ ID NO: 470; SEQ ID NO: 471

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31348571	NM_004006.2:c.9148C > T	NP_003997.1:p.Gln3050Ter	not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 132	SEQ ID NO: 472; SEQ ID NO: 473; SEQ ID NO: 474; SEQ ID NO: 475
X	32485072	NM_004006.2:c.2650C > T	NP_003997.1:p.Gln884Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy; not provided	G > A	nonsense	ABE	SEQ ID NO: 133	SEQ ID NO: 476; SEQ ID NO: 477; SEQ ID NO: 478; SEQ ID NO: 479
X	32463444	NM_004006.2:c.3427C > T	NP_003997.1:p.Gln1143Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 134	SEQ ID NO: 480; SEQ ID NO: 481; SEQ ID NO: 482
X	32565742	NM_004006.2:c.1952G > A	NP_003997.1:p.Trp651Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B	C > T	nonsense	ABE	SEQ ID NO: 135	SEQ ID NO: 483; SEQ ID NO: 484; SEQ ID NO: 485; SEQ ID NO: 486
X	31147421	NM_004006.2:c.10651C > T	NP_003997.1:p.Gln3551Ter	not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 136	
X	31178700	NM_004006.2:c.10192C > T	NP_003997.1:p.Gln3398Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 137	
X	31178751	NM_004006.2:c.10141C > T	NP_003997.1:p.Arg3381Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 138	
X	31180437	NM_004006.2:c.10019G > A	NP_003997.1:p.Cys3340Tyr	Duchenne muscular dystrophy, mental retardation, and absence of erg b-wave; DUCHENNE MUSCULAR DYSTROPHY, MENTAL RETARDA- TION, AND ABSENCE OF ERG B-WAVE	C > T	missense variant	ABE	SEQ ID NO: 139	
X	31182784	NM_004006.2:c.9928C > T	NP_003997.1:p.Gln3310Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 140	
X	31209497	NM_004006.2:c.9563 + 1G > A		Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 141	

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31223062	NM_004006.2:c.9346C > T	NP_003997.1:p.Gln3116Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 142	
X	31478330	NM_004006.2:c.8713C > T	NP_003997.1:p.Arg2905Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 143	
X	31478995	NM_004006.2:c.8656C > T	NP_003997.1:p.Gln2886Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 144	
X	31627681	NM_004006.2:c.82,09C > T	NP_003997.1:p.Gln2737Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 145	
X	31679575	NM_004006.2:c.7672C > T	NP_003997.1:p.Gln2558Ter	not provided; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 146	
X	31836729	NM_004006.2:c.7189C > T	NP_003997.1:p.Gln2,397Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	5 prime UTR variant	ABE	SEQ ID NO: 147	
X	31929718	NM_004006.2:c.6790C > T	NP_003997.1:p.Gln2264Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	5 prime UTR variant	ABE	SEQ ID NO: 148	
X	32287536	NM_004006.2:c.6283C > T	NP_003997.1:p.Arg2095Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 149	
X	32287596	NM_004006.2:c.6223C > T	NP_003997.1:p.Gln2075Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 150	
X	32287680	NM_004006.2:c.6139C > T	NP_003997.1:p.Gln2047Ter	Elevated serum creatine phosphokinase	G > A	nonsense	ABE	SEQ ID NO: 151	
X	32287702	NM_004006.2:c.6118 - 1G > A		Duchenne muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 152	
X	32342171	NM_004006.2:c.5851C > T	NP_003997.1:p.Gln1951Ter	not provided; not provided	G > A	nonsense	ABE	SEQ ID NO: 153	
X	32345975	NM_004006.2:c.5554C > T	NP_003997.1:p.Gln1852Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 154	

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32345978	NM_004006.2:c.5551C > T	NP_003997.1:p.Gln1851Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 155	
X	32346023	NM_004006.2:c.5506C > T	NP_003997.1:p.Gln1836Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 156	
X	32364602	NM_004006.2:c.5134C > T	NP_003997.1:p.Gln1712Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 157	
X	32364605	NM_004006.2:c.5131C > T	XP_003997.1:p.Gln1711Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 158	
X	32365175	NM_004006.2:c.4870C > T	NP_003997.1:p.Gln1624Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 159	
X	32389536	NM_004006.2:c.4483C > T	NP_003997.1:p.Gln1495Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 160	
X	32389644	NM_004006.2:c.4375C > T	NP_003997.1:p.Arg1459Ter	Dilated cardiomyopathy 3B; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 161	
X	32454778	NM_004006.2:c.3487C > T	NP_003997.1:p.Gln1163Ter	not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 162	
X	32463438	NM_004006.2:c.3432 + 1G > A		Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 163	
X	32464585	NM_004006.2:c.3276 + 1G > A		Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 164	
X	32464603	NM_004006.2:c.3259C > T	XP_003997.1:p.Gln1087Ter	not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 165	
X	32468509	NM_004006.2:c.3151C > T	NP_003997.1:p.Arg1051Ter	Dilated cardiomyopathy 3B; Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 166	

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32468704	NM_004006.2:c.2956C > T	XP_003997.1:p.Gln986Ter	Duchenne muscular dystrophy; Becker muscular dystrophy; Dilated cardiomyopathy 3B; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 167	
X	32472310	NM_004006.2:c.2804 - 1G > A		Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 168	
X	32484925	NM_004006.2:c.2797C > T	NP_003997.1:p.Gln933Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 169	
X	32484964	NM_004006.2:c.2758C > T	NP_003997.1:p.Gln920Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 170	
X	32491276	NM_004006.2:c.2622 + 1G > A		Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	C > T	splice donor variant	ABE	SEQ ID NO: 171	
X	32491344	NM_004006.2:c.2555G > A	NP_003997.1:p.Trp852Ter	not provided; Not Provided	C > T	nonsense	ABE	SEQ ID NO: 172	
X	32491480	NM_004006.2:c.2419C > T	NP_003997.1:p.Gln807Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 173	
X	32501803	NM_004006.2:c.2332C > T	NP_003997.1:p.Gln778Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 174	
X	32545158	NM_004006.2:c.2168 + 1G > A		not provided; Not Provided; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 175	
X	32545190	NM_004006.2:c.2137C > T	NP_003997.1:p.Gln713Ter	Dilated cardiomyopathy 3B; Dilated cardiomyopathy 3B	G > A	nonsense	ABE	SEQ ID NO: 176	

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32565704	NM_004006.2:c.1990C > T	NP_003997.1:p.Gln664Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 177	
X	32565782	NM_004006.2:c.1912C > T	NP_003997.1:p.Gln638Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 178	
X	32573796	NM_004006.2:c.1653G > A	NP_003997.1:p.Trp551Ter	Duchenne muscular dystrophy; not provided	C > T	nonsense	ABE	SEQ ID NO: 179	
X	32595870	NM_004006.2:c.1489C > T	NP_003997.1:p.Gln497Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 180	
X	32614320	NM_004006.2:c.1465C > T	NP_003997.1:p.Gln489Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 181	
X	32614397	NM_004006.2:c.1388G > A	NP_003997.1:p.Trp463Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 182	
X	32644131	NM_004006.2:c.1331 + 1G > A		Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 183	
X	32644139	NM_004006.2:c.1324C > T	NP_003997.1:p.Gln442Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 184	
X	32644202	NM_004006.2:c.1261C > T	NP_003997.1:p.Gln421Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 185	
X	32645020	NM_004006.2:c.1093C > T	NP_003997.1:p.Gln365Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 186	

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
	32809492	NM_004006.2:c.649 + 1G > A		Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 187	
X	32816641	NM_004006.2:c.358 - 1G > A		Becker muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 188	
X	32823297	NM_004006.2:c.355C > T	NP_003997.1:p.Gln119Ter	Duchenne muscular dystrophy; Duchenne muscular dystrophy	G > A	5 prime UTR variant	ABE	SEQ ID NO: 189	
X	32849727	NM_004006.2:c.186 + 1G > A		Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 190	
X	32849736	NM_004006.2:c.178C > T	NP_003997.1:p.Gln60Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; not provided	G > A	5 prime UTR variant	ABE	SEQ ID NO: 191	
X	33020138	NM_004006.2:c.93 + 1G > A		Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice donor variant	ABE	SEQ ID NO: 192	
X	32595756	NM_004006.2:c.1602 + 1G > A		not provided; Not Provided	C > T	splice donor variant	ABE	SEQ ID NO: 193	SEQ ID NO: 487
X	31206668	NM_004006.2:c.9564 - 1G > A		Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 194	SEQ ID NO: 488; SEQ ID NO: 489; SEQ ID NO: 490; SEQ ID NO: 491
X	32217062	NM_004006.2:c.6292C > T	NP_003997.1:p.Arg2098Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 195	SEQ ID NO: 492
X	32644314	NM_004006.2:c.1150 - 1G > A		Duchenne muscular dystrophy; not provided; Not Provided	C > T	splice acceptor variant	ABE	SEQ ID NO: 196	SEQ ID NO: 493; SEQ ID NO: 494; SEQ ID NO: 495
X	32849821	NM_004006.2:c.94 - 1G > A		Duchenne muscular dystrophy; Duchenne muscular dystrophy	C > T	splice acceptor variant	ABE	SEQ ID NO: 197	SEQ ID NO: 496
X	31496871	NM_004006.2:c.8464C > T	NP_003997.1:p.Gln2822Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 198	SEQ ID NO: 497; SEQ ID NO: 498; SEQ ID NO: 499

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32380626	NM_004006.2:c.4729C > T	NP_003997.1:p.Arg1577Ter	Duchenne muscular dystrophy; not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 199	SEQ ID NO: 500; SEQ ID NO: 501
X	32645052	NM_004006.2:c.1061G > A	NP_003997.1:p.Trp354Ter	Duchenne muscular dystrophy	C > T	nonsense	ABE	SEQ ID NO: 200	SEQ ID NO: 502; SEQ ID NO: 503; SEQ ID NO: 504; SEQ ID NO: 505
X	32491414	NM_004006.2:c.2485C > T	NP_003997.1:p.Gln829Ter	not specified	G > A	nonsense	ABE	SEQ ID NO: 201	SEQ ID NO: 506; SEQ ID NO: 507; SEQ ID NO: 508; SEQ ID NO: 509; SEQ ID NO: 510
X	31169519	NM_004020.3:c.2843 + 9150C > T	NP_003997.1:p.Gln3493Ter	Duchenne muscular dystrophy; Becker muscular dystrophy	G > A	intron variant	ABE	SEQ ID NO: 202	SEQ ID NO: 511; SEQ ID NO: 512; SEQ ID NO: 513; SEQ ID NO: 514
X	32364704	NM_004006.2:c.5032C > T	NP_003997.1:p.Gln1678Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 203	SEQ ID NO: 515; SEQ ID NO: 516
X	32545295	NM_004006.2:c.2032C > T	NP_003997.1:p.Gln678Ter	Dilated cardiomyopathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 204	SEQ ID NO: 517; SEQ ID NO: 518; SEQ ID NO: 519; SEQ ID NO: 520; SEQ ID NO: 521; SEQ ID NO: 522; SEQ ID NO: 523
X	31178784	NM_004006.2:c.10108C > T	NP_003997.1:p.Arg3370Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 205	SEQ ID NO: 524; SEQ ID NO: 525; SEQ ID NO: 526; SEQ ID NO: 527
X	32342123	NM_004006.2:c.5899C > T	NP_003997.1:p.Arg1967Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomyopathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 206	SEQ ID NO: 528; SEQ ID NO: 529

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	31729736	NM_004006.2:c.7555G > A	NP_003997.1:p.Asp2519Asn	not provided; Not Provided; Duchenne muscular dystrophy	C > T	missense variant	ABE	SEQ ID NO: 207	SEQ ID NO: 530; SEQ ID NO: 531; SEQ ID NO: 532
X	33174335	NM_004006.2:c.31 + 36947G > A		Duchenne muscular dystrophy; Becker muscular dystrophy; Dilated cardiomy- opathy 3B	C > T	intron variant	ABE	SEQ ID NO: 208	SEQ ID NO: 533; SEQ ID NO: 534
X	32472247	NM_004006.2:c.2866C > T	NP_003997.1:p.Gln956Ter	Dilated cardiomy- opathy 3B; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 209	SEQ ID NO: 535; SEQ ID NO: 536; SEQ ID NO: 537; SEQ ID NO: 538
X	32389496	NM_004006.2:c.4518 + 5G > A		Dilated cardiomy- opathy 3B; Duchenne muscular dystrophy	C > T	intron variant	ABE	SEQ ID NO: 210	SEQ ID NO: 539; SEQ ID NO: 540; SEQ ID NO: 541
X	32438372	NM_004006.2:c.3940C > T	NP_003997.1:p.Arg1314Ter	Becker muscular dystrophy; BECKER MUSCULAR DYSTROPHY; not provided; Not Provided; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 211	SEQ ID NO: 542; SEQ ID NO: 543
X	31206663	NM_004006.2:c.9568C > T	NP_003997.1:p.Arg3190Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 212	SEQ ID NO: 544; SEQ ID NO: 545; SEQ ID NO: 546; SEQ ID NO: 547
X	31729634	NM_004006.2:c.7657C > T	NP_003997.1:p.Arg2553Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 213	SEQ ID NO: 548
X	31507280	NM_004006.2:c.8390 + 1G > A		not provided; Not Provided	C > T	splice donor variant	ABE	SEQ ID NO: 214	SEQ ID NO: 549; SEQ ID NO: 550; SEQ ID NO: 551; SEQ ID NO: 552

TABLE 5-continued

chr	pos	hgvs_c	hgvs_p	all_traits	vari- ants	vari- ant type	re- pair editor	refer- ence_seq	sgRNA_re- pair
X	32823316	NM_004006.2:c.336G > A	NP_003997.1:p.Trp112Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	C > T	5 prime UTR variant	ABE	SEQ ID NO: 215	SEQ ID NO: 553; SEQ ID NO: 554; SEQ ID NO: 555
X	32816565	NM_004006.2:c.433C > T	NP_003997.1:p.Arg145Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; not provided; Not Provided; Becker muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 216	SEQ ID NO: 556; SEQ ID NO: 557; SEQ ID NO: 558; SEQ ID NO: 559
X	32463576	NM_004006.2:c.3295C > T	NP_003997.1:p.Gln1099Ter	Dilated cardiomy- opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 217	SEQ ID NO: 560; SEQ ID NO: 561
X	32844794	NM_004006.2:c.253C > T	NP_003997.1:p.Gln85Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY	G > A	5 prime UTR variant	ABE	SEQ ID NO: 218	SEQ ID NO: 562
X	32468539	NM_004006.2:c.312C > T	NP_003997.1:p.Gln1041Ter	Duchenne muscular dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B	G > A	nonsense	ABE	SEQ ID NO: 219	SEQ ID NO: 563; SEQ ID NO: 564; SEQ ID NO: 565; SEQ ID NO: 566
X	32390103	NM_004006.2:c.4312C > T	NP_003997.1:p.Gln1438Ter	Duchenne muscular dystrophy	G > A	nonsense	ABE	SEQ ID NO: 220	SEQ ID NO: 567; SEQ ID NO: 568
X	32342264	NM_004006.2:c.575C > T	NP_003997.1:p.Gln1920Ter	not provided; Not Provided	G > A	nonsense	ABE	SEQ ID NO: 221	SEQ ID NO: 569

SEQUENCES

SEQ ID NO: 1 (pCMV-ABEmaxNG)
ATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCCGCTG
GCATTATGCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGT
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GTGCTGGTGCACAATAAGTAGTGATCGGAGAGGGATGGAACAGGCCAATCGGCC
GCCACGACCCCTACCGCACACGAGAGATCATGGCACTGAGGCAGGGAGGCGCTGGTC
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- continued

SEQUENCES

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 CAGGCACAAGCGAGTCCGCCACACCAGAGAGCTCCGGCGGCTCCTCCGGAGGATCC
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SEQ ID NO: 10 (pLKO-puro-2A-mdx⁴^{cr}-EGFP)

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SEQUENCES

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SEQ ID NO: 12, SpCas9 (D10A)-NG: Protein sequence:
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SEQ ID NO: 13, xCas9 (3.7)-D10A: Protein sequence:
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SEQ ID NO: 14, SpCas9 (D10A)-NGX: Protein sequence:
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SEQ ID NO: 15, SpCas9 (D10A)-NGA: Protein sequence:

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NGLFGNLI ALSGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
KNLSDAIILSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIEFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPH
QIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRI PYVVGPLARGNSRFAMWTRKSEETI
TPWNFEVVDKGSASQSFIERMTNFDKNLPNEKVL PKHSLLEYFTVYNELTKVKYVI
EGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNAS
LGTYYHLLKI IKDKDFLDNEENEDILEDIVLTLTLEFEDREMIERLKYAHFLDDKVMKQ
LKRRRYTGWRLSRKLINGIRDQSGKTI LDFLKS DGFANRNFMQLIHDDSLTFKEDIQ
KAQVSGQGDLSLHEHIANLAGSPA I KKGILQTVKVVDLKVVMGRHKPENIV IEMAREN
QTTQKGQKNSRERMKRI E EGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVD
QELDINRLSDYDVHDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMNYW
RQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKQVLVETRQITKHVAQILDSRMNTK
YDENDKLIREVKVITLKSCLVSDFRKDFQFYKVR E INNYHHAHDAYLNAVVG TALIKK
YPKLESEFVYGDYKVDVRKMIAKSEQEIGKATAKYFFYSNIMNPFKTEITLANGEIRK
RPLIETNGETGEI VWDKGRDFATVRKVL SMPQVNI VKKTEVQTGGFSKESIRPKRNSDK
LIARKKDWDPKKYGGFVSPTVAYSVLVAKVEKGKSKKLKSVKELLGITIMERS SFEK
NPIDFLEAKGYKEVKKDLIIKL PKYSLFELENGRKRMLASARFLQKGNELALPSKYVNF
LYLASHYEKLKGPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVL SAYN
KHRDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKQYRSTKEVL DATLIHQ SITGLY
ETRIDLSQLGGD

SEQ ID NO: 16, SpCas9 (D10A)-NGC: Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSI KKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEI FSNEMAKVDDSFHRL EESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRLIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLFQLVQTYNQLFEEENPINASGVDAKAIL SARLSKSRRL ENLIAQLPGEKK
NGLFGNLI ALSGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
KNLSDAIILSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIEFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPH
QIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRI PYVVGPLARGNSRFAMWTRKSEETI
TPWNFEVVDKGSASQSFIERMTNFDKNLPNEKVL PKHSLLEYFTVYNELTKVKYVT
EGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNAS
LGTYYHLLKI IKDKDFLDNEENEDILEDIVLTLTLEFEDREMIERLKYAHFLDDKVMKQ
LKRRRYTGWRLSRKLINGIRDQSGKTI LDFLKS DGFANRNFMQLIHDDSLTFKEDIQ
KAQVSGQGDLSLHEHIANLAGSPA I KKGILQTVKVVDLKVVMGRHKPENIV IEMAREN
QTTQKGQKNSRERMKRI E EGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVD
QELDINRLSDYDVHDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMNYW
RQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKQVLVETRQITKHVAQILDSRMNTK
YDENDKLIREVKVITLKSCLVSDFRKDFQFYKVR E INNYHHAHDAYLNAVVG TALIKK
YPKLESEFVYGDYKVDVRKMIAKSEQEIGKATAKYFFYSNIMNPFKTEITLANGEIRK
RPLIETNGETGEI VWDKGRDFATVRKVL SMPQVNI VKKTEVQTGGFSKESIRPKRNSDK
LIARKKDWDPKKYGGFVSPTVAYSVLVAKVEKGKSKKLKSVKELLGITIMERS SFEK
NPIDFLEAKGYKEVKKDLIIKL PKYSLFELENGRKRMLASARFLQKGNELALPSKYVNF
LYLASHYEKLKGPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVL SAYN
KHRDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKEYRSTKEVL DATLIHQ SITGLY
ETRIDLSQLGGD

SEQ ID NO: 17, SpCas9 (D10A)-NGX-NGA: Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSI KKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEI FSNEMAKVDDSFHRL EESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRLIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLFQLVQTYNQLFEEENPINASGVDAKAIL SARLSKSRRL ENLIAQLPGEKK
NGLFGNLI ALSGLTPNFKSNFDLAEDTKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
KNLSDAIILSDILRVNTEITKAPLSASMIKLYDEHHQDLTLLKALVRQQLPEKYKEIEFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGIIPHQ
IHLGELHAILRRQEDFYFPLKDNREKIEKILTFRI PYVVGPLARGNSRFAMWTRKSEETIT
PWNFEVVDKGSASQSFIERMTNFDKNLPNEKVL PKHSLLEYFTVYNELTKVKYVTE
GMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASL

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SEQUENCES

GTYHDLKKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKYAHLFDDKVMKQL
 KRRRYTGWRLSRKLINGIRDKQSGKTIIDFLKSDGFANRNFIQLIHDDSLTFKEDIQKA
 QVSGQDLSLHEHIANLAGSPAIIKKGILQTVKVVDLKVVMGRHKPENIVIMARENQTT
 QKGQKNSRERMKRIEKGKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQEL
 DINRLSDYDVIDHIVPQSFLKDDSIDNKVLTSDKNRGKSDNVPSEEVVKMKNYWRQL
 LNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDE
 NDKLIREVKVITLKSCLVSDFRKDFQFYKREINNYHHAHDAYLNAVVGITALIKKYPKL
 ESEFVYGDYKVDVVRKMIAKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIE
 TNGETGEIVWDKGRDFATVRKVLSPQVNI VKKTEVQTGGFSKESIRPKRNSDKLIARK
 KDWDPKKYGGFVSPTVAYSVLVVAKEVGKSKKLKSVKELLGITIMERSSEKPNIDFL
 EAKGYKEVKKDLIIKLPKYSLEFENGRKRMLASARFLQKGNELALPSKYVNFYLYASH
 YEKLKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDK
 PIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKQYRSTKEVL DATLIHQSI TGLYETRIDL
 SQLGGD

SEQ ID NO: 18, SpCas9 (D10A)-NGX-NGC: Protein sequence:
 MDKKYSIGLAIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDS
 GETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHE
 RHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTDKADLRILIYALAHMIKFRGHFLIEGDL
 NPDNSDVKLFTQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRLENLIAQLPGEKK
 NGLFGNLI ALSGLTPNFKSNFDLAEDTKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
 KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLKALVRQQLPEKYKEIFFD
 QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGIIPHQ
 IHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAMWTRKSEETIT
 PWNFEKVVDKGASQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTVYNELTKVKYVTE
 GMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASL
 GTYHDLKKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKYAHLFDDKVMKQL
 KRRRYTGWRLSRKLINGIRDKQSGKTIIDFLKSDGFANRNFIQLIHDDSLTFKEDIQKA
 QVSGQDLSLHEHIANLAGSPAIIKKGILQTVKVVDLKVVMGRHKPENIVIMARENQTT
 QKGQKNSRERMKRIEKGKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQEL
 DINRLSDYDVIDHIVPQSFLKDDSIDNKVLTSDKNRGKSDNVPSEEVVKMKNYWRQL
 LNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDE
 NDKLIREVKVITLKSCLVSDFRKDFQFYKREINNYHHAHDAYLNAVVGITALIKKYPKL
 ESEFVYGDYKVDVVRKMIAKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIE
 TNGETGEIVWDKGRDFATVRKVLSPQVNI VKKTEVQTGGFSKESIRPKRNSDKLIARK
 KDWDPKKYGGFVSPTVAYSVLVVAKEVGKSKKLKSVKELLGITIMERSSEKPNIDFL
 EAKGYKEVKKDLIIKLPKYSLEFENGRKRMLASARFLQKGNELALPSKYVNFYLYASH
 YEKLKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDK
 PIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKEYRSTKEVL DATLIHQSI TGLYETRIDL
 SQLGGD

SEQ ID: 19, SpCas9 (D10A)-NG-loop: Protein sequence:
 MDKKYSIGLAIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDS
 GETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHE
 RHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTDKADLRILIYALAHMIKFRGHFLIEGDL
 NPDNSDVKLFTQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRLENLIAQLPGEKK
 NGLFGNLI ALSGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
 KNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPEKYKEIFFD
 QSKNGYAGYVGDKKLRSSKLATEEEFYKFIKPILEKMDGTEELLVKLNREDLLRK
 QRTFDNGSI PHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRF
 AWMTRKSEETITPWNFEVVDKGASQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTV
 YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSV
 EISGVEDRFNASLGTYHDLKKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKY
 AHLFDDKVMKQLKRRRYTGWRLSRKLINGIRDKQSGKTIIDFLKSDGFANRNFMQLI
 HDDSLTFKEDIQKAQVSGQDLSLHEHIANLAGSPAIIKKGILQTVKVVDLKVVMGRHK
 PENIVIMARENQTTQKGQKNSRERMKRIEKGKELGSQILKEHPVENTQLQNEKLYLY
 YLQNGRDMYVDQELDINRLSDYDVIDHIVPQSFLKDDSIDNKVLTSDKNRGKSDNVP
 SEEVVKMKNYWRQLLNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKH
 VAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKREINNYHHAHDAY
 LNAVVGITALIKKYPKLESEFVYGDYKVDVVRKMIAKSEQEIGKATAKYFFYSNIMNFF
 KTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSPQVNI VKKTEVQTGGFS
 KESIRPKRNSDKLIARKKDWDPKKYGGFVSPTVAYSVLVVAKEVGKSKKLKSVKELL
 GITIMERSSEKPNIDFLEAKGYKEVKKDLIIKLPKYSLEFENGRKRMLASARFLQKGN
 ELALPSKYVNFYLYASHYEKLKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILAD
 ANLDKVL SAYNKHDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKQYRSTKEVL
 DATLIHQSI TGLYETRIDLSQLGGD

SEQ ID NO: 20, SpCas9 (D10A)-NGX-loop: Protein sequence:
 MDKKYSIGLAIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDS
 GETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHE
 RHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTDKADLRILIYALAHMIKFRGHFLIEGDL
 NPDNSDVKLFTQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRLENLIAQLPGEKK
 NGLFGNLI ALSGLTPNFKSNFDLAEDTKLQLSKDTYDDDLNLLAQIGDQYADLFLAA
 KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLKALVRQQLPEKYKEIFFD
 QSKNGYAGYVGDKKLRSSKLATEEEFYKFIKPILEKMDGTEELLVKLNREDLLRK
 QRTFDNGI PHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRF
 WMTKSEETITPWNFEKVVDKGASQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTV

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SEQUENCES

NELTKVKYVTEGMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEI
SGVEDRFNASLGTYYHDLKKI IKDKDFLDNEENEDILEDIVLTLTLFEDREMIERLKTYYA
HLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQKSGKTILDFLKSDGFANRNFQIQLIHD
DSLTFPKEDIQKAQVSGQDLSLHEHIANLAGSPAIIKKGILQTVKVVDELVKVMGRHKPEN
IVIEMARENQTTQKQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLYLYYLQ
NGRDMYVDQELDINRLSDYDVHIVPQSFLKDDSDNKVLT RSDKNRGKSDNVPSEEV
VKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRLVETROITKHVAQ
ILDSRMNTKYDENDKLIREV KVI TLKSKLVSDFRKDFQFYKVR EINNYYHHAHDAYLNA
VVG TALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNPFKTEI
TLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVL SMPQVNI VKKTEVQTGGFSKESI
RPKRNSDKLIAR KDWDPKKYGGFVSPTVAYSVLVVAKEKGKSKKLKSVKELLGITI
MERS SF EKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELA
LPSKYVNFLYLASHYEKLKSGPEDNEQQLFVEQHKHYLDEII EQISEFSKRVILADANL
DKVLSAYNKHRDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKVYRSTKEVL DATL
IHQSITGLYETRIDLSQLGGD

SEQ ID NO: 21, ScCas9 (D10A): Protein sequence:
MEKKYSIGLAIGTNSVGWAVITDDYKVP SKKFKVLGNTNRKSIKKNLMGALLFD
SGETA EATRLKRTARRRYTRRKNRIRYLQEIFANEMAKLDDSFQORLEESFLVEEDKKN
ERHPIFGNLADEVAHRYNYPTIYHLRKKLADSP EKA DLR LIYLA LAHIIKPRGHFLIEGKL
NAENSDVAKLFYQLIQTYNQLF EESPLDEIEVDAKGILSARLSKSKRLEKLI AVFPNEKK
NGLFGNIIALALGLTPNFKSNFDLTEDAKLQLSKDTYDDDLDELGGIQDGQYADLFSAA
KNLSDAIILSDILRSNSEVTKAPLSASMVKRYDEHHQDLAL LKTLVRQQFP EKYAEIFK
DDTKNGYAGYVVGIGIKHRKRTTKLATQE EYFKFIKPILEKMDGAEELLAKLNRDDLRLK
QRTFDNGSIPHQIHLKELHAILRRQE EFPFLKENREKIEKILTFRIPIYYGPLARGNSRFA
WLTRKSEEAITPWNFE EVDK GASAQSFIERMTNFDEQLPNKKVLPKHSLLY EYFTVYN
ELTKVKYVTERMRKPEFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEIIG
VEDRFNASLGTYYHDLKKI IKDKDFLDNEENEDILEDIVLTLTLFEDREMIERLKTYYAHLF
DDKVMKQLKRRHYTGWGRLSRKMINGIRDQKSGKTILDFLKSDGF SNRNFQIQLIHDDS
LTFKEEIEKAQVSGQDLSLHEQIADLAGSPAIIKKGILQTVKIVDELVKVMGHKPENIVIE
MARENQTTTKGLQQSRERKRIE EGIKELESQILKENPVENTQLQNEKLYLYYLQNGRD
MYVDQELDINRLSDYDVHIVPQSFIKDDSDNKVLT RSVENRGKSDNVPSEEVVKM
KNYWRQLLNAKLITQRKFDNLTKAERGGLSEADKAGFIKRLVETROITKHVARILDSR
MNTKRDKNDKPIREV KVI TLKSKLVSDFRKDFQLYKVRDINNYHHAHDAYLNAVVG
TALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGKATAKRFYYSNIMNPFKTEVKLAN
GEIRKRPLIETNGETGEIVWNKEKDFATVRKVLAMPQVNI VKKTEVQTGGFSKESI LSK
RESAKLIPRKKGWDTRKYGGFVSPTVAYSILVVAKEKGKAKKLKSVKVLVGITIMEK
GSYEKDP IGFLEAKGYKDIKKELIFKLPKYSLFELENGRRRMLASATELQKANELVLV PQH
LVRLLYYTQNISATGSSNNLGYIEQHREEFKEIFEKIIDFSEKYILKNKVN SNLKSSPDEQF
AVSDSILLNSFVSLK YTSFGASGGFTFLDLVDVKQGR LR YQT VTEVL DATLIYQSITGL
YETRTDLSQLGGD

SEQ ID NO: 22, TadA-TadA*: Protein sequence:
SEVEFSHEYWMRHALTLAKRAWDEREVPVGAVLVHNNRVIGEGWNRPIGRHD
PTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCVMCAGAMIHSRIGRVVFGARDAKT
GAAGSLMDVLVHHPGMNHRVEITEGILADECAALLSDFFRMRRQEIKAKKKAQSSTD SG
GSSGGSSGSETPGTSESATPESSGGSSGSSSEVEFSHEYWMRHALTLAKRARDEREVPV
GAVLVLNNRVIGEGWNRAI GLHDPTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCV
MCAGAMIHSRIGRVVFGVRNAKTGAAGSLMDVLHYPGMNH RVEITEGILADECAALL
CYFFRMPRQVFNAQKKAQSSTD

SEQ ID: 23, TadA* (A56G_V82G): Protein sequence:
SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDP
TGHA EIMALRQGGLVMQNYRLIDATLYGT FEP CVMCAGAMIHSRIGRVVFGVRNAKT
GAAGSLMDVLHYPGMNH RVEITEGILADECAALLCYFFRMPRQVFNAQKKAQSSTD

SEQ ID NO: 24, Gp41-1-N: Protein sequence:
CLDLK TQVQTPQGMKEISNIQVGDVLVSN TGYNEVLNVEPKSKKSYKITLEDG
KEIICSEHLFPQTQTEGMMNISGGLKEGMCLYVKE

SEQ ID NO: 25, Gp41-1-C: Protein sequence:
MMLKKILKIEELDERELIDIEVSGNHLFYANDILTHNS

SEQ ID NO: 26, Cfa-N: Protein sequence:
CLSYDTEILTV EYGF LPIGKIVEERIECTVYTVDKNGFVYTQPIAQWHRGEQEV
FEYCLEDGSIIRATKDHKFM TTDGQMLPIDEIFERGLDLKQVDGLPN

SEQ ID NO: 27, Cfa-C: Protein sequence:
MVKISRKSLGTQNVYDIGVEKDHNFLLKNGLVASN

SEQ ID NO: 28, meCMV:
tegagCGCGTGATGAGAGCAGCCACTACGGGTCTAGGCTGCCCATGTAAGGAG
GCAAGGCTGGGGACCCGAGATGCCTGGTTATAATTAACCCAGACATGTGGCTG
CCCCCCCCCCCCAACACCTGCTGCCTGCTAAAAATAACCTGTCCCTGGTGGCCctgc
atgcccACTCACGGGGATTTC AAGTCTCCACCCCATGACGCTCAATGGGAGTTTGT

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SEQUENCES

TGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGAC
GCAAATGGCGGTAGGCGGTGTACGGTGGGAGGTCTATATAAGCAGAGCTGGTTTAG
TGAAACCGTCAGATC

SEQ ID NO: 29, SEMHP1:

CCCTTCAGATTAAAAATAACTGAGGTAAGGGCCTGGGTAGGGGAGGTGGTG
TGAGACGCTCCTGTCTCTCTCTATCTGCCCATCGGCCCTTTGGGGAGGAGGAATGT
GCCCAAGGACTAAAAAAGGCCATGGAGCCAGAGGGGCGAGGGCAACAGACCTTT
CATGGGCAAACTTGGGGCCCTGCTGactgtaGATGAGAGCAGCCACTACGGGTCTAG
GCTGCCCATGTAAAGAGGCAAGGCCCTGGGGACACCCGAGATGCCTGGTTATAATTA
ACCCAGACATGTGGCTGCCCCCCCCCCCCAACCTGCTGCCTGTCTAAAAATAACC
CTGTCCCTGGTGGcctgcatgCCCTCCCTGGGGACAGCCCCCTCTGGCTAGTCACACCC
TGTAGGCTCCTCTATATAACCAGGGGCACAGGGGCTGCCCTcatttctACCACCACCTCC
ACAGCACAGACAGACTCAGGAGCCAGC

SEQ ID NO: 30, Wild Type SpCas9 (D10A): Nucleotide sequence:

ATGGACAAGAAGTACAGCATCGGCCTGGCCATCGGCACCAACTCTGTGGGCT
GGGCGGTGATCACCGACGAGTACAAGGTGCCCAGCAAGAAATTCAGGTGCTGGGC
AACACCCGACCGGCACAGCATCAAGAAGAACCTGATCGGAGCCCTGCTGTTTCGACAG
CGCGAAGACAGCGAGGCCACCCGGCTGAAGAGAACCAGCAGAGAGATACACC
AGACGGAAGAACCAGGATCTGCTATCTGCAAGAGATCTTCAGCAACGAGATGGCCAA
GGTGGACGACAGCTTCTTCCACAGACTGGAAGAGTCTCTCTGGTGGAAAGAGGATA
AGAAGCAGGAGCGGCACCCCATCTTCGGCAACATCGTGGACGAGGTGGCCTACCA
GAGAAGTACCCCCACCATCTACCACTGAGAAAGAACTGGTGGACAGCACCGACA
AGGCCGACCTCGGGTGTATCTATCTGGCCCTGGGCCACATGATCAAGTTCGGGGC
CACTTCTGTATCGAGGGCGACCTGAACCCGACACAGCGACGTGGCAAGCTGTT
CATCCAGCTGGTGCAGACCTACAACAGCTGTTTCGAGGAAAACCCCATCAACGCCA
GCGGCGTGGACGCCAAGGCCATCTCTGTCTGCCAGACTGAGCAAGAGCAGACGGCTG
GAAAATCTGATCGCCAGCTGCCCGCGAGAGAAGAATGGCCTGTTTCGGAACCT
GATTGCCCTGAGCCTGGGCCTGACCCCCAATTCAAGAGCAACTTCGACCTGGCCG
AGGATGCCAACTGACGCTGAGCAAGGACACCTACGACGACGACCTGGACAACT
GCTGGCCAGATCGGCGACCAAGTACGCGACCTGTTCTGGCCGCAAGAACCTGT
CCGACGCCATCTGTCTGAGCGACATCTTGAGAGTGAACACCGAGATCACCAAGGCC
CCCCTGAGCGCTCTATGATCAAGAGATACGACGAGCACCACAGGACCTGACCT
GCTGAAGCTCTCGTGCAGCAGCTGCTGAGAGTACAAAGAGATTTTCTTCG
ACCAGAGCAAGAACGGCTACGCGGCTACATTGACGCGGAGCCAGCCAGGAAGA
GTTCTACAAGTTTCATCAAGCCCATCTGGAAAAGATGGACGGCACCGGGAAGTGC
TCGTGAAGCTGAACAGAGAGGACCTGCTGCGGAAGCAGCGGACCTTCGCAACCG
CAGCATCCCCACAGATCCACCTGGGAGAGCTGCACGCCATTCTGCGGCGGAGG
AAGATTTTACCATTTCTGAAGGACAACCGGGAAGAGATCGAGAAGATCTTGACC
TTCCGCATCCCTACTACGTGGGCCCTTGCCAGGGGAAACAGCAGATTCGCTG
GATGACCAGAAAGAGCGAGGAAACCATCACCCCTGGAATTCGAGGAAGTGGTG
GACAAGGGCGCTTCCGCCCAGAGCTTATCGAGCGGATGACCAACTTCGATAAGAA
CCTGCCAACGAGAAGGTGCTGCCCAAGCACAGCCTGCTGTACGAGTACTTCACCG
TGTATAACGAGCTGACCAAAGTGAATACGTGACCGAGGGAATGAGAAAGCCCGC
CTTCTGAGCGGCGAGCAGAAAAGGCCATCGTGGACCTGCTGTTCAAGACCAACC
GGAAAGTGACCTGAAGCAGCTGAAGAGGACTACTTCAAGAAAATCGAGTGCTTC
GACTCCGTGGAATCTCCGGCTGGAAGATCGGTTCAACGCCCTCCCTGGGCACATA
CCACGATCTGTGAAAATATCAAGGACAAGGACTTCTTGGAATAGAGGAAAACG
AGGACATTTCTGGAAGATATCGTGTGACCTGACACTGTTTGAGGACAGAGAGAT
ATCGAGGAACGGCTGAAAACCTATGCCACCTGTTTCGACGACAAAGTGATGAAGCA
GCTGAAGCGGCGGAGATACACCGCTGGGGCAGGCTGAGCCGGAAGCTGATCAAC
GGCATCCGGGACAAGCAGTCCGGCAAGACAATCTTGATTCTCTGAAGTCCGAGG
CTTCGCCAACAGAACTTCATGCAAGCTGATCCACGACGACAGCTGACCTTTAAG
AGGACATCCAGAAAAGCCAGGTGTCCGGCCAGGGCGATAGCTGCACGAGCACATT
GCCAATCTGGCCGGCAGCCCCGCCATTAAAGAGGGCATCTGCAGACAGTGAAGGT
GGTGGACGAGCTCGTGAAGTGATGGGCCGACACAGCCGAGAACATCGTGATC
GAAATGGCCAGAGAGAACCAGACCACCCAGAGGGACAGAGAACAGCCGCGAG
AGAAAGAGCGGATCGAAGAGGGCATCAAAGAGCTGGGCAGCCAGATCTTGAAAG
AACACCCCGTGGAAAACACCCAGCTGCAGAACGAGAAGCTGTACCTGTACTACCTG
CAGAATGGGCGGATATGTACGTGACCAAGGAACTGGACATCAACCGCTGTCCGA
CTACGATGTGGAACCATATCTGTGCTCAGAGCTTTCTGAAGGACGACTCCATCGACA
ACAAGGTGCTGACCAAGCGACAAAGAACCGGGGCAAGAGCGACAACGTGCCCTC
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CAGATCACAAAGCACGTTGGCAGATCTTGACTCCCGGATGAACACTAAGTACGA
CGAGAATGACAAGCTGATCCGGGAAGTGAAGTGTATCACCTGAAGTCCAAGCTGG
TGTCCGATTTCCGGAAGGATTTCCAGTTTACAAAGTGCGCGAGATCAACCACTAC
ACCACGCCCAAGCAGCTTACCTGAACGCGCTGTGGGAACCGCCTGATCAAAAAG
TACCCTAAGCTGGAAGCGAGTTCTGTGTACGGCGACTACAAGGTGTACGACGTGCG
GAAGATGATCGCAAGAGCGAGCAGGAAATCGGCAAGGCTACCGCAAGTACTTCT
TCTACAGCAACATCATGAACTTTTCAGACCGAGATTACCTGGCCAACGGCGAG
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ATCGTGAAGAAAGACCGAGGTGCAGACAGCGGCTTCAGCAAGAGTCTATCCgGCC
CAAGAGGAACAGCGATAAGCTGATCGCCAGAAGAAGGACTGGGACCTAAGAAG
TACGGCGGCTTCgtCAGCCCCACCGTGGCTATTCTGTGTGGTGGTGGCCAAAGTG
GAAAAGGGCAAGTCCAAGAACTGAAGAGTGTGAAGAGCTGTGGGGATCACCA
TCATGGAAGAAGCAGCTTCGAGAAGAATCCATCGACTTCTGGAAGCCAGGGCG
TACAAAGAAGTGAAGAAAGGACCTGATCATCAAGCTGCCTAAGTACTCCCTGTTTGA
GCTGGAAACCGGCCGAAGAGAATGCTGGCCTCTGCCGCTtCTGCGAAGGGGAA
CGAAGTGGCCCTGCCCTCAAAATATGTGAATCTCTGTACTTGGCCAGCCACTATGA
GAAGCTGAAGGCTCCCCGAGGATAATGAGCAGAAACAGCTGTTTGTGGAACAGC
ACAAGCACTACTTGGACGAGATCATCGAGCAGATCAGCGAGTTTCTCAAGAGAGTG

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SEQUENCES

ATCCTGGCCGACGCTAATCTGGACAAAGTGCTGTCCGCCTACAACAAGCACCGGGA
TAAGCCCATCAGAGAGCAGGCCGAGAATATCATCCACCTGTTTACCCTGACCAATC
TGGGAGCCCCCTcggGCTTCAAGTACTTTGACACCACCATCGACCGGAAGgtGTACcgg
AGCACCAAGAGGTGCTGGACGCCACCTGTATCCACAGAGCATCACCGGCTGTGA
CGAGACACGGATCGACCTGTCTAGCTGGGAGGTGAC

SEQ ID NO: 39, SpCas9 (D10A)-NGX-loop: Nucleotide sequence:
ATGGACAAGAAGTACAGCATCGGCCTGGCCATCGGCACCAACTCTGTGGGCT
GGGCCGTGATCACCAGCAGTACAAGGTGCCCAGCAAGAAATTCAGGTGCTGGGC
AACACCGACCGGCACAGCATCAAGAAGAACCTGATCGGAGCCCTGCTGTTTCGACAG
CGCGGAAACAGCCGAGGCCACCCGGCTGAAGAGAACCAGCAGAGAGATACACC
AGACGGAAAGAACCGGATCTGCTATCTGCAAGAGATCTTCAGCAACGAGATGGCCAA
GGTGGACGACAGCTTCTTCCACAGACTGGAAGAGTCCTTCTGGTGGAAAGAGGATA
AGAAGCAGGAGCGGCACCCCATCTTCGGCAACATCGTGGACGAGGTGGCCTACCAC
GAGAAGTACCCCAACATCTACCACCTGAGAAAGAACTGGTGGACAGCACCGGACA
AGGCCGACCTGCGGTGATCTATCTGGCCCTGGCCACATGATCAAGTTCCGGGGC
CACTTCTGTATCGAGGGCGACCTGAACCCCGACAACAGCAGCTGGACAAGCTGTT
CATCCAGCTGGTGCAGACCTACAACAGCTGTTTCGAGGAAAACCCCATCAACGCCA
GGCGCGTGGACGCCAAGGCCATCCTGTCTGCCAGACTGAGCAAGAGCAGACGGCTG
GAAAATCTGATCGCCAGCTGCCCGCGAGAGAAGAATGGCTGTTTCGGAACCT
GATTGCCCTGAGCCTGGGCTGACCCCAACTTCAAGAGCAACTTCGACCTGGCCG
AGGATACCAAACTGCAGCTGAGCAAGGACACCTACGACGACGACCTGGACAACCT
GCTGGCCAGATCGGCGACAGTACGCGGACCTGTTCTGGCCGCAAGAACCTGT
CCGACGCCATCCTGCTGAGCGACATCCTGAGAGTGAACACCGAGATCACCAAGGCC
CCCCTGAGCGCCTCTATGATCAAGCTGTACGACGAGCACCACAGGACCTGACCT
GCTGAAAGCTCTCGTGGCGGACGCTGCTGAGAAGTACAAGAGATTTTCTTCG
ACCAGAGCAAGAACGGCTACGCGCGCTACgtgggagcagcaagaagctgcgcaagcgcagctctaaactg
gccacagagGAAGAGTTTACAAAGTTATCAAGCCCATCCTGGAAAAGATGGACGGCAC
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AGATCTGACCTTCGCGATCCCTACTACGTGGGCCCTTGGCCAGGGGAAACAGC
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CCTTTAAAGAGGACATCCAGAAAGCCAGGTGTCGGCCAGGGCGATAGCTGCAC
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AGTGAAGGTGGTGGACGAGCTCGTGAAGTGTGGGCCGGCACAAGCCGAGAAC
ATCGTGATCGAAATGGCCAGAGAGAACCAGACACCCAGAGGGACAGAGAACA
GCCCGGAGAGAAATGAAGCGGATCGAAGAGGGCATCAAAGAGCTGGGACGCGAGAT
CCTGAAGAAACACCCCGTGGAAAACACCCAGCTGCAGAACGAGAAGCTGTACCTGT
ACTACCTGCAGAAATGGCGGGATATGTACGTGGACAGGAACCTGACATCAACCG
CTGTCCGATACGATGTGGACATATCTGTGCTCAGAGCTTCTGAAGGACGACTCC
ATCGACAACAAGGTGCTGACCAAGCGACAAGAACCAGGGCAAGAGCGACAACG
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GAAGATGATCGCCAAGAGCGAGCAGGAAATCGGCAAGGCTACCGCAAGTACTTCT
TCTACAGCAACATCATGAACTTTTTCAAGACCAGATTACCTGGCCAACGGCGAG
ATCCGGAAGCGGCTTCTGATCGAGACAACCGGCAAAACCGGGAGATCGTGTGGG
ATAAGGGCCGGGATTTTGCCACCGTGGGAAAGTGTGAGCATGCCCCAAGTGAAT
ATCGTGAAGAAAGACCGAGGTGAGACAGGCGGCTTCAGCAAGAGTCTATCCGCC
CAAGAGGAACAGCGATAAGCTGATCGCCAGAAAGAGACTGGGACCTAAGAAG
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GAAAAGGGCAAGTCCAAGAACTGAAGAGTGTGAAGAGCTGCTGGGGATCACCA
TCATGGAAGAAAGAGCTTCGAGAAGAATCCATCGACTTTCTGGAAGCCAAGGGC
TACAAGAAAGTGAAGAAAGGACCTGATCATCAAGCTGCCTAAGTACTCCTGTTCGA
GCTGGAAAACGGCCGGAAGAGAATGCTGGCCTCTGCCGCTTCTGTCAGAGGGAAA
CGAAGCTGGCCCTGCCCTCCAAATATGTGAACCTTCTGTACCTGGCCAGCCACTATGA
GAAGCTGAAGGGCTCCCCGAGGATAATGAGCAGAAACAGCTGTTTGTGGAACAGC
ACAAGCATACTCGACGAGATCATCGAGCAGATCAGCGAGTTCTCAAGAGAGTG
ATCCTGGCCGACGCTAATCTGGACAAGTGCTGTCCGCCTACAACAAGCACCGGGA

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SEQUENCES

TAAGCCCATCAGAGAGCAGGCCGAGAATATCATCCACCTGTTTACCCTGACCAATC
TGGGAGCCCCCTcgGCCCTTCAAGTACTTTGACACCACCATCGACCGGAAGgtGTACcgg
AGCACCAAGAGGTGCTGGACGCCACCTGATCCACAGAGCATCACCGGCTGTGA
CGAGACACGGATCGACCTGTCTCAGCTGGGAGGTGAC

SEQ ID NO: 40, ScCas9 (D10A): Nucleotide sequence:

ATGGAGAAGAAGTACAGCATCGGCTGGCCATCGGCACCAACAGCGTGGG
TGGCCGTGATCACCGACGACTACAAGGTGCCAGCAAGAAGTTCAAGGTGCTGGG
CAACACCAACCGCAAGAGCATCAAGAAGAACCTGATGGGCGCCCTGCTGTTTCGACA
GGGGCGAGACCGCCGAGGCCACCCGCTGAAGCGCACCGCCCGCCGCTACACC
CGCCGCAAGAACCGCATCCGCTACCTGCAGGAGATATTGCCCAACGAGATGGCCAA
GCTGGACGACAGCTTCTTCCAGCGCCTGGAGGAGAGCTTCTGTTGGAGGAGGACA
AGAAGAACGAGCGCCACCCATCTTCGGCAACCTGGCCGACGAGGTGGCCTACCAC
CGCAACTACCCACCATCTACCACCTGCGCAAGAAGCTGGCCGACAGCCCGAGAA
GGCCGACCTGCGCCTGATCTACCTGGCCCTGGCCACATCATCAAGTTCCGCGGCCA
CTTCTGATCGAGGGCAAGCTGAACGCGGAGAACAGCGACGTGGCCAAAGCTGTTCT
ACCAGCTGATCCAGACCTACAACAGCTGTTTCGAGGAGAGCCCCCTGGACGAGATC
GAGGTGGACGCCAAGGCGATCCTGAGCGCCCGCTGAGCAAGAGCAAGCGCCTGG
AGAAGCTGATCGCCGTGTTCCCAACGAGAAGAAGAAGCGCCCTGTTCCGCAACATC
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GGACGCCAAGCTGCAGCTGAGCAAGGACACCTACGACGACGACCTGGACGAGCTG
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CGACGCCATCTGCTGAGCGACATCTGCGCAGCAACAGCGAGGTGACCAAGGCC
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GCACCACCAAGCTGGCCACCCAGGAGGAGTTCACAAGTTTCATCAAGCCCATCTG
GAGAAGATGGACGGCGCGAGGAGCTGCTGGCCAAAGCTGAACCGCGACGACCTGC
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GAGCTGCACGCCATCTGCGCGCGCAGGAGGAGTTCACCCCTTCTGAGGAGAGAA
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TGGCCCGCGGCAACAGCGCTTCCGCTGCTGACCCGCAAGAGCGAGGAGGCCATC
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CCTGTTTCGACGACAAGGTGATGAAGCAGCTGAAGCGCGCCACTACACCGGCTGGG
GCCGCTGAGCGCGAAGATGATCAACCGCATCCGCGACAAGCAGAGCGGCAAGAC
CATCTGGACTTCTTGAAGAGCGACGGCTTCAGCAACCGCAACTTCATGACGTGA
TCCACGACGACGAGCTGACCTTCAAGGAGGAGATCGAGAAGGCCAGGTGAGCGG
CCAGGGCGAGCAGCTGCACGAGCAGATCGCCGACCTGGCCGCGAGCCCGCCATCA
AGAAGGGCATCTGCGAGACGTGAAGATCGTGGACGAGCTGGTGAAGGTGATGGG
CCACAAGCCCGAGAACATCGTATCGAGATGGCCCGCGAGAACCAGACCACCC
AAGGGCTGCAGCAGAGCGCGAGCGCAAGAAGCGCATCGAGGAGGGCATCAAGG
AGCTGGAGAGCCAGATCTGAAGGAGAACCCTGGAGAACACCCAGCTGCAGAA
CGAGAAGCTGTACTGTACTACCTGCAGAACGGCGCGACATGTACGTGGACAGG
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TTCATCAAGGACGACGAGCATCGACAACAGGTGCTGACCCGCGAGCTGGAGAACC
CGGCAAGAGCGACAACGTGCCAGCGAGGAGGTGGTGAAGAAGATGAAGAAGTAC
TGGCGCCAGCTGCTGAACGCCAAGCTGATCACCCAGCGCAAGTTTCGACAACCTGAC
CAAGGCGAGCGCGCGGCTGAGCGAGGCGACAAAGGCCGGCTTCATCAAGCGC
CAGCTGGTGGAGACCCGCGAGATCACCAAGCAGCTGGCCCGCATCTGGACAGCGG
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GCGCGACATCAACAACCTACCACCGCCACGACGCTTACCTGAACGCGGTGGTGG
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AGGCCACCGCCAAAGCGCTTCTTCTACAGCAACATCATGAAGTCTTCAAGACCGAG
GTGAAGCTGGCCAAAGCGAGATCCGCAAGCGCCCTGATCGAGACCAACGGCG
AGACCGGCGAGGTGGTGGAAACAGGAGAAGGACTTCGCAACCGTGCAGAAAGT
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TCAGCAAGGAGAGCATCTGAGCAAGCGCGAGAGCGCAAGCTGATCCCCCGCAA
GAAGGGCTGGGACACCCGAAGTACGGCGGCTTCGGCAGCCCCACCGTGGCCTACA
GCATCTTGGTGGTGGCCAAAGGTGGAGAAGGGCAAGGCCAAGAAGTGAAGAGCGT
GAAGGTGCTGGTGGGCATCACCATCATGGAGAAGGGCAGCTACGAGAAGGACCCC
ATCGGCTTCTTGGAGGCGAAGGGCTACAAGGACATCAAGAAGGAGCTGATCTCAA
GCTGCCCAAGTACAGCTGTTTCGAGCTGGAGAACGGCCGCGCGCATGCTGGCCA
GCGCCACCGAGCTGCAGAAGGCCAACGAGCTGGTGTGCCCGAGCACCTGGTGC
CTGCTGTACTACACCCAGAACATCAGCGCCACACCGCGAGCAACAACCTGGGCTA
CATCGAGCAGCACCGCGAGGAGTTCAGGAGATATTGAGAAGATCATCGACTTCA
CGAGAAGTACATCTGTAAGAACAAGGTGAACAGCAACCTGAAGAGCAGCTTCGA
CGAGCAGTTCCGCGTGAAGCAGCATCTGCTGAGCAACAGCTTCGTGAGCCTGC

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SEQUENCES

TGAAGTACACCAGCTTCGGCGCCAGCGCGGCTTACCTTCTGGACCTGGACGTG
AAGCAGGGCCGCTGCGCTACAGACCGTGACCGAGGTGCTGGACGCCACCTGAT
CTACCAGAGCATACCGGCTGTACGAGACCCGACCGACCTGAGCCAGCTGGGCG
GCGAC

SEQ ID NO: 41, TadA-TadA* Nucleotide sequence:

TCTGAAGTCGAGTTTAGCCACGAGTATTGGATGAGGCACGCACTGACCTGG
CAAAGCGAGCATGGGATGAAAGAGAAGTCCCGTGGGCGCCGTGCTGGTGACAA
CAATAGAGTGATCGGAGAGGGATGGAACAGGCCAATCGGCCGCCACGACCTACC
GCACACGCAGAGATCATGGCACTGAGGCGAGGAGGCTGGTATGCAGAATTACCG
CTGTATCGATGCCACCTGTATGTGACACTGGAGCCATGCGTGATGTGCGCAGGAG
CAATGATCCACAGCAGGATCGGAAGAGTGGTGTTCGGAGCACGGGACGCCAAGAC
CGGCGCAGCAGGCTCCCTGATGGATGTGCTGCACACCCCGGCATGAACCAACGGG
TGGAGATCACAGAGGGAATCCTGGCAGACGAGTGCGCCGCCCTGCTGAGCGATTTC
TTTAGAATCGGAGACAGGAGATCAAGGCCCAGAAGAAGGCACAGAGCTCCACCG
ACTCTGGAGGATCTAGCGGAGGATCCTCTGGAAGCGAGACACAGGCCACAAGCGA
GTCCGCCACACAGAGAGCTCCGCGCGCTCCTCCGAGGATCCTCTGAGGTGGAGT
TTTCCACGAGTACTGGATGAGACATGCCCTGACCTGGCCAAGAGGGCACGCGAT
GAGAGGGAGGTGCCTGTGGGAGCGGTGCTGGTGCTGAACAATAGAGTGATCGGCG
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GCCCTGAGACAGGGCGGCTGGTATGCAGAACTACAGACTGATTGACGCCACCT
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CGGCGCGTGGTGTGTTGGCGTGAGGAACGCAAAAACCGCGCCGCGCAGGCTCCCTGA
TGGACGTGTGCACTACCCCGGCATGAATCACCGCGTCGAAATTACCGAGGGAATC
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TTCAATGCTCAGAAGAAGGCCAGAGCTCCACCGAC

SEQ ID: 42, TadA* (A56G V82G) Nucleotide sequence:

TCTGAGGTGGAGTTTCCACGAGTACTGGATGAGACATGCCCTGACCTGG
CCAAGAGGGCACGCGATGAGAGGGAGGTGCCTGTGGGAGCGGTGCTGGTGCTGAA
CAATAGAGTGATCGGCGAGGGCTGGAACAGAGCCATCGGCTGACAGCCCAACA
GgCCATGCCGAATATATGGCCCTGAGgCAGGCGCGCTGGTCAATGCAGAACTACAG
ACTGATTGACGCCACCTGTACGgGACATTGAGCCTTGCGTGATGTGCGCCGGCGC
CATGATCCACTCTAGGATCGGCGCGGTGGTGTGTCGCTGAGGAACGCAAAAACCG
GCGCGCAGGCTCCCTGATGGACGTGCTGCACTACCCCGGCATGAATCACCGCGTC
GAAATTACCGAGGGAATCCTGGCAGATGAATGTGCCGCCCTGCTGTGCTATTTCTTT
CGGATGCCTAGACAGGTGTTCAATGCTCAGAAGAAGGCCAGAGCTCCACCGAC

SEQ ID NO: 43, Gp41-1-N: Nucleotide sequence:

TGTTTGGATCTGAAAACGCAAGTTCAAACGCCACAGGGTATGAAAAGAAATAT
CCAATATACAGGTCGGCGATCTCGTCTGTCTAACACTGGCTATAACGAGGTGCTGA
ATGTATTTCCAAAAAGCAAGAAAAAAGTTACAAGATAACTCTGGAAGATGGAAA
AGAAATATCTGTTCTGAGGAGCATCTGTTCCGACCCAAACAGGGGAGATGAATA
TCAGTGGCGGTCTCAAAGAGGGTATGTGTTGTATGTCAAGGaataa

SEQ ID NO: 44, Gp41-1-C: Nucleotide sequence:

ATGATGCTCAAGAAGATCCTCAAGATTGAAGAGTTGGACGAGCGCGAGCTT
ATAGACATAGAAGTCAGTGGTAATCACCTTTCTACGCAATGACATTTGACTCAC
AACTCC

SEQ ID NO: 45, Cfa-N: Nucleotide sequence:

TGTCTCAGTTATGAcACCGAAATCCTGACAGTCGAGTATGGATtTCTGCCGATC
GGCAAGATTGTGGAGgAGAGAATTGAATGTACGGTCTATAcgTGCACAAgAATGGTt
tCgTCTACACCCAACCAaTTGCTCAATGGCAtaATCGAGGGGAGCAGGAGGTGTTTGA
GTATTGCCCTGGAGGACGGGTCAaTCATTAGAGCTACAAAGGACCATAAGTTTATGAC
AacCGATGGTCAAATGTGCCGATAGATGAAATATTTCGAAAGGgACTGGATCTTAa
GCaAGTCGATggCCTTCCAaac

SEQ ID NO: 46, Cfa-C, Nucleotide sequence:

ATGgTcAagATTatCagcCGCAAAATcCTTGGGGAcACAGAATGTATATGACATCG
GCGTGGAAaaGGATCACAATTTTctgCTGAAGAATGGTcTtgTTGCTTccAat

SEQ ID NO: 47, Tad A,

SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDP
TAHAEIMALRQGGVLVMQNYRLIDATLYVTFEPVCMCAGAMIHSRIGRVVFGVRNAKT
GAAGSLMDVLHYPGMNRVEITEGILADECAALLCYFFRMPRQVFNAQKKAQSSD

SEQ ID NO: 48

TGAGTATCATCGTGTGAAGCTgAGGGGACGAGGCAGGCCTATAA

SEQ ID NO: 49

AAAAACATCAACTTCAGCCATCCATTTCTTCAGGGTTTGTATGTG

SEQ ID NO: 50

TATATCATAATGAAAACGCCGCCATTTCTCAACAGATCTGTCAA

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SEQUENCES

SEQ ID NO: 51
CTGTGTGAAATGGCTGCAAATCgATGGTTGAGCTCTGAGATTGG

SEQ ID NO: 52
GTTCTGCTTTTGCTACTACTCACGTTTCCATGTTGTCCCCCTCTA

SEQ ID NO: 53
ATTTTATGGCCTTTTGCAACTCgACCAGAAAAAAGCAGCTTTGG

SEQ ID NO: 54
TGAGGAGATCGCCACGGGCTGCCAGGATCCCTTGATCACCTCAG

SEQ ID NO: 55
GGCTGCTCTGTGAGAAATATTCgTACAGTCTCAAGAGTACTCATG

SEQ ID NO: 56
GTGTAGGCATAGCTCTTGAATCgAGGCTTAGGGGAAGAAGTTCTC

SEQ ID NO: 57
CCTGTTCTTCAGTAAGACGTTGCCATTTGAGAAGGATGTCTTGTA

SEQ ID NO: 58
GCCATTTTAGGCTTTTACTTACTTGTCTGTAGCTCTTCTCTCT

SEQ ID NO: 59
CTAGTTTCTCACACATGACACACCTGTTCTTCAGTAAGACGTTGC

SEQ ID NO: 60
GTGAAGTTGATTACATTAACCTgTGGATAATTACGAGTTGATTGT

SEQ ID NO: 61
ttttgtatATCTGAGTTAAACTgCTCCAATTCCTTCAAAGGAATG

SEQ ID NO: 62
TCTGCAATATAAGCTGCCAACTgCTTGTCATGAATGTGAGGGAC

SEQ ID NO: 63
GGACTGGGGTTCCAGTCTCATCCAGTCTAGGAAGAGGGCCGCTTC

SEQ ID NO: 64
AGTCGTTGTGTGGCTGACTGCTgGCAAACCACACTATTCCAGTCA

SEQ ID NO: 65
ATTTGtgtctttctgagaaactgttcagCTTCTGTTAGCCACTGA

SEQ ID NO: 66
AAGAACCCAGCGGTCTTCTGTCCATCTACAGATGTTGCCCATCG

SEQ ID NO: 67
AGACTTTTCCGAAGTTCACCTCACTTGAAGTTCATGTTATCCAA

SEQ ID NO: 68
TACCTGTTGGCACATGTGATCCCACTGAGTGTTAAGTTCTTTGAG

SEQ ID NO: 69
CAAAGGGCCTTTCGAGTCTTCgGAGTTTCATGGCAGTCCTATAA

SEQ ID NO: 70
ATGGTTAATGTCTAACCTTTATCCACTGGAGATTTGTCTGCTTGA

SEQ ID NO: 71
CAAAATAATCTGACCTTAAGTTgTTCTTCCAAAGCAGCAGTTGCG

SEQ ID NO: 72
CATCTACAGATGTTTGCCCATCgATCTCCCAATACCTGGAGAAGA

SEQ ID NO: 73
AAGGTGTTCTTGTACTTCATCCCACTGATTCTGAATTCTTTCAac

SEQ ID NO: 74
GAAGGTGTTCTTGTACTTCATCCCACTGATTCTGAATTCTTTCAa

SEQ ID NO: 75
GCTATGCTTTGAATTTTAAATCgTTCATTTGAGGTTGAAGATCT

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SEQUENCES

SEQ ID NO: 76
TCTAGGAGGCGCCTCCCATCCTgTAGGTCACCTGAAGAGGTTCTCA

SEQ ID NO: 77
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SEQ ID NO: 78
TGAGCATGCTTTACCAGGATCTgTTCCCTTGTGGTCACCGTAGTT

SEQ ID NO: 79
CTACTGTATAGGGACCCCTCCTTCCATGACTCAAGCTTGGCTCTGG

SEQ ID NO: 80
CAGCTTCTTCCTTAGCTTCCAGCCATTGTGTTGAATCCTTTAACA

SEQ ID NO: 81
AGCTGCCCAAGGTCTTTTATTgAGCTTCAATTTCTCCTTGTTTC

SEQ ID NO: 82
TGCCAGTAACAACCTCACAATTgTGCAAAGTTGAGTCTTCGAAAC

SEQ ID NO: 83
AGAGTAACAGTCTGAGTAGGAGctaaaatattttgggtttttgca

SEQ ID NO: 84
CTTCAGCAAAAAAGTACTCACgCAGAATCTACTGGCCAGAAGTT

SEQ ID NO: 85
ACATCTACATTTGTCTGCCACTgGCGGAGGTCTTTGGCCAACTGC

SEQ ID NO: 86
CTGAGATAGTATAGGCCACTTTgTTGCTCTTGACAGAACTTTGT

SEQ ID NO: 87
TAGCTGTCCTTTACACACTTTACCTGTTGAGAATAGTGCATTTGA

SEQ ID NO: 88
GCCTGGGCTTCCTGAGGCATTgAGCTGCGTCCACCTTGTCTGCA

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TGCACTGGCAGGTAGCCCATTCgGGGATGCTTCGCAAAATACCTT

SEQ ID NO: 90
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SEQ ID NO: 93
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SEQ ID NO: 94
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SEQ ID NO: 95
AGAATGGGATCCAGTATACTTAcAGGCTCCAATAGTGGTCAGTCC

SEQ ID NO: 96
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SEQ ID NO: 97
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SEQ ID NO: 98
CTGAACTTCTCAGCTTTTCTCgCTCTATGGCCTGCAGCATGAGA

SEQ ID NO: 99
AGATTTAACCACCTCTCTGCTCgGGAGGTGACAGCTATCCAGTTA

SEQ ID NO: 100
tgttttatctttatttcctctCgCTTCTCTCATCTGTGATTCTT

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SEQUENCES

SEQ ID NO: 101
CCAGCTGGGAGGAGAGCTTCTTcCAGCGTCCCTCAATTCTTCAA

SEQ ID NO: 102
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SEQ ID NO: 103
AAGTAAACGGTTTACCGCCTTcACTCAGAGCTCAGATCTTCTAA

SEQ ID NO: 104
aaatagaaaaattagatgacttgccaaaggtcacaaaGGTAACTG

SEQ ID NO: 105
TTGACTTTCTCGAGGTGATCTTgGAGAGAGTCAATGAGGAGATCG

SEQ ID NO: 106
TCTAAAATCATCTTACTTTCTTgTAGACGCTGCTCAAATTTGGCT

SEQ ID NO: 107
GAATTGACCCTGACTTGTTCTTgTTCTAGATCTTCTTGAAGCACc

SEQ ID NO: 108
TGGATGGCTTCAATGCTCACTTgTTGAGGCAAACTTGAAGAGT

SEQ ID NO: 109
AACAGTCCTCTACTTCTTCCCAcCAAAGCATTTTGAAGGTAT

SEQ ID NO: 110
AGGCCTCCTTTCTGGCATAGAccttccacaaaacaaacacacaaa

SEQ ID NO: 111
TTTGGTTTCTGACTGCTGGACCcATGTCCTGATGGCACTCATGGT

SEQ ID NO: 112
ATCTTACTTTCTGTAGACGCTgCTCAAATTTGGCTGGTTTCTGG

SEQ ID NO: 113
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SEQ ID NO: 114
CTGTTCAAGTTGTTCTGAGGCTTgTTTGATGCTATCTGCATTAAcA

SEQ ID NO: 115
CAGCATTAATATACACGACTTAcATCTGTACTTGCTTCCAAATG

SEQ ID NO: 116
TCATGACTTGTCAAATCAGATTgGATTTTCTGTTGGGAGGATAGC

SEQ ID NO: 117
ATCTGCTCCAATTGTTGTAGCTgATTATAGAAAGCGATGATGTTG

SEQ ID NO : 118
CAAATTTGCTCTCAATTTCCCGcCAGCGCTTGCTGAGCTGGATCT

SEQ ID NO: 119
CATTCAAAGCCAGGCCATCAGAcCAGCTGGTGGTGAAGTTGATTA

SEQ ID NO: 120
ttcatcTCTTCAACTGCTTTCTgTAATTCATCTGGAGTTTATAT

SEQ ID NO: 121
ATTGAAAGCTAGAAAGTACATAcGGCCAGTTTTTGAAGACTTGAT

SEQ ID NO: 122
TTCAAATACTGGCCAATACTTAcAGCAAAGGGCCTTCTGCAGTCT

SEQ ID NO: 123
GTTGTCTGTGTTAGTGATGGCTgAGTGGTGGTGACAGCCTGTGAA

SEQ ID NO: 124
TCATCAGCCTGCCTCTTGTACTgATACCACTGATGAGAAATTTCT

SEQ ID NO: 125
TACTGTATAGGGACCCTCCTTcATGACTCAAGCTTGGCTCTGGC

SEQ ID NO: 126
ATGTTGAATGCATGTTCCAGTCgTTGTGTGGCTGACTGCTGGCAA

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SEQUENCES

SEQ ID NO: 127
TGCCATTTGAGAAGGATGTCTTgTAAAAGAACCAGCGGTCTTCT

SEQ ID NO: 128
GAGACTTTTTCCGAAGTTCACTcCACTTGAAGTTCATGTTATCCA

SEQ ID NO: 129
TTAGCAACTGGCAGAATTCGATcCACCGGCTGTTCAAGTTGTTCTG

SEQ ID NO: 130
TTGCCACATCTACATTTGTCTGcCACTGGCGAGGTCTTTGGCCA

SEQ ID NO: 131
TCCCATTCAGCCTAGTGCAGAGcCACTGGTAGTTGGTGGTTAGAG

SEQ ID NO: 132
GACTTACTGGAAAGAAAGTGCTgAGATGCTGGACCAAAGTCCCTG

SEQ ID NO: 133
TTTAATCGTTCAATTTGAGGTTgAAGATCTGATAGCCGGTTGACT

SEQ ID NO: 134
AAAGAGATTGTCTATACCTGTTgGCACATGTGATCCCACTGAGTG

SEQ ID NO: 135
AAGTTTTTGGACTAAATTATCCcAACACCGGGCAAAGTTATCCAG

SEQ ID NO: 136
AGCTCAGCATCCCGGGGACTCTgGGGAGAGGTGGGCATCATTTCA

SEQ ID NO: 137
TTGTCCCCCTCTAAGACAGTCTgCACTGGCAGGTAGCCCATTCTGG

SEQ ID NO: 138
TTCGCAAAATACCTTTTGGTTcGAAATTTGTTTTTTAGTACCTTG

SEQ ID NO: 139
TGCAACTCGACCAGAAAAAAGcAGCTTTGGCAGATGTCATAATT

SEQ ID NO: 140
TTGCAGATGTTACATTTGGCCTgATGCTTGGCAGTTTCTGCAGCA

SEQ ID NO: 141
AAATAAAAACATGCCATACGTAcGTATCATAAACATTGAGCAGCC

SEQ ID NO: 142
TACTTACAGCAAAGGGCCTTCTgCAGTCTTCGGAGTTTCATGGCA

SEQ ID NO: 143
TTGACCTCCTCAGCCTGCTTTcGTAGAAGCCGAGTGACATTCTGG

SEQ ID NO: 144
ATTCAATTACCTCTGGGCTCCTgGTAGAGTTTCTCTAGTCCTTCC

SEQ ID NO: 145
AATGCCTGACTTACTTGCCATTgTTTCATCAGCTCTTTACTCCC

SEQ ID NO: 146
TGTACTTCATCCCACTGATTCTgAATTCTTTCActagaataaaa

SEQ ID NO: 147
TGCTTCATTACCTTCACTGGCTgAGTGGCTGGTTTTCTTGTAC

SEQ ID NO: 148
TTTAATTGTTTGAGAATTCCCTgGCGCAGGGGCAACTCTCCACC

SEQ ID NO: 149
atatgtgttaCCTACCCTTGTCgGTCCTGTacattttggttaact

SEQ ID NO: 150
TCAAGCTGGGAGAGAGCTTCCTgTAGCTTCACCTTTCCACAGGC

SEQ ID NO: 151
ATGTCAATCCGACCTGAGCTTTgTTGTAGACTATCTTTATATTc

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SEQUENCES

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SEQ ID NO: 153
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SEQ ID NO: 154
TTATGTTTTGTCTGTAACAGCTgctgttttatctttatttcctct

SEQ ID NO: 155
TGTTTTGTCTGTAACAGCTgctgttttatctttatttcctctCGC

SEQ ID NO: 156
TTTCTCTCATCTGTGATTCTTTgTTGTAAGTTGTCTCCTCTTTGC

SEQ ID NO: 157
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SEQ ID NO: 158
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SEQ ID NO: 159
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SEQ ID NO: 160
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SEQ ID NO: 164
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SEQ ID NO: 174
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SEQ ID NO: 175
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SEQ ID NO: 176
ATTTCAGAATCCACAGTAATCTgCCTCTCTTttggggaggtggt

SEQ ID NO: 177
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SEQUENCES

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SEQ ID NO: 180
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SEQ ID NO: 181
CTACCTTATGTTGTTGTACTTgGCGTTTTAGGTCTTCAAGATCA

SEQ ID NO: 182
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SEQ ID NO: 183
attaaaaacaaataaggacTTAcTTGCTTTGTTTTTCCATGCTAG

SEQ ID NO: 184
caaataaggacTTACTTGCTTgTTTTTCCATGCTAGCTACCCTG

SEQ ID NO: 185
TTTAGGAGATTCATCTGCTCTgtacttcagtttcttcatcttct

SEQ ID NO: 186
ACATCATTAGAAATCTCTCCTTgTGCTTGCAATGTGTCTCAGCA

SEQ ID NO: 187
TGGTAGTCCAGAAATTTACCAAcCTTCAGGATCGAGTAGTTTCTC

SEQ ID NO: 188
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SEQ ID NO: 189
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SEQ ID NO: 190
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SEQ ID NO: 191
aataagtcaCATACCAGTTTTTgCCCTGTCAGGCCTTCGAGGAGG

SEQ ID NO: 192
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SEQ ID NO: 193
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SEQ ID NO: 195
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SEQ ID NO: 197
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SEQ ID NO: 198
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SEQ ID NO: 199
GTCAAGACATTCAATTCCTTTTcCATCTTACGGGACAATTTCAAG

SEQ ID NO: 200
TGTGTCCTCAGCAGAAAGAAGCcACGATAATACTTCTTCTAAAGC

SEQ ID NO: 201
TAGAAAGCGATGATGTTGTTCTgATACTCCAGCCAGTTAAGTCTC

SEQ ID NO: 202
CTCTCTAAGGAAATCAAGATCTgGGCAGGACTACGAGGCTGGCTC

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SEQUENCES

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SEQ ID NO: 204
ACTGTTTCATTACAGTTGTCTgTGTTAGTGATGGCTGAGTGGTG

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TTAGTACCTTGCGAAAGTCTCgAACATCTTCTCCTGATGTAGTC

SEQ ID NO: 206
ATTTGTGCAAAGTTGAGTCTTCgAAACTGAGCAAATTTGCTCTCA

SEQ ID NO: 207
TGGGAGCGCCTCTGTTCCAAATcCTGCATTGTTGCCTGTAAGAAC

SEQ ID NO: 208
GTCTCCTATGAACTCGAGAAGCcGCAAaccaaggaagagaaaga

SEQ ID NO: 209
GAGAGTTTGGTTTCTGACTGCTgGACCCATGTCCTGATGGCACTC

SEQ ID NO: 210
TGCGTATTTGCCACCAGAAATAcATACCACACAATGATTTAGCTG

SEQ ID NO: 211
TTTGGGTATCCTCTGAATGTcGCATCAAATTTTCAAGTGACTGA

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SEQ ID NO: 214
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SEQ ID NO: 215
CCTGCCAGTGGAGGATTATATTCcCAAATCAAACCAAGAGTCAGTT

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SEQ ID NO: 221 TTCagctcctcttcttcttcttgcaaTTCGGATCAATTTCTCTAT

SEQ ID NO: 222 AAAGCTaAGGGGACGAGGCAGGC

SEQ ID NO: 223 GAAAGCTaAGGGGACGAGGCAGG

SEQ ID NO: 224 AAATaGATGGCTGAAGTTGATGT

SEQ ID NO: 225 GAAATaGATGGCTGAAGTTGATG

SEQ ID NO: 226 AAGAAATaGATGGCTGAAGTTGA

SEQ ID NO: 227 CTGAAGAAATaGATGGCTGAAGTTGAT

SEQ ID NO: 228 AAATaGCGGCGTTTTCATTATGA

SEQ ID NO: 229 GAGAAATaGCGGCGTTTTCATTATGAT

SEQ ID NO: 230 AAATCaATGGTTGAGCTCTGAGA

SEQ ID NO: 231 GCAAATCaATGGTTGAGCTCTGA

SEQ ID NO: 232 TGCAAATCaATGGTTGAGCTCTGAGAT

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SEQUENCES

SEQ ID NO: 233 AACaTGAGTAGTAGCAAAAGCAG
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 SEQ ID NO: 235 GGAAACaTGAGTAGTAGCAAAAG
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 SEQ ID NO: 237 AACTCaACCAGAAAAAAGCAGC
 SEQ ID NO: 238 CAACTCaACCAGAAAAAAGCAG
 SEQ ID NO: 239 AAGGGATCCTGaCAGCCCGTGGGCGAT
 SEQ ID NO: 240 AATATTCaTACAGTCTCAAGAGT
 SEQ ID NO: 241 AATCaAGGCTTAGGGGAAGAAGT
 SEQ ID NO: 242 GAATCaAGGCTTAGGGGAAGAAG
 SEQ ID NO: 243 TGAATCaAGGCTTAGGGGAAGAA
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 SEQ ID NO: 245 CTTGAATCaAGGCTTAGGGGAAGAAGT
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 SEQ ID NO: 247 AAATGaCAACGTCTTACTGA AGA
 SEQ ID NO: 248 CAAATGaCAACGTCTTACTG AAG
 SEQ ID NO: 249 TCAAATGaCAACGTCTTACT GAA
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 SEQ ID NO: 254 AGAACAGaTGTGTCATGTGT GAG
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 SEQ ID NO: 256 AACCTaTGGATAATTACGAG TTG
 SEQ ID NO: 257 TTAACCTaTGGATAATTACG AGT
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 SEQ ID NO: 266 GCCAACTaCTTGTCATGAAT TGT
 SEQ ID NO: 267 ACTaGATGAGACTGGAACCC CAG
 SEQ ID NO: 268 TAGACTaGATGAGACTGGAAC CCCAGT
 SEQ ID NO: 269 ACTGCTaGCAAACCACTAT TCCAGT
 SEQ ID NO: 270 agaaactattcaGCTTCTGT TAG

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SEQUENCES

SEQ ID NO: 271 AGATaGACAGAAGACCGCTG GGT
 SEQ ID NO: 272 TAGATaGACAGAAGACCGCT GGG
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 SEQ ID NO: 274 TGTAGATaGACAGAAGACCG CTG
 SEQ ID NO: 275 CTGTAGATaGACAGAAGACCG CTGGGT
 SEQ ID NO: 276 AGTaGAGTGAACCTTCGGAAA AAG
 SEQ ID NO: 277 AAGTaGAGTGAACCTTCGGAA AAA
 SEQ ID NO: 278 CAAGTaGAGTGAACCTTCGGA AAA
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 SEQ ID NO: 282 CTCAGTaGGATCACATGTGC CAA
 SEQ ID NO: 283 TCAGTaGGATCACATGTGCCA ACAGGT
 SEQ ID NO: 284 AGTCTTCaGAGTTTCATGGC AGT
 SEQ ID NO: 285 CTGCAGTCTTCaGAGTTTCAT GGCAGT
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 SEQ ID NO: 287 AGTTaTTCTTCCAAAGCAGC AGT
 SEQ ID NO: 288 AAGTTaTTCTTCCAAAGCAG CAG
 SEQ ID NO: 289 TTAAGTTaTTCTTCCAAAGC AGC
 SEQ ID NO: 290 CTTAAGTTaTTCTTCCAAAGC AGCAGT
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 SEQ ID NO: 294 CCCATCaATCTCCCAATACC TGG
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 SEQ ID NO: 306 CCATCCTaTAGGTCACTGAA GAG
 SEQ ID NO: 307 TCCCATCCTaTAGGTCACTGA AGAGGT
 SEQ ID NO: 308 ATCTaTTCCACCAGGGCCTG AGC
 SEQ ID NO: 309 ACCATCTaTTCCACCAGGGC CTG

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SEQUENCES

SEQ ID NO: 310 ATCTaTTCACCAGGGCCTGA GCTGAT
SEQ ID NO: 311 ATCTaTTCCTTGTGGTCAC CGT
SEQ ID NO: 312 GATCTaTTCCTTGTGGTCA CCG
SEQ ID NO: 313 GATCTaTTCCTTGTGGTCAC CGTAGT
SEQ ID NO: 314 ATGaAAGGAGGGTCCCTATA CAG
SEQ ID NO: 315 GTCATGaAAGGAGGGTCCCTA TACAGT
SEQ ID NO: 316 ATGaCTGGAAGCTAAGGAAG AAG
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SEQ ID NO: 320 CACAATGaCTGGAAGCTAAG GAA
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SEQ ID NO: 325 CACaCAGAATCTACTGGCCA GAA
SEQ ID NO: 326 CTCACaCAGAATCTACTGGC CAG
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SEQUENCES

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 SEQ ID NO: 352 CTCCTCaAAGTGCCTGTGTGC AATAGT
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SEQUENCES

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 SEQ ID NO: 392 TCACCTaTTGAGGCAAACT TGG
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SEQUENCES	
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SEQ ID NO: 431	AGCGCTGaCGGGAAATTGAGA GCAAAT
SEQ ID NO: 432	GCTGaTCTGATGGCCTGGCT TTG
SEQ ID NO: 433	AGCTGaTCTGATGGCCTGGCT TTGAAT
SEQ ID NO: 434	GCTTCTaTAATTCATCTGG AGT
SEQ ID NO: 435	AACTGCTTTCTaTAATTCATC TGGAGT
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SEQ ID NO: 447	CCAGTCaTTGTGTGGCTGAC TGC
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SEQ ID NO: 453	GTGaAGTGAACCTTCGAAAA AGT
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SEQ ID NO: 458	TCAAGTGaAGTGAACCTTCGGA AAAAGT
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SEQ ID NO: 460	GGTGaATCGAATTCTGCCAG TTG
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Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosed invention belongs. Publications cited herein and the materials for which they are cited are specifically incorporated by reference.

Those skilled in the art will appreciate that numerous changes and modifications can be made to the preferred embodiments of the invention and that such changes and modifications can be made without departing from the spirit of the invention. It is, therefore, intended that the appended claims cover all such equivalent variations as fall within the true spirit and scope of the invention.

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

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<210> SEQ ID NO 10
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<213> ORGANISM: Artificial Sequence
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu
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Lys Arg Thr Ala Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys
65     70     75     80
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser
85     90     95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys
100    105    110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr
115    120    125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp
130    135    140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His
145    150    155    160
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro
165    170    175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr
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Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn
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Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp
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Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His	Pro	545

Val	Glu	Asn	Thr	Gln			Leu	Gln	Asn	Glu	Lys	Leu	Tyr	Leu	Tyr	Leu	Tyr	Leu		
				805							810							815		
Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	Glu	Leu	Asp	Ile	Asn	Arg					
				820							825							830		
Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	Val	Pro	Gln	Ser	Phe	Leu	Lys					
				835							840							845		
Asp	Asp	Ser	Ile	Asp	Asn	Lys	Val	Leu	Thr	Arg	Ser	Asp	Lys	Asn	Arg					
				850							855							860		
Gly	Lys	Ser	Asp	Asn	Val	Pro	Ser	Glu	Glu	Val	Val	Lys	Lys	Met	Lys					
				865							870							875		
Asn	Tyr	Trp	Arg	Gln	Leu	Leu	Asn	Ala	Lys	Leu	Ile	Thr	Gln	Arg	Lys					
				885							890							895		
Phe	Asp	Asn	Leu	Thr	Lys	Ala	Glu	Arg	Gly	Gly	Leu	Ser	Glu	Leu	Asp					
				900							905							910		
Lys	Ala	Gly	Phe	Ile	Lys	Arg	Gln	Leu	Val	Glu	Thr	Arg	Gln	Ile	Thr					
				915							920							925		
Lys	His	Val	Ala	Gln	Ile	Leu	Asp	Ser	Arg	Met	Asn	Thr	Lys	Tyr	Asp					
				930							935							940		
Glu	Asn	Asp	Lys	Leu	Ile	Arg	Glu	Val	Lys	Val	Ile	Thr	Leu	Lys	Ser					
				945							950							955		
Lys	Leu	Val	Ser	Asp	Phe	Arg	Lys	Asp	Phe	Gln	Phe	Tyr	Lys	Val	Arg					
				965							970							975		
Glu	Ile	Asn	Asn	Tyr	His	His	Ala	His	Asp	Ala	Tyr	Leu	Asn	Ala	Val					
				980							985							990		
Val	Gly	Thr	Ala	Leu	Ile	Lys	Lys	Tyr	Pro	Lys	Leu	Glu	Ser	Glu	Phe					
				995							1000							1005		
Val	Tyr	Gly	Asp	Tyr	Lys	Val	Tyr	Asp	Val	Arg	Lys	Met	Ile	Ala						
				1010							1015							1020		
Lys	Ser	Glu	Gln	Glu	Ile	Gly	Lys	Ala	Thr	Ala	Lys	Tyr	Phe	Phe						
				1025							1030							1035		
Tyr	Ser	Asn	Ile	Met	Asn	Phe	Phe	Lys	Thr	Glu	Ile	Thr	Leu	Ala						
				1040							1045							1050		
Asn	Gly	Glu	Ile	Arg	Lys	Arg	Pro	Leu	Ile	Glu	Thr	Asn	Gly	Glu						
				1055							1060							1065		
Thr	Gly	Glu	Ile	Val	Trp	Asp	Lys	Gly	Arg	Asp	Phe	Ala	Thr	Val						
				1070							1075							1080		
Arg	Lys	Val	Leu	Ser	Met	Pro	Gln	Val	Asn	Ile	Val	Lys	Lys	Thr						
				1085							1090							1095		
Glu	Val	Gln	Thr	Gly	Gly	Phe	Ser	Lys	Glu	Ser	Ile	Leu	Pro	Lys						
				1100							1105							1110		
Arg	Asn	Ser	Asp	Lys	Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp	Asp	Pro						
				1115							1120							1125		
Lys	Lys	Tyr	Gly	Gly	Phe	Asp	Ser	Pro	Thr	Val	Ala	Tyr	Ser	Val						
				1130							1135							1140		
Leu	Val	Val	Ala	Lys	Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys	Leu	Lys						
				1145							1150							1155		
Ser	Val	Lys	Glu	Leu	Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg	Ser	Ser						
				1160							1165							1170		
Phe	Glu	Lys	Asn	Pro	Ile	Asp	Phe	Leu	Glu</											

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1205	1210	1215
Glu Leu Gln Lys Gly Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val		
1220	1225	1230
Asn Phe Leu Tyr Leu Ala Ser His Tyr Glu Lys Leu Lys Gly Ser		
1235	1240	1245
Pro Glu Asp Asn Glu Gln Lys Gln Leu Phe Val Glu Gln His Lys		
1250	1255	1260
His Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys		
1265	1270	1275
Arg Val Ile Leu Ala Asp Ala Asn Leu Asp Lys Val Leu Ser Ala		
1280	1285	1290
Tyr Asn Lys His Arg Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn		
1295	1300	1305
Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Ala Ala		
1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Arg Tyr Thr Ser		
1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr		
1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp		
1355	1360	1365

<210> SEQ ID NO 12

<211> LENGTH: 1309

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 12

Glu Ala Thr Arg Leu Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg		
1	5	10 15
Lys Asn Arg Ile Cys Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala		
20	25	30
Lys Val Asp Asp Ser Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val		
35	40	45
Glu Glu Asp Lys Lys His Glu Arg His Pro Ile Phe Gly Asn Ile Val		
50	55	60
Asp Glu Val Ala Tyr His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg		
65	70	75 80
Lys Lys Leu Val Asp Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr		
85	90	95
Leu Ala Leu Ala His Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu		
100	105	110
Gly Asp Leu Asn Pro Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln		
115	120	125
Leu Val Gln Thr Tyr Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala		
130	135	140
Ser Gly Val Asp Ala Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser		
145	150	155 160
Arg Arg Leu Glu Asn Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn		
165	170	175
Gly Leu Phe Gly Asn Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn		
180	185	190
Phe Lys Ser Asn Phe Asp Leu Ala Glu Asp Ala Lys Leu Gln Leu Ser		

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195	200	205
Lys Asp Thr Tyr Asp Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly 210 215 220		
Asp Gln Tyr Ala Asp Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala 225 230 235 240		
Ile Leu Leu Ser Asp Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala 245 250 255		
Pro Leu Ser Ala Ser Met Ile Lys Arg Tyr Asp Glu His His Gln Asp 260 265 270		
Leu Thr Leu Leu Lys Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr 275 280 285		
Lys Glu Ile Phe Phe Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile 290 295 300		
Asp Gly Gly Ala Ser Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile 305 310 315 320		
Leu Glu Lys Met Asp Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg 325 330 335		
Glu Asp Leu Leu Arg Lys Gln Arg Thr Phe Asp Asn Gly Ser Ile Pro 340 345 350		
His Gln Ile His Leu Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu 355 360 365		
Asp Phe Tyr Pro Phe Leu Lys Asp Asn Arg Glu Lys Ile Glu Lys Ile 370 375 380		
Leu Thr Phe Arg Ile Pro Tyr Tyr Val Gly Pro Leu Ala Arg Gly Asn 385 390 395 400		
Ser Arg Phe Ala Trp Met Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro 405 410 415		
Trp Asn Phe Glu Glu Val Val Asp Lys Gly Ala Ser Ala Gln Ser Phe 420 425 430		
Ile Glu Arg Met Thr Asn Phe Asp Lys Asn Leu Pro Asn Glu Lys Val 435 440 445		
Leu Pro Lys His Ser Leu Leu Tyr Glu Tyr Phe Thr Val Tyr Asn Glu 450 455 460		
Leu Thr Lys Val Lys Tyr Val Thr Glu Gly Met Arg Lys Pro Ala Phe 465 470 475 480		
Leu Ser Gly Glu Gln Lys Lys Ala Ile Val Asp Leu Leu Phe Lys Thr 485 490 495		
Asn Arg Lys Val Thr Val Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys 500 505 510		
Ile Glu Cys Phe Asp Ser Val Glu Ile Ser Gly Val Glu Asp Arg Phe 515 520 525		
Asn Ala Ser Leu Gly Thr Tyr His Asp Leu Leu Lys Ile Ile Lys Asp 530 535 540		
Lys Asp Phe Leu Asp Asn Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile 545 550 555 560		
Val Leu Thr Leu Thr Leu Phe Glu Asp Arg Glu Met Ile Glu Glu Arg 565 570 575		
Leu Lys Thr Tyr Ala His Leu Phe Asp Asp Lys Val Met Lys Gln Leu 580 585 590		
Lys Arg Arg Arg Tyr Thr Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile 595 600 605		
Asn Gly Ile Arg Asp Lys Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu 610 615 620		

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Lys Ser Asp Gly Phe Ala Asn Arg Asn Phe Met Gln Leu Ile His Asp
 625 630 635 640
 Asp Ser Leu Thr Phe Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly
 645 650 655
 Gln Gly Asp Ser Leu His Glu His Ile Ala Asn Leu Ala Gly Ser Pro
 660 665 670
 Ala Ile Lys Lys Gly Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu
 675 680 685
 Val Lys Val Met Gly Arg His Lys Pro Glu Asn Ile Val Ile Glu Met
 690 695 700
 Ala Arg Glu Asn Gln Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu
 705 710 715 720
 Arg Met Lys Arg Ile Glu Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile
 725 730 735
 Leu Lys Glu His Pro Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu
 740 745 750
 Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met Tyr Val Asp Gln Glu
 755 760 765
 Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val Asp His Ile Val Pro
 770 775 780
 Gln Ser Phe Leu Lys Asp Asp Ser Ile Asp Asn Lys Val Leu Thr Arg
 785 790 795 800
 Ser Asp Lys Asn Arg Gly Lys Ser Asp Asn Val Pro Ser Glu Glu Val
 805 810 815
 Val Lys Lys Met Lys Asn Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu
 820 825 830
 Ile Thr Gln Arg Lys Phe Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly
 835 840 845
 Leu Ser Glu Leu Asp Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu
 850 855 860
 Thr Arg Gln Ile Thr Lys His Val Ala Gln Ile Leu Asp Ser Arg Met
 865 870 875 880
 Asn Thr Lys Tyr Asp Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val
 885 890 895
 Ile Thr Leu Lys Ser Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln
 900 905 910
 Phe Tyr Lys Val Arg Glu Ile Asn Asn Tyr His His Ala His Asp Ala
 915 920 925
 Tyr Leu Asn Ala Val Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys
 930 935 940
 Leu Glu Ser Glu Phe Val Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg
 945 950 955 960
 Lys Met Ile Ala Lys Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys
 965 970 975
 Tyr Phe Phe Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr
 980 985 990
 Leu Ala Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly
 995 1000 1005
 Glu Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr
 1010 1015 1020
 Val Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys
 1025 1030 1035

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Thr	Glu	Val	Gln	Thr	Gly	Gly	Phe	Ser	Lys	Glu	Ser	Ile	Arg	Pro
1040						1045					1050			
Lys	Arg	Asn	Ser	Asp	Lys	Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp	Asp
1055						1060					1065			
Pro	Lys	Lys	Tyr	Gly	Gly	Phe	Val	Ser	Pro	Thr	Val	Ala	Tyr	Ser
1070						1075					1080			
Val	Leu	Val	Val	Ala	Lys	Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys	Leu
1085						1090					1095			
Lys	Ser	Val	Lys	Glu	Leu	Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg	Ser
1100						1105					1110			
Ser	Phe	Glu	Lys	Asn	Pro	Ile	Asp	Phe	Leu	Glu	Ala	Lys	Gly	Tyr
1115						1120					1125			
Lys	Glu	Val	Lys	Lys	Asp	Leu	Ile	Ile	Lys	Leu	Pro	Lys	Tyr	Ser
1130						1135					1140			
Leu	Phe	Glu	Leu	Glu	Asn	Gly	Arg	Lys	Arg	Met	Leu	Ala	Ser	Ala
1145						1150					1155			
Arg	Phe	Leu	Gln	Lys	Gly	Asn	Glu	Leu	Ala	Leu	Pro	Ser	Lys	Tyr
1160						1165					1170			
Val	Asn	Phe	Leu	Tyr	Leu	Ala	Ser	His	Tyr	Glu	Lys	Leu	Lys	Gly
1175						1180					1185			
Ser	Pro	Glu	Asp	Asn	Glu	Gln	Lys	Gln	Leu	Phe	Val	Glu	Gln	His
1190						1195					1200			
Lys	His	Tyr	Leu	Asp	Glu	Ile	Ile	Glu	Gln	Ile	Ser	Glu	Phe	Ser
1205						1210					1215			
Lys	Arg	Val	Ile	Leu	Ala	Asp	Ala	Asn	Leu	Asp	Lys	Val	Leu	Ser
1220						1225					1230			
Ala	Tyr	Asn	Lys	His	Arg	Asp	Lys	Pro	Ile	Arg	Glu	Gln	Ala	Glu
1235						1240					1245			
Asn	Ile	Ile	His	Leu	Phe	Thr	Leu	Thr	Asn	Leu	Gly	Ala	Pro	Arg
1250						1255					1260			
Ala	Phe	Lys	Tyr	Phe	Asp	Thr	Thr	Ile	Asp	Arg	Lys	Val	Tyr	Arg
1265						1270					1275			
Ser	Thr	Lys	Glu	Val	Leu	Asp	Ala	Thr	Leu	Ile	His	Gln	Ser	Ile
1280						1285					1290			
Thr	Gly	Leu	Tyr	Glu	Thr	Arg	Ile	Asp	Leu	Ser	Gln	Leu	Gly	Gly
1295						1300					1305			

Asp

<210> SEQ ID NO 13

<211> LENGTH: 1368

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 13

Met	Asp	Lys	Lys	Tyr	Ser	Ile	Gly	Leu	Ala	Ile	Gly	Thr	Asn	Ser	Val
1				5					10					15	

Gly	Trp	Ala	Val	Ile	Thr	Asp	Glu	Tyr	Lys	Val	Pro	Ser	Lys	Lys	Phe
		20					25						30		

Lys	Val	Leu	Gly	Asn	Thr	Asp	Arg	His	Ser	Ile	Lys	Lys	Asn	Leu	Ile
	35					40						45			

Gly	Ala	Leu	Leu	Phe	Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr	Arg	Leu
	50					55					60				

Lys	Arg	Thr	Ala	Arg	Arg	Arg	Tyr	Thr	Arg	Arg	Lys	Asn	Arg	Ile	Cys
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65	70	75	80
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser	85	90	95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys	100	105	110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr	115	120	125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp	130	135	140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His	145	150	155
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro	165	170	175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr	180	185	190
Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala	195	200	205
Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn	210	215	220
Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn	225	230	235
Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe	245	250	255
Asp Leu Ala Glu Asp Thr Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp	260	265	270
Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp	275	280	285
Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp	290	295	300
Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser	305	310	315
Met Ile Lys Leu Tyr Asp Glu His His Gln Asp Leu Thr Leu Leu Lys	325	330	335
Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe	340	345	350
Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser	355	360	365
Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile Leu Glu Lys Met Asp	370	375	380
Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg Glu Asp Leu Leu Arg	385	390	395
Lys Gln Arg Thr Phe Asp Asn Gly Ile Ile Pro His Gln Ile His Leu	405	410	415
Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu Asp Phe Tyr Pro Phe	420	425	430
Leu Lys Asp Asn Arg Glu Lys Ile Glu Lys Ile Leu Thr Phe Arg Ile	435	440	445
Pro Tyr Tyr Val Gly Pro Leu Ala Arg Gly Asn Ser Arg Phe Ala Trp	450	455	460
Met Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro Trp Asn Phe Glu Lys	465	470	475
Val Val Asp Lys Gly Ala Ser Ala Gln Ser Phe Ile Glu Arg Met Thr	485	490	495

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Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser
			500					505					510		
Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys
		515					520					525			
Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Asp	Gln
	530					535					540				
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr
545					550					555					560
Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp
			565						570					575	
Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly
		580						585					590		
Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp
		595				600						605			
Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr
	610					615					620				
Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala
625					630					635					640
His	Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr
			645						650						655
Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp
		660						665					670		
Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	Asp	Phe	Leu	Lys	Ser	Asp	Gly	Phe
		675				680						685			
Ala	Asn	Arg	Asn	Phe	Ile	Gln	Leu	Ile	His	Asp	Asp	Ser	Leu	Thr	Phe
	690					695					700				
Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu
705					710					715					720
His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly
			725					730						735	
Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu	Val	Lys	Val	Met	Gly
		740						745					750		
Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu	Met	Ala	Arg	Glu	Asn	Gln
		755				760						765			
Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	Ser	Arg	Glu	Arg	Met	Lys	Arg	Ile
	770					775					780				
Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His	Pro
785					790					795					800
Val	Glu	Asn	Thr	Gln	Leu	Gln	Asn	Glu	Lys	Leu	Tyr	Leu	Tyr	Tyr	Leu
			805						810					815	
Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	Glu	Leu	Asp	Ile	Asn	Arg
		820						825					830		
Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	Val	Pro	Gln	Ser	Phe	Leu	Lys
		835					840					845			
Asp	Asp	Ser	Ile	Asp	Asn	Lys	Val	Leu	Thr	Arg	Ser	Asp	Lys	Asn	Arg
		850				855						860			
Gly	Lys	Ser	Asp	Asn	Val	Pro	Ser	Glu	Glu	Val	Val	Lys	Lys	Met	Lys
865					870					875					880
Asn	Tyr	Trp	Arg	Gln	Leu	Leu	Asn	Ala	Lys	Leu	Ile	Thr	Gln	Arg	Lys
			885						890					895	
Phe	Asp	Asn	Leu	Thr	Lys	Ala	Glu	Arg	Gly	Gly	Leu	Ser	Glu	Leu	Asp
			900					905						910	

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Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr
 915 920 925

Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp
 930 935 940

Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser
 945 950 955 960

Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg
 965 970 975

Glu Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val
 980 985 990

Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe
 995 1000 1005

Val Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala
 1010 1015 1020

Lys Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe
 1025 1030 1035

Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala
 1040 1045 1050

Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu
 1055 1060 1065

Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val
 1070 1075 1080

Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr
 1085 1090 1095

Glu Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Leu Pro Lys
 1100 1105 1110

Arg Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro
 1115 1120 1125

Lys Lys Tyr Gly Gly Phe Asp Ser Pro Thr Val Ala Tyr Ser Val
 1130 1135 1140

Leu Val Val Ala Lys Val Glu Lys Gly Lys Ser Lys Lys Leu Lys
 1145 1150 1155

Ser Val Lys Glu Leu Leu Gly Ile Thr Ile Met Glu Arg Ser Ser
 1160 1165 1170

Phe Glu Lys Asn Pro Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys
 1175 1180 1185

Glu Val Lys Lys Asp Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu
 1190 1195 1200

Phe Glu Leu Glu Asn Gly Arg Lys Arg Met Leu Ala Ser Ala Gly
 1205 1210 1215

Val Leu Gln Lys Gly Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val
 1220 1225 1230

Asn Phe Leu Tyr Leu Ala Ser His Tyr Glu Lys Leu Lys Gly Ser
 1235 1240 1245

Pro Glu Asp Asn Glu Gln Lys Gln Leu Phe Val Glu Gln His Lys
 1250 1255 1260

His Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys
 1265 1270 1275

Arg Val Ile Leu Ala Asp Ala Asn Leu Asp Lys Val Leu Ser Ala
 1280 1285 1290

Tyr Asn Lys His Arg Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn
 1295 1300 1305

Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Ala Ala

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1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Arg Tyr Thr Ser		
1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr		
1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp		
1355	1360	1365

<210> SEQ ID NO 14
 <211> LENGTH: 1368
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 14

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val		
1	5	10
Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe		
20	25	30
Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile		
35	40	45
Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu		
50	55	60
Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys		
65	70	75
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser		
85	90	95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys		
100	105	110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr		
115	120	125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp		
130	135	140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His		
145	150	155
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro		
165	170	175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr		
180	185	190
Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala		
195	200	205
Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn		
210	215	220
Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn		
225	230	235
Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe		
245	250	255
Asp Leu Ala Glu Asp Thr Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp		
260	265	270
Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp		
275	280	285
Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp		
290	295	300
Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser		

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305	310	315	320
Met Ile Lys Leu Tyr Asp Glu His His Gln Asp Leu Thr Leu Leu Lys	325	330	335
Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe	340	345	350
Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser	355	360	365
Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile Leu Glu Lys Met Asp	370	375	380
Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg Glu Asp Leu Leu Arg	385	390	395
Lys Gln Arg Thr Phe Asp Asn Gly Ile Ile Pro His Gln Ile His Leu	405	410	415
Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu Asp Phe Tyr Pro Phe	420	425	430
Leu Lys Asp Asn Arg Glu Lys Ile Glu Lys Ile Leu Thr Phe Arg Ile	435	440	445
Pro Tyr Tyr Val Gly Pro Leu Ala Arg Gly Asn Ser Arg Phe Ala Trp	450	455	460
Met Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro Trp Asn Phe Glu Lys	465	470	475
Val Val Asp Lys Gly Ala Ser Ala Gln Ser Phe Ile Glu Arg Met Thr	485	490	495
Asn Phe Asp Lys Asn Leu Pro Asn Glu Lys Val Leu Pro Lys His Ser	500	505	510
Leu Leu Tyr Glu Tyr Phe Thr Val Tyr Asn Glu Leu Thr Lys Val Lys	515	520	525
Tyr Val Thr Glu Gly Met Arg Lys Pro Ala Phe Leu Ser Gly Asp Gln	530	535	540
Lys Lys Ala Ile Val Asp Leu Leu Phe Lys Thr Asn Arg Lys Val Thr	545	550	555
Val Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp	565	570	575
Ser Val Glu Ile Ser Gly Val Glu Asp Arg Phe Asn Ala Ser Leu Gly	580	585	590
Thr Tyr His Asp Leu Leu Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp	595	600	605
Asn Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile Val Leu Thr Leu Thr	610	615	620
Leu Phe Glu Asp Arg Glu Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala	625	630	635
His Leu Phe Asp Asp Lys Val Met Lys Gln Leu Lys Arg Arg Tyr	645	650	655
Thr Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp	660	665	670
Lys Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe	675	680	685
Ala Asn Arg Asn Phe Ile Gln Leu Ile His Asp Asp Ser Leu Thr Phe	690	695	700
Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp Ser Leu	705	710	715
His Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly	725	730	735

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Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu Val Lys Val Met Gly
 740 745 750
 Arg His Lys Pro Glu Asn Ile Val Ile Glu Met Ala Arg Glu Asn Gln
 755 760 765
 Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu Arg Met Lys Arg Ile
 770 775 780
 Glu Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile Leu Lys Glu His Pro
 785 790 795 800
 Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu
 805 810 815
 Gln Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg
 820 825 830
 Leu Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Lys
 835 840 845
 Asp Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg
 850 855 860
 Gly Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys
 865 870 875 880
 Asn Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys
 885 890 895
 Phe Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp
 900 905 910
 Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr
 915 920 925
 Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp
 930 935 940
 Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser
 945 950 955 960
 Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg
 965 970 975
 Glu Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val
 980 985 990
 Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe
 995 1000 1005
 Val Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala
 1010 1015 1020
 Lys Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe
 1025 1030 1035
 Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala
 1040 1045 1050
 Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu
 1055 1060 1065
 Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val
 1070 1075 1080
 Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr
 1085 1090 1095
 Glu Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Arg Pro Lys
 1100 1105 1110
 Arg Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro
 1115 1120 1125
 Lys Lys Tyr Gly Gly Phe Val Ser Pro Thr Val Ala Tyr Ser Val
 1130 1135 1140

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Leu Val	Val Ala Lys Val	Glu Lys Gly Lys Ser	Lys Lys Leu Lys
1145		1150	1155
Ser Val	Lys Glu Leu Leu	Gly Ile Thr Ile Met	Glu Arg Ser Ser
1160		1165	1170
Phe Glu	Lys Asn Pro Ile	Asp Phe Leu Glu Ala	Lys Gly Tyr Lys
1175		1180	1185
Glu Val	Lys Lys Asp Leu	Ile Ile Lys Leu Pro	Lys Tyr Ser Leu
1190		1195	1200
Phe Glu	Leu Glu Asn Gly	Arg Lys Arg Met Leu	Ala Ser Ala Arg
1205		1210	1215
Phe Leu	Gln Lys Gly Asn	Glu Leu Ala Leu Pro	Ser Lys Tyr Val
1220		1225	1230
Asn Phe	Leu Tyr Leu Ala	Ser His Tyr Glu Lys	Leu Lys Gly Ser
1235		1240	1245
Pro Glu	Asp Asn Glu Gln	Lys Gln Leu Phe Val	Glu Gln His Lys
1250		1255	1260
His Tyr	Leu Asp Glu Ile	Ile Glu Gln Ile Ser	Glu Phe Ser Lys
1265		1270	1275
Arg Val	Ile Leu Ala Asp	Ala Asn Leu Asp Lys	Val Leu Ser Ala
1280		1285	1290
Tyr Asn	Lys His Arg Asp	Lys Pro Ile Arg Glu	Gln Ala Glu Asn
1295		1300	1305
Ile Ile	His Leu Phe Thr	Leu Thr Asn Leu Gly	Ala Pro Arg Ala
1310		1315	1320
Phe Lys	Tyr Phe Asp Thr	Thr Ile Asp Arg Lys	Val Tyr Arg Ser
1325		1330	1335
Thr Lys	Glu Val Leu Asp	Ala Thr Leu Ile His	Gln Ser Ile Thr
1340		1345	1350
Gly Leu	Tyr Glu Thr Arg	Ile Asp Leu Ser Gln	Leu Gly Gly Asp
1355		1360	1365

<210> SEQ ID NO 15

<211> LENGTH: 1368

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 15

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val			
1	5	10	15
Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe			
20	25	30	
Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile			
35	40	45	
Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu			
50	55	60	
Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys			
65	70	75	80
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser			
85	90	95	
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys			
100	105	110	
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr			
115	120	125	

His 130	Lys	Tyr	Pro	Thr	Ile 135	Tyr	His	Leu	Arg	Lys 140	Lys	Leu	Val	Asp
Ser 145	Thr	Asp	Lys	Ala	Asp 150	Leu	Arg	Leu	Ile	Tyr 155	Ala	Leu	Ala	His 160
Met	Ile	Lys	Phe	Arg 165	Gly	His	Phe	Leu	Ile 170	Glu	Gly	Asp	Leu	Pro 175
Asp	Asn	Ser	Asp 180	Val	Asp	Lys	Leu	Phe	Ile 185	Gln	Leu	Val	Gln	Tyr 190
Asn	Gln	Leu	Phe 195	Glu	Glu	Asn	Pro	Ile	Asn 200	Ala	Ser	Gly 205	Val	Ala 210
Lys	Ala	Ile	Leu 210	Ser	Ala	Arg 215	Leu	Ser	Lys 220	Ser	Arg	Arg	Leu	Asn 225
Leu	Ile	Ala	Gln 225	Leu	Pro 230	Gly	Glu	Lys	Lys 235	Asn	Gly	Leu	Phe	Asn 240
Leu	Ile	Ala	Leu 245	Ser	Leu	Gly	Leu	Thr	Pro 250	Asn	Phe	Lys	Ser	Phe 255
Asp	Leu	Ala	Glu 260	Asp	Ala	Lys	Leu	Gln	Leu 265	Ser	Lys	Asp	Thr	Asp 270
Asp	Asp	Leu	Asp 275	Asn	Leu	Leu	Ala	Gln	Ile 280	Gly	Asp	Gln	Tyr	Asp 285
Leu	Phe	Leu	Ala 290	Ala	Lys	Asn 295	Leu	Ser	Asp 300	Ala	Ile	Leu	Leu	Asp 305
Ile	Leu	Arg	Val 310	Asn	Thr 315	Glu	Ile	Thr	Lys 320	Ala	Pro	Leu	Ser	Ser 325
Met	Ile	Lys	Arg 325	Tyr	Asp	Glu	His	His	Gln 330	Asp	Leu	Thr	Leu	Lys 335
Ala	Leu	Val	Arg 340	Gln	Gln	Leu	Pro	Glu	Lys 345	Tyr	Lys	Glu	Ile	Phe 350
Asp	Gln	Ser	Lys 355	Asn	Gly	Tyr	Ala 360	Gly	Tyr	Ile	Asp	Gly 365	Gly	Ser 370
Gln	Glu	Glu	Phe 375	Tyr	Lys	Phe 380	Ile	Lys	Pro 385	Ile	Leu	Glu	Lys	Met 390
Gly	Thr	Glu	Glu 395	Leu	Leu 400	Val	Lys	Leu	Asn 405	Arg	Glu	Asp	Leu	Arg 410
Lys	Gln	Arg	Thr 415	Phe	Asp	Asn	Gly	Ser	Ile 420	Pro	His	Gln	Ile	Leu 425
Gly	Glu	Leu	His 430	Ala	Ile	Leu	Arg	Arg	Gln 435	Glu	Asp	Phe	Tyr	Phe 440
Leu	Lys	Asp	Asn 445	Arg	Glu	Lys	Ile	Glu	Lys 450	Ile	Leu	Thr	Phe	Arg 455
Pro	Tyr	Tyr	Val 460	Gly	Pro	Leu	Ala	Arg	Gly 465	Asn	Ser	Arg	Phe	Trp 470
Met	Thr	Arg	Lys 475	Ser	Glu	Glu	Thr	Ile	Thr 480	Pro	Trp	Asn	Phe	Glu 485
Val	Val	Asp	Lys 490	Gly	Ala	Ser	Ala	Gln	Ser 495	Phe	Ile	Glu	Arg	Thr 500
Asn	Phe	Asp	Lys 505	Asn	Leu	Pro	Asn	Glu	Lys 510	Val	Leu	Pro	Lys	Ser 515
Leu	Leu	Tyr	Glu 520	Tyr	Phe	Thr	Val	Tyr	Asn 525	Glu	Leu	Thr	Lys	Lys 530
Tyr	Val	Thr	Glu 535	Gly	Met	Arg	Lys	Pro	Ala 540	Phe	Leu	Ser	Gly	Gln 545
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Thr

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545	550	555	560
Val Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp	565	570	575
Ser Val Glu Ile Ser Gly Val Glu Asp Arg Phe Asn Ala Ser Leu Gly	580	585	590
Thr Tyr His Asp Leu Leu Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp	595	600	605
Asn Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile Val Leu Thr Leu Thr	610	615	620
Leu Phe Glu Asp Arg Glu Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala	625	630	635
His Leu Phe Asp Asp Lys Val Met Lys Gln Leu Lys Arg Arg Arg Tyr	645	650	655
Thr Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp	660	665	670
Lys Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe	675	680	685
Ala Asn Arg Asn Phe Met Gln Leu Ile His Asp Asp Ser Leu Thr Phe	690	695	700
Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp Ser Leu	705	710	715
His Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly	725	730	735
Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu Val Lys Val Met Gly	740	745	750
Arg His Lys Pro Glu Asn Ile Val Ile Glu Met Ala Arg Glu Asn Gln	755	760	765
Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu Arg Met Lys Arg Ile	770	775	780
Glu Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile Leu Lys Glu His Pro	785	790	795
Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu	805	810	815
Gln Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg	820	825	830
Leu Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Lys	835	840	845
Asp Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg	850	855	860
Gly Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys	865	870	875
Asn Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys	885	890	895
Phe Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp	900	905	910
Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr	915	920	925
Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp	930	935	940
Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser	945	950	955
Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg	965	970	975

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Glu Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val	980	985	990
Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe	995	1000	1005
Val Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala	1010	1015	1020
Lys Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe	1025	1030	1035
Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala	1040	1045	1050
Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu	1055	1060	1065
Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val	1070	1075	1080
Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr	1085	1090	1095
Glu Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Arg Pro Lys	1100	1105	1110
Arg Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro	1115	1120	1125
Lys Lys Tyr Gly Gly Phe Val Ser Pro Thr Val Ala Tyr Ser Val	1130	1135	1140
Leu Val Val Ala Lys Val Glu Lys Gly Lys Ser Lys Lys Leu Lys	1145	1150	1155
Ser Val Lys Glu Leu Leu Gly Ile Thr Ile Met Glu Arg Ser Ser	1160	1165	1170
Phe Glu Lys Asn Pro Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys	1175	1180	1185
Glu Val Lys Lys Asp Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu	1190	1195	1200
Phe Glu Leu Glu Asn Gly Arg Lys Arg Met Leu Ala Ser Ala Arg	1205	1210	1215
Phe Leu Gln Lys Gly Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val	1220	1225	1230
Asn Phe Leu Tyr Leu Ala Ser His Tyr Glu Lys Leu Lys Gly Ser	1235	1240	1245
Pro Glu Asp Asn Glu Gln Lys Gln Leu Phe Val Glu Gln His Lys	1250	1255	1260
His Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys	1265	1270	1275
Arg Val Ile Leu Ala Asp Ala Asn Leu Asp Lys Val Leu Ser Ala	1280	1285	1290
Tyr Asn Lys His Arg Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn	1295	1300	1305
Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Arg Ala	1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Gln Tyr Arg Ser	1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr	1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp	1355	1360	1365

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<210> SEQ ID NO 16
<211> LENGTH: 1368
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 16

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val
1          5          10          15

Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe
20          25          30

Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile
35          40          45

Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu
50          55          60

Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys
65          70          75          80

Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser
85          90          95

Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys
100         105         110

His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr
115         120         125

His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp
130         135         140

Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His
145         150         155         160

Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro
165         170         175

Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr
180         185         190

Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala
195         200         205

Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn
210         215         220

Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn
225         230         235         240

Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe
245         250         255

Asp Leu Ala Glu Asp Ala Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp
260         265         270

Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp
275         280         285

Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp
290         295         300

Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser
305         310         315         320

Met Ile Lys Arg Tyr Asp Glu His His Gln Asp Leu Thr Leu Leu Lys
325         330         335

Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe
340         345         350

Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser
355         360         365

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Gln 370	Glu	Phe	Tyr	Lys	Phe 375	Ile	Lys	Pro	Ile	Leu 380	Glu	Lys	Met	Asp
Gly 385	Thr	Glu	Glu	Leu	Leu 390	Val	Lys	Leu	Asn	Arg 395	Glu	Asp	Leu	Leu 400
Lys	Gln	Arg	Thr	Phe 405	Asp	Asn	Gly	Ser	Ile 410	Pro	His	Gln	Ile	His 415
Gly	Glu	Leu	His 420	Ala	Ile	Leu	Arg	Arg	Gln 425	Glu	Asp	Phe	Tyr 430	Pro 435
Leu	Lys	Asp 435	Asn	Arg	Glu	Lys	Ile 440	Glu	Lys	Ile	Leu	Thr 445	Phe	Arg 450
Pro	Tyr 450	Tyr	Val	Gly	Pro	Leu 455	Ala	Arg	Gly	Asn	Ser 460	Arg	Phe	Ala 465
Met 465	Thr	Arg	Lys	Ser	Glu 470	Glu	Thr	Ile	Thr	Pro 475	Trp	Asn	Phe	Glu 480
Val	Val	Asp	Lys 485	Gly	Ala	Ser	Ala	Gln	Ser 490	Phe	Ile	Glu	Arg	Met 495
Asn	Phe	Asp 500	Lys	Asn	Leu	Pro	Asn	Glu 505	Lys	Val	Leu	Pro	Lys 510	His 515
Leu	Leu	Tyr 515	Glu	Tyr	Phe	Thr	Val 520	Tyr	Asn	Glu	Leu	Thr 525	Lys	Val 530
Tyr 530	Val	Thr	Glu	Gly	Met	Arg 535	Lys	Pro	Ala	Phe	Leu 540	Ser	Gly	Glu 545
Lys 545	Lys	Ala	Ile	Val	Asp 550	Leu	Leu	Phe	Lys	Thr 555	Asn	Arg	Lys	Val 560
Val	Lys	Gln	Leu	Lys 565	Glu	Asp	Tyr	Phe	Lys 570	Lys	Ile	Glu	Cys	Phe 575
Ser	Val	Glu	Ile 580	Ser	Gly	Val	Glu	Asp 585	Arg	Phe	Asn	Ala	Ser 590	Leu 595
Thr	Tyr	His 595	Asp	Leu	Leu	Lys	Ile 600	Ile	Lys	Asp	Lys	Asp 605	Phe	Leu 610
Asn 610	Glu	Glu	Asn	Glu	Asp	Ile 615	Leu	Glu	Asp	Ile	Val 620	Leu	Thr	Leu 625
Leu 625	Phe	Glu	Asp	Arg	Glu 630	Met	Ile	Glu	Glu	Arg 635	Leu	Lys	Thr	Tyr 640
His	Leu	Phe	Asp 645	Asp	Lys	Val	Met	Lys	Gln 650	Leu	Lys	Arg	Arg	Arg 655
Thr	Gly	Trp	Gly 660	Arg	Leu	Ser	Arg	Lys 665	Leu	Ile	Asn	Gly	Ile 670	Arg 675
Lys	Gln	Ser 675	Gly	Lys	Thr	Ile	Leu 680	Asp	Phe	Leu	Lys	Ser 685	Asp	Gly 690
Ala 690	Asn	Arg	Asn	Phe	Met	Gln 695	Leu	Ile	His	Asp	Asp 700	Ser	Leu	Thr 705
Lys 705	Glu	Asp	Ile	Gln	Lys 710	Ala	Gln	Val	Ser	Gly 715	Gln	Gly	Asp	Ser 720
His	Glu	His	Ile 725	Ala	Asn	Leu	Ala	Gly	Ser 730	Pro	Ala	Ile	Lys	Lys 735
Ile	Leu	Gln	Thr 740	Val	Lys	Val	Val	Asp 745	Glu	Leu	Val	Lys 750	Val	Met 755
Arg	His	Lys 755	Pro	Glu	Asn	Ile	Val 760	Ile	Glu	Met	Ala	Arg 765	Glu	Asn 770
Thr 770	Thr	Gln	Lys	Gly	Gln	Lys 775	Asn	Ser	Arg	Glu	Arg 780	Met	Lys	Arg 785
Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His 790

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785	790	795	800
Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu	805	810	815
Gln Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg	820	825	830
Leu Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Lys	835	840	845
Asp Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg	850	855	860
Gly Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys	865	870	875
Asn Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys	885	890	895
Phe Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp	900	905	910
Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr	915	920	925
Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp	930	935	940
Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser	945	950	955
Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg	965	970	975
Glu Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val	980	985	990
Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe	995	1000	1005
Val Tyr Gly Asp Tyr Lys Val Tyr Asp Val Arg Lys Met Ile Ala	1010	1015	1020
Lys Ser Glu Gln Glu Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe	1025	1030	1035
Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala	1040	1045	1050
Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu	1055	1060	1065
Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val	1070	1075	1080
Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr	1085	1090	1095
Glu Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Arg Pro Lys	1100	1105	1110
Arg Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro	1115	1120	1125
Lys Lys Tyr Gly Gly Phe Val Ser Pro Thr Val Ala Tyr Ser Val	1130	1135	1140
Leu Val Val Ala Lys Val Glu Lys Gly Lys Ser Lys Lys Leu Lys	1145	1150	1155
Ser Val Lys Glu Leu Leu Gly Ile Thr Ile Met Glu Arg Ser Ser	1160	1165	1170
Phe Glu Lys Asn Pro Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys	1175	1180	1185
Glu Val Lys Lys Asp Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu	1190	1195	1200

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Phe Glu  Leu Glu Asn Gly Arg  Lys Arg Met Leu Ala  Ser Ala Arg
1205                                1210                1215

Phe Leu  Gln Lys Gly Asn Glu  Leu Ala Leu Pro Ser  Lys Tyr Val
1220                                1225                1230

Asn Phe  Leu Tyr Leu Ala Ser  His Tyr Glu Lys Leu  Lys Gly Ser
1235                                1240                1245

Pro Glu  Asp Asn Glu Gln Lys  Gln Leu Phe Val Glu  Gln His Lys
1250                                1255                1260

His Tyr  Leu Asp Glu Ile Ile  Glu Gln Ile Ser Glu  Phe Ser Lys
1265                                1270                1275

Arg Val  Ile Leu Ala Asp Ala  Asn Leu Asp Lys Val  Leu Ser Ala
1280                                1285                1290

Tyr Asn  Lys His Arg Asp Lys  Pro Ile Arg Glu Gln  Ala Glu Asn
1295                                1300                1305

Ile Ile  His Leu Phe Thr Leu  Thr Asn Leu Gly Ala  Pro Arg Ala
1310                                1315                1320

Phe Lys  Tyr Phe Asp Thr Thr  Ile Asp Arg Lys Glu  Tyr Arg Ser
1325                                1330                1335

Thr Lys  Glu Val Leu Asp Ala  Thr Leu Ile His Gln  Ser Ile Thr
1340                                1345                1350

Gly Leu  Tyr Glu Thr Arg Ile  Asp Leu Ser Gln Leu  Gly Gly Asp
1355                                1360                1365

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<210> SEQ ID NO 17

<211> LENGTH: 1368

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 17

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Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val
1      5              10              15

Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe
20     25              30

Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile
35     40              45

Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu
50     55              60

Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys
65     70              75              80

Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser
85     90              95

Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys
100    105             110

His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr
115    120             125

His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp
130    135             140

Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His
145    150             155             160

Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro
165    170             175

Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr
180    185             190

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Asn	Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala	Ser	Gly	Val	Asp	Ala			
		195																
Lys	Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser	Arg	Arg	Leu	Glu	Asn			
		210																
Leu	Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn	Gly	Leu	Phe	Gly	Asn			
		225						230								240		
Leu	Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn	Phe	Lys	Ser	Asn	Phe			
					245										255			
Asp	Leu	Ala	Glu	Asp	Thr	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp			
					260						265						270	
Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp			
		275																
Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp			
		290																
Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser			
		305						310								320		
Met	Ile	Lys	Leu	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Lys			
					325										335			
Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Phe			
					340						345						350	
Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile	Asp	Gly	Gly	Ala	Ser			
		355																
Gln	Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp			
		370						375								380		
Gly	Thr	Glu	Glu	Leu	Leu	Val	Lys	Leu	Asn	Arg	Glu	Asp	Leu	Leu	Arg			
		385						390								395	400	
Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ile	Ile	Pro	His	Gln	Ile	His	Leu			
					405										410			415
Gly	Glu	Leu	His	Ala	Ile	Leu	Arg	Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Phe			
					420						425						430	
Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile			
		435																
Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp			
		450						455								460		
Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Lys			
		465						470								475	480	
Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr			
					485										490			495
Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser			
					500						505						510	
Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys			
		515																
Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Asp	Gln			
		530						535								540		
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr			
		545						550								555	560	
Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp			
					565										570			575
Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly			
					580						585						590	
Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp			
		595																

Asn 610	Glu	Asn	Glu	Asp	Ile 615	Leu	Glu	Asp	Ile	Val 620	Leu	Thr	Leu	Thr
Leu 625	Phe	Glu	Asp	Arg	Glu 630	Met	Ile	Glu	Glu	Arg 635	Leu	Lys	Thr	Tyr
His	Leu	Phe	Asp	Asp 645	Lys	Val	Met	Lys	Gln 650	Leu	Lys	Arg	Arg 655	Tyr
Thr	Gly	Trp	Gly 660	Arg	Leu	Ser	Arg	Lys 665	Leu	Ile	Asn	Gly	Ile 670	Arg
Lys	Gln	Ser	Gly 675	Lys	Thr	Ile	Leu	Asp 680	Phe	Leu	Lys	Ser 685	Asp	Gly
Ala 690	Asn	Arg	Asn	Phe	Ile	Gln 695	Leu	Ile	His	Asp 700	Asp	Ser	Leu	Thr
Lys 705	Glu	Asp	Ile	Gln	Lys 710	Ala	Gln	Val	Ser	Gly 715	Gln	Gly	Asp	Ser
His	Glu	His	Ile 725	Ala	Asn	Leu	Ala	Gly	Ser 730	Pro	Ala	Ile	Lys 735	Lys
Ile	Leu	Gln	Thr 740	Val	Lys	Val	Val	Asp 745	Glu	Leu	Val	Lys	Val 750	Met
Arg	His	Lys 755	Pro	Glu	Asn	Ile 760	Val	Ile	Glu	Met	Ala 765	Arg	Glu	Asn
Thr 770	Thr	Gln	Lys	Gly	Gln	Lys 775	Asn	Ser	Arg	Glu	Arg 780	Met	Lys	Arg
Glu 785	Glu	Gly	Ile	Lys	Glu 790	Leu	Gly	Ser	Gln	Ile 795	Leu	Lys	Glu	His
Val	Glu	Asn	Thr 805	Gln	Leu	Gln	Asn	Glu	Lys 810	Leu	Tyr	Leu	Tyr 815	Tyr
Gln	Asn	Gly 820	Arg	Asp	Met	Tyr	Val	Asp 825	Gln	Glu	Leu	Asp	Ile 830	Asn
Leu	Ser 835	Asp	Tyr	Asp	Val	Asp 840	His	Ile	Val	Pro	Gln	Ser 845	Phe	Leu
Asp 850	Asp	Ser	Ile	Asp	Asn	Lys 855	Val	Leu	Thr	Arg	Ser 860	Asp	Lys	Asn
Gly 865	Lys	Ser	Asp	Asn	Val 870	Pro	Ser	Glu	Glu	Val 875	Val	Lys	Lys	Met
Asn	Tyr	Trp	Arg 885	Gln	Leu	Leu	Asn	Ala	Lys 890	Leu	Ile	Thr	Gln 895	Arg
Phe	Asp	Asn	Leu 900	Thr	Lys	Ala	Glu	Arg 905	Gly	Gly	Leu	Ser	Glu 910	Leu
Lys	Ala 915	Gly	Phe	Ile	Lys	Arg	Gln 920	Leu	Val	Glu	Thr 925	Arg	Gln	Ile
Lys 930	His	Val	Ala	Gln	Ile 935	Leu	Asp	Ser	Arg	Met	Asn 940	Thr	Lys	Tyr
Glu 945	Asn	Asp	Lys	Leu	Ile 950	Arg	Glu	Val	Lys 955	Val	Ile	Thr	Leu	Lys
Lys	Leu	Val	Ser 965	Asp	Phe	Arg	Lys	Asp 970	Phe	Gln	Phe	Tyr	Lys 975	Val
Glu	Ile	Asn	Asn 980	Tyr	His	His	Ala	His 985	Asp	Ala	Tyr	Leu	Asn 990	Ala
Val	Gly 995	Thr	Ala	Leu	Ile	Lys	Lys	Tyr 1000	Pro	Lys	Leu	Glu	Ser 1005	Glu
Val 1010	Tyr	Gly	Asp	Tyr	Lys	Val 1015	Tyr	Asp	Val	Arg	Lys 1020	Met	Ile	Ala
Lys	Ser	Glu	Gln	Glu	Ile	Gly	Lys	Ala	Thr	Ala	Lys	Tyr	Phe	Phe

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1025	1030	1035
Tyr Ser Asn Ile Met Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala		
1040	1045	1050
Asn Gly Glu Ile Arg Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu		
1055	1060	1065
Thr Gly Glu Ile Val Trp Asp Lys Gly Arg Asp Phe Ala Thr Val		
1070	1075	1080
Arg Lys Val Leu Ser Met Pro Gln Val Asn Ile Val Lys Lys Thr		
1085	1090	1095
Glu Val Gln Thr Gly Gly Phe Ser Lys Glu Ser Ile Arg Pro Lys		
1100	1105	1110
Arg Asn Ser Asp Lys Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro		
1115	1120	1125
Lys Lys Tyr Gly Gly Phe Val Ser Pro Thr Val Ala Tyr Ser Val		
1130	1135	1140
Leu Val Val Ala Lys Val Glu Lys Gly Lys Ser Lys Lys Leu Lys		
1145	1150	1155
Ser Val Lys Glu Leu Leu Gly Ile Thr Ile Met Glu Arg Ser Ser		
1160	1165	1170
Phe Glu Lys Asn Pro Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys		
1175	1180	1185
Glu Val Lys Lys Asp Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu		
1190	1195	1200
Phe Glu Leu Glu Asn Gly Arg Lys Arg Met Leu Ala Ser Ala Arg		
1205	1210	1215
Phe Leu Gln Lys Gly Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val		
1220	1225	1230
Asn Phe Leu Tyr Leu Ala Ser His Tyr Glu Lys Leu Lys Gly Ser		
1235	1240	1245
Pro Glu Asp Asn Glu Gln Lys Gln Leu Phe Val Glu Gln His Lys		
1250	1255	1260
His Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys		
1265	1270	1275
Arg Val Ile Leu Ala Asp Ala Asn Leu Asp Lys Val Leu Ser Ala		
1280	1285	1290
Tyr Asn Lys His Arg Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn		
1295	1300	1305
Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Arg Ala		
1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Gln Tyr Arg Ser		
1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr		
1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp		
1355	1360	1365

<210> SEQ ID NO 18

<211> LENGTH: 1368

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 18

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val

-continued

1	5	10	15
Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe	20	25	30
Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile	35	40	45
Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu	50	55	60
Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys	65	70	75
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser	85	90	95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys	100	105	110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr	115	120	125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp	130	135	140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His	145	150	155
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro	165	170	175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr	180	185	190
Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala	195	200	205
Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn	210	215	220
Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn	225	230	235
Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe	245	250	255
Asp Leu Ala Glu Asp Thr Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp	260	265	270
Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp	275	280	285
Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp	290	295	300
Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser	305	310	315
Met Ile Lys Leu Tyr Asp Glu His His Gln Asp Leu Thr Leu Lys	325	330	335
Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe	340	345	350
Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser	355	360	365
Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile Leu Glu Lys Met Asp	370	375	380
Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg Glu Asp Leu Leu Arg	385	390	395
Lys Gln Arg Thr Phe Asp Asn Gly Ile Ile Pro His Gln Ile His Leu	405	410	415
Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu Asp Phe Tyr Pro Phe	420	425	430

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Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile
	435						440					445			
Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp
	450					455					460				
Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Lys
465					470					475				480	
Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr
			485						490					495	
Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser
		500						505					510		
Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys
	515						520					525			
Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Asp	Gln
	530					535					540				
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr
545					550					555				560	
Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp
			565						570					575	
Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly
		580						585					590		
Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp
		595					600					605			
Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr
	610					615					620				
Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala
625					630					635					640
His	Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr
			645						650					655	
Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp
		660						665					670		
Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	Asp	Phe	Leu	Lys	Ser	Asp	Gly	Phe
		675					680					685			
Ala	Asn	Arg	Asn	Phe	Ile	Gln	Leu	Ile	His	Asp	Asp	Ser	Leu	Thr	Phe
	690					695				700					
Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu
705					710					715					720
His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly
			725					730					735		
Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu	Val	Lys	Val	Met	Gly
		740						745					750		
Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu	Met	Ala	Arg	Glu	Asn	Gln
		755					760					765			
Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	Ser	Arg	Glu	Arg	Met	Lys	Arg	Ile
	770				775						780				
Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His	Pro
785					790					795					800
Val	Glu	Asn	Thr	Gln	Leu	Gln	Asn	Glu	Lys	Leu	Tyr	Leu	Tyr	Tyr	Leu
			805						810					815	
Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	Glu	Leu	Asp	Ile	Asn	Arg
		820						825					830		
Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	Val	Pro	Gln	Ser	Phe	Leu	Lys
	835						840						845		

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Asp	Asp	Ser	Ile	Asp	Asn	Lys	Val	Leu	Thr	Arg	Ser	Asp	Lys	Asn	Arg
850						855						860			
Gly	Lys	Ser	Asp	Asn	Val	Pro	Ser	Glu	Glu	Val	Val	Lys	Lys	Met	Lys
865					870					875					880
Asn	Tyr	Trp	Arg	Gln	Leu	Leu	Asn	Ala	Lys	Leu	Ile	Thr	Gln	Arg	Lys
				885					890					895	
Phe	Asp	Asn	Leu	Thr	Lys	Ala	Glu	Arg	Gly	Gly	Leu	Ser	Glu	Leu	Asp
			900					905					910		
Lys	Ala	Gly	Phe	Ile	Lys	Arg	Gln	Leu	Val	Glu	Thr	Arg	Gln	Ile	Thr
		915					920					925			
Lys	His	Val	Ala	Gln	Ile	Leu	Asp	Ser	Arg	Met	Asn	Thr	Lys	Tyr	Asp
		930					935					940			
Glu	Asn	Asp	Lys	Leu	Ile	Arg	Glu	Val	Lys	Val	Ile	Thr	Leu	Lys	Ser
945					950					955					960
Lys	Leu	Val	Ser	Asp	Phe	Arg	Lys	Asp	Phe	Gln	Phe	Tyr	Lys	Val	Arg
				965					970						975
Glu	Ile	Asn	Asn	Tyr	His	His	Ala	His	Asp	Ala	Tyr	Leu	Asn	Ala	Val
				980					985						990
Val	Gly	Thr	Ala	Leu	Ile	Lys	Lys	Tyr	Pro	Lys	Leu	Glu	Ser	Glu	Phe
				995				1000					1005		
Val	Tyr	Gly	Asp	Tyr	Lys	Val	Tyr	Asp	Val	Arg	Lys	Met	Ile	Ala	
	1010					1015						1020			
Lys	Ser	Glu	Gln	Glu	Ile	Gly	Lys	Ala	Thr	Ala	Lys	Tyr	Phe	Phe	
	1025					1030						1035			
Tyr	Ser	Asn	Ile	Met	Asn	Phe	Phe	Lys	Thr	Glu	Ile	Thr	Leu	Ala	
	1040					1045						1050			
Asn	Gly	Glu	Ile	Arg	Lys	Arg	Pro	Leu	Ile	Glu	Thr	Asn	Gly	Glu	
	1055					1060						1065			
Thr	Gly	Glu	Ile	Val	Trp	Asp	Lys	Gly	Arg	Asp	Phe	Ala	Thr	Val	
	1070					1075						1080			
Arg	Lys	Val	Leu	Ser	Met	Pro	Gln	Val	Asn	Ile	Val	Lys	Lys	Thr	
	1085					1090						1095			
Glu	Val	Gln	Thr	Gly	Gly	Phe	Ser	Lys	Glu	Ser	Ile	Arg	Pro	Lys	
	1100					1105						1110			
Arg	Asn	Ser	Asp	Lys	Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp	Asp	Pro	
	1115					1120						1125			
Lys	Lys	Tyr	Gly	Gly	Phe	Val	Ser	Pro	Thr	Val	Ala	Tyr	Ser	Val	
	1130					1135						1140			
Leu	Val	Val	Ala	Lys	Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys	Leu	Lys	
	1145					1150						1155			
Ser	Val	Lys	Glu	Leu	Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg	Ser	Ser	
	1160					1165						1170			
Phe	Glu	Lys	Asn	Pro	Ile	Asp	Phe	Leu	Glu	Ala	Lys	Gly	Tyr	Lys	
	1175					1180						1185			
Glu	Val	Lys	Lys	Asp	Leu	Ile	Ile	Lys	Leu	Pro	Lys	Tyr	Ser	Leu	
	1190					1195						1200			
Phe	Glu	Leu	Glu	Asn	Gly	Arg	Lys	Arg	Met	Leu	Ala	Ser	Ala	Arg	
	1205					1210						1215			
Phe	Leu	Gln	Lys	Gly	Asn	Glu	Leu	Ala	Leu	Pro	Ser	Lys	Tyr	Val	
	1220					1225						1230			
Asn	Phe	Leu	Tyr	Leu	Ala	Ser	His	Tyr	Glu	Lys	Leu	Lys	Gly	Ser	
	1235					1240						1245			
Pro	Glu	Asp	Asn	Glu	Gln	Lys	Gln	Leu	Phe	Val	Glu	Gln	His	Lys	

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1250	1255	1260
His Tyr Leu Asp Glu Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys		
1265	1270	1275
Arg Val Ile Leu Ala Asp Ala Asn Leu Asp Lys Val Leu Ser Ala		
1280	1285	1290
Tyr Asn Lys His Arg Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn		
1295	1300	1305
Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Arg Ala		
1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Glu Tyr Arg Ser		
1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr		
1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp		
1355	1360	1365

<210> SEQ ID NO 19

<211> LENGTH: 1378

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 19

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val		
1	5	10
Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe		
20	25	30
Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile		
35	40	45
Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu		
50	55	60
Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys		
65	70	75
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser		
85	90	95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys		
100	105	110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr		
115	120	125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp		
130	135	140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His		
145	150	155
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro		
165	170	175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr		
180	185	190
Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala		
195	200	205
Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn		
210	215	220
Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn		
225	230	235
Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe		

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245								250				255					
Asp	Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp		
			260				265						270				
Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp		
			275				280						285				
Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp		
			290				295						300				
Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser		
305						310						315			320		
Met	Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Lys		
			325						330						335		
Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Phe		
			340						345						350		
Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Val	Gly	Ala	Asp	Lys	Lys		
			355						360						365		
Leu	Arg	Lys	Arg	Ser	Ser	Lys	Leu	Ala	Thr	Glu	Glu	Glu	Phe	Tyr	Lys		
			370						375						380		
Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp	Gly	Thr	Glu	Glu	Leu	Leu		
385						390						395			400		
Val	Lys	Leu	Asn	Arg	Glu	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp		
			405						410						415		
Asn	Gly	Ser	Ile	Pro	His	Gln	Ile	His	Leu	Gly	Glu	Leu	His	Ala	Ile		
			420						425						430		
Leu	Arg	Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Phe	Leu	Lys	Asp	Asn	Arg	Glu		
			435						440						445		
Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro		
			450						455						460		
Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp	Met	Thr	Arg	Lys	Ser	Glu		
465						470						475			480		
Glu	Thr	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly	Ala		
			485						490						495		
Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr	Asn	Phe	Asp	Lys	Asn	Leu		
			500						505						510		
Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser	Leu	Leu	Tyr	Glu	Tyr	Phe		
			515						520						525		
Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys	Tyr	Val	Thr	Glu	Gly	Met		
			530						535						540		
Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Glu	Gln	Lys	Lys	Ala	Ile	Val	Asp		
545						550						555			560		
Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr	Val	Lys	Gln	Leu	Lys	Glu		
			565						570						575		
Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp	Ser	Val	Glu	Ile	Ser	Gly		
			580						585						590		
Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly	Thr	Tyr	His	Asp	Leu	Leu		
			595						600						605		
Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp	Asn	Glu	Glu	Asn	Glu	Asp		
			610						615						620		
Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr	Leu	Phe	Glu	Asp	Arg	Glu		
625						630						635			640		
Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala	His	Leu	Phe	Asp	Asp	Lys		
			645						650						655		
Val	Met	Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr	Thr	Gly	Trp	Gly	Arg	Leu		
			660						665						670		

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Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp Lys Gln Ser Gly Lys Thr	675	680	685
Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe Ala Asn Arg Asn Phe Met	690	695	700
Gln Leu Ile His Asp Asp Ser Leu Thr Phe Lys Glu Asp Ile Gln Lys	705	710	715
Ala Gln Val Ser Gly Gln Gly Asp Ser Leu His Glu His Ile Ala Asn	725	730	735
Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly Ile Leu Gln Thr Val Lys	740	745	750
Val Val Asp Glu Leu Val Lys Val Met Gly Arg His Lys Pro Glu Asn	755	760	765
Ile Val Ile Glu Met Ala Arg Glu Asn Gln Thr Thr Gln Lys Gly Gln	770	775	780
Lys Asn Ser Arg Glu Arg Met Lys Arg Ile Glu Glu Gly Ile Lys Glu	785	790	795
Leu Gly Ser Gln Ile Leu Lys Glu His Pro Val Glu Asn Thr Gln Leu	805	810	815
Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met	820	825	830
Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val	835	840	845
Asp His Ile Val Pro Gln Ser Phe Leu Lys Asp Asp Ser Ile Asp Asn	850	855	860
Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly Lys Ser Asp Asn Val	865	870	875
Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr Trp Arg Gln Leu	885	890	895
Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe Asp Asn Leu Thr Lys	900	905	910
Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp Lys Ala Gly Phe Ile Lys	915	920	925
Arg Gln Leu Val Glu Thr Arg Gln Ile Thr Lys His Val Ala Gln Ile	930	935	940
Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp Glu Asn Asp Lys Leu Ile	945	950	955
Arg Glu Val Lys Val Ile Thr Leu Lys Ser Lys Leu Val Ser Asp Phe	965	970	975
Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg Glu Ile Asn Asn Tyr His	980	985	990
His Ala His Asp Ala Tyr Leu Asn Ala Val Val Gly Thr Ala Leu Ile	995	1000	1005
Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe Val Tyr Gly Asp Tyr	1010	1015	1020
Lys Val Tyr Asp Val Arg Lys Met Ile Ala Lys Ser Glu Gln Glu	1025	1030	1035
Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe Tyr Ser Asn Ile Met	1040	1045	1050
Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala Asn Gly Glu Ile Arg	1055	1060	1065
Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu Thr Gly Glu Ile Val	1070	1075	1080

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Trp	Asp	Lys	Gly	Arg	Asp	Phe	Ala	Thr	Val	Arg	Lys	Val	Leu	Ser
1085						1090					1095			
Met	Pro	Gln	Val	Asn	Ile	Val	Lys	Lys	Thr	Glu	Val	Gln	Thr	Gly
1100						1105					1110			
Gly	Phe	Ser	Lys	Glu	Ser	Ile	Arg	Pro	Lys	Arg	Asn	Ser	Asp	Lys
1115						1120					1125			
Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp	Asp	Pro	Lys	Lys	Tyr	Gly	Gly
1130						1135					1140			
Phe	Val	Ser	Pro	Thr	Val	Ala	Tyr	Ser	Val	Leu	Val	Val	Ala	Lys
1145						1150					1155			
Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys	Leu	Lys	Ser	Val	Lys	Glu	Leu
1160						1165					1170			
Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg	Ser	Ser	Phe	Glu	Lys	Asn	Pro
1175						1180					1185			
Ile	Asp	Phe	Leu	Glu	Ala	Lys	Gly	Tyr	Lys	Glu	Val	Lys	Lys	Asp
1190						1195					1200			
Leu	Ile	Ile	Lys	Leu	Pro	Lys	Tyr	Ser	Leu	Phe	Glu	Leu	Glu	Asn
1205						1210					1215			
Gly	Arg	Lys	Arg	Met	Leu	Ala	Ser	Ala	Arg	Phe	Leu	Gln	Lys	Gly
1220						1225					1230			
Asn	Glu	Leu	Ala	Leu	Pro	Ser	Lys	Tyr	Val	Asn	Phe	Leu	Tyr	Leu
1235						1240					1245			
Ala	Ser	His	Tyr	Glu	Lys	Leu	Lys	Gly	Ser	Pro	Glu	Asp	Asn	Glu
1250						1255					1260			
Gln	Lys	Gln	Leu	Phe	Val	Glu	Gln	His	Lys	His	Tyr	Leu	Asp	Glu
1265						1270					1275			
Ile	Ile	Glu	Gln	Ile	Ser	Glu	Phe	Ser	Lys	Arg	Val	Ile	Leu	Ala
1280						1285					1290			
Asp	Ala	Asn	Leu	Asp	Lys	Val	Leu	Ser	Ala	Tyr	Asn	Lys	His	Arg
1295						1300					1305			
Asp	Lys	Pro	Ile	Arg	Glu	Gln	Ala	Glu	Asn	Ile	Ile	His	Leu	Phe
1310						1315					1320			
Thr	Leu	Thr	Asn	Leu	Gly	Ala	Pro	Arg	Ala	Phe	Lys	Tyr	Phe	Asp
1325						1330					1335			
Thr	Thr	Ile	Asp	Arg	Lys	Val	Tyr	Arg	Ser	Thr	Lys	Glu	Val	Leu
1340						1345					1350			
Asp	Ala	Thr	Leu	Ile	His	Gln	Ser	Ile	Thr	Gly	Leu	Tyr	Glu	Thr
1355						1360					1365			
Arg	Ile	Asp	Leu	Ser	Gln	Leu	Gly	Gly	Asp					
1370						1375								

<210> SEQ ID NO 20
 <211> LENGTH: 1378
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 20

Met Asp Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val
 1 5 10 15

Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe
 20 25 30

Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile
 35 40 45

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Gly	Ala	Leu	Leu	Phe	Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr	Arg	Leu
50						55					60				
Lys	Arg	Thr	Ala	Arg	Arg	Arg	Tyr	Thr	Arg	Arg	Lys	Asn	Arg	Ile	Cys
65				70						75				80	
Tyr	Leu	Gln	Glu	Ile	Phe	Ser	Asn	Glu	Met	Ala	Lys	Val	Asp	Asp	Ser
			85					90					95		
Phe	Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val	Glu	Glu	Asp	Lys	Lys
		100						105					110		
His	Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val	Asp	Glu	Val	Ala	Tyr
	115						120					125			
His	Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg	Lys	Lys	Leu	Val	Asp
	130					135					140				
Ser	Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr	Leu	Ala	Leu	Ala	His
145					150					155					160
Met	Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu	Gly	Asp	Leu	Asn	Pro
			165					170						175	
Asp	Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln	Leu	Val	Gln	Thr	Tyr
			180					185					190		
Asn	Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala	Ser	Gly	Val	Asp	Ala
	195						200					205			
Lys	Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser	Arg	Arg	Leu	Glu	Asn
	210					215					220				
Leu	Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn	Gly	Leu	Phe	Gly	Asn
225					230					235					240
Leu	Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn	Phe	Lys	Ser	Asn	Phe
			245						250					255	
Asp	Leu	Ala	Glu	Asp	Thr	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp
	260						265						270		
Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp
	275						280					285			
Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp
	290					295					300				
Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser
305					310					315					320
Met	Ile	Lys	Leu	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Lys
			325						330					335	
Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Phe
		340						345					350		
Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Val	Gly	Ala	Asp	Lys	Lys
		355					360					365			
Leu	Arg	Lys	Arg	Ser	Ser	Lys	Leu	Ala	Thr	Glu	Glu	Glu	Phe	Tyr	Lys
	370					375					380				
Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp	Gly	Thr	Glu	Glu	Leu	Leu
385					390					395					400
Val	Lys	Leu	Asn	Arg	Glu	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp
			405						410					415	
Asn	Gly	Ile	Ile	Pro	His	Gln	Ile	His	Leu	Gly	Glu	Leu	His	Ala	Ile
		420					425						430		
Leu	Arg	Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Phe	Leu	Lys	Asp	Asn	Arg	Glu
	435						440					445			
Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro
	450					455					460				
Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp	Met	Thr	Arg	Lys	Ser	Glu

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465	470	475	480
Glu Thr Ile Thr Pro Trp Asn Phe Glu Lys Val Val Asp Lys Gly Ala			
485		490	495
Ser Ala Gln Ser Phe Ile Glu Arg Met Thr Asn Phe Asp Lys Asn Leu			
500		505	510
Pro Asn Glu Lys Val Leu Pro Lys His Ser Leu Leu Tyr Glu Tyr Phe			
515		520	525
Thr Val Tyr Asn Glu Leu Thr Lys Val Lys Tyr Val Thr Glu Gly Met			
530		535	540
Arg Lys Pro Ala Phe Leu Ser Gly Asp Gln Lys Lys Ala Ile Val Asp			
545	550	555	560
Leu Leu Phe Lys Thr Asn Arg Lys Val Thr Val Lys Gln Leu Lys Glu			
565		570	575
Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp Ser Val Glu Ile Ser Gly			
580		585	590
Val Glu Asp Arg Phe Asn Ala Ser Leu Gly Thr Tyr His Asp Leu Leu			
595	600	605	
Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp Asn Glu Glu Asn Glu Asp			
610	615	620	
Ile Leu Glu Asp Ile Val Leu Thr Leu Thr Leu Phe Glu Asp Arg Glu			
625	630	635	640
Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala His Leu Phe Asp Asp Lys			
645	650	655	
Val Met Lys Gln Leu Lys Arg Arg Arg Tyr Thr Gly Trp Gly Arg Leu			
660	665	670	
Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp Lys Gln Ser Gly Lys Thr			
675	680	685	
Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe Ala Asn Arg Asn Phe Ile			
690	695	700	
Gln Leu Ile His Asp Asp Ser Leu Thr Phe Lys Glu Asp Ile Gln Lys			
705	710	715	720
Ala Gln Val Ser Gly Gln Gly Asp Ser Leu His Glu His Ile Ala Asn			
725	730	735	
Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly Ile Leu Gln Thr Val Lys			
740	745	750	
Val Val Asp Glu Leu Val Lys Val Met Gly Arg His Lys Pro Glu Asn			
755	760	765	
Ile Val Ile Glu Met Ala Arg Glu Asn Gln Thr Thr Gln Lys Gly Gln			
770	775	780	
Lys Asn Ser Arg Glu Arg Met Lys Arg Ile Glu Glu Gly Ile Lys Glu			
785	790	795	800
Leu Gly Ser Gln Ile Leu Lys Glu His Pro Val Glu Asn Thr Gln Leu			
805	810	815	
Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met			
820	825	830	
Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val			
835	840	845	
Asp His Ile Val Pro Gln Ser Phe Leu Lys Asp Asp Ser Ile Asp Asn			
850	855	860	
Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly Lys Ser Asp Asn Val			
865	870	875	880
Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr Trp Arg Gln Leu			
885	890	895	

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Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe Asp Asn Leu Thr Lys
 900 905 910

Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp Lys Ala Gly Phe Ile Lys
 915 920 925

Arg Gln Leu Val Glu Thr Arg Gln Ile Thr Lys His Val Ala Gln Ile
 930 935 940

Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp Glu Asn Asp Lys Leu Ile
 945 950 955 960

Arg Glu Val Lys Val Ile Thr Leu Lys Ser Lys Leu Val Ser Asp Phe
 965 970 975

Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg Glu Ile Asn Asn Tyr His
 980 985 990

His Ala His Asp Ala Tyr Leu Asn Ala Val Val Gly Thr Ala Leu Ile
 995 1000 1005

Lys Lys Tyr Pro Lys Leu Glu Ser Glu Phe Val Tyr Gly Asp Tyr
 1010 1015 1020

Lys Val Tyr Asp Val Arg Lys Met Ile Ala Lys Ser Glu Gln Glu
 1025 1030 1035

Ile Gly Lys Ala Thr Ala Lys Tyr Phe Phe Tyr Ser Asn Ile Met
 1040 1045 1050

Asn Phe Phe Lys Thr Glu Ile Thr Leu Ala Asn Gly Glu Ile Arg
 1055 1060 1065

Lys Arg Pro Leu Ile Glu Thr Asn Gly Glu Thr Gly Glu Ile Val
 1070 1075 1080

Trp Asp Lys Gly Arg Asp Phe Ala Thr Val Arg Lys Val Leu Ser
 1085 1090 1095

Met Pro Gln Val Asn Ile Val Lys Lys Thr Glu Val Gln Thr Gly
 1100 1105 1110

Gly Phe Ser Lys Glu Ser Ile Arg Pro Lys Arg Asn Ser Asp Lys
 1115 1120 1125

Leu Ile Ala Arg Lys Lys Asp Trp Asp Pro Lys Lys Tyr Gly Gly
 1130 1135 1140

Phe Val Ser Pro Thr Val Ala Tyr Ser Val Leu Val Val Ala Lys
 1145 1150 1155

Val Glu Lys Gly Lys Ser Lys Lys Leu Lys Ser Val Lys Glu Leu
 1160 1165 1170

Leu Gly Ile Thr Ile Met Glu Arg Ser Ser Phe Glu Lys Asn Pro
 1175 1180 1185

Ile Asp Phe Leu Glu Ala Lys Gly Tyr Lys Glu Val Lys Lys Asp
 1190 1195 1200

Leu Ile Ile Lys Leu Pro Lys Tyr Ser Leu Phe Glu Leu Glu Asn
 1205 1210 1215

Gly Arg Lys Arg Met Leu Ala Ser Ala Arg Phe Leu Gln Lys Gly
 1220 1225 1230

Asn Glu Leu Ala Leu Pro Ser Lys Tyr Val Asn Phe Leu Tyr Leu
 1235 1240 1245

Ala Ser His Tyr Glu Lys Leu Lys Gly Ser Pro Glu Asp Asn Glu
 1250 1255 1260

Gln Lys Gln Leu Phe Val Glu Gln His Lys His Tyr Leu Asp Glu
 1265 1270 1275

Ile Ile Glu Gln Ile Ser Glu Phe Ser Lys Arg Val Ile Leu Ala
 1280 1285 1290

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Asp Ala Asn Leu Asp Lys Val Leu Ser Ala Tyr Asn Lys His Arg
 1295 1300 1305
 Asp Lys Pro Ile Arg Glu Gln Ala Glu Asn Ile Ile His Leu Phe
 1310 1315 1320
 Thr Leu Thr Asn Leu Gly Ala Pro Arg Ala Phe Lys Tyr Phe Asp
 1325 1330 1335
 Thr Thr Ile Asp Arg Lys Val Tyr Arg Ser Thr Lys Glu Val Leu
 1340 1345 1350
 Asp Ala Thr Leu Ile His Gln Ser Ile Thr Gly Leu Tyr Glu Thr
 1355 1360 1365
 Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp
 1370 1375

<210> SEQ ID NO 21
 <211> LENGTH: 1375
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 21

Met Glu Lys Lys Tyr Ser Ile Gly Leu Ala Ile Gly Thr Asn Ser Val
 1 5 10 15
 Gly Trp Ala Val Ile Thr Asp Asp Tyr Lys Val Pro Ser Lys Lys Phe
 20 25 30
 Lys Val Leu Gly Asn Thr Asn Arg Lys Ser Ile Lys Lys Asn Leu Met
 35 40 45
 Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu
 50 55 60
 Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Arg
 65 70 75 80
 Tyr Leu Gln Glu Ile Phe Ala Asn Glu Met Ala Lys Leu Asp Asp Ser
 85 90 95
 Phe Phe Gln Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys
 100 105 110
 Asn Glu Arg His Pro Ile Phe Gly Asn Leu Ala Asp Glu Val Ala Tyr
 115 120 125
 His Arg Asn Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Ala Asp
 130 135 140
 Ser Pro Glu Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His
 145 150 155 160
 Ile Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Lys Leu Asn Ala
 165 170 175
 Glu Asn Ser Asp Val Ala Lys Leu Phe Tyr Gln Leu Ile Gln Thr Tyr
 180 185 190
 Asn Gln Leu Phe Glu Glu Ser Pro Leu Asp Glu Ile Glu Val Asp Ala
 195 200 205
 Lys Gly Ile Leu Ser Ala Arg Leu Ser Lys Ser Lys Arg Leu Glu Lys
 210 215 220
 Leu Ile Ala Val Phe Pro Asn Glu Lys Lys Asn Gly Leu Phe Gly Asn
 225 230 235 240
 Ile Ile Ala Leu Ala Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe
 245 250 255
 Asp Leu Thr Glu Asp Ala Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp
 260 265 270

Asp	Asp	Leu	Asp	Glu	Leu	Leu	Gly	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp
		275					280					285			
Leu	Phe	Ser	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp
	290					295					300				
Ile	Leu	Arg	Ser	Asn	Ser	Glu	Val	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser
305				310						315					320
Met	Val	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Ala	Leu	Leu	Lys
				325					330					335	
Thr	Leu	Val	Arg	Gln	Gln	Phe	Pro	Glu	Lys	Tyr	Ala	Glu	Ile	Phe	Lys
			340					345					350		
Asp	Asp	Thr	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Val	Gly	Ile	Gly	Ile	Lys
		355					360					365			
His	Arg	Lys	Arg	Thr	Thr	Lys	Leu	Ala	Thr	Gln	Glu	Glu	Phe	Tyr	Lys
		370				375					380				
Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp	Gly	Ala	Glu	Glu	Leu	Leu
385				390						395					400
Ala	Lys	Leu	Asn	Arg	Asp	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp
				405					410					415	
Asn	Gly	Ser	Ile	Pro	His	Gln	Ile	His	Leu	Lys	Glu	Leu	His	Ala	Ile
			420					425					430		
Leu	Arg	Arg	Gln	Glu	Glu	Phe	Tyr	Pro	Phe	Leu	Lys	Glu	Asn	Arg	Glu
		435					440					445			
Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro
		450				455					460				
Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp	Leu	Thr	Arg	Lys	Ser	Glu
465				470						475					480
Glu	Ala	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly	Ala
				485					490					495	
Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr	Asn	Phe	Asp	Glu	Gln	Leu
			500					505					510		
Pro	Asn	Lys	Lys	Val	Leu	Pro	Lys	His	Ser	Leu	Leu	Tyr	Glu	Tyr	Phe
		515					520					525			
Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys	Tyr	Val	Thr	Glu	Arg	Met
		530				535						540			
Arg	Lys	Pro	Glu	Phe	Leu	Ser	Gly	Glu	Gln	Lys	Lys	Ala	Ile	Val	Asp
545					550					555					560
Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr	Val	Lys	Gln	Leu	Lys	Glu
				565					570					575	
Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp	Ser	Val	Glu	Ile	Ile	Gly
			580					585					590		
Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly	Thr	Tyr	His	Asp	Leu	Leu
		595					600					605			
Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp	Asn	Glu	Glu	Asn	Glu	Asp
		610				615					620				
Ile	Leu														

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690	695	700
Gln Leu Ile His Asp Asp Ser Leu Thr Phe Lys Glu Glu Ile Glu Lys 705 710 715 720		
Ala Gln Val Ser Gly Gln Gly Asp Ser Leu His Glu Gln Ile Ala Asp 725 730 735		
Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly Ile Leu Gln Thr Val Lys 740 745 750		
Ile Val Asp Glu Leu Val Lys Val Met Gly His Lys Pro Glu Asn Ile 755 760 765		
Val Ile Glu Met Ala Arg Glu Asn Gln Thr Thr Thr Lys Gly Leu Gln 770 775 780		
Gln Ser Arg Glu Arg Lys Lys Arg Ile Glu Glu Gly Ile Lys Glu Leu 785 790 795 800		
Glu Ser Gln Ile Leu Lys Glu Asn Pro Val Glu Asn Thr Gln Leu Gln 805 810 815		
Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met Tyr 820 825 830		
Val Asp Gln Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp Val Asp 835 840 845		
His Ile Val Pro Gln Ser Phe Ile Lys Asp Asp Ser Ile Asp Asn Lys 850 855 860		
Val Leu Thr Arg Ser Val Glu Asn Arg Gly Lys Ser Asp Asn Val Pro 865 870 875 880		
Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr Trp Arg Gln Leu Leu 885 890 895		
Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe Asp Asn Leu Thr Lys Ala 900 905 910		
Glu Arg Gly Gly Leu Ser Glu Ala Asp Lys Ala Gly Phe Ile Lys Arg 915 920 925		
Gln Leu Val Glu Thr Arg Gln Ile Thr Lys His Val Ala Arg Ile Leu 930 935 940		
Asp Ser Arg Met Asn Thr Lys Arg Asp Lys Asn Asp Lys Pro Ile Arg 945 950 955 960		
Glu Val Lys Val Ile Thr Leu Lys Ser Lys Leu Val Ser Asp Phe Arg 965 970 975		
Lys Asp Phe Gln Leu Tyr Lys Val Arg Asp Ile Asn Asn Tyr His His 980 985 990		
Ala His Asp Ala Tyr Leu Asn Ala Val Val Gly Thr Ala Leu Ile Lys 995 1000 1005		
Lys Tyr Pro Lys Leu Glu Ser Glu Phe Val Tyr Gly Asp Tyr Lys 1010 1015 1020		
Val Tyr Asp Val Arg Lys Met Ile Ala Lys Ser Glu Gln Glu Ile 1025 1030 1035		
Gly Lys Ala Thr Ala Lys Arg Phe Phe Tyr Ser Asn Ile Met Asn 1040 1045 1050		
Phe Phe Lys Thr Glu Val Lys Leu Ala Asn Gly Glu Ile Arg Lys 1055 1060 1065		
Arg Pro Leu Ile Glu Thr Asn Gly Glu Thr Gly Glu Val Val Trp 1070 1075 1080		
Asn Lys Glu Lys Asp Phe Ala Thr Val Arg Lys Val Leu Ala Met 1085 1090 1095		
Pro Gln Val Asn Ile Val Lys Lys Thr Glu Val Gln Thr Gly Gly 1100 1105 1110		

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Phe Ser  Lys Glu Ser Ile Leu  Ser Lys Arg Glu Ser  Ala Lys Leu
1115                      1120                      1125

Ile Pro  Arg Lys Lys Gly Trp  Asp Thr Arg Lys Tyr  Gly Gly Phe
1130                      1135                      1140

Gly Ser  Pro Thr Val Ala Tyr  Ser Ile Leu Val Val  Ala Lys Val
1145                      1150                      1155

Glu Lys  Gly Lys Ala Lys Lys  Leu Lys Ser Val Lys  Val Leu Val
1160                      1165                      1170

Gly Ile  Thr Ile Met Glu Lys  Gly Ser Tyr Glu Lys  Asp Pro Ile
1175                      1180                      1185

Gly Phe  Leu Glu Ala Lys Gly  Tyr Lys Asp Ile Lys  Lys Glu Leu
1190                      1195                      1200

Ile Phe  Lys Leu Pro Lys Tyr  Ser Leu Phe Glu Leu  Glu Asn Gly
1205                      1210                      1215

Arg Arg  Arg Met Leu Ala Ser  Ala Thr Glu Leu Gln  Lys Ala Asn
1220                      1225                      1230

Glu Leu  Val Leu Pro Gln His  Leu Val Arg Leu Leu  Tyr Tyr Thr
1235                      1240                      1245

Gln Asn  Ile Ser Ala Thr Thr  Gly Ser Asn Asn Leu  Gly Tyr Ile
1250                      1255                      1260

Glu Gln  His Arg Glu Glu Phe  Lys Glu Ile Phe Glu  Lys Ile Ile
1265                      1270                      1275

Asp Phe  Ser Glu Lys Tyr Ile  Leu Lys Asn Lys Val  Asn Ser Asn
1280                      1285                      1290

Leu Lys  Ser Ser Phe Asp Glu  Gln Phe Ala Val Ser  Asp Ser Ile
1295                      1300                      1305

Leu Leu  Ser Asn Ser Phe Val  Ser Leu Leu Lys Tyr  Thr Ser Phe
1310                      1315                      1320

Gly Ala  Ser Gly Gly Phe Thr  Phe Leu Asp Leu Asp  Val Lys Gln
1325                      1330                      1335

Gly Arg  Leu Arg Tyr Gln Thr  Val Thr Glu Val Leu  Asp Ala Thr
1340                      1345                      1350

Leu Ile  Tyr Gln Ser Ile Thr  Gly Leu Tyr Glu Thr  Arg Thr Asp
1355                      1360                      1365

Leu Ser  Gln Leu Gly Gly Asp
1370                      1375

```

<210> SEQ ID NO 22

<211> LENGTH: 364

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 22

```

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
1          5              10              15

Leu Ala Lys Arg Ala Trp Asp Glu Arg Glu Val Pro Val Gly Ala Val
20          25              30

Leu Val His Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Pro Ile
35          40              45

Gly Arg His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
50          55              60

Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr
65          70              75              80

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Val	Thr	Leu	Glu	Pro	Cys	Val	Met	Cys	Ala	Gly	Ala	Met	Ile	His	Ser
				85					90					95	
Arg	Ile	Gly	Arg	Val	Val	Phe	Gly	Ala	Arg	Asp	Ala	Lys	Thr	Gly	Ala
		100						105					110		
Ala	Gly	Ser	Leu	Met	Asp	Val	Leu	His	His	Pro	Gly	Met	Asn	His	Arg
		115					120					125			
Val	Glu	Ile	Thr	Glu	Gly	Ile	Leu	Ala	Asp	Glu	Cys	Ala	Ala	Leu	Leu
	130					135					140				
Ser	Asp	Phe	Phe	Arg	Met	Arg	Arg	Gln	Glu	Ile	Lys	Ala	Gln	Lys	Lys
145					150					155				160	
Ala	Gln	Ser	Ser	Thr	Asp	Ser	Gly	Gly	Ser	Ser	Gly	Gly	Ser	Ser	Gly
				165				170						175	
Ser	Glu	Thr	Pro	Gly	Thr	Ser	Glu	Ser	Ala	Thr	Pro	Glu	Ser	Ser	Gly
			180					185					190		
Gly	Ser	Ser	Gly	Gly	Ser	Ser	Glu	Val	Glu	Phe	Ser	His	Glu	Tyr	Trp
		195					200					205			
Met	Arg	His	Ala	Leu	Thr	Leu	Ala	Lys	Arg	Ala	Arg	Asp	Glu	Arg	Glu
	210					215					220				
Val	Pro	Val	Gly	Ala	Val	Leu	Val	Leu	Asn	Asn	Arg	Val	Ile	Gly	Glu
225					230					235					240
Gly	Trp	Asn	Arg	Ala	Ile	Gly	Leu	His	Asp	Pro	Thr	Ala	His	Ala	Glu
				245					250					255	
Ile	Met	Ala	Leu	Arg	Gln	Gly	Gly	Leu	Val	Met	Gln	Asn	Tyr	Arg	Leu
		260						265					270		
Ile	Asp	Ala	Thr	Leu	Tyr	Val	Thr	Phe	Glu	Pro	Cys	Val	Met	Cys	Ala
		275					280					285			
Gly	Ala	Met	Ile	His	Ser	Arg	Ile	Gly	Arg	Val	Val	Phe	Gly	Val	Arg
		290				295					300				
Asn	Ala	Lys	Thr	Gly	Ala	Ala	Gly	Ser	Leu	Met	Asp	Val	Leu	His	Tyr
305					310					315					320
Pro	Gly	Met	Asn	His	Arg	Val	Glu	Ile	Thr	Glu	Gly	Ile	Leu	Ala	Asp
				325					330					335	
Glu	Cys	Ala	Ala	Leu	Leu	Cys	Tyr	Phe	Phe	Arg	Met	Pro	Arg	Gln	Val
			340					345					350		
Phe	Asn	Ala	Gln	Lys	Lys	Ala	Gln	Ser	Ser	Thr	Asp				
		355					360								

<210> SEQ ID NO 23

<211> LENGTH: 166

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 23

Ser	Glu	Val	Glu	Phe	Ser	His	Glu	Tyr	Trp	Met	Arg	His	Ala	Leu	Thr
1				5					10					15	
Leu	Ala	Lys	Arg	Ala	Arg	Asp	Glu	Arg	Glu	Val	Pro	Val	Gly	Ala	Val
		20						25					30		
Leu	Val	Leu	Asn	Asn	Arg	Val	Ile	Gly	Glu	Gly	Trp	Asn	Arg	Ala	Ile
		35				40						45			
Gly	Leu	His	Asp	Pro	Thr	Gly	His	Ala	Glu	Ile	Met	Ala	Leu	Arg	Gln
		50				55					60				
Gly	Gly	Leu	Val	Met	Gln	Asn	Tyr	Arg	Leu	Ile	Asp	Ala	Thr	Leu	Tyr
65					70					75					80

Gly	Thr	Phe	Glu	Pro	Cys	Val	Met	Cys	Ala	Gly	Ala	Met	Ile	His	Ser	
				85					90					95		
Arg	Ile	Gly	Arg	Val	Val	Phe	Gly	Val	Arg	Asn	Ala	Lys	Thr	Gly	Ala	
				100					105					110		
Ala	Gly	Ser	Leu	Met	Asp	Val	Leu	His	Tyr	Pro	Gly	Met	Asn	His	Arg	
				115					120					125		
Val	Glu	Ile	Thr	Glu	Gly	Ile	Leu	Ala	Asp	Glu	Cys	Ala	Ala	Leu	Leu	
				130					135					140		
Cys	Tyr	Phe	Phe	Arg	Met	Pro	Arg	Gln	Val	Phe	Asn	Ala	Gln	Lys	Lys	
				145					150					155		
Ala	Gln	Ser	Ser	Thr	Asp											
				165												

```
<210> SEQ ID NO 24
<211> LENGTH: 88
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
```

<400> SEQUENCE: 24

[illegible]

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<210> SEQ ID NO 25
<211> LENGTH: 38
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 25

Met Met Leu Lys Lys Ile Leu Lys Ile Glu Glu Leu Asp Glu Arg Glu
1 5 10 15
Leu Ile Asp Ile Glu Val Ser Gly Asn His Leu Phe Tyr Ala Asn Asp
20 25 30
Ile Leu Thr His Asn Ser
35

```
<210> SEQ ID NO 26
<211> LENGTH: 102
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
```

<400> SEQUENCE: 26

Cys Leu Ser Tyr Asp Thr Glu Ile Leu Thr Val Glu Tyr Gly Phe Leu
1 5 10 15

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Pro Ile Gly Lys Ile Val Glu Glu Arg Ile Glu Cys Thr Val Tyr Thr
 20 25 30

Val Asp Lys Asn Gly Phe Val Tyr Thr Gln Pro Ile Ala Gln Trp His
 35 40 45

Asn Arg Gly Glu Gln Glu Val Phe Glu Tyr Cys Leu Glu Asp Gly Ser
 50 55 60

Ile Ile Arg Ala Thr Lys Asp His Lys Phe Met Thr Thr Asp Gly Gln
 65 70 75 80

Met Leu Pro Ile Asp Glu Ile Phe Glu Arg Gly Leu Asp Leu Lys Gln
 85 90 95

Val Asp Gly Leu Pro Asn
 100

<210> SEQ ID NO 27
 <211> LENGTH: 36
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 27

Met Val Lys Ile Ile Ser Arg Lys Ser Leu Gly Thr Gln Asn Val Tyr
 1 5 10 15

Asp Ile Gly Val Glu Lys Asp His Asn Phe Leu Leu Lys Asn Gly Leu
 20 25 30

Val Ala Ser Asn
 35

<210> SEQ ID NO 28
 <211> LENGTH: 353
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 28

tcgagcgcgt gatgagagca gccactacgg gtctaggctg cccatgtaag gagggcaaggc	60
ctgggggacac ccgagatgcc tggttataat taaccacagac atgtgggtgc ccccccccc	120
ccaacacctg ctgcctgcta aaaataaccc tgtccctggg gccctgcac gccactcac	180
ggggatttcc aagtctccac cccattgacg tcaatgggag tttgttttg caccaaaatc	240
aacgggactt tccaaaatgt cgtaacaact ccgccccatt gacgcaaatg ggcggtaggc	300
gtgtacggtg ggaggtctat ataagcagag ctggtttagt gaaccgtcag atc	353

<210> SEQ ID NO 29
 <211> LENGTH: 483
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 29

cccttcagat taaaaataac tgaggtaagg gcctgggtag gggaggtggg gtgagacgct	60
cctgtctctc ctctatctgc ccatcgggcc tttggggagg aggaatgtgc ccaaggacta	120
aaaaaaggcc atggagccag aggggagagg gcaacagacc tttcatgggc aaaccttggg	180
gccttgctga ctgtagatga gacgagccac tacgggtcta ggctgcccac gtaaggaggc	240
aaggcctggg gacacccgag atgcctgggt ataattaacc cagacatgtg gctgcccccc	300

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ccccccaaac acctgctgcc tgctaaaaat aacctgtgcc ctggtggccc tgcattgcct 360
ccctggggac agccctctct ggctagtcac accctgtagg ctctctata taaccaggg 420
gcacaggggc tgcctcatt ctaccaccac ctccacagca cagacagaca ctgaggagcc 480
agc 483

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<210> SEQ ID NO 30
<211> LENGTH: 4104
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 30

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atggacaaga agtacagcat cggcctggcc atcggcacca actctgtggg ctgggccgtg 60
atcacccgac agtacaaggt gccagcaag aattcaagg tgctgggcaa caccgaccgg 120
cacagcatca agaagaacct gatcggagcc ctgctgttcg acagcggcga aacagccgag 180
gccaccggc tgaagagaac cgccagaaga agatacacca gacggaagaa ccgcatctgc 240
tatctgcaag agatcttcag caacgagatg gccagggtgg acgacagctt ctccacaga 300
ctggaagagt ccttctcgtt ggaagaggat aagaagcacg agcggcacc catcttcggc 360
aacatcgtgg acgaggtggc ctaccacgag aagtacccca ccatctacca cctgagaaag 420
aaactgggtg acagcaccga caaggccgac ctgcggtga tctatctggc cctggcccac 480
atgatcaagt tccggggcca ctctctgac gagggcgacc tgaaccocga caacagcgac 540
gtggacaagc tgttcacca gctggtgcag acctacaacc agctgttcga ggaaaacccc 600
atcaacgcca cggcgtgga cgccaaggcc atcctgtctg ccagactgag caagagcaga 660
cggctggaat atctgatcgc ccagctgccc ggcgagaaga agaattggct gttcggaaac 720
ctgattgccc tgagcctggg cctgaccccc aacttcaaga gcaacttcga cctggccgag 780
gatgccaac tgcagctgag caaggacacc tacgacgacg acctggacaa cctgctggcc 840
cagatcggcg accagtacgc cgacctgtt ctggccgcca agaacctgtc cgacgccatc 900
ctgctgagcg acatcctgag agtgaacacc gagatcacca agggccccc gagcgctct 960
atgatcaaga gatacgacga gcaccaccag gacctgacct tgctgaaagc tctcgtgcgg 1020
cagcagctgc ctgagaagta caaagagatt ttcttcgacc agagcaagaa cggctacgcc 1080
ggctacattg acggcggagc cagccaggaa gagttctaca agttcatca gccatcctg 1140
gaaaagatgg acggcaccga ggaactgtc gtgaagctga acagagagga cctgctgcgg 1200
aagcagcgga ccttcgacaa cggcagcacc cccaccaga tccacctggg agagctgcac 1260
gccattctgc ggcggcagga agatttttac ccattcctga aggacaacc ggaaaagatc 1320
gagaagatcc tgaccttcg catcccctac tacgtgggcc ctctggccag gggaaacagc 1380
agattcgctt ggatgaccag aaagagcgag gaaaccatca cccctgggaa cttcgaggaa 1440
gtggtggaca agggcgcttc cgcccagagc ttcctcgagc ggatgaccaa cttcgataag 1500
aacctgcccc acgagaaggt gctgccccaa cacagcctgc tgtacgagta ctccaccgtg 1560
tataacgagc tgaccaaagt gaaatacgtg accgagggaa tgagaaagcc cgccttctg 1620
agcggcgagc agaaaaaggc catcgtggac ctgctgttca agaccaacc gaaagtgacc 1680
gtgaagcagc tgaaagagga ctacttcaag aaaatcgagt gcttcgactc cgtggaaatc 1740
tccggcgtgg aagatcggtt caacgcctcc ctgggcacat accacgatct gctgaaaatt 1800
atcaaggaca aggacttctt ggacaatgag gaaaacgagg acattctgga agatctgtg 1860

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ctgaccctga cactgtttga ggacagagag atgatcgagg aacggctgaa aacctatgcc	1920
cacctgttcg acgacaaaagt gatgaagcag ctgaagcggc ggagatacac cggtcggggc	1980
aggctgagcc ggaagctgat caacggcatc cgggacaagc agtccggcaa gacaatcctg	2040
gatttctcga agtccgacgg cttcgccaac agaaacttca tgcagctgat ccacgacgac	2100
agcctgacct ttaaagagga catccagaaa gcccaggtgt ccggccaggc cgatagcctg	2160
cacgagcaca ttgccaatct ggcggcgagc cccgccatta agaagggcat cctgcagaca	2220
gtgaaggtgg tggacgagct cgtgaaagt atgggccggc acaagccga gaacatcgtg	2280
atcgaaatgg ccagagagaa ccagaccacc cagaaggagc agaagaacag ccgcgagaga	2340
atgaagcggg tcgaagaggg catcaaagag ctgggcagcc agatcctgaa agaacacccc	2400
gtggaaaaca ccagctgca gaacgagaag ctgtacctgt actacctgca gaatggcgcg	2460
gatatgtacg tggaccagga actggacatc aaccggctgt ccgactacga tgtggaccat	2520
atcgtgcctc agagctttct gaaggacgac tccatcgaca acaaggtgct gaccagaagc	2580
gacaagaacc ggggcaagag cgacaacgtg cctccgaag aggtcgtgaa gaagtgaag	2640
aactactggc ggcagctgct gaacgccaag ctgattaccc agagaaagt cgacaatctg	2700
accaaggccg agagaggcgg cctgagcgaa ctggataagg ccggcttcat caagagacag	2760
ctggtggaag cccggcagat cacaagcac gtggcacaga tcctggactc ccggtgaac	2820
actaagtacg acgagaatga caagctgac cggaagtga aagtgatcac cctgaagtcc	2880
aagctggtgt ccgatttcg gaaggatttc cagttttaca aagtgcgca gatcaacaac	2940
taccaccacg cccacgacgc ctacctgaac gccgtcgtgg gaaccgcct gatcaaaaag	3000
taccctaagc tggaaagcga gttcgtgtac ggcgactaca aggtgtacga cgtgcggaag	3060
atgatcgcca agagcgagca ggaatcggc aaggctaccg ccaagtactt cttctacagc	3120
aacatcatga actttttcaa gaccgagatt accctggcca acggcgagat ccggaagcgg	3180
cctctgatcg agacaaacgg cgaaccggg gagatcgtgt gggataaggc ccgggatttt	3240
gccaccgtgc ggaagtgtct gagcatgccc caagtgaata tcgtgaaaaa gaccgaggtg	3300
cagacaggcg gcttcagcaa agagtctatc ctgccaaga ggaacagcga taagctgac	3360
gccagaaaga aggactggga ccctaagaag tacggcggct tcgacagccc caccgtggcc	3420
tattctgtgc tgggtgtggc caaagtggaa aagggcaagt ccaagaaact gaagagtgtg	3480
aaagagctgc tggggatcac catcatggaa agaagcagct tcgagaagaa tcccatcgac	3540
tttctggaag ccaagggcta caaagaagtg aaaaaggacc tgatcatcaa gctgcctaag	3600
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cagaagggaa acgaactggc cctgcctccc aaatatgtga acttctgta cctggccagc	3720
cactatgaga agctgaaggc ctccccgag gataatgagc agaaacagct gtttgtggaa	3780
cagcacaagc actacctgga cgagatcatc gagcagatca gcgagttctc caagagagtg	3840
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cccatcagag agcaggccga gaatatcatc cacctgttta ccctgaccaa tctgggagcc	3960
cctgccgcct tcaagtactt tgacaccacc atcgaccgga agaggtagac cagcaccaaa	4020
gaggtgctgg acgccacct gatccaccag agcatcaccg gcctgtacga gacacggatc	4080
gacctgtctc agctgggagg tgac	4104

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<211> LENGTH: 4104
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 31
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atcaccgacg agtacaaggt gccagcaag aaattcaagg tgctgggcaa caccgaccgg      120
cacagcatca agaagaacct gatcggagcc ctgctgttcg acagcggcga aacagccgag      180
gccacccggc tgaagagaac cgccagaaga agatacacca gacggaagaa ccgcatctgc      240
tatctgcaag agatcttcag caacgagatg gccaaagggtg acgacagctt ctccacaga      300
ctggaagagt ccttctcgtg ggaagaggat aagaagcacg agcggcaccc catcttcggc      360
aacatcgtgg acgagggtgg ctaccacgag aagtaccccc ccatctacca cctgagaaag      420
aaactggtgg acagcacoga caaggccgac ctgcgggtga tctatctggc cctggcccac      480
atgatcaagt tccggggcca ctctctgatc gagggcgacc tgaacccga caacagcgac      540
gtggacaagc tgttcacca gctggtgcag acctacaacc agctgttcga ggaaaacccc      600
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gatgccaac tgcagctgag caaggacacc tacgacgacg acctggacaa cctgctggcc      840
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cagcagctgc ctgagaagta caaagagatt ttcttcgacc agagcaagaa cggtacgcc      1080
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aagcagcgga ccttcgacaa cggcagcatc cccaccaga tccacctggg agagctgcac      1260
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gagaagatcc tgaccttcg catcccttac tacgtgggccc ctctggccag gggaaacagc      1380
agattcgctt ggatgaccag aaagacgag gaaaccatca cccctggaa cttcgaggaa      1440
gtggtggaca agggcgcttc cgcccagagc ttcacgagc ggatgaccaa cttcgataag      1500
aacctgcccc acgagaaggt gctgccccag cacagcctgc tgtacgagta cttcacctg      1560
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gatttcctga agtccgacgg cttcgccaac agaaacttca tgcagctgat ccacgacgac      2100
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<210> SEQ ID NO 32

<211> LENGTH: 4048

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 32

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<210> SEQ ID NO 33

<211> LENGTH: 4104

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 33

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ctggaagagt ccttctgggt ggaagaggat aagaagcacg agcggcacc catcttcggc 360
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<210> SEQ ID NO 34

<211> LENGTH: 4104

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 34

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<210> SEQ ID NO 35
<211> LENGTH: 4104
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<210> SEQ ID NO 36

<211> LENGTH: 4104

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 36

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<210> SEQ ID NO 37

<211> LENGTH: 4104

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 37

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<210> SEQ ID NO 38
<211> LENGTH: 4134
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<210> SEQ ID NO 39

<211> LENGTH: 4134

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 39

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<210> SEQ ID NO 40

<211> LENGTH: 4125

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 40

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gagagccaga tcttgaagga gaacccctg gagaaccccc agctgcagaa cgagaagctg	2460
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cgcttgagcg actacgacgt ggaccacatc gtgccccaga gcttcatcaa ggacgacagc	2580
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agcgaggagg tggatgaagaa gatgaagaac tactggcgcc agctgtgaa cgccaagctg	2700
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cagttcgccg tgagcgacag catcctgctg agcaacagct tcgtgagcct gctgaagtac	3960
accagcttcg gcgcccagcg cggttcacc ttcctggacc tggacgtgaa gcagggccgc	4020
ctgcgctacc agaccgtgac cgagggtgctg gacgccaccc tgatctacca gagcatcacc	4080
ggcctgtacg agaccgcac cgacctgagc cagctgggcg gcgac	4125

<210> SEQ ID NO 41
 <211> LENGTH: 1092
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 41

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gcattgggatg aaagagaagt ccccggtggc gccgtgctgg tgcacaacaa tagagtgatc	120
ggagagggat ggaacaggcc aatcgccgc caccaccta ccgcacacgc agagatcatg	180
gcactgaggc agggaggcct ggtcatgcag aattaccgcc tgatcgatgc caccctgtat	240
gtgacactgg agccatcgct gatgtgcgca ggagcaatga tccacagcag gatcggaaga	300
gtggtgttcg gagcacggga cgccaagacc ggcgcagcag gctccctgat ggatgtgctg	360
caccaccccg gcatgaacca ccgggtggag atcacagagg gaatcctggc agacgagtgc	420
gccgcctcgc tgagcgattt ctttagaatg cggagacagg agatcaaggc ccagaagaag	480

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gcacagagct ccacogactc tggaggatct agcggaggat cctctggaag cgagacacca 540
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gtggagtttt cccacgagta ctggatgaga catgccctga ccctggccaa gagggcacgc 660
gatgagaggg aggtgcctgt gggagccgtg ctggtgctga acaatagagt gatcggcgag 720
ggctggaaca gagccatcgg cctgcacgac ccaacagccc atgccgaaat tatggcctg 780
agacagggcg gcctggctcat gcagaactac agactgattg acgccaccct gtacgtgaca 840
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tttggcgtga ggaacgcaaa aaccggcgcc gcaggctccc tgatggacgt gctgcactac 960
cccgcatga atcaccgcgt cgaattacc gaggaatcc tggcagatga atgtgccgcc 1020
ctgtgtgtct atttttttcg gatgcctaga caggtgttca atgctcagaa gaaggcccag 1080
agctccaccg ac 1092

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<210> SEQ ID NO 42
<211> LENGTH: 498
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 42
tctgaggtgg agttttccca cgagtactgg atgagacatg ccctgaccct ggccaagagg 60
gcacgcgatg agagggaggt gcctgtggga gccgtgctgg tgctgaacaa tagagtgatc 120
ggcgagggct ggaacagagc catcggcctg cagcacccaa caggccatgc cgaattatg 180
gccctgaggc agggcggcct ggtcatgcag aactacagac tgattgacgc caccctgtac 240
gggacattcg agccttgcgt gatgtgcgcc ggcgccatga tccactctag gatcggccgc 300
gtggtgtttg gcgtgaggaa cgcaaaaacc ggcgcgcgag gctccctgat ggacgtgctg 360
cactaccccg gcataatca ccgcgtcgaa attaccgagg gaatcctggc agatgaatgt 420
gccgccctgc tgtctatatt ctttcggatg cctagacagg tgttcaatgc tcagaagaag 480
gcccagagct ccacgcag 498

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<210> SEQ ID NO 43
<211> LENGTH: 267
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 43
tgtttgatc tgaacacgca agttcaaacg ccacagggtg tgaagaaat atccaatata 60
caggtcggcg atctcgtctt gtctaact gcctataacg aggtgctgaa tgtatttcca 120
aaaagcaaga aaaaagttta caagataact ctggaagatg gaaaagaaat tatctgttct 180
gaggagcatc tgtttccgac ccaaacaggg gagatgaata tcagtggcgg tctcaaagag 240
ggatatgtgt tgatatgtcaa ggaataa 267

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<210> SEQ ID NO 44
<211> LENGTH: 114
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 44

atgatgctca agaagatcct caagattgaa gagttggacg agcgcgagct tatagacata 60
gaagtcagtg gtaatcacct tttctacgca aatgacattt tgactcacia ctcc 114

<210> SEQ ID NO 45

<211> LENGTH: 306

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 45

tgtctcagtt atgacaccga aatcctgaca gtcgagtatg gatttctgcc gatcggaag 60
attgtggagg agagaattga atgtacggtc tatacggtcg acaagaatgg ttcgtctac 120
acccaaccaa ttgctcaatg gcataatcga ggggagcagg aggtgtttga gtattgcctg 180
gaggacgggt caatcattag agctacaaaag gaccataagt ttatgacaac cgatgggtcaa 240
atgctgccga tagatgaaat attcgaaagg ggactggatc ttaagcaagt cgatggcctt 300
ccaaac 306

<210> SEQ ID NO 46

<211> LENGTH: 108

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 46

atggtaaga ttatcagccg caaatccttg gggacacaga atgtatatga catcggcgtg 60
gaaaaggatc acaattttct gctgaagaat ggtcttggtg cttccaat 108

<210> SEQ ID NO 47

<211> LENGTH: 166

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 47

Ser Glu Val Glu Phe Ser His Glu Tyr Trp Met Arg His Ala Leu Thr
1 5 10 15
Leu Ala Lys Arg Ala Arg Asp Glu Arg Glu Val Pro Val Gly Ala Val
20 25 30
Leu Val Leu Asn Asn Arg Val Ile Gly Glu Gly Trp Asn Arg Ala Ile
35 40 45
Gly Leu His Asp Pro Thr Ala His Ala Glu Ile Met Ala Leu Arg Gln
50 55 60
Gly Gly Leu Val Met Gln Asn Tyr Arg Leu Ile Asp Ala Thr Leu Tyr
65 70 75 80
Val Thr Phe Glu Pro Cys Val Met Cys Ala Gly Ala Met Ile His Ser
85 90 95
Arg Ile Gly Arg Val Val Phe Gly Val Arg Asn Ala Lys Thr Gly Ala
100 105 110
Ala Gly Ser Leu Met Asp Val Leu His Tyr Pro Gly Met Asn His Arg
115 120 125
Val Glu Ile Thr Glu Gly Ile Leu Ala Asp Glu Cys Ala Ala Leu Leu
130 135 140

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Cys Tyr Phe Phe Arg Met Pro Arg Gln Val Phe Asn Ala Gln Lys Lys
145 150 155 160

Ala Gln Ser Ser Thr Asp
165

<210> SEQ ID NO 48

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 48

tgagtatcat cgtgtgaaag ctgaggggac gaggcaggcc tataa 45

<210> SEQ ID NO 49

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 49

aaaaacatca acttcagcca tccatttctt cagggtttgt atgtg 45

<210> SEQ ID NO 50

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 50

tatatcataa tgaaaacgcc gccatttctc aacagatctg tcaaa 45

<210> SEQ ID NO 51

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 51

ctgtgtgaaa tggctgcaaa tcgatgggtg agctctgaga tttgg 45

<210> SEQ ID NO 52

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 52

gttctgcttt tgctactact cacgtttcca tgttgtcccc ctcta 45

<210> SEQ ID NO 53

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 53

attttatggc cttttgcaac tcgaccagaa aaaaagcagc tttgg 45

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<210> SEQ ID NO 54
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 54

tgaggagatc gccacgggc tgccaggatc ccttgatcac ctacg 45

<210> SEQ ID NO 55
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 55

ggctgctctg tcagaaatat tcgtacagtc tcaagagtac tcatg 45

<210> SEQ ID NO 56
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 56

gtgtaggcat agctcttgaa tcgaggctta ggggaagaag ttctc 45

<210> SEQ ID NO 57
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 57

cctgttcttc agtaagacgt tgccatttga gaaggatgct ttgta 45

<210> SEQ ID NO 58
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 58

gccatttttag gctttttact tacttgtctg tagctcttct tctct 45

<210> SEQ ID NO 59
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 59

ctagtttctc acacatgaca cacctgttct tcagtaagac gttgc 45

<210> SEQ ID NO 60
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 60

gtgaagttga ttacattaac ctgtggataa ttacgagttg attgt 45

<210> SEQ ID NO 61

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 61

ttttgtatat ctgagttaaa ctgctccaat tccttcaaag gaatg 45

<210> SEQ ID NO 62

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 62

tctgcaatat aagctgcaa ctgcttgta atgaatgtga gggac 45

<210> SEQ ID NO 63

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 63

ggactggggt tccagtctca tccagtctag gaagagggcc gcttc 45

<210> SEQ ID NO 64

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 64

agtcgttggtg tggctgactg ctggcaaacc acactattcc agtca 45

<210> SEQ ID NO 65

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 65

atttgtgtct ttctgagaaa ctgttcagct tctgttagcc actga 45

<210> SEQ ID NO 66

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 66

aagaaccag cggtctcttg tccatctaca gatgtttgcc catcg 45

<210> SEQ ID NO 67

<211> LENGTH: 45

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 67

agacttttttc cgaagttcac tccacttgaa gttcatgtta tccaa 45

<210> SEQ ID NO 68
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 68

tacctgttgg cacatgtgat cccactgagt gttaagttct ttgag 45

<210> SEQ ID NO 69
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 69

caaaagggcct tctgcagtct tcggagtttc atggcagtc tataa 45

<210> SEQ ID NO 70
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 70

atggttaatg tctaaccttt atccactgga gatttgtctg cttga 45

<210> SEQ ID NO 71
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 71

caaaataatc tgaccttaag ttgttcttcc aaagcagcag ttgcg 45

<210> SEQ ID NO 72
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 72

catctacaga tgtttgccca tcgatctccc aatacctgga gaaga 45

<210> SEQ ID NO 73
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 73

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aaagtggtct tgtacttcat cccactgatt ctgaattctt tcaac 45

<210> SEQ ID NO 74
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 74

gaagtggttc ttgtacttca tcccactgat tctgaattct ttaa 45

<210> SEQ ID NO 75
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 75

gctatgcttt gaatttttaa tcgttcaatt tgaggttgaa gatct 45

<210> SEQ ID NO 76
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 76

tctaggaggc gcctcccatc ctgtaggtca ctgaagaggt tctca 45

<210> SEQ ID NO 77
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 77

gtgtaattac cattcaccat ctgttccacc agggcctgag ctgat 45

<210> SEQ ID NO 78
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 78

tgagcatgct ttaccaggat ctgttccctt gtggtcaccg tagtt 45

<210> SEQ ID NO 79
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 79

ctactgtata gggaccctcc ttccatgact caagcttggc tctgg 45

<210> SEQ ID NO 80
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 80

cagcttcttc cttagcttcc agccattgtg ttgaatcctt taaca 45

<210> SEQ ID NO 81
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 81

agctgccc aa ggtcttttat ttgagcttca atttctcctt gtttc 45

<210> SEQ ID NO 82
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 82

tgccagtaac aactcacaat ttgtgcaaag ttgagtcctc gaaac 45

<210> SEQ ID NO 83
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 83

agagtaacag tctgagtagg agctaaaata ttttgggttt ttgca 45

<210> SEQ ID NO 84
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 84

cttcagcaaa aaaagtactc acgcagaatc tactggccag aagtt 45

<210> SEQ ID NO 85
<211> LENGTH: 45
<212> TYPE: DNA
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<400> SEQUENCE: 85

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<210> SEQ ID NO 86
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 86

ctgagatagt ataggccact ttgttgctct tgcagagaac tttgt 45

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<210> SEQ ID NO 87
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 87

tagctgtcct ttacacactt tacctgttga gaatagtgc tttga 45

<210> SEQ ID NO 88
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 88

gcctgggctt cctgaggeat ttgagctgcg tccaccttgt ctgca 45

<210> SEQ ID NO 89
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 89

tgcactggca ggtagcccat tcggggatgc ttgcacaaat acctt 45

<210> SEQ ID NO 90
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 90

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<210> SEQ ID NO 91
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 91

cgtcaggctg gcgtcaaaact taccgagtg caatattcca ccatg 45

<210> SEQ ID NO 92
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 92

tcacccaaaa gtgtgtcagc ctgaatgac cactttgtga tgtgg 45

<210> SEQ ID NO 93
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 93

tgtagccaca ccagaagttc ctgcagagaa aggtgcagac gcttc 45

<210> SEQ ID NO 94

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 94

tgtaaggatt tttcagtctc ctgggcagac tggatgctct gttca 45

<210> SEQ ID NO 95

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 95

agaatgggat ccagtatact tacaggctcc aatagtggtc agtcc 45

<210> SEQ ID NO 96

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 96

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<210> SEQ ID NO 97

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 97

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<210> SEQ ID NO 98

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<213> ORGANISM: Artificial Sequence

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 99

agatttaacc actcttctgc tcgggaggtg acagctatcc agtta 45

<210> SEQ ID NO 100

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<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 100

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<210> SEQ ID NO 101
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 101

ccagctggga ggagagcttc ttccagcgtc cctcaatttc ttcaa 45

<210> SEQ ID NO 102
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 102

acacagcttc tgagcgagta atccagctgt gaagttcagt tatat 45

<210> SEQ ID NO 103
<211> LENGTH: 45
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 103

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<210> SEQ ID NO 104
<211> LENGTH: 45
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 104

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<210> SEQ ID NO 105
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 105

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<210> SEQ ID NO 106
<211> LENGTH: 45
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 106

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tctaaaaatca tcttactttc ttgtagacgc tgctcaaaat tggct 45

<210> SEQ ID NO 107
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 107

gaattgaccc tgacttggtc ttgttctaga tcttcttgaa gcacc 45

<210> SEQ ID NO 108
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 108

tggaatggcctt caatgctcac ttgttgaggc aaaacttgga agagt 45

<210> SEQ ID NO 109
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 109

aacagtcttc tactttcttc caccaaagca ttttgaaaag tgtat 45

<210> SEQ ID NO 110
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 110

aggcctcctt tctggcatag accttcacaa aaacaaacaa acaaa 45

<210> SEQ ID NO 111
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 111

tttggtttct gactgctgga cccatgtcct gatggcactc atggt 45

<210> SEQ ID NO 112
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 112

atcttacttt cttgtagacg ctgctcaaaa ttggctggtt tctgg 45

<210> SEQ ID NO 113
<211> LENGTH: 45
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 113

agatttttca cttatcttca tacctcttca tgtagttccc tccaa 45

<210> SEQ ID NO 114
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 114

ctgttcagtt gttctgaggc ttgtttgatg ctatctgcat taaca 45

<210> SEQ ID NO 115
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 115

cagcattaat atacacgact tacatctgta ctgtcttccc aaatg 45

<210> SEQ ID NO 116
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 116

tcatgacttg tcaaatcaga ttggattttc tggtgggagg atagc 45

<210> SEQ ID NO 117
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 117

atctgctcca attgtttag ctgattatag aaagcgatga tgttg 45

<210> SEQ ID NO 118
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 118

caaatttgct ctcaatttcc cgccagcgct tgetgagctg gatct 45

<210> SEQ ID NO 119
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 119

cattcaaagc caggccatca gaccagctgg tggatgaagt gatta 45

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<210> SEQ ID NO 120
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 120

ttcatctctt caactgcttt ctgtaattca tctggagttt tatat 45

<210> SEQ ID NO 121
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 121

attgaaagct agaaagtaca tacggccagt ttttgaagac ttgat 45

<210> SEQ ID NO 122
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 122

ttcaaatact ggccaatact tacagcaaag ggccttctgc agtct 45

<210> SEQ ID NO 123
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 123

gttgtctgtg ttagtgatgg ctgagtggtg gtgacagcct gtgaa 45

<210> SEQ ID NO 124
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 124

tcatagcct gcctcttgta ctgataccac tgatgagaaa tttct 45

<210> SEQ ID NO 125
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 125

tactgtatag ggacctcct tccatgactc aagcttggtc ctggc 45

<210> SEQ ID NO 126
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 126

atgttgaatg catgttccag tcgttgtgtg gctgactgct ggcaa 45

<210> SEQ ID NO 127

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 127

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<210> SEQ ID NO 128

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 128

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<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 129

ttagcaactg gcagaattcg atccaccggc tgttcagttg ttctg 45

<210> SEQ ID NO 130

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 130

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<210> SEQ ID NO 131

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 131

tcccattcag cctagtgcag agccactggc agttgggtgg tagag 45

<210> SEQ ID NO 132

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 132

gacttactgg aaagaaagtg ctgagatgct ggaccaaagt ccctg 45

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<210> SEQ ID NO 133
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 133

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<210> SEQ ID NO 134
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 134

aaagagattg tctatacctg ttggcacatg tgatcccact gagtg 45

<210> SEQ ID NO 135
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 135

aagtttttgg actaaattat cccaacaccg ggcaaagtta tccag 45

<210> SEQ ID NO 136
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 136

agctcagcat cccgggggact ctggggagag gtgggcatca tttca 45

<210> SEQ ID NO 137
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 137

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<210> SEQ ID NO 138
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 138

ttcgcaaaat accttttgggt tcgaaatttg ttttttagta ccttg 45

<210> SEQ ID NO 139
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 139

tgcaactcga ccagaaaaaa agcagctttg gcagatgtca taatt 45

<210> SEQ ID NO 140

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 140

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<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<400> SEQUENCE: 141

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<210> SEQ ID NO 142

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 142

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<210> SEQ ID NO 143

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 143

ttgacctcct cagcctgctt tcgtagaagc cgagtgcacat tctgg 45

<210> SEQ ID NO 144

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 144

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<210> SEQ ID NO 145

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 145

aatgcctgac ttacttgcca ttgtttcacc agctctttta ctecc 45

<210> SEQ ID NO 146

<211> LENGTH: 45

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 146

tgtacttcat cccactgatt ctgaattctt tcaactagaa taaaa 45

<210> SEQ ID NO 147
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 147

tgcttcatta ccttcaactgg ctgagtggct gggttttccct tgtac 45

<210> SEQ ID NO 148
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 148

tttaattggt tgagaattcc ctggcgcagg ggcaactctt ccacc 45

<210> SEQ ID NO 149
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 149

atatgtgtta cctacccttg tcggtccttg tacattttgt taact 45

<210> SEQ ID NO 150
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 150

tcaagctggg agagagcttc ctgtagcttc accctttcca caggc 45

<210> SEQ ID NO 151
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 151

atgtcaatcc gacctgagct ttgttgtaga ctatctttta tattu 45

<210> SEQ ID NO 152
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 152

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gttgtagact atcttttata ttctgtaata taaaaatttt aaaac 45

<210> SEQ ID NO 153
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 153

tcccgcacgc gcttgctgag ctggatctga gttggctcca ctgcc 45

<210> SEQ ID NO 154
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 154

ttatgttttg tctgtaacag ctgctgtttt atctttattt cctct 45

<210> SEQ ID NO 155
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 155

tgttttgtct gtaacagctg ctgttttata tttatttcct ctgcg 45

<210> SEQ ID NO 156
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 156

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<210> SEQ ID NO 157
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 157

accttaagca cgtcttcttt ttgctggggt ttctttttct ctgat 45

<210> SEQ ID NO 158
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 158

ttaagcacgt cttctttttg ctgggggttc ttttctctg attca 45

<210> SEQ ID NO 159
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 159

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<210> SEQ ID NO 160
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 160

agctgtgact gtactacttc ctgttccaca ctctttgttt ccaat 45

<210> SEQ ID NO 161
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 161

ttggctggtt tctggaataa tcgaaacttc atggagacat cttgt 45

<210> SEQ ID NO 162
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 162

tgcatctctg atagatcttt ctggaggctt acagttttct ccaaa 45

<210> SEQ ID NO 163
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 163

acagtgaag agattgtcta tacctgttgg cacatgtgat ccac 45

<210> SEQ ID NO 164
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 164

aaggcatcat ataaaaatct tactctgcac tgttccagct gcttt 45

<210> SEQ ID NO 165
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 165

cttactctgc actgtttcag ctgctttttt agaatttctg aatcc 45

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<210> SEQ ID NO 166
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 166

tcttgaatta cctgaatttt tcggagttaa ttcatttgc cctct 45

<210> SEQ ID NO 167
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 167

tgttgctctt gcagagaact ttgtaaagcc taaaaaaca ttttt 45

<210> SEQ ID NO 168
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 168

attgtgtgca aagtgtcaaa aactttatca aaagggaaaa aagaa 45

<210> SEQ ID NO 169
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 169

taggtctttt acttacttgt ctgtagctct ttctctctgg cctgc 45

<210> SEQ ID NO 170
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 170

gcctgcacat cagaaaagac ttgcttaaaa tgatttgtaa aggcc 45

<210> SEQ ID NO 171
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 171

atggaaggag aagagattct taccttaca atttttaact gactt 45

<210> SEQ ID NO 172
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 172

ggtggtgggt tggattttca accagttttc agcagtagtt gtcac 45

<210> SEQ ID NO 173

<211> LENGTH: 45

<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 173

tcgateccacc ggctgttcag ttgttctgag gcttgtttga tgcta 45

<210> SEQ ID NO 174

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 174

tgagctgac tgctggcatc ttgcagtttt ctgaacttct cagct 45

<210> SEQ ID NO 175

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 175

ataaaagctt aagatgctct caccttttcc taatttcaga atcca 45

<210> SEQ ID NO 176

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 176

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<211> LENGTH: 45

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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<400> SEQUENCE: 178

tccagccatg cttccgtctt ctgggtcact gacttattct tcaagt 45

<210> SEQ ID NO 179

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<211> LENGTH: 45
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 179

gaaggatgtc ttgtaaaaga acccagcggt cttctgtcca tctac 45

<210> SEQ ID NO 180
<211> LENGTH: 45
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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 180

tggtcttggt ctagatcttc ttgaagcacc tgaaagataa aatgt 45

<210> SEQ ID NO 181
<211> LENGTH: 45
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 181

cctaccttat gttgtgtgtac ttggcgtttt aggtcttcaa gatca 45

<210> SEQ ID NO 182
<211> LENGTH: 45
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 182

tctttcttct gtttttggtta gccagtcatt caactcttcc agttt 45

<210> SEQ ID NO 183
<211> LENGTH: 45
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 183

attaaaaaca aataaggact tacttgcttt gtttttccat gctag 45

<210> SEQ ID NO 184
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 184

caaataagga cttacttgct ttgtttttcc atgctagcta ccctg 45

<210> SEQ ID NO 185
<211> LENGTH: 45
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 185

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tttaggagat tcatctgctc ttgtacttca gtttcttcat cttct 45

<210> SEQ ID NO 186
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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 186

acatcattag aaatctctcc ttgtgcttgc aatgtgtcct cagca 45

<210> SEQ ID NO 187
<211> LENGTH: 45
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<400> SEQUENCE: 187

tggtagtcca gaaatttacc aaccttcagg atcgagtagt ttctc 45

<210> SEQ ID NO 188
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<220> FEATURE:
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<400> SEQUENCE: 188

tatttttcat tacatttttg acctacatgt ggaaataaat ttcca 45

<210> SEQ ID NO 189
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<400> SEQUENCE: 189

ccattcatca ggattcttac ctgccagtgg aggattatat tccaa 45

<210> SEQ ID NO 190
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 190

tttcttaaaa ataagtcaca taccagtttt tgccctgtca ggccct 45

<210> SEQ ID NO 191
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 191

aataagtcac ataccagttt ttgccctgtc aggccttcga ggagg 45

<210> SEQ ID NO 192
<211> LENGTH: 45
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 192

gtaaagtaac aaaccattct taccttagaa aattgtgcat ttacc 45

<210> SEQ ID NO 193
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 193

tttactaagc aaaataatct gaccttaagt tgttcttcca aagca 45

<210> SEQ ID NO 194
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 194

acggatcctc cctgttcgct ccctattatg aagaatcaaa gcaga 45

<210> SEQ ID NO 195
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 195

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<210> SEQ ID NO 196
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 196

ctgtcaaadc catcatgtac ccctgacaaa gaaggaagtt aacaa 45

<210> SEQ ID NO 197
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 197

tctcaatatg ctgcttccca aactgaaatt aaaaaaata cactc 45

<210> SEQ ID NO 198
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 198

cttaattcat catctttcag ctgtagccac accagaagtt cctgc 45

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<210> SEQ ID NO 199
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 199

gtcaagacat tcatttcctt tcgcatctta cgggacaatt tcaag 45

<210> SEQ ID NO 200
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 200

tgtgtcctca gcagaaagaa gccacgataa tacttcttct aaagc 45

<210> SEQ ID NO 201
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 201

tagaaagcga tgatgttggt ctgatactcc agccaggtaa gtctc 45

<210> SEQ ID NO 202
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 202

ctctctaagg aaatcaagat ctgggcagga ctacgaggct ggctc 45

<210> SEQ ID NO 203
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 203

tcaaaagttt ccatgtgttt ctggtattcc ttaattgtac agaga 45

<210> SEQ ID NO 204
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 204

actgtttcca ttacagttgt ctgtgttagt gatggctgag tgggtg 45

<210> SEQ ID NO 205
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 205

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<210> SEQ ID NO 206
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 206

attttgtgcaa agttgagtct tcgaaactga gcaaatttgc tctca 45

<210> SEQ ID NO 207
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 207

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<210> SEQ ID NO 208
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 208

gtctcctatg aactcgagaa gccgcaaac caaggaagag aaaga 45

<210> SEQ ID NO 209
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 209

gagagtttgg tttctgactg ctggacccat gtctctgatgg cactc 45

<210> SEQ ID NO 210
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 210

tgcgtatttg ccaccagaaa tacataccac acaatgattt agctg 45

<210> SEQ ID NO 211
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 211

tttgggttat cctctgaatg tcgcatcaaa ttttcaagtg actga 45

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<210> SEQ ID NO 212
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 212

aggacacgga tcttcctgt tcgtcccta ttatgaagaa tcaaa 45

<210> SEQ ID NO 213
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 213

tgcttgtaa aaaacttact tcgatccga atgattgttc tagcc 45

<210> SEQ ID NO 214
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 214

ttttgtcca catcttttcc tacctaagt tgagagactt tttcc 45

<210> SEQ ID NO 215
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 215

cctgccagtg gaggattata ttccaaatca aaccaagagt cagtt 45

<210> SEQ ID NO 216
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 216

ggataattac gagttgattg tcggaccag ctcaggagaa tcttt 45

<210> SEQ ID NO 217
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 217

tttagactgg gctgaattgt ctgaatatca ctgactaaaa gctaa 45

<210> SEQ ID NO 218
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 218

gtactactta cattattgtt ctgcaaaacc cgcagtgcc tgggtg 45

<210> SEQ ID NO 219

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 219

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<210> SEQ ID NO 220

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 220

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<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 221

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<210> SEQ ID NO 222

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 222

aaagctaagg ggacgaggca ggc 23

<210> SEQ ID NO 223

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 223

gaaagctaag gggacgaggc agg 23

<210> SEQ ID NO 224

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 224

aaatagatgg ctgaagttga tgt 23

<210> SEQ ID NO 225

<211> LENGTH: 23

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 225

gaaatagatg gctgaagttg atg 23

<210> SEQ ID NO 226
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 226

aagaaataga tggctgaagt tga 23

<210> SEQ ID NO 227
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 227

ctgaagaaat agatggctga agttgat 27

<210> SEQ ID NO 228
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 228

aaatagcggc gttttcatta tga 23

<210> SEQ ID NO 229
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 229

gagaaatagc ggcgttttca ttatgat 27

<210> SEQ ID NO 230
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 230

aaatcaatgg ttgagctctg aga 23

<210> SEQ ID NO 231
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 231

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gcaaatcaat ggttgagctc tga 23

<210> SEQ ID NO 232
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 232

tgcaaatcaa tgggtgagct ctgagat 27

<210> SEQ ID NO 233
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 233

aacatgagta gtagcaaaag cag 23

<210> SEQ ID NO 234
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 234

gaaacatgag tagtagcaaa agc 23

<210> SEQ ID NO 235
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 235

ggaaacatga gtagtagcaa aag 23

<210> SEQ ID NO 236
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 236

tggaaacatg agtagtagca aaa 23

<210> SEQ ID NO 237
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 237

aactcaacca gaaaaaaagc agc 23

<210> SEQ ID NO 238
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 238

caactcaacc agaaaaaaag cag 23

<210> SEQ ID NO 239
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 239

aagggatcct gacagcccggt gggcgat 27

<210> SEQ ID NO 240
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 240

aatattcata cagtctcaag agt 23

<210> SEQ ID NO 241
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 241

aatcaaggct taggggaaga agt 23

<210> SEQ ID NO 242
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 242

gaatcaaggc ttaggggaag aag 23

<210> SEQ ID NO 243
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 243

tgaatcaagg cttaggggaa gaa 23

<210> SEQ ID NO 244
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 244

ttgaatcaag gcttagggga aga 23

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<210> SEQ ID NO 245
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 245

cttgaatcaa ggcttagggg aagaagt 27

<210> SEQ ID NO 246
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 246

aatgacaacg tcttactgaa gaa 23

<210> SEQ ID NO 247
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 247

aaatgacaac gtcttactga aga 23

<210> SEQ ID NO 248
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 248

caaatgacaa cgtcttactg aag 23

<210> SEQ ID NO 249
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 249

tcaaatgaca acgtcttact gaa 23

<210> SEQ ID NO 250
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 250

acaaataagt aaaaagccta aaa 23

<210> SEQ ID NO 251
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 251

acagatgtgt catgtgtgag aaa 23

<210> SEQ ID NO 252

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 252

aacagatgtg tcatgtgtga gaa 23

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<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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gaacagatgt gtcatgtgtg aga 23

<210> SEQ ID NO 254

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 254

agaacagatg tgtcatgtgt gag 23

<210> SEQ ID NO 255

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 255

acctatggat aattacgagt tga 23

<210> SEQ ID NO 256

<211> LENGTH: 23

<212> TYPE: DNA

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ttaacctatg gataattacg agt 23

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aactactcca attccttcaa agg 23

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aactacttgt caatgaatgt gag 23

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caagtagagt gaacttcgga aaa 23

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agtgaataaaa ggtagacat taa 23

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agttattctt ccaaagcagc agt 23

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gcccataat ctccaatac ctg 23

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attttaactc ctactcagac tgt 23

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<223> OTHER INFORMATION: Synthetic Construct

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 335

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<212> TYPE: DNA

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 336

cccattcagg gatgcttcgc aaa 23

<210> SEQ ID NO 337

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<400> SEQUENCE: 337

cccagaagta attgaatgtg gaa 23

<210> SEQ ID NO 338
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gccagaagt aattgaatgt gga 23

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<400> SEQUENCE: 340

actccgataa gtttgacgcc agc 23

<210> SEQ ID NO 341
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<400> SEQUENCE: 341

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<400> SEQUENCE: 342

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cctacagaga aaggtgcaga cgc 23

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tcctacagag aaaggtgcag acg 23

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gttcctacag agaaaggtgc aga 23

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cctaggcaga ctggatgctc tgt 23

<210> SEQ ID NO 348
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tcctaggcag actggatgct ctg 23

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<400> SEQUENCE: 350

ctcaaagtc ctgtgtgcaa tag 23

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ctcctcaaag tgcctgtgtg caa 23

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tttctctca aagtgcctgt gtgcaat 27

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ctcactcact caccctgcaa agg 23

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cctctcactc actcaccctg caa 23

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ctcactctat ggctgcagc atg 23

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cagctgaatt actcgctcag aag 23

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acagctgaat tactcgctca gaa 23

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ctgagtagaa ggcggtaa ac cgt 23

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tctgagtaga aggcggtaaa ccg 23

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cttaccaaag gtcacaaagg taa 23

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gacttaccaa aggtcacaaa ggt 23

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<400> SEQUENCE: 381

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atcttagaga gagtcaatga gga 23

<210> SEQ ID NO 383

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gatcttagag agagtcaatg agg 23

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tgatcttaga gagagtcaat gag 23

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cttatagacg ctgctcaaaa ttg 23

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tttcttatag acgctgctca aaa 23

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ctttcttata gacgctgctc aaa 23

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tactttctta tagacgctgc tcaaaat 27

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cttattctag atcttcttga agc 23

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cttattgagg caaaacttgg aag 23

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tcacttattg aggcaaaact tgg 23

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ctcacttatt gaggcaaaac ttg 23

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acttattgag gcaaaacttg gaagagt 27

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ctttgatggg aagaagtaga gga 23

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<400> SEQUENCE: 396

gctttgatgg gaagaagtag agg 23

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ggaaagtcta tgccagaaag gag 23

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tggaaagtct atgccagaaa gga 23

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<400> SEQUENCE: 401

gtggaaagtc tatgccagaa agg 23

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tgtggaaagt ctatgccaga aag 23

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gacataggtc cagcagtcag aaa 23

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<400> SEQUENCE: 404

ggacataggt ccagcagtcg gaa 23

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<400> SEQUENCE: 405

aggacatagg tccagcagtc aga 23

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<400> SEQUENCE: 406

gacgctactc aaaattggct ggt 23

<210> SEQ ID NO 407
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<400> SEQUENCE: 407

agacgctact caaaattggc tgg 23

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tgtagacgct actcaaaatt ggctggt 27

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gagatatgaa gataagtgaa aaa 23

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<212> TYPE: DNA

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aagagatatg aagataagtg aaa 23

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<211> LENGTH: 23

<212> TYPE: DNA

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<220> FEATURE:

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tgaagagata tgaagataag tga 23

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<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 414

gaagagatat gaagataagt gaaaaat 27

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<212> TYPE: DNA

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<223> OTHER INFORMATION: Synthetic Construct

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gaggcttatt tgaatgtatc tgc 23

<210> SEQ ID NO 416

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<211> LENGTH: 23
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<400> SEQUENCE: 416

gatataagtc gtgtatatta atg 23

<210> SEQ ID NO 417
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cagatataag tcgtgtatat taa 23

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gtacagatat aagtcgtgta tattaat 27

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<400> SEQUENCE: 419

gattagattt tctgttgga gga 23

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agattagatt ttctgttggg agg 23

<210> SEQ ID NO 421
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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 421

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tcagattaga ttttctgttg gga 23

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tcagattaga ttttctgttg ggaggat 27

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gctaattata gaaagcgatg atg 23

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<400> SEQUENCE: 425

tagctaatta tagaaagcga tga 23

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<212> TYPE: DNA
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<400> SEQUENCE: 426

gtagctaatt atagaaagcg atg 23

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<400> SEQUENCE: 427

ttgtagctaa ttatagaaag cgatgat 27

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gctgacggga aattgagagc aaa 23

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<400> SEQUENCE: 429

cgctgacggg aaattgagag caa 23

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<400> SEQUENCE: 430

agcgctgacg ggaaattgag agc 23

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<400> SEQUENCE: 431

agcgctgacg ggaaattgag agcaaat 27

<210> SEQ ID NO 432
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gctgatctga tggcctggct ttg 23

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<212> TYPE: DNA
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<400> SEQUENCE: 433

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 434

gctttctata attcatctgg agt 23

<210> SEQ ID NO 435
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<400> SEQUENCE: 435

aactgctttc tataattcat ctggagt 27

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<400> SEQUENCE: 436

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27

<210> SEQ ID NO 437
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ggccctttgc tataagtatt ggccagt

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23

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gatggctaag tgggtgtgac agc

23

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 440

gtactaatac cactgatgag aaa

23

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<400> SEQUENCE: 441

tgtactaata ccactgatga gaa

23

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 442

ttgtactaat accactgatg aga 23

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<400> SEQUENCE: 443

cttgtactaa taccactgat gagaaat 27

<210> SEQ ID NO 444
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gtcatagaag gagggtcctc atacagt 27

<210> SEQ ID NO 445
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gtcattgtgt ggctgactgc tgg 23

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agtcattgtg tggctgactg ctg 23

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gtcttataaa agaaccagc ggt 23

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cagtgacaga caaatgtaga tgt 23

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gagggttaaag atctgatagc cgg 23

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tgagggtaaa gatctgatag ccg 23

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taagatcaga ttatttgct tag 23

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atttcctttc acatcttacg ggacaat 27

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atcgtagctt cttctgctg agg 23

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<400> SEQUENCE: 513

agatctaggc aggactacga ggc 23

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aagatctagg caggactacg agg 23

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<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 523

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 527

gtctcaaaca tcttctcctg atgtagt 27

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<400> SEQUENCE: 528

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<400> SEQUENCE: 529

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<400> SEQUENCE: 531

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<400> SEQUENCE: 532

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ttgcagcttc togagttcat agg 23

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<213> ORGANISM: Artificial Sequence

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 539

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tgttcatccc ctattatgaa gaa 23

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attagatagg aaaagatgtg gag 23

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<400> SEQUENCE: 574

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ctttcaaccc gaacggagac 20

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<400> SEQUENCE: 598
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<223> OTHER INFORMATION: Synthetic Construct

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 601

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<210> SEQ ID NO 602
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 602

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 603

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 610

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<210> SEQ ID NO 612
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<210> SEQ ID NO 613
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<210> SEQ ID NO 620

<211> LENGTH: 27

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<212> TYPE: DNA
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<400> SEQUENCE: 620

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 622

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<220> FEATURE:
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<400> SEQUENCE: 624

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<211> LENGTH: 23
<212> TYPE: DNA
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<400> SEQUENCE: 625

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<211> LENGTH: 23
<212> TYPE: DNA
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<400> SEQUENCE: 626

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gggaagctgg gtgaatggag cga 23

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gatccaggtg ctgcagaagg gat 23

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<220> FEATURE:
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<210> SEQ ID NO 630
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<212> TYPE: DNA
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What is claimed is:

1. A fusion protein comprising a Cas9 nickase and a monomeric adenine deaminase,

wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when compared to SEQ ID NO: 11, and

wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.

2. The fusion protein of claim 1, wherein the Cas9 nickase comprises the amino acid sequence of SEQ ID NO: 15.

3. The fusion protein of claim 1, wherein the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.

4. The fusion protein of claim 1, wherein the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.

5. The fusion protein of claim 1, wherein the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 23.

6. A system for base editing comprising:

a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase;

wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a monomeric adenine deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when compared to SEQ ID NO: 11, and

wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.

7. The system of claim 6, wherein the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-573 of the amino acid sequence of SEQ ID NO: 15 and the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence of SEQ ID NO: 15.

8. The system of claim 6, wherein the intein-N is a Cfa intein-N or a gp41-1 intein-N.

9. The system of claim 6, wherein the intein-C is a Cfa intein-C or a gp41-1 intein-C.

10. The system of claim 6, wherein the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).

11. A composition comprising:

a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and

a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein first nucleotide sequence further comprises a nucleotide sequence encoding a monomeric adenine deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase,

wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when compared to SEQ ID NO: 11, and

wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.

12. A method of treating a genetic disease in a subject, comprising administering a therapeutically effective amount of the composition of claim 11.

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13. The method of claim **12**, wherein the genetic disease is a muscular dystrophy or spinal muscular atrophy.

14. The method of claim **13**, wherein the muscular dystrophy is Duchenne muscular dystrophy.

15. The method of claim **13**, wherein the muscular dystrophy is dysferlinopathy.

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